GNBF5020

The Beginner's Guide

to Molecular Biology and Genetics

Introduction to Molecular Biology and Genetics · Dr. Tony Wong · Lectures 1-5, 12 Sep - 17 Oct 2026

Written for a reader with zero biology and zero chemistry. This book starts at the atom and builds up to the exam: eight parts, thirteen chapters, fourteen full-page diagrams, a 671-entry master glossary, the key numbers, and the exam hooks that say what Dr. Wong will actually test.

Every technical term is defined the moment it is used. Technical terms stay in English, because that is how you will be examined.

> **About this edition**
>
> **Part 0 (Read this first) + Chapters 1 and 2 (Foundations)** Course: GNBF5020 Introduction to Molecular Biology and Genetics. Lecturer: Dr. Tony Wong. This file is the front door of one continuous book. It ends where Chapter 3 begins.

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Video guides

Sixteen terms in this book carry the most exam weight, and each one has a short video guide: **English narration, Traditional Chinese subtitles**, about a minute each. The MP4 files live in the project folder **_video/assets/** - the full path is printed next to every link below, so you can still find a file after this PDF has been moved. **_video/index.html** plays all Sixteen on one page.

Video	Term	Chapter	Video file
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PART 0

Read this first

0.1 How to use this guide

This guide is written for a reader who has **zero prior knowledge of biology and zero prior knowledge of chemistry**. That is not a polite assumption. It is the stated audience.

You do not need to know what an atom is. You do not need to know what a cell is. You do not need to have met the word "gene" before. This book starts at the very bottom and builds up.

Here is how the book is put together.

- **Part 0** is the page you are reading now. It tells you the one rule, what the course covers, and how to read the rest.
- **Part 1 (Foundations, Chapters 1 to 7)** teaches the background the lectures assumed you already had. Chapters 1 and 2 are here, in this file. Chapter 3 is proteins, Chapter 4 is DNA and chromosomes, Chapter 5 is the cell, Chapter 6 is how a cell actually works, Chapter 7 shows you where the plain-English glossary is.
- **Parts 2 to 6** follow the five lectures in order. One part per lecture.
- **Part 7** is the exam pack: the master glossary, the five questions Dr. Wong named as exam-type, the key numbers, the "will not be examined" list, and a self-test.

Every chapter follows the same shape. Learn the shape once and you never have to hunt for anything again.

Section	What it is	How long it takes you
n.0 What this chapter is about	Three to five sentences saying why the lecturer taught this.	1 minute
n.1 Words to know before we start	A table of the terms the chapter uses, in plain English.	3 minutes
n.2 The big picture	The chapter told as a story, in flowing prose. This part stands on its own.	5 minutes
n.3 Walkthrough	Topic by topic, following the order the lecturer taught.	The main work
n.4 The N things to remember	The short list you will actually be able to recite in the exam.	5 minutes
n.5 Check yourself	Five questions, then five answers that show the reasoning.	10 minutes

If you read only one thing in each chapter, read the Big Picture. That is the whole orientation in one sentence.

Two house rules run through everything below.

First, **every technical term is defined the moment it is used**. When you meet a new word, the sentence you are in also tells you what it means. You never have to look anything up.

Second, **technical terms stay in English**. The Chinese edition of this book translates the explanation and keeps the term in English. That is deliberate, because that is how you will be examined.

0.2 The one rule: write every answer in English

[EXAM HOOK] THE ONE RULE THAT MATTERS MOST

Dr. Wong said this three separate times across the lectures. He said it on day one, and he said he gave zero marks because of it.

> **Lecture 1**
> *"every assignments and exam should be done in English, even this is a Chinese University of Hong Kong."*

> **Lecture 1**
> *"I'm not kidding, because last year there are some student they answered the question in Chinese, and I have no choice that I give the zero mark, because everything should be in English."*

> **Lecture 4**
> *"you have to answer in English, although this is a CUHK, haha, okay."*

He was joking the third time. He was not joking the first two.

What this means for you, in plain terms.

1. **The midterm is graded in English.** Answer in English, even in complete sentences that are short and simple. Simple English scores better than no English.
2. **A Chinese answer scores zero**, not "partial credit for the idea". This is the specific thing he said he did last year.
3. **Write full sentences.** His advice for the short questions is to defend your reasoning. A page of bullet points is not an answer.
4. **Keep the technical terms in English** inside those sentences. Write "the KRAS mutation means cetuximab will not work" in English, not in Chinese with one English word in it.
5. **This book is in English for that reason.** It is the language you will be graded in, so this is the language to revise in.

One more sentence from him, because it tells you what he is actually grading:

> **Lecture 1**
> *"multiple choice is for you to pass the exam. But if you want to get a full mark or a better result, you'd better perform well for the short questions, which emphasize on data analysis and also the scenarios questions."*

So: the multiple choice questions (MCQs) are the floor. The short questions are where the marks are. Both are in English.

0.3 What this course is, and how the five lectures build on each other

GNBF5020 is an introduction to molecular biology and genetics. The name sounds like two subjects. It is really one subject with two halves.

The first half is **molecular biology**. What are living things made of? How do cells take in fuel and turn it into usable energy? How do they build machines out of small pieces? How do they switch those machines on and off?

The second half is **genetics**. How is the recipe for those machines stored? How is it copied? How is it repaired? How is it changed, and what happens when it is changed wrongly?

Dr. Wong set the course goal himself, on the first lecture.

His stated goal for the course: to translate genomics and molecular biology into **clinical and real-world applications** - diagnostic, prognostic and predictive biomarkers.

A **biomarker** is a measurable thing in a sample - blood, tissue, urine - that tells you something about disease. A **diagnostic biomarker** tells you the patient has the disease. A **prognostic biomarker** tells you how the disease is likely to go. A **predictive biomarker** tells you which drug will work. He came back to this framing in the first two lectures.

There are also two running themes, which he put on his overview slide (Slide 4) next to a portrait of Gregor Mendel, a multi-generation family tree, and a comparison of sperm and egg formation.

- **Diversity** - genetic variation. Why do we look different from each other? How do new combinations of genes arise? How do tumours become genetically different from one another?
- **Stability** - reliable inheritance. How is the recipe copied without errors? How is damage repaired? How does the cell remember what it was doing last week?

Every lecture touches both. The memorable ones are the moments where they collide. Recombination creates diversity but can also break a gene. Epigenetic marks give stability but can lock a cell into the wrong state.

Here is the map of the five lectures.

No.	Date	Title	What you will be able to do after it
1	12 Sep 2026	Introduction to Biomolecules	Name the four classes of biomolecule, and explain how cancer changes cell metabolism. Read a metabolic or proteomic dataset and see a biomarker.
2	19 Sep 2026	Basic Cell Biology	Use the core tools (FACS, microscopy, FISH, IHC). Explain what organelles do and what breaks when they fail. Run the cell cycle, apoptosis, stem cells, CAR-T and liquid biopsy.
3	3 Oct 2026	DNA and Chromatin Structure	Explain how an enormously long piece of DNA fits inside a nucleus. Read a karyotype. Understand chromatin, methylation, the histone code, telomeres, copy number, driver and passenger.
4	10 Oct 2026	DNA Replication, Repair and Recombination	Follow the replication fork. Match each kind of DNA damage to its repair pathway. Use PCR, cloning and CRISPR. Read a clinical trial design.
5	17 Oct 2026	Control of Gene Expression	Explain how the cell decides what to make and when. Work with the lac operon, eukaryotic promoters, splicing, nuclear receptors, signalling pathways and microRNA.

Here is the same map as one chain, which is the version to keep in your head.

1. **Lecture 1 asks: what are we made of?** Four families of molecules. Sugar, fat, protein, DNA. Each one is also a fuel, a building material, or a machine. This is the vocabulary for the whole course.
2. **Lecture 2 asks: where does the work happen?** Inside a cell. A cell is a factory. It has a control room (the nucleus), power stations (mitochondria), a factory floor (ribosomes), and a set of tools (the methods: FACS, microscopy, IHC, FISH).
3. **Lecture 3 asks: where is the recipe, and how is it switched on and off?** DNA is the recipe. Chromatin is the filing cabinet and the on/off switch. Methylation and histone marks are the labels on the folders.
4. **Lecture 4 asks: how is the recipe copied and repaired?** This is the maintenance half. Copying has errors, damage happens, repair systems fix most of it, and when they fail you get cancer. This is also where CRISPR comes from.
5. **Lecture 5 asks: how does the cell control the output?** Having the recipe is not enough. The cell decides when to use it, which version to use, and when to stop. This is the control half, and it is where most drug targets live.

Chapters 1 and 2 of this book sit underneath that entire chain. Chapter 1 gives you the chemistry - atoms, bonds, water, energy, ATP. Chapter 2 gives you the four molecule families from Lecture 1, but taught from zero.

0.4 How the marks work, and how to read the exam-hook boxes

First, how the marks work. From the exam-focus notes built from his own statements:

Item	What he said	Where
Midterm weight	**30 %** of the final grade. The final exam is **40 %**. Two "super-easy" assignments make up the other **30 %**.	Lecture 1, Slide 2
Duration	**1.5 hours (90 minutes)**	Lecture 5
Part A	**Multiple choice, single best option only.** Last year students lost marks by trying to select multiple options.	Lectures 4, 5
Part B	**Data-analysis or scenario questions.** Choose one (Lecture 4) or two out of three (Lecture 5).	Lectures 4, 5
Language	**English only. Zero marks for Chinese answers.**	Lectures 1, 4, 5

There is one inconsistency you should know about, because he flagged it himself. In Lecture 4 he described "about 25 questions, one mark each" with two short questions where you choose one. In Lecture 5, which was later and was written on his own slide, he described "20 multiple choice questions, and two out of three short question ... 20 marks ... short question is account for 10 marks ... 1.5 hours".

Plan for the Lecture 5 version:

20 MCQ (20 marks) + choose 2 of 3 short questions (10 marks) = 30 marks, in 90 minutes.

Either way the ratio is identical, so the strategy is identical. MCQs are pure recall and they get you over the line. The scenario questions are where the remaining marks are.

He named only **two families** of scenario question, and they are worth practising:

1. **A pedigree or genetic-counselling report.** "If you are a genetic consultant and I give you a medical report, how would you advise the client whether he or she will have a baby with a certain genetic disease?"
2. **A survival dataset.** "I give you some survival data, and then you tell me how this survival data can really benefit the treatment of the patient."

Second, how to read the boxes in this book. There are two kinds of marker, and the difference between them is the whole point.

EXAM HOOK	means Dr. Wong's own words, his own numbers, or something he said he would ask. Where his exact wording matters, you will get his exact wording in quotation marks.

- **He did not flag this as examinable** means he never mentioned that point as a possible question. It is **not** a claim that the topic is off the paper. It is a statement that he gave you no signal, so do not spend your time memorising it.

Everything that is not in a hook is **background explanation**. That is teaching, not invention. In plain terms, it is the part of science the lectures assumed you already had. Chapter 1 of this book is almost entirely that kind of material, because he told the class not to worry about it:

> **Lecture 1**
> *"Please don't memorize all these chemical structure ... They would not be included in my exam."*

So you will notice that Chapter 1 says he did not flag this very often. That is honest, and it is good news: it means the chemistry is something you understand once rather than memorise.

Third, the numbers. Every number Dr. Wong stated is reproduced exactly, and followed immediately by a plain-English sentence saying what it means and why it matters. Numbers are the easiest MCQ fodder there is, and they are also the easiest thing to get wrong by a factor of ten.

Fourth, a habit worth forming now. He said he is emphasising on how you analyse the scientific questions, not on memorised facts. The exam notes turn that into a four-move shape - state the observation with the number, name the mechanism, interpret the clinical meaning, then state the action or the limit. That is a sensible reading of what he said, not a quotation of it, and it is a good answer shape to practise. Read the answer templates in Part 7 before you sit the exam.

0.5 A sixty-second orientation

If you read only one thing in each chapter, read the Big Picture. That sentence is the entire navigation system of this book.

Here is the sixty-second version of the whole course.

1. **Everything is atoms.** Atoms join into bonds, bonds make molecules, molecules join into long chains called macromolecules, and macromolecules fill a cell. A cell is a factory, and the factory runs on energy.
2. **There are four families of macromolecule.** Carbohydrates (sugar and starch), lipids (fat and membranes), proteins (the machines), nucleic acids (the information). Everything else in the course is a detail of one of these four.
3. **Information flows one way.** DNA is the recipe. mRNA is a photocopy of one recipe. A ribosome reads the photocopy and builds a protein, which is a folded machine made from a bead necklace. This is the central dogma, and Lecture 5 is about how the cell controls it.
4. **Most of the course is about damage and repair.** DNA gets damaged. Repair systems fix most of it. When repair fails, you get mutations, and when the mutation hits the wrong word, you get cancer. Mutation means a typo. A driver is a typo in an important word. A passenger is a typo on page 900.
5. **Almost everything is a drug target or a biomarker.** If you can measure it, you can test for it. If you can switch it on or off, you can drug it. That is why the course is built around clinical application rather than around taxonomy.
6. **And you answer in English, in full sentences, with the number, the mechanism, the clinical meaning, and the limit.** Zero marks for a Chinese answer. He said so three times.

PART 1

Foundations: biology and chemistry from zero

IN THIS PART

Chapter 1 · Chemistry from zero: atoms to molecules

Chapter 2 · The four kinds of biomolecule

Chapter 3 · Proteins: what they are and how they work [the folded machines that do the cell's work]

Chapter 4 · DNA, genes and chromosomes [the recipe, the printed manual, and the typo]

Chapter 5 · The cell and its parts

Chapter 6 · How a cell actually works

Chapter 7 · The plain-English glossary

About this part

Part 1 - Foundations - Chapters 3 and 4 - Dr. Tony Wong

Part 1 - Foundations (continued) - Chapters 5, 6 and 7

Course: GNBF5020 Introduction to Molecular Biology and Genetics - **Instructor:** Dr. Tony Wong - **Dates:** Sep-Oct 2026

This file continues Part 1 of the beginner guide. Chapter 5 gives you the cell and its parts. Chapter 6 shows you the cell at work. Chapter 7 shows you where the big alphabetical glossary lives - the one that turns every technical word in this book into plain English.

CHAPTER 1

Chemistry from zero: atoms to molecules

1.0 What this chapter is about

You cannot understand a cell, a gene or a drug without knowing what atoms and bonds are. This chapter builds that from nothing.

It starts with the smallest piece of matter, and ends with the question that drives the whole course: where does a cell get its energy, and how does it spend it?

Dr. Wong did not teach this material. He taught the biology on top of it and assumed you already had it. He even told you to skip the details. That is why this chapter exists.

1.1 Words to know before we start

Term	In plain English
atom	The smallest piece of an element that still behaves like that element. The unit that everything is built from.
proton	A particle with a positive charge. The number of protons decides which element an atom is.
neutron	A particle with no charge, sitting in the nucleus. It adds mass without changing the element.
electron	A particle with a negative charge that moves around the nucleus. Electrons do the bonding.
electron shell	A fixed energy layer around the nucleus. Each shell holds a set number of electrons.
covalent bond	Two atoms sharing a pair of electrons. This is the bond that builds most of biology.
ionic bond	A strong attraction between atoms with opposite charges, with no sharing.
hydrogen bond	A weak, temporary tug between a slightly positive hydrogen and a slightly negative atom.
molecule	Two or more atoms joined together. A molecule has a shape and its own properties.
functional group	A small cluster of atoms on a molecule that gives that molecule its chemical personality and its reactivity.
solvent	The liquid that something else dissolves in. For a cell, that liquid is water.
pH	A number from 0 to 14 that says how acidic a liquid is. 7 is neutral, below 7 is acidic, above 7 is alkaline.
ATP	Adenosine triphosphate. The molecule a cell spends as energy, the way you spend money.
oxidation and reduction	Losing electrons (oxidation) and gaining electrons (reduction). They always happen together.

1.2 The big picture

Everything you will meet in this course - a protein, a piece of DNA, a sugar, a cell - is built out of atoms. Start there and molecular biology stops being frightening. It becomes an assembly problem whose rules you can learn.

An **atom** is the smallest piece of an element that still behaves like that element. "Element" means a substance made of only one kind of atom. Carbon is an element, and carbon is the one this whole course is built on.

Every atom has a dense centre called the **atomic nucleus**, holding **protons** and **neutrons**. Protons carry a positive charge, and their number is the atom's identity: six protons means carbon. Neutrons carry no charge and mostly add bulk. Around the nucleus orbit **electrons**, which carry a negative charge. They occupy **shells**, fixed energy layers, and the outermost shell decides what an atom can do.

Here is the rule that does most of the work. An atom is happy when its outer shell is full. For carbon a full shell means four electrons in the outer layer. Carbon has four, so it wants four more, and will share with one atom here and two atoms there. That single fact is why carbon can build chains, rings, branches and cages. It is why there is organic chemistry at all, and why there is you.

Three kinds of attraction hold atoms together, and you must keep them apart. A **covalent bond** is two atoms sharing a pair of electrons, so both get a fuller outer shell. This is the bond that builds molecules: in water, in sugar, in a protein backbone. An **ionic bond** is a strong pull between atoms with opposite charges, with nothing shared - sodium and chlorine. A **hydrogen bond** is much weaker: the tug between a slightly positive hydrogen and a slightly negative atom such as oxygen. They break and reform on demand, which is exactly what makes them useful. They are the folding pins in proteins, the pairing rules in DNA, and the reason water behaves so usefully.

When atoms join they make a **molecule**. One oxygen and two hydrogens make one water molecule. A molecule has a definite shape, and the shape decides what it can touch. Water is not a straight line. It is bent, because the oxygen pulls the two hydrogens to one side. That bend makes water **polar**, one end slightly positive and the other slightly negative. Polar attracts polar and ignores non-polar. That asymmetry explains why water dissolves salt and sugar while oil refuses to mix, and why a phospholipid builds a membrane.

Life needs water for four structural reasons. It is the **solvent** in which every reaction happens, and a **participant** in the chemistry of proteins and ATP. Its **hydrogen bonds** give large molecules their three-dimensional shape. And its **high specific heat** keeps body temperature steady.

Acidity is measured on the **pH** scale, from 0 to 14. Seven is neutral, below seven is acidic, above seven is alkaline. Each step is a ten-fold jump, so the scale is logarithmic. pH matters here for one reason: it changes protein shape. Charge attracts charge, so pH changes which bits of a protein stick to which. His slides use this. He showed the **deamidation** pathway, in which an asparagine (Asn) side chain becomes an aspartate (Asp) side chain at high pH, and the mass moves from 114.04 Da to 115.03 Da. One unit of mass in a drug can change how well it binds its target.

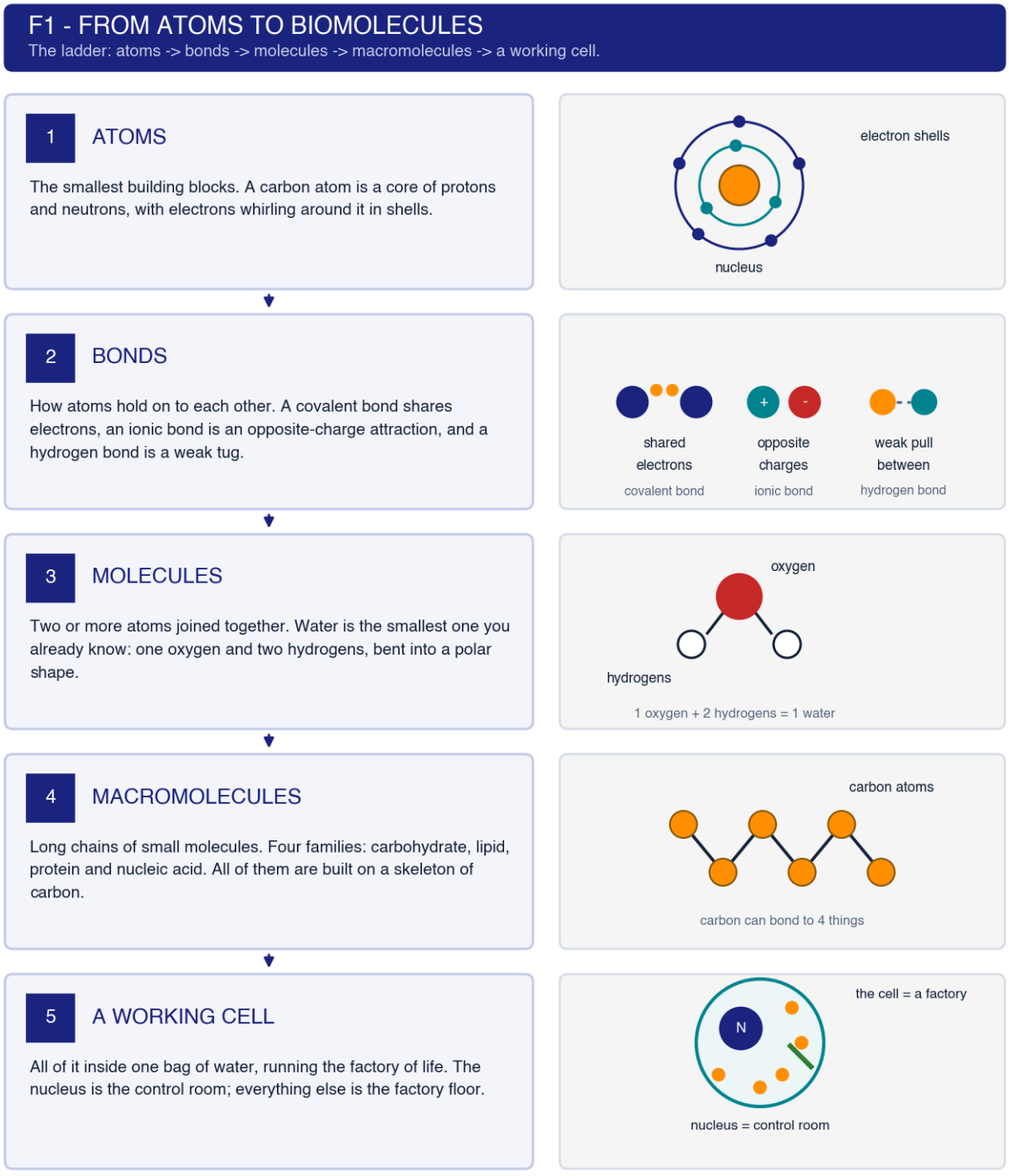
Now the part that drives everything else: **energy**. In a cell, energy is not a vague warmth. It is the capacity to do work - to move something, build something, change a chemical. And work is almost always paid for in one currency: **ATP**, adenosine triphosphate. ATP is the spendable unit. A kinase takes a phosphate from ATP and sticks it onto a protein; a phosphatase takes it off again. That one switch is how the cell turns things on and off. In his Hippo pathway example, MST1 and LATS1 add phosphate to YAP, and phosphorylated YAP is pushed out of the nucleus so the cell stops dividing. Remove it and YAP walks in and the cell grows.

Finally, **oxidation and reduction**. Oxidation means losing electrons. Reduction means gaining electrons. They are two halves of one transfer and always happen together. A useful handle: oxidation often means losing hydrogen, reduction often means gaining it. When a cell burns sugar with oxygen, the sugar is oxidised, the oxygen reduced, and the released energy is captured as ATP.

That is the whole chapter. Atoms want to be satisfied. Bonds satisfy them. Molecules with shapes make membranes. Water holds the reaction together. pH tunes the machinery. ATP pays for the work. And electrons moving is what energy is.

1.3 Walkthrough

Figure F1 - From atoms to biomolecules



The ladder, in five rungs: atoms, bonds, molecules, macromolecules, and finally a working cell. Each rung on the right carries the picture for that step - an atom with its shells, the three ways atoms hold on to each other, one water molecule, a chain of carbons, and a cell that is a factory.

1.3.1 The building block of everything - atom, proton, neutron, electron

WHAT IT IS	An **atom** is the smallest piece of an element that still behaves like that element. Its centre, the atomic nucleus, holds **protons** (positive) and **neutrons** (no charge). **Electrons** (negative) orbit outside it in shells. The count of protons is the element's identity: 6 protons is carbon, 8 is oxygen, 1 is hydrogen.
HOW IT WORKS / WHY IT MATTERS	Every molecule in this course is built from atoms, so every mechanism in this course is really a story about electrons. Change an electron and you change the chemistry. That is the whole reason chemistry belongs in a molecular biology course. Also: the atoms in your body are not special. They are the same carbon, oxygen, hydrogen and nitrogen that are in the air and in a textbook. What is special is which atoms are joined to which, and in what order.
EXAM HOOK	He did not flag this as examinable. He did say on Lecture 1: "Please don't memorize all these chemical structure ... They would not be included in my exam." Learn the idea so the rest of the course makes sense. Do not memorise atomic numbers.

1.3.2 The layers around the nucleus - electron shells

WHAT IT IS	**Electrons** do not float anywhere. They sit in **electron shells**, which are fixed energy layers around the nucleus. Each shell holds a maximum number of electrons: two in the innermost, then eight in the second.
HOW IT WORKS / WHY IT MATTERS	The outermost shell, the valence shell, decides what an atom can bond to. An atom with a full outer shell is stable and does not react. An atom with a partly filled outer shell is looking for a partner. This single rule explains why carbon is so useful: it has four electrons in its outer shell and needs four more, so it always has four hands to hold on to other atoms.
EXAM HOOK	He did not flag shell structure as examinable. No number he stated in the exam-focus document depends on it.

1.3.3 How atoms hold on to each other - covalent, ionic and hydrogen bonds

WHAT IT IS	A **covalent bond** is two atoms sharing a pair of electrons. An **ionic bond** is a strong attraction between oppositely charged atoms, with no sharing. A **hydrogen bond** is a weak, temporary attraction between a slightly positive hydrogen and a slightly negative atom such as oxygen.
HOW IT WORKS / WHY IT MATTERS	All three appear in the course, and they do different jobs. Covalent bonds are strong and permanent; they build the skeleton of every molecule. Ionic bonds hold salts and are used in batteries and in mineral bone. Hydrogen bonds are weak but reversible, and reversibility is the point: they let a protein fold one way and then change shape, they let the two strands of DNA unzip so the cell can copy them, and they let the base pairs of the double helix be swapped for the wrong letter.
EXAM HOOK	He did state the hydrogen bond numbers, and they are the numbers most likely to appear as an MCQ stem. **Guanine pairs with cytosine using 3 hydrogen bonds. Adenine pairs with thymine using 2.** In plain English: a GC-rich piece of DNA needs more heat to pull the two strands apart, because there are more cross-links holding them together. That is why melting temperature depends on GC content. His slide gave the direction, not a number: more GC content means a higher melting temperature, and that is what matters when you design a PCR primer. The exam notes also carry a '~40 deg C drop' and a '60 deg C versus 55 deg C' board example, and those two do not reconcile with each other - so learn the direction and the reason, and do not put a number on it.

1.3.4 Two or more atoms joined - molecule

WHAT IT IS	A **molecule** is two or more atoms joined together. The atoms are held in a fixed arrangement, and the molecule has its own shape and its own properties. It is not a bag of separate atoms any more. Water is one oxygen and two hydrogens. Glucose is six carbons, twelve hydrogens and six oxygens.
HOW IT WORKS / WHY IT MATTERS	The shape of a molecule decides everything it does, because shape decides which parts touch what. This is the idea you will meet again as "structure determines function" in proteins, and it is the same idea again when a DNA base pairs with its partner. Dr. Wong's slides use the two simplest sugars as examples: glucose and fructose have exactly the same formula, C6H12O6, yet they behave differently because their atoms are arranged differently. Glucose forms a six-membered ring. Fructose forms a five-membered ring. Same parts, different shape, different job.
EXAM HOOK	He did not flag the formula or the ring shapes as examinable, and he explicitly excluded memorising chemical structures. The idea that shape decides function is examinable in the protein form, and that is Chapter 3.

1.3.5 The element that builds life - carbon

WHAT IT IS	**Carbon** is the element at the centre of all organic chemistry. It has four electrons in its outer shell, so it forms four covalent bonds. It bonds to itself in chains, rings and branches, and it bonds to hydrogen, oxygen, nitrogen and sulphur.
HOW IT WORKS / WHY IT MATTERS	Because carbon can make chains, and chains can be any length and any shape, one element can build an enormous variety of different molecules. Sugars, fats, proteins and DNA are all carbon-based. That is why we call them "organic" molecules even though there is no organic chemistry of the supermarket kind involved. A useful mental picture: carbon is the steel of biology, and the other atoms are the rivets and panels.
EXAM HOOK	He did not flag carbon itself as examinable. It matters here because it is the reason the four molecule families in Chapter 2 can all exist at the same time.

1.3.6 The reactive part of a molecule - functional groups

WHAT IT IS	A **functional group** is a small cluster of atoms on a molecule that gives that molecule its chemical personality. Common ones are the hydroxyl group (an oxygen with a hydrogen attached, written -OH), the carboxyl group (a carbon with two oxygens, written -COOH), the amino group (written -NH2) and the phosphate group (written -PO4).
HOW IT WORKS / WHY IT MATTERS	The long carbon chain of a molecule is mostly a handle. The interesting chemistry happens at the functional groups, and it is the functional group that tells you how a molecule will behave. A carboxyl group makes an acid. An amino group can accept a hydrogen, which is how amino acids join into proteins. A phosphate group carries charge, which is why adding and removing phosphates is the cell's favourite on/off switch. The famous exception is carbon dioxide: it is what you are left with after a carboxyl group is removed, so it has no carbon chain of its own.
EXAM HOOK	He did not flag functional groups as examinable, and he excluded memorising chemical structures. But he did build one functional group into a clinical teaching point: the phosphate group, used by kinases and phosphatases. That is covered next.

1.3.7 The liquid life runs in - water

WHAT IT IS	**Water** (H2O) is one oxygen and two hydrogens. Because oxygen is much bigger and pulls the shared electrons harder, the molecule is bent and **polar**: one end is slightly negative, the other slightly positive.
HOW IT WORKS / WHY IT MATTERS	Four jobs, all structural. (1) It is the **solvent** - the liquid everything dissolves in, so every reaction in the cell happens in it. (2) It is a **reactant** in key reactions, including the reactions that make and spend ATP. (3) Its **hydrogen bonds** give proteins their three-dimensional shape, because they hold one part of a chain against another. (4) Its **high specific heat** buffers temperature changes, so the cell does not swing wildly when it is working. Water is also the reason a phospholipid membrane works, because a polar molecule dissolves in polar water and a non-polar one does not. That is the entire trick behind the bilayer.
EXAM HOOK	He did not teach water as chemistry. He used it as scenery on the slides, for example when he showed the phosphatase in the phosphorylation switch cartoon using water to release inorganic phosphate. The examinable version is the membrane consequence, and that is the next chapter.

1.3.8 How acidic a liquid is - pH

WHAT IT IS	**pH** is a number that says how acidic a liquid is, running from 0 to 14. Seven is neutral, lower is acidic, higher is alkaline. Each step is a ten-fold change in acidity, so the scale is logarithmic, not linear.
HOW IT WORKS / WHY IT MATTERS	pH changes the charge on molecules, and charge decides which parts of a protein stick together. So pH changes protein shape, and protein shape decides function. His slide on **deamidation** makes this concrete: the side chain of an asparagine (Asn, 114.04 Da) is converted first into a succinimide intermediate (Asu, 97.02 Da), then into either aspartate (Asp, 115.03 Da) or isoaspartate (isoAsp, 115.03 Da). This is a therapeutic protein liability, because a change like that can alter how well a drug binds its target.
EXAM HOOK	He did not list deamidation among the numbers or named question types in the exam-focus document, but it is the only worked example of a chemical change on a drug in Lecture 1, so it is worth knowing. What he did flag: the amino-acid R-group table is **not** in the exam. His words: "don't memorize this ... definitely not include in the exam."

1.3.9 What a cell spends - energy and metabolism

WHAT IT IS	**Energy** in a cell means the capacity to do work: move something against a gradient, build a structure, or drive a chemical reaction. **Metabolism** is the whole set of chemical reactions in a cell, and the reactions that release energy are **catabolic**, while the reactions that spend energy to build are **anabolic**.
HOW IT WORKS / WHY IT MATTERS	The trade is simple and it runs the whole course. Catabolism breaks big molecules down into small ones and harvests energy. Anabolism spends that energy to build big molecules. A cell that is growing is running anabolism hard, and it needs a constant supply of both fuel and raw material. This is exactly the argument behind the **Warburg effect**, his Lecture 1 cancer case: a proliferating cancer cell diverts most of its pyruvate to lactate even when oxygen is available, which yields less ATP but leaves the half-broken sugar molecules behind as perfect building blocks for nucleotides, lipids and amino acids. The inefficient switch is a growth strategy.
EXAM HOOK	He used the Warburg effect as the entry point to Lecture 1, because it is the reason metabolomics became a useful way to find cancer biomarkers. He did **not** ask students to memorise the details of the glycolysis pathway. On his comparison slide, a quiescent normal cell with oxygen makes about **36 ATP molecules per glucose** by oxidative phosphorylation, and only **2 ATP molecules per glucose** by anaerobic glycolysis without oxygen; a proliferating cancer cell makes about **4 ATP molecules per glucose** by aerobic glycolysis either way. Those three numbers were on the slide, but they are not in his list of stated numbers, so treat them as background rather than as exam fodder.

1.3.10 The cell's rechargeable battery - ATP

WHAT IT IS	**ATP** (adenosine triphosphate) is a molecule made of adenine, ribose sugar and three phosphate groups. The bond between the last two phosphates is the one the cell actually spends.
HOW IT WORKS / WHY IT MATTERS	A **kinase** (an enzyme that adds a phosphate, using ATP) takes the terminal phosphate from ATP and attaches it to a protein. The added phosphate carries charge, and the mutual repulsion of the charges changes the protein's shape. A **phosphatase** removes it again, using water and releasing inorganic phosphate, and the protein snaps back. So ATP is not just energy for building; it is also the currency of the cell's fastest on/off switch. His slide showed it as two Pac-Man shapes, one adding the phosphate and one taking it away.
EXAM HOOK	ATP is one of the two things he named when he listed what biomolecules are for: they serve as "disease biomarkers, energy sources, structural components, and signalling molecules". He did not ask you to draw ATP. He did build it into the Hippo pathway example, where MST1 and LATS1 phosphorylate YAP and push it out of the nucleus. That pathway is examinable, and it is Lecture 1 topic 9.

1.3.11 Losing and gaining electrons - oxidation and reduction

WHAT IT IS	**Oxidation** is losing electrons. **Reduction** is gaining electrons. They are two halves of one transfer, so they always occur together. In biological systems, oxidation very often means losing hydrogen, and reduction very often means gaining hydrogen.
HOW IT WORKS / WHY IT MATTERS	Energy capture is an oxidation-reduction reaction. When a cell burns sugar with oxygen, the sugar gives up electrons and hydrogen and the oxygen accepts them. The released energy is trapped as ATP. Reverse the idea and you get the other half of the course: when repair enzymes work by adding or removing hydrogen and electrons, they are running the same chemistry in a smaller box. It also explains a piece of vocabulary you will meet in metabolism: pyruvate can be converted to lactate, or it can go on to acetyl-CoA with carbon dioxide released.
EXAM HOOK	He did not teach oxidation and reduction as a named concept in Lecture 1, and the phrase does not appear in his exam-focus list. He used the underlying chemistry in the Warburg slide, where pyruvate is converted to lactate in one direction and to acetyl-CoA with release of carbon dioxide in the other. Learn the plain definition here so that when you meet the words in Lecture 2 (oxidative phosphorylation in mitochondria) they mean something.

1.4 The 12 things to remember

1. An atom's identity is set by its number of protons. Electrons do the bonding.
2. Electrons sit in shells. The outermost shell decides what an atom can bond to.
3. A covalent bond is shared electrons. An ionic bond is opposite charges attracting. A hydrogen bond is a weak tug.
4. Hydrogen bonds are weak but reversible, and reversibility is why proteins change shape and DNA can be copied.
5. A molecule has a shape, and the shape decides what it can touch and what it can do.
6. Carbon has four hands. It bonds to itself and to everything else, which is why one element builds four families of molecules.
7. Functional groups are where the chemistry happens. The long carbon chain is only the handle.
8. Water is polar, and that single fact explains why it dissolves salt, shapes proteins, and lets a membrane exist.
9. pH changes charge, charge changes shape, shape changes function. The scale is logarithmic.
10. Energy in a cell is ATP, and ATP is the spendable unit. A phosphate added or removed is the cell's fastest switch.
11. Oxidation is losing electrons; reduction is gaining them; they always happen together.
12. He said do not memorise chemical structures. Understand the logic and move on.

1.5 Check yourself

Question 1. A water molecule is drawn as a bent shape, not a straight line. Why does that matter to a cell?

Answer. The bend comes from the oxygen atom being larger and pulling the shared electrons more strongly than hydrogen does. That makes the whole molecule **polar**, meaning one end is slightly positive and the other slightly negative. Polar attracts polar. So water can hold salt and sugar in solution while oil stays separate. It also means that a phospholipid, which has a polar head and a non-polar tail, packs itself into a membrane with the heads facing the water and the tails hiding from it. Without the bend, water would behave like oil, and membranes and cells would not exist. **The reasoning to hold on to: shape causes polarity, polarity causes chemistry, chemistry causes the cell.**

Question 2. Two DNA sequences are identical in length, but one has more guanine-cytosine pairs than the other. Which one needs more heat to unzip, and why?

> **Answer.** The GC-rich one needs more heat. Dr. Wong's number is that G pairs with C using **3 hydrogen bonds** while A pairs with T using **2**. So the GC-rich sequence has more cross-links holding the two strands together, and more energy is needed to break them. That is why melting temperature depends on GC content, and why it matters when designing a PCR primer - a primer with the wrong melting temperature will not anneal properly. **The reasoning: more hydrogen bonds means more energy is needed to separate the strands.**

Question 3. A drug is a protein. The manufacturing process makes it slightly alkaline, and the drug then binds its target less well than it used to. Using only Chapter 1, what has probably happened?

> **Answer.** pH has changed the shape of the protein. The pH scale is logarithmic, so a small numerical change is a large chemical change. A change in pH alters the charge on certain side chains, and since opposite charges attract, that shifts which parts of the protein stick to which. He showed exactly this mechanism with the **deamidation** pathway: an asparagine side chain (114.04 Da) is converted through a succinimide intermediate (97.02 Da) into aspartate (115.03 Da) or isoaspartate (115.03 Da) at high pH. A changed side chain means a changed shape, and a changed shape can mean a changed binding site. **The reasoning: pH to charge to shape to function. That chain is the whole answer.**

Question 4. A cell has plenty of glucose and plenty of oxygen, yet it is not making much ATP. Meanwhile it is growing and dividing fast. What might explain this, and did Dr. Wong talk about it?

> **Answer.** The cell may be running the Warburg pattern: sending most of its pyruvate to lactate even though oxygen is present. On his comparison slide, a quiescent normal cell with oxygen makes about 36 ATP molecules per glucose by oxidative phosphorylation, while a proliferating cancer cell makes about 4 ATP molecules per glucose by aerobic glycolysis either way. That looks wasteful, but the half-broken sugar is left behind as raw material for nucleotides, lipids and amino acids, and a cell dividing a billion times needs building material more than it needs fuel. Dr. Wong did use this as his entry point to Lecture 1, because it is the reason metabolomics became a useful biomarker-discovery tool. He did **not** ask you to memorise the glycolysis pathway. **The reasoning: the "inefficient" choice is a growth strategy, not a mistake.**

Question 5. A kinase puts a phosphate on a protein and the protein changes shape. A phosphatase takes it away and the protein changes back. Where does the phosphate come from, and what does that make ATP?

> **Answer.** The phosphate comes from ATP. A kinase uses ATP to attach a phosphate group to a protein. The added phosphate carries charge, and the mutual repulsion of the added charges is what changes the protein's shape. The phosphatase reverses it, using water and releasing inorganic phosphate. This is why ATP is described as the cell's spendable unit of energy, and why the addition and removal of a single phosphate is the cell's fastest on-off switch. It is exactly the mechanism behind his Hippo pathway example, where MST1 and LATS1 phosphorylate YAP so that YAP is pushed out of the nucleus and the cell stops dividing. **The reasoning: ATP is not just fuel for building. It is also the currency of signalling.**

CHAPTER 2

The four kinds of biomolecule

2.0 What this chapter is about

Every living thing is made from four families of large molecules. Lecture 1 opened with them and the rest of the course is built on them.

This chapter teaches all four from zero: what they are made of, what shape they take, what job they do in the cell, and how their failure causes disease.

Two of the four - proteins and nucleic acids - get only a preview here, because Chapters 3 and 4 give them their own full chapters. This chapter is the map, and you will return to it every time a new molecule appears in the lectures.

2.1 Words to know before we start

Term	In plain English
biomolecule	A molecule that a living thing makes and uses. Carbohydrates, lipids, proteins and nucleic acids are the four families.
carbohydrate	A molecule built from carbon, hydrogen and oxygen, often with the formula CH2O. Sugars and starches.
monosaccharide	The simplest carbohydrate: a single sugar unit, such as glucose.
disaccharide	Two sugar units joined together, such as sucrose, which is table sugar.
polysaccharide	A long chain of sugar units joined together, such as glycogen, starch or cellulose.
glycosidic bond	The oxygen bridge that joins one sugar to the next.
lipid	A fat-like molecule that does not dissolve in water. It stores energy or builds membranes.
fatty acid	A long chain of carbons with a small acidic end. It is the building block of most lipids.
saturated / unsaturated fatty acid	Saturated means the carbon chain carries no double bonds and is straight. Unsaturated means it has one or more, which makes it kinked.
triglyceride	Three fatty acids joined to a glycerol. The body's energy store.
phospholipid	A lipid with a water-loving head and a water-fearing tail. This is what membranes are made of.
amphipathic	Having one end that likes water and one end that avoids it. The key word for understanding a cell membrane.
bilayer	Two layers of phospholipid, heads out, tails in. The wall of the cell.
nucleotide	A base plus a sugar plus a phosphate. It is the building block of DNA and RNA, and it is also ATP.

2.2 The big picture

Here is the whole of molecular biology in one idea: living things are made from four families of molecule, each has a natural shape, and that shape assigns it a job. When the job fails, you get disease. That is the logic of this chapter, and of the course.

A **biomolecule** is simply a molecule that a living thing makes and uses. Dr. Wong's own summary was that biomolecules serve four purposes: as **disease biomarkers** (a measurable molecule in blood, tissue or urine that tells you something about disease), as **energy sources**, as **structural components**, and as **signalling molecules**. Keep those four roles in mind, because every clinical example below sits in one of them.

The first family is the **carbohydrates**, built from carbon, hydrogen and oxygen. A single sugar unit is a **monosaccharide**, and the two you will meet most are glucose and fructose. Two units joined make a **disaccharide**, such as sucrose, which is table sugar. Long chains make a **polysaccharide**. The units are joined by a **glycosidic bond**, an oxygen bridge from one sugar to the next. Here is the interesting part: the same chain of

glucose can be built in different shapes, and each shape has a completely different job. **Glycogen** is a branched chain: compact, and branched so the cell can release sugar from many points at once, which is what you want when fuel is needed urgently. **Starch** is the plant version, and it is what you eat. **Cellulose** is an unbranched chain with the bonds flipped. It forms long straight fibres, and that is what plant cell walls are made of - which is why you cannot digest it and a cow can. So the job of a carbohydrate is set by its shape: energy currency, energy storage, or structure.

The second family is the **lipids**, and lipids are the odd ones out because they do not dissolve in water. Their building block is the **fatty acid**: a long chain of carbons with a small acidic end. If the chain carries no double bonds it is a **saturated fatty acid**, and it is straight and tightly packable. If it carries one or more double bonds it is an **unsaturated fatty acid**, and each double bond puts a kink in the chain so it cannot pack tightly. That difference is why a saturated fat is solid at room temperature and an unsaturated oil is liquid. Three fatty acids joined to a small three-carbon molecule called glycerol make a **triglyceride**, and that is the body's energy store: compact, waterproof and calorie-dense. The other major lipid is the **phospholipid**, which is **amphipathic** - a water-loving head and a water-fearing tail. This one property is why the cell has a membrane. Put amphipathic molecules in water and they arrange themselves, without any instruction, into a **bilayer**: heads out into the water, tails tucked away from it. Your cell wall is a self-assembled oil slick, and it is a wall because the water on both sides pushes it that way. **Cholesterol** is a small lipid that slots into the middle of the bilayer and controls how fluid it is, so the membrane works at every temperature. In the blood, fats cannot travel alone, so they are packaged into **lipoproteins** - particles of fat wrapped in protein, acting as delivery vans.

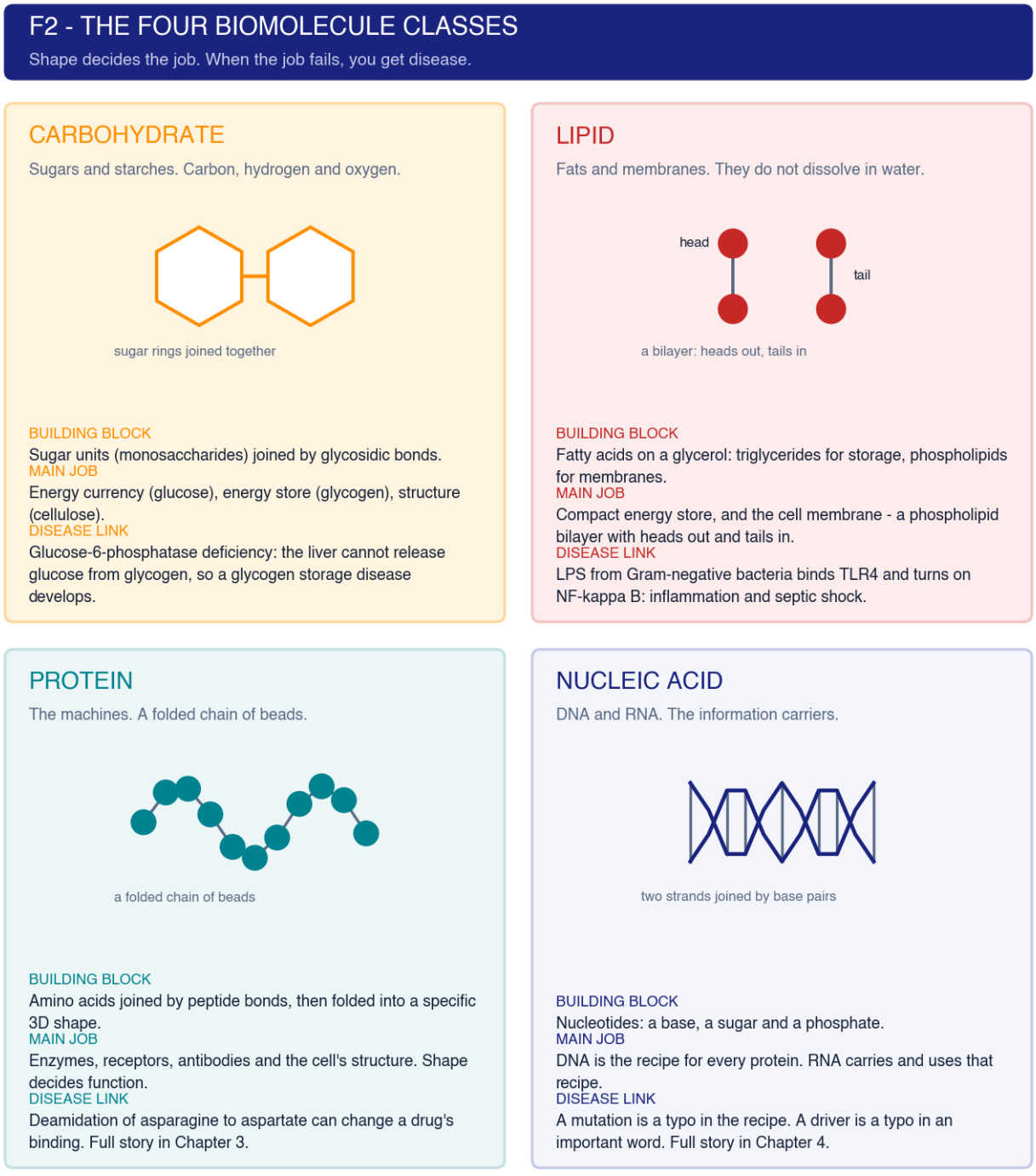
The third family is the **proteins**, and this is the family that does the work. A protein is a chain of **amino acids** (the 20 small building blocks from which every protein is made) joined by **peptide bonds** (the strong links between them), and the chain then folds into a specific three-dimensional shape. The folding is not decoration. The shape is the machine. Enzymes that speed up reactions, receptors that receive signals, antibodies that catch invaders and the cell's structural framework are all folded protein. Chapter 3 gives the full hierarchy, from the sequence of beads up to the assembled multi-part machine, and the methods for studying it.

The fourth family is the **nucleic acids**: DNA and RNA. A **nucleotide** is a nitrogenous base plus a sugar plus a phosphate; remove the phosphate and you have a **nucleoside**. Long chains of nucleotides make the double helix of DNA and the working forms of RNA. DNA is the recipe; RNA is a working copy or a tool. Chapter 4 covers the helix, the packing and what a mutation is.

One last idea carries through everything: **the shape of a molecule is set by its chemistry, and its chemistry is set by its shape.** A glycosidic bond flipped one way makes cellulose; flipped the other, glycogen. A double bond makes a fatty acid bend. A phosphate added to a protein changes what the protein does.

2.3 Walkthrough

Figure F2 - The four biomolecule classes



Four panels. For each class: a tiny schematic of the shape, the building block it is made from, the main job it does in the cell, and the disease link Dr. Wong used.

Figure F2 is the map for this whole chapter. If you can redraw it from memory, you can follow Lecture 1.

2.3.1 Molecules that living things make - biomolecule

WHAT IT IS	A **biomolecule** is a molecule synthesised naturally in a living system. Dr. Wong's own list has four: carbohydrates, lipids, proteins and nucleic acids. Everything else in the course is a detail, an example, or a method applied to one of these four.
HOW IT WORKS / WHY IT MATTERS	His slide listed the subcategories, and they matter because they are the vocabulary of the exams. Carbohydrates run monosaccharide, disaccharide, polysaccharide. Lipids run fatty acid, triglyceride, phospholipid. Proteins run amino acid, peptide, polypeptide, protein complex. Nucleic acids run nucleotide, DNA, RNA, mRNA, tRNA, microRNA, lncRNA. The reason this matters clinically is his central point about the course: the link between genotype and phenotype is mediated through biomolecular processes, so a measurement of a biomolecule in blood is a readout of what is happening inside cells.
EXAM HOOK	He did not frame "what is a biomolecule" as a standalone question, but he did use the four classes as the spine of Lecture 1, and the whole of Lecture 1 topic 1. He also stated clearly that chemical structures are **not** examinable.

2.3.2 Sugar and starch - carbohydrates

WHAT IT IS	**Carbohydrates** are molecules of carbon, hydrogen and oxygen, often with the formula CH2O. A single unit is a **monosaccharide** (glucose, fructose). Two units joined are a **disaccharide** (sucrose). Long chains are a **polysaccharide** (glycogen, starch, cellulose), and the units are joined by a **glycosidic bond**.
HOW IT WORKS / WHY IT MATTERS	Three jobs, decided by shape. **Glucose** is the cell's energy currency, the molecule that gets broken down to make ATP. **Glycogen** is the storage form, a branched chain so that it can be broken down fast and compactly. **Cellulose** is the structural form, a straight unbranched chain with the bonds oriented differently, which is what makes plant cell walls strong and why you cannot digest it. His slide showed the difference precisely: glycogen has **alpha-1,4** linkages along the chain with **alpha-1,6** branch points, and cellulose has the same sugar but a flipped bond orientation.
EXAM HOOK	He taught a disease that follows directly from the biochemistry. **Glucose-6-phosphatase deficiency** means the liver cannot release free glucose from glycogen, so the hepatocyte has an energy crisis and the patient develops a glycogen storage disease. His slide traced the whole path: glycogen, broken down by glycogen phosphorylase, to glucose-1-phosphate, to glucose-6-phosphate, and then only liver and kidney have the phosphatase to release free glucose. He did not flag this as a named exam-type question, and he did exclude "metabolic pathway details" from the exam, so do not memorise the enzyme names. The concept to keep is that a missing enzyme produces a metabolic block and a disease.

2.3.3 Fat and membranes - lipids

WHAT IT IS	**Lipids** do not dissolve in water. The building block is the **fatty acid**: a long carbon chain with a small acidic end. A **saturated fatty acid** has no double bonds and is straight. An **unsaturated fatty acid** has one or more double bonds and is kinked. Three fatty acids plus glycerol make a **triglyceride**, the energy store. A **phospholipid** is a lipid with a water-loving head and a water-fearing tail, so it is **amphipathic**.
HOW IT WORKS / WHY IT MATTERS	Two jobs, and the second one is the elegant one. Triglycerides are compact energy storage: fat has more energy per gram than carbohydrate. Phospholipids build the **bilayer**. Because they are amphipathic, in water they spontaneously arrange with the heads out and the tails in, and that self-assembly needs no energy and no builder. This is why every cell on earth has the same basic membrane. **Cholesterol** sits within the bilayer and controls its fluidity, so it stays workable in the cold and does not turn fluid in the heat. In blood, fats are carried as **lipoproteins**, particles of lipid wrapped in protein so they can move in plasma, which is water-based.
EXAM HOOK	Two things he did emphasise. First, **LPS**. The lipopolysaccharide of Gram-negative bacteria is a lipid that binds **TLR4**, and that triggers **NF-kappa B**, which turns on inflammation, and in overwhelming amounts causes septic shock - he noted on the slide that this is a common cause of death among surgical patients in intensive care. Second, and more exam-adjacent: lipids are also **signalling molecules**, not just structural ones. His slide 19 showed phospholipid signalling: PI becomes PI(4)P, then PI(4,5)P by ATP-driven phosphorylation, and the enzyme PLC splits it into DAG, which stays in the membrane and activates PKC, and soluble IP3, which binds the IP3 receptor on the endoplasmic reticulum and releases calcium. A phosphatase then inactivates IP3. He did not flag the second pathway as exam-type, but the first one, LPS to TLR4 to NF-kappa B, runs through both Lecture 1 and Lecture 5 and you will meet it again.

2.3.4 The machines - proteins (preview)

WHAT IT IS	A **protein** is a chain of **amino acids** joined by **peptide bonds**. Dr. Wong's slide listed the categories as amino acid, peptide, polypeptide, protein complex. The chain then folds into a specific three-dimensional shape.
HOW IT WORKS / WHY IT MATTERS	Shape decides function, and that is the whole of Chapter 3. The chain builds up in levels: primary (the sequence of beads), secondary (local folds such as alpha-helix and beta-sheet), tertiary (the overall three-dimensional fold), quaternary (several folded chains assembled together). A **domain** is a part of a protein that folds and works on its own. Two disease and drug hooks from Lecture 1 are worth previewing here. The first is a **therapeutic protein liability**: the deamidation reaction, where an asparagine becomes an aspartate or isoaspartate and the mass shifts by about one unit, which can change how well the drug binds its target. The second is the **antibody**, which he described as having Fab and Fc halves, disulphide bridges, and a set of hypervariable loops called CDR1, CDR2 and CDR3 that make each antibody different - the basis of antibody diversity in Lecture 1 topic 8.
EXAM HOOK	Two exclusions to note. The amino-acid R-group table is **not** in the exam: "don't memorize this ... definitely not include in the exam". And antibody-drug conjugate structures are **not** in the midterm: "all this would not be included in the midterm. I'm just want you to know more". The concepts stay; the structures do not.

2.3.5 The information carriers - nucleic acids (preview)

WHAT IT IS	A **nucleotide** is a nitrogenous base plus a sugar plus a phosphate. Without the phosphate it is a **nucleoside**. The bases are **A, T, C, G and U** - adenine, thymine, cytosine, guanine and uracil. Long chains of nucleotides form DNA and the several kinds of RNA.
HOW IT WORKS / WHY IT MATTERS	DNA holds the recipes; RNA does the work. His slide listed the RNA types: mRNA, tRNA, microRNA and lncRNA, and he later added siRNA. The history matters too, because he used it to make a logical point. Griffith found that something transformed one strain of pneumococcus into another. Avery, MacLeod and McCarty showed that the transforming principle was destroyed by **DNase** and not by **proteases**, which proved the hereditary material is DNA and not protein. Watson, Crick and Franklin produced the double-helix model, and that model is what makes PCR possible.
EXAM HOOK	He stated the pairing numbers: **G-C uses 3 hydrogen bonds, A-T uses 2**. He also stated the PCR amplification figure of roughly **2 to the n** copies after n cycles, and the codon detail that the start codon is AUG and the stop codons are UAA, UAG and UGA. The base-pairing numbers are the ones most likely to turn up as an MCQ stem, because they connect straight to PCR primer design. The full structural story of DNA is Chapter 4, and the replication machinery is Lecture 4.

2.4 The 12 things to remember

1. A biomolecule is a molecule a living thing makes. There are four families, and they cover everything in this course.
2. Carbohydrates, lipids, proteins and nucleic acids are the four. Everything else is an example or a method.
3. A carbohydrate is built from sugar units joined by glycosidic bonds. Same units, different shape, completely different job.
4. Glucose is the energy currency. Glycogen is the branched storage form. Cellulose is the unbranched structural form.
5. A missing enzyme creates a metabolic block. Glucose-6-phosphatase deficiency is his worked example, and it produces a glycogen storage disease.
6. Lipids do not dissolve in water. That single fact explains storage fat, blood lipoproteins, and the reason a membrane can exist.
7. Saturated fatty acid means straight. Unsaturated means kinked. Straight packs tight, so it is solid.
8. A triglyceride is three fatty acids on a glycerol. It is the compact energy store.
9. A phospholipid is amphipathic, so it self-assembles into a bilayer. Heads out, tails in. That is the cell wall.
10. Cholesterol controls membrane fluidity. Lipoproteins deliver fat through blood.
11. LPS is a lipid. It binds TLR4, turns on NF-kappa B, and drives inflammation and septic shock.
12. A protein is a folded bead necklace, and DNA is a recipe. Full treatment in Chapters 3 and 4.

2.5 Check yourself

Question 1. Why does a cell membrane form by itself, with no energy and no machinery to build it?

> **Answer.** Because the lipid that membranes are built from is **amphipathic**. A **phospholipid** has a water-loving head and a water-fearing tail. Put those molecules in water - which is exactly what a cell is, a bag of water - and the cheapest arrangement for each molecule puts the head in the water and the tail away from it. When a whole sheet of them does this together, you get a **bilayer**: heads facing out on both sides, tails packed in the middle. The cell does not spend energy building this. It gets the bilayer for free from the chemistry of the molecule and the polarity of the water around it. **The reasoning: amphipathic plus polar water forces the arrangement.**

Question 2. A plant cell wall and a storage glycogen granule are both made of glucose units joined by glycosidic bonds. Why can a cow eat one and not the other?

> **Answer.** Because the bonds point in different directions. Glycogen uses **alpha-1,4** linkages along the chain with **alpha-1,6** branch points, and that branching makes it compact and quickly breakable - it is an energy store, so the animal needs to release its glucose fast. Cellulose uses the same sugar but a flipped bond orientation, which makes straight, strong, parallel fibres - ideal for a plant cell wall. The cow has bacteria in its stomach carrying the enzyme that can cut cellulose's bonds. You do not, so cellulose passes through you as fibre. **The reasoning: same building block, different bond geometry, different job, different animal that can use it.**

Question 3. A patient has a Gram-negative bacterial infection and their blood pressure collapses. Dr. Wong's lecture gives a specific molecular chain for this. What is it, and why does a lipid appear in it?

> **Answer.** The chain is **LPS to TLR4 to NF-kappa B to inflammation**, and in overwhelming amounts of endotoxin in the blood this causes septic shock, which he noted on the slide is a common cause of death among surgical patients in intensive care. The lipid appears because **LPS**, lipopolysaccharide, is itself a lipid - it is the outer membrane material of Gram-negative bacteria. Its lipid portion is what TLR4 recognises. So the same chapter of biology you just learned - lipids as membrane building blocks - turns out to be the recognition signal that starts the whole inflammatory cascade. **The reasoning: the thing that is recognised by the immune system is a lipid, so the lipid chapter and the signalling chapter are the same chapter.**

Question 4. A colleague says a protein drug has changed behaviour between two batches made under slightly different conditions. Based on Chapter 2, what are the three candidate explanations you would check?

> **Answer.** Check the shape of the protein first, because shape decides function, and then work backwards. (1) **pH.** A change in pH alters the charge on side chains, which alters which parts of the protein stick to which. He showed exactly this as the **deamidation** pathway, where an asparagine (114.04 Da) is converted through a succinimide (97.02 Da) into aspartate or isoaspartate (115.03 Da) at high pH, and that can change how well the drug binds its target. (2) **Oxidation.** If a molecule has lost electrons, or gained hydrogen, its chemistry has changed. (3) **Contamination or truncation** - a protein that was cut short, or a batch that was not pure. Note that he explicitly excluded the amino-acid R-group table from the exam, so you would not be asked to recite the table. You would be asked to reason about which mechanism explains the change. **The reasoning: identify the class of chemical change, then reason from chemistry to shape to function.**

Question 5. Dr. Wong says a nucleotide is a nucleoside plus a phosphate. What useful things does that one statement let you predict?

> **Answer.** It tells you where the negative charge lives, and negative charge is what makes molecules bind to each other and what makes DNA stable. A chain of nucleotides is a **sugar-phosphate backbone** with bases sticking out, and the bases are what pair up. It also tells you that the phosphate group is the reactive handle, which is why the same phosphate group shows up as ATP, where it is the spendable unit of energy, and in phosphorylation, where adding and removing it is the cell's on-off switch. And because the backbone carries the charge, the two strands of the double helix have to run in opposite directions - the antiparallel arrangement, with one strand 5-prime to 3-prime and the other 3-prime to 5-prime, which is the fact Lecture 4 builds the whole replication fork around. **The reasoning: one structural statement, and it explains backbone geometry, charge, energy currency, and replication directionality.**

CHAPTER 3

Proteins: what they are and how they work [the folded machines that do the cell's work]

3.0 What this chapter is about

Your body is built mostly from water and from a few very large molecules. The busiest of those large molecules are proteins. A protein starts as a long chain of small building blocks, and then it folds itself into one precise shape. That shape is the reason the protein can do its one job well, whether that job is copying a piece of DNA, cutting a sugar in half, or grabbing a germ and holding it still. Dr. Wong teaches proteins early in Lecture 1, right after the simple biomolecules, because the rest of the course is about what these machines do and how we measure them. This chapter gives you the vocabulary you need before he uses it.

3.1 Words to know before we start

Term	In plain English
protein	A large folded machine, made of a chain of small building blocks called amino acids. It does its one job because of its shape.
amino acid	One of the small building blocks of a protein. There are 20 common kinds, and they all have the same middle part with a different side piece.
R-group (side chain)	The variable side piece of an amino acid. It is what gives each amino acid its own size, charge and behaviour.
peptide bond	The chemical link that joins one amino acid to the next. A chain of them is a polypeptide.
polypeptide	A chain of amino acids joined by peptide bonds. A long folded polypeptide is a protein.
primary structure	The exact order of the amino acids in a chain. This is the protein's own "one-letter" code.
secondary structure	Two regular local shapes that the chain takes up: the alpha-helix and the beta-sheet. Both are held by hydrogen bonds.
tertiary structure	The full 3D shape of one folded chain.
quaternary structure	The way two or more folded chains (called subunits) come together into one working unit.
domain	A stretch of a protein that folds on its own and does a job by itself.
hydrogen bond	A weak sideways attraction between a slightly positive and a slightly negative atom. Many of them together hold a protein's shape.
enzyme	A protein that makes a chemical reaction go faster. Almost every tool in this course is an enzyme.
codon	A group of three letters in mRNA. It tells the ribosome which amino acid to add next.
antibody	A protein shaped like a pair of tongs. It sticks to one specific target, called an antigen.

3.2 The big picture

A protein is a machine. Like any machine, it has parts, and the parts only work if they are arranged the right way. The best way to picture a protein is as a bead necklace that folds itself.

Start with the beads. A **protein** is a chain of **amino acids**, and there are only 20 common kinds. Yet these 20 units, in different orders and lengths, make every protein in your body. Each amino acid has the same central carbon with the same attachments: an amino group, a hydrogen atom, a carboxyl group, and one variable piece called the **R-group**. The R-group is the personality of the amino acid. Some are tiny. Some carry a negative charge. Some can link to each other. The 20 amino acids are the same 20 backbones wearing 20 different hats, and the hat decides how each bead behaves.

Joining the beads together is simple chemistry. The carboxyl group of one amino acid reacts with the amino group of the next, and a **peptide bond** forms, releasing a water molecule. Because every amino acid has both an amino

end and a carboxyl end, they join head to tail into a long **polypeptide**. The chain always has a beginning and an end. The first amino acid keeps a free amino group, so that end is the **N-terminus**. The last keeps a free carboxyl group, so that end is the **C-terminus**. Proteins are always written from the N-terminus to the C-terminus, which is why a position number like "position 114" is meaningful.

Now the folding. A chain of amino acids is floppy, but it does not stay floppy. Two forces tidy it up. First, the **backbone** settles into two regular local shapes. The **alpha-helix** is a spring, like a corkscrew. The **beta-sheet** is a flat, pleated sheet, made when the chain folds back on itself and lies alongside its own earlier part. Both are held in place by **hydrogen bonds**, weak sideways attractions between a slightly positive hydrogen atom and a slightly negative oxygen atom. One hydrogen bond is too weak to matter. Dozens, all pulling the same way, hold a shape steady. Together these two regular shapes are the **secondary structure**.

Second, the whole chain folds into one overall 3D shape, the **tertiary structure**. Now the R-groups decide where things go. R-groups that dislike water hide inside the protein. R-groups that like water stay on the outside. Two cysteine R-groups can form a strong **disulfide bond**, which acts like a staple across the fold. The final shape is the working shape.

Sometimes one chain is not enough. When two or more folded chains come together, the assembly is the **quaternary structure**, and each chain in it is a **subunit**. Haemoglobin is the classic example. Antibodies are another. Inside a single chain there is one more layer: a **domain** is a stretch that folds by itself and does one job.

Why does all this matter? Because the shape is the function. A small chemical change can alter the shape, and therefore alter the job. Dr. Wong uses this idea directly. A **phosphorylation** switch works this way: a **kinase** adds a phosphate group using ATP, and a **phosphatase** removes it again. Adding that charged phosphate pushes the protein open. Removing it lets the protein close. Open and closed, on and off, in seconds. The same logic explains why a single chemical change on a therapeutic protein, such as the **deamidation** of one asparagine into aspartate, can change how well a drug binds and how well it works.

Some proteins are **enzymes**. An enzyme makes a reaction go faster by lowering the energy hill it has to climb. The enzyme has a pocket called the **active site**, and the molecule it acts on is the **substrate**. Almost every tool in this course is an enzyme: helicase, primase, polymerase, ligase, kinase, phosphatase, and many more.

So how does a cell build a protein? It copies the gene into **mRNA**, a working copy of the recipe. A **ribosome** reads that mRNA three letters at a time. Three letters is a **codon**, and each codon names one amino acid. With four letters there are 4 x 4 x 4 = 64 possible codons, and 64 is more than enough for 20 amino acids, so most have more than one spelling.

Finally, how do we know what a protein looks like? Four methods answer four different questions. **Edman degradation** reads the sequence from the N-terminus, one amino acid at a time. **LC-MS/MS** cuts the protein into pieces, weighs them and reads their sequences, so it tells you which proteins are in a sample and exactly what they are. **SDS-PAGE** sorts proteins by size. **X-ray crystallography** works out the exact 3D atomic shape. Using the wrong one answers nothing at all.

3.3 Walkthrough

Figure F3 - How a protein folds



From a chain of beads, to the alpha-helix and the beta-sheet, to one folded 3D shape, to an assembled multi-subunit machine. Every level is labelled in plain language, so the figure can be read on its own.

3.3.1 Amino acids: the small units every protein is built from

- **What it is.** An **amino acid** is a small molecule with a central carbon atom. That carbon carries four things: an amino group (written NH2), a hydrogen atom, a carboxyl group (written COOH), and one variable piece called the **R-group**. Every one of the 20 common amino acids has this same central carbon and these same three fixed groups. Only the R-group changes. A **protein** is a long chain of these units joined together.

- **How it works / Why it matters.** Because each amino acid carries both an amino group and a carboxyl group, two amino acids can react with each other. That is why proteins are chains and not rings, and why every protein has a definite beginning and end. The beginning keeps the free amino group, and it is called the **N-terminus** (N for nitrogen). The end keeps the free carboxyl group, and it is called the **C-terminus** (C for carbon). Chains are always read and written from the N-terminus to the C-terminus, which is what makes a position number such as "residue 114" meaningful. This numbering is not decoration. In one of Dr. Wong's own examples, the asparagine at position 114 is exactly the residue that goes wrong. The mass of that residue changes by about one dalton as the reaction happens, and the two possible end products are mirror-image twins with the same mass but different chemistry. A dalton is the mass unit used for molecules, named after John Dalton.

- **Exam hook.** Dr. Wong showed the generic amino acid on a slide, with the central carbon carrying NH2, H, R and COOH, and used exactly this structure when he explained the deamidation pathway. He did not set out to examine the structure diagram itself. The concept you need is the one sentence: same backbone for all 20, different R-group.

3.3.2 The twenty common amino acids and the R-group idea

- **What it is.** There are 20 **amino acids** that build essentially every protein in every living thing. They differ only in their **R-group**, the side chain hanging off the central carbon. The R-group can be tiny (glycine has only a hydrogen atom), bulky (tryptophan has a large ring), negatively charged (aspartate and glutamate), positively charged (lysine and arginine), or able to form a special bond (cysteine forms disulfide bridges). Group the 20 by their R-groups and the list becomes short: uncharged and water-shy, uncharged and water-loving, negatively charged, positively charged.

- **How it works / Why it matters.** The R-group decides everything local. Dr. Wong's point in Lecture 1 is that R-group chemistry dictates how flexible a stretch of chain is. Glycine, with the smallest possible R-group, is the most flexible amino acid, because there is almost nothing beside the backbone. Bulky or charged R-groups such as tryptophan and arginine make the backbone rigid, because those groups are physically awkward to move. So the same chain of beads can be floppy in one place and stiff in another, depending only on which beads are next to each other. Two of the 20 have names you will meet again, and they behave differently. Cysteine is the one that makes disulfide bonds. Proline is different from the others because its side chain loops back and joins its own backbone, which makes it a kink: proline is where the chain bends sharply, and the chain cannot spin freely there either.

The 20 also fall into four families, and the family a residue belongs to decides how it behaves in water. Nine are uncharged and water-shy: glycine, alanine, valine, leucine, isoleucine, methionine, phenylalanine, proline and tryptophan. Six are uncharged but water-loving: serine, threonine, cysteine, tyrosine, asparagine and glutamine. Two carry a negative charge: aspartate and glutamate. Three carry a positive charge: lysine, arginine and histidine. The two charged families are called acidic and basic. Remember this grouping, because it is the reason a protein folds the way it does: the water-shy residues hide inside, and the water-loving and charged ones stay on the outside where they can meet water and other molecules.

- **Exam hook.** Dr. Wong displayed the full table of 20 standard amino acids, with three-letter and one-letter codes and each R-group drawn out, on a slide of Lecture 1. He then told the class not to memorise it: *"don't memorize this... definitely not include in the exam"* (Lecture 1). He excluded the chemical structures on the same grounds: *"Please don't memorize all these chemical structure... They would not be included in my exam"* (Lecture 1). So do not learn the table. Learn the idea it exists: 20 types, one shared backbone, one variable R-group, and the R-group sets both charge and bulk. He also said this will not be examined.

3.3.3 Peptide bonds: how the beads are joined

- **What it is.** A **peptide bond** is the covalent link that joins one amino acid to the next. It forms when the carboxyl group of one amino acid reacts with the amino group of the next, and a molecule of water is released in the process. A chain of amino acids joined by peptide bonds is a **polypeptide**. A short polypeptide is often called a peptide. A long one that folds into a working shape is a protein.

- **How it works / Why it matters.** The peptide bond is strong, but it is also flat and rather rigid. Each amino acid sits in a plane, and the chain can turn only at the joints between them. This restriction is not a nuisance; it is the reason regular shapes are possible at all. If the joints were free, the chain would be a random spaghetti and would never settle into a helix or a sheet. The water released at each joining is worth remembering too, because it is the same kind of reaction that joins sugars into starch and glycogen, and it is one of the reasons water matters so much to cells.

- **Exam hook.** He did not flag peptide-bond chemistry as an examinable topic in any of the five lectures. The idea to hold on to is that the chain is directional, and that its ends are named for the groups they keep free.

3.3.4 Primary structure: the order of the beads

- **What it is.** The **primary structure** of a protein is its sequence: the exact order in which the amino acids appear, read from the N-terminus to the C-terminus. It is the flat, one-dimensional description. Every later level of structure is built on top of it and is determined by it.

- **How it works / Why it matters.** The order of the beads decides the fold, and the fold decides the job. Change one bead in the middle of a chain and you have changed the packing of everything around it. This is the link between this chapter and Chapter 4: a change in the DNA sequence is a change in the primary structure, and a change in the primary structure is usually a change in the protein's behaviour. It is also why reading a protein's sequence is such a big deal in the clinic. If you know the sequence, you can predict the shape, and therefore predict what the protein does and whether a drug will fit into it.

- **Exam hook.** He did not flag "the primary structure is the sequence" as a numbered exam item, but it is the spine of the protein hierarchy he taught in Lecture 1, and it is the answer to the common question "which level of structure is the sequence?"

3.3.5 Secondary structure: the alpha-helix and the beta-sheet

- **What it is. Secondary structure** is the set of regular local shapes that the backbone of a chain takes up. There are two important ones. The **alpha-helix** is a right-handed corkscrew, about 3.6 amino acids per turn. The **beta-sheet** is a flat, pleated sheet, made when the chain folds back on itself and runs alongside its own earlier section, either in the same direction or in the opposite direction.

- **How it works / Why it matters.** Both shapes are held in place by **hydrogen bonds**. A hydrogen bond is a weak attraction between a hydrogen atom that is slightly positive and an oxygen atom that is slightly negative. In a protein, the carbonyl oxygen of the backbone on one turn hydrogen-bonds to the backbone N-H group a few residues further along. On his Lecture 1 slide of the hierarchy, Dr. Wong drew the beta-sheet with the bonds drawn as C=O to H-N dashes, exactly this. Because a hydrogen bond is weak on its own, a single one would not hold anything. But the helix can line up hydrogen bonds down its whole length, and the sheet can line them up across its whole width, and then the sum of many weak attractions becomes a strong hold. The parts of the chain that are in neither regular shape are called loops or coils, and they are usually the most mobile parts.

- **Exam hook.** He did not flag the alpha-helix and beta-sheet as separate exam items, and he did not ask you to remember the 3.6 residues per turn. What he did stress, in Lecture 3, is the neighbouring principle: **histone-DNA binding is non-covalent**, so chromatin can open and close on demand. Do not confuse the two - a protein is switched on and off by adding or removing a phosphate, not by breaking hydrogen bonds.

3.3.6 Tertiary structure: one chain, one 3D shape

- **What it is.** The **tertiary structure** is the complete three-dimensional shape of a single folded polypeptide chain. It is the second big folding step, after the local helices and sheets.

- **How it works / Why it matters.** The R-groups decide the fold. R-groups that fear water tuck into the inside of the protein, away from the water around it. R-groups that carry a charge or like water stay on the outside, where they can interact with water and with other molecules. On top of that, two cysteine R-groups can form a **disulfide bond**, a strong chemical link between two parts of the chain that acts like a staple. Dr. Wong pointed out in Lecture 1 that disulfide bonds are what stabilise the tertiary and quaternary structure of a protein. The result is a single, specific, stable shape, and that shape is the working shape of the protein.

- **Exam hook.** Disulfide bonds stabilising tertiary and quaternary structure is his own statement from Lecture 1, and the accompanying point is the one that matters for drug design: in a therapeutic protein, any chemical change that disturbs the fold is a liability. He did not ask you to memorise the folding rules.

3.3.7 Quaternary structure: several chains, one machine

- **What it is.** The **quaternary structure** is the arrangement of two or more separate folded chains, each called a **subunit**, into one functional assembly. Not every protein has one. A single-chain protein stops at tertiary structure.

- **How it works / Why it matters.** Putting subunits together gives a machine extra capacity and extra control. Haemoglobin, which carries oxygen in your blood, is built from four subunits. Antibodies are built from four. Because the subunits sit next to each other, one subunit can detect a signal and the other can act on it, so the protein can switch itself on and off. This is also where the therapeutic industry takes hold: an antibody-drug conjugate uses one part of the antibody to find the target and another part to carry the drug payload.

- **Exam hook.** Dr. Wong named haemoglobin and antibodies as the examples of quaternary structure in Lecture 1. He did not flag the term itself as an exam item; the point to take away is that a multi-chain protein has more than one folded shape, and that the way the pieces come together can be regulated.

3.3.8 Domains: the reusable working parts

- **What it is.** A **domain** is a stretch of a protein that folds into a stable shape by itself and carries out a recognisable job. Domains are the modular building blocks of larger proteins.

- **How it works / Why it matters.** Domains let one protein do several jobs, and let the cell re-use the same module in different proteins. The clearest example in this course comes in Lecture 5, with nuclear receptors. Every nuclear receptor has the same three-part architecture: an N-terminal regulatory domain, a **DBD**, the DNA-binding domain, and an **LBD**, the ligand-binding domain. The DBD and the LBD are separate domains with separate jobs, and separating them is exactly how you work out which part of the receptor a drug acts on. Antibody structure gives the same lesson in a different way. The antibody has an antigen-binding part and a recruiting part, each a separate domain with a separate job.

- **Exam hook.** Dr. Wong defined a domain in Lecture 1 as an independently folding functional unit, and he returned to the idea as the DBD versus LBD pair - one of the classic confusable pairs the exam notes list as **DBD vs LBD: DNA-binding domain versus ligand-binding domain**. The DBD versus LBD distinction is the exam-facing version of the domain idea.

3.3.9 Why shape decides function

- **What it is.** The general rule of protein chemistry: a protein does its job by having a particular three-dimensional shape. Change the shape and you change the job. Small chemical changes to a side chain are enough to do it.

- **How it works / Why it matters.** Dr. Wong uses two worked examples from Lecture 1, and both are worth knowing. The first is the phosphorylation switch. A **kinase** uses ATP to attach a phosphate group to a protein, and a **phosphatase** uses water to remove it. The phosphate carries charge, and the mutual repulsion between

two charged groups pushes the protein open. Removing the phosphate lets the protein close again. So a protein can be switched on and off within seconds, using nothing but the addition and removal of one small chemical group. His Hippo pathway example is built on this: when the kinases MST1 and LATS1 phosphorylate YAP, YAP is held outside the nucleus and the cell stops dividing. When that phosphorylation is lost, unphosphorylated YAP walks into the nucleus and switches on proliferation, and in double-knockout mice the liver grows enormously. The second example is **deamidation**, a chemical change in a therapeutic protein. At high pH, an asparagine can lose ammonia and become a succinimide, and then take on water to become aspartate or its mirror-image twin iso-aspartate. On the numbers he used, asparagine is 114.04 daltons, the succinimide intermediate is 97.02, and both final products are 115.03. So one side chain quietly changes chemistry, and the protein's binding and its potency can change with it.

- **Exam hook.** Deamidation of asparagine to aspartate at high pH, altering drug efficacy and binding, is his own example of a therapeutic protein liability in Lecture 1. The phosphorylation switch is the mechanism he drew on a slide of Lecture 1. He did not ask you to memorise the dalton figures.

3.3.10 Enzymes: proteins that speed up reactions

- **What it is.** An **enzyme** is a protein that makes a chemical reaction go faster. It does not get used up by the reaction, so one enzyme molecule can do the job many times.

- **How it works / Why it matters.** Every reaction has an energy hill to climb, called the **activation energy**. An enzyme lowers that hill, so the reaction runs faster at body temperature. The enzyme holds its reacting partner in a pocket called the **active site**, and the molecule being changed is the **substrate**. The fit between the pocket and the substrate is specific, which is why each enzyme acts on one type of substrate. Almost every tool in this course is an enzyme: helicase, topoisomerase, primase, polymerase, exonuclease, ligase, kinase, phosphatase, histone deacetylase, DNA glycosylase and the caspases. When the lectures mention an enzyme by name, you can assume it is a protein with a shape, and that blocking or changing its shape changes the chemistry.

- **Exam hook.** He did not teach enzyme kinetics or active-site chemistry as an examinable topic. Enzymes are background here, and the reason for the background is simple: every machine in the rest of the course is a protein with a pocket, so you cannot follow the lectures without the idea.

3.3.11 How a protein is made and folded

- **What it is.** Making a protein is a four-step factory process. First, the gene is **transcribed**, which means the DNA recipe is copied into **mRNA**, a working copy. Second, mRNA leaves the nucleus and goes to a **ribosome**, a molecular machine that reads it and joins amino acids together. Third, the ribosome reads the mRNA three letters at a time, and each group of three letters is a **codon** that names one amino acid. Fourth, the finished chain folds itself into its working shape.

- **How it works / Why it matters.** The codon arithmetic is worth doing carefully, because it is the kind of sentence students trip over. There are four letters in RNA: A, U, C and G. A codon is three letters long, so the number of possible codons is 4 x 4 x 4 = 64. There are only 20 amino acids, so 64 codons is comfortably enough. Most amino acids are named by more than one codon, and a few have only one. One codon, AUG, does double duty: it starts the chain and it codes for methionine. Three codons, UAA, UAG and UGA, are stop signals, and they mean "the protein ends here". Because the same amino acid can be spelled in several ways, different organisms prefer different spellings, and this preference is called **codon usage**. That matters in industry. When a company makes a human protein in a **CHO cell**, a hamster cell line used to manufacture medicines, the human gene may need rewriting so that the host cell reads it more happily. That rewriting is **codon optimisation**, and Dr. Wong showed a western blot of the result, comparing the original sequence against optimised versions for two proteins. After the chain is built, cells use helper proteins called **chaperones** to keep the new chain from tangling and to help it fold correctly. A folded protein can then be modified chemically, and these modifications are called **post-translational modifications**. Phosphorylation is the one that matters most in this course. Finally, many proteins carry a short address label at the N-terminus that tells the cell where to send them, which is the signal peptide you will meet in Lecture 2.

- **Exam hook.** The central dogma, the start codon AUG and the stop codons UAA, UAG and UGA come straight from his Lecture 1 slides on the central dogma. The codon-usage point is his: for CHO cells, *"when you use the CHO cell, you need to do the codon optimization"*, and he showed the western blot with FALVAC-1 at 38.7 kd and PTP4A3 at 24 kd, each compared against a GAPDH loading control at 39 kd. The loading control is the band that tells you the experiment worked. He did not ask you to memorise the codon table.

3.3.12 Antibodies: proteins shaped to grab things

- **What it is.** An **antibody** is a protein shaped like a Y, or like a pair of tongs. The tips grip a target, and the target is called an **antigen**. The exact patch of the antigen that the antibody touches is the **epitope**.

- **How it works / Why it matters.** An antibody is built from two **heavy chains** and two **light chains**, held together by disulfide bridges. Each arm of the Y is called a **Fab** fragment, the antigen-binding fragment. The stem is the **Fc** fragment, and it is the part that recruits other parts of the immune system. On his Lecture 1 slide he gave the masses, and they make the architecture easy to remember: the smallest piece, the variable fragment or **Fv**, is about 25 kDa; a **Fab** is about 50 kDa; and the **Fc** is about 50 kDa. At the tip of each arm sits the **variable region**, which contains three hypervariable loops called **CDR1**, **CDR2** and **CDR3**. Those three loops are the gripping surface. Because they can take so many shapes, the immune system can build antibodies against so many different things. Dr. Wong put a number on that: somatic recombination produces billions of antibody variants. The raw material is a set of gene segments held in the DNA, and he gave their counts: about 39 copies of the V segment, 27 of the D segment and 6 of the J segment. The factory that shuffles them is the subject of Lecture 4, where we will meet V(D)J recombination and the NHEJ repair machinery that carries it out. For now, the shape is the point: the tips are shaped to fit one target, the stem is shaped to talk to the rest of the immune system.

- **Exam hook.** Antibody anatomy, the masses and the three CDR loops are all from his Lecture 1 slides. He also used the antibody shape as the delivery vehicle for antibody-drug conjugates in Lecture 1 and again in Lecture 4. The antibody-diversity chain (V(D)J recombination, billions of variants) links directly into Lecture 4, where a defect in the repair machinery that builds the repertoire causes severe combined immunodeficiency, which he named as one of his five exam-type questions.

3.3.13 The four standard ways we study proteins, and what each one actually tells you

- **What it is.** Four techniques come up again and again when proteins are measured. Each answers a different question, and the exam loves to test whether you know which is which.

- **How it works / Why it matters. Edman degradation** is the classical method. It chemically peels off one amino acid at a time from the N-terminus and identifies each one, so it gives you the sequence read straight from the start of the chain. Its limitation is that it works on a single, clean chain, so a real protein has to be broken into pieces first. **LC-MS/MS** is the modern workhorse. The protein is cut into smaller peptides with an enzyme, the peptides are separated by liquid chromatography, and each one is weighed by mass spectrometry. The mass tells you which amino acids are in the piece, and breaking the piece further tells you their order. Matching the result against a database identifies the protein. Crucially, LC-MS/MS also tells you **how much** of each protein is present, which is why it is the method behind every proteomics biomarker in this course. **SDS-PAGE** is the sorting method. The protein is treated with the detergent SDS, which coats every protein with the same negative charge, and then it is run through a gel under an electric field. Small proteins travel further than large ones, so the gel reads out like a set of stairs. It tells you the sizes of the proteins and roughly how many are there. It cannot tell you the sequence. **X-ray crystallography** is the shape method. The protein is turned into a crystal, bombarded with X-rays, and the pattern of scattered rays is used to work out where every atom sits. It tells you the exact 3D atomic structure, which is the thing that explains function.

- **How the methods fit together.** These four are not alternatives. They answer four different questions about the same protein, so a real study usually uses more than one of them. The next block, 3.3.14, follows one protein all the way from a cell sample to a patient blood test, using every one of these methods in turn.

- **Exam hook.** Edman degradation as the classical method and LC-MS/MS as the modern one are his own words from Lecture 1, and 2D-PAGE separating by charge and then mass is the workflow he showed. The X-ray connection comes from Lecture 3: the nucleosome core particle was solved at 2.8 angstrom resolution, and he noted that below 2.0 angstrom is "very excellent". He also made the practical point that histones are difficult to crystallise because of their heavy positive charge. So the method-to-question mapping is worth being able to state: Edman gives sequence from the N-terminus, LC-MS/MS gives identity and amount, SDS-PAGE gives size, X-ray gives atomic shape.

3.3.14 Putting the methods together: how a protein becomes a blood test

- **What it is.** A **biomarker** is a molecule you can measure in a sample from a patient to say something about their disease. A protein can become a biomarker, but only if the lab can find it, identify it, and show that its level differs between sick and healthy people. That is why the methods in 3.3.13 are chained together rather than used one at a time.

- **How it works / Why it matters.** Dr. Wong's worked example is **vimentin**, a protein of the cytoskeleton, in liver cancer. Here is the chain, step by step. Step one: get a cell sample. Step two: run the two-dimensional gel. **Isoelectric focusing** sorts the proteins by charge along one axis, using a pH gradient running from 4.0 to 9.0, and SDS-PAGE sorts them by size along the other, from 220 kDa down to 10 kDa. So every protein lands in its own spot of a grid. Step three: cut out the spot that changed, and cut the protein inside it into small peptides with an enzyme. Step four: run the peptides through LC-MS/MS. The mass of each peptide identifies the protein, and the size of the peak tells you how much of it is there. Step five: check that the result is real, using other tools: a western blot, immunohistochemistry, or qPCR on the RNA. Step six: ask whether the finding is worth anything, by consulting the Human Protein Atlas, which records where in the body the protein is normally found, and whether its level predicts survival in a given cancer. Step seven: move the measurement into something you can take from a patient. The vimentin finding in liver cancer was eventually measured in serum, using ELISA, a method that detects one specific protein using an antibody. That final step is what turns a laboratory finding into a blood test. The disease groups were separated by a real margin: controls averaged 176.0 nanograms per millilitre, plus or minus 61.3; patients with tumours of 2 cm or less averaged 271.5, plus or minus 165.9; and patients with larger tumours averaged 231.3, plus or minus 132.3.

- **Exam hook.** The two-dimensional workflow, the Human Protein Atlas and the vimentin case are all his own Lecture 1 material, and the serum vimentin figures are on his slide. He also gave the worked example a clinical twist worth remembering: the tumour tissue next to a tumour is not the same thing as normal tissue. He warned that adjacent tissue is *"biochemically different from truly normal tissue"* and should not be treated as healthy. He did not ask you to memorise the workflow step by step. The exam-facing idea is the logic: separate, identify, quantify, validate, then make it measurable in a patient.

3.4 The twelve things to remember

1. A protein is a folded machine, and its shape is the reason it can do its job.
2. There are only 20 common amino acids, and they all share the same backbone. Only the R-group differs.
3. The R-group sets charge and bulk, and therefore sets how rigid the chain is at that point. Glycine is the most flexible; bulky or charged R-groups such as tryptophan and arginine are the most rigid.
4. Amino acids join head to tail by peptide bonds, releasing water. The chain has a named start (the N-terminus) and a named end (the C-terminus).
5. Primary structure is the order of the beads. Secondary structure is the alpha-helix and the beta-sheet, held by hydrogen bonds.
6. Tertiary structure is the whole folded chain, held by the packing of the R-groups and stapled by disulfide bonds.
7. Quaternary structure is several subunits working together, as in haemoglobin and antibodies.
8. A domain is a part of a protein that folds and works on its own. Nuclear receptor DBD versus LBD is the exam version of this.
9. A codon is three letters of mRNA. Four letters give 4 x 4 x 4 = 64 codons, which is more than enough for 20 amino acids. AUG starts the chain; UAA, UAG and UGA stop it.
10. Enzymes lower the activation energy of a reaction by holding the substrate in a pocket called the active site.
11. Four methods, four different answers: Edman gives sequence from the N-terminus, LC-MS/MS gives identity and amount, SDS-PAGE gives size, X-ray crystallography gives atomic shape.
12. Antibodies are Y-shaped, built from two heavy and two light chains, with an antigen-binding Fab and a recruiting Fc, and three hypervariable loops, CDR1, CDR2 and CDR3, that do the gripping.

3.5 Check yourself

Q1. Someone tells you that a protein's function is decided by its shape, and asks you to explain why that is not just a slogan.

A1. Start with the observation that a protein is a chain of amino acids with one variable piece, the R-group. Because the R-groups differ in charge and bulk, they cannot all sit comfortably in the same place, so the chain has to fold in a particular way. That fold puts some R-groups inside, away from water, and others outside, where they can reach out. Next, name the mechanism: the fold is held by hydrogen bonds in the backbone and by hydrophobic packing of the R-groups, and it is reinforced by disulfide bonds between cysteines. Then the interpretation: because the shape determines which surfaces are available, it determines which molecule the protein can bind and what it can do. Dr. Wong's own examples are the phosphorylation switch, where adding and removing one charged phosphate opens and closes the protein, and deamidation, where one asparagine turns into aspartate and the protein's binding and potency change with it. So the slogan is shorthand for a causal chain: chemical change, then change of shape, then change of job.

Q2. Why does a codon have to be three letters long, and why is 64 enough?

A2. Do the arithmetic. There are four letters in RNA. A codon of three letters can therefore be any combination of three choices from four, so 4 x 4 x 4 = 64. Now compare with what has to be encoded: 20 amino acids, plus start and stop signals. Sixty-four is more than enough, so the code is redundant. Most amino acids are named by more than one codon, and only a few are named by a single one. That redundancy has a practical consequence which Dr. Wong used in Lecture 1: because the same amino acid can be spelled in several ways, different organisms prefer different spellings, so a gene may have to be rewritten, or codon optimised, when it is expressed in a CHO cell. The answer to "why three" is simply that three is the shortest word that fits 20 amino acids comfortably.

Q3. A colleague ran a sample on SDS-PAGE, saw four bands, and says "I now know what proteins are in the sample." What is wrong with the statement, and what should the colleague have used?

A3. Separate what the method can and cannot do. SDS-PAGE treats the proteins with SDS, which gives them the same negative charge, and then separates them by size in a gel. So the four bands tell you there are about four abundant protein sizes. They cannot tell you the identity of any of them, because a band of the right size could be any protein of that size. To identify them you need a method that reads the molecule itself. That is LC-MS/MS: cut the protein into peptides, separate them by liquid chromatography, weigh them, and match the fragments against a database. If you also want the shape, you need X-ray crystallography. So the correction is not "SDS-PAGE is wrong" but "SDS-PAGE answers the size question, and you have used it for an identity question".

Q4. You are told that the deamidation of asparagine to aspartate is a "liability" for a therapeutic protein. Explain what that word means.

A4. The observation first: the change is chemical and happens on a side chain, without the protein being rebuilt. The mechanism: at high pH, an asparagine can lose ammonia to become a succinimide intermediate, and then take on water to become aspartate or iso-aspartate. On Dr. Wong's figures, asparagine is 114.04 daltons, the intermediate is 97.02, and both products are 115.03, so you can even see the change on a mass measurement. Then the clinical meaning: because the shape of a protein carries out its job, and the shape depends on the chemistry of the side chains, a side chain that quietly changes chemistry can change how well the protein binds its target and therefore how well the drug works. A liability is a weakness you have to watch, test for and control, not a defect that makes the protein useless. And the limit of the claim: he did not say that every deamidation ruins a protein; he said it can alter efficacy and binding, which is why you monitor it.

Q5. Why is an antibody a good shape to explain quaternary structure, and why does Dr. Wong return to it in Lecture 4?

A5. First the structure: an antibody is built from four chains, two heavy and two light, joined by disulfide bridges. So it has more than one folded chain, and the way those chains come together is the quaternary structure. The parts are also named for what they do: the two arms are Fab, the antigen-binding fragments, and the stem is Fc, which recruits other immune cells. Second the reason it returns in Lecture 4: the enormous variety of antibodies, billions of variants, is produced by shuffling gene segments, and that shuffling is carried out by the same NHEJ repair machinery the cell uses to fix broken DNA. Dr. Wong made the NHEJ chain one of his five named exam-type questions, with the clinical endpoint being severe combined immunodeficiency when the machinery fails. So the antibody is where protein structure, protein function and DNA repair all meet.

CHAPTER 4

DNA, genes and chromosomes [the recipe, the printed manual, and the typo]

4.0 What this chapter is about

DNA is the recipe book of the cell, and this chapter opens the book. You will learn what the letters are, how two of them are always stuck together, and how a single letter is copied into a working copy. Then you will meet the two words that carry most of the course: a **gene**, which is one recipe, and a **chromosome**, which is the printed manual that holds thousands of recipes and packs them into something small enough to fit inside a nucleus. Finally you will meet the **mutation**, which is a typo. Dr. Wong teaches this in Lecture 1 as the chemistry and history of nucleic acids, then spends Lectures 3 and 4 on how DNA is packed and how it is copied, so this chapter is the bridge to both.

4.1 Words to know before we start

Term	In plain English
nucleotide	The building block of DNA. It is a base plus a sugar plus a phosphate group.
base	The letter part of a nucleotide. DNA has four: A, T, G and C.
sugar-phosphate backbone	The two long rails of the DNA ladder, made of alternating sugar and phosphate.
double helix	Two such rails wound around each other, with the bases pairing across the middle like ladder rungs.
base pairing	The strict rule for the rungs. Adenine pairs with thymine, and guanine pairs with cytosine.
hydrogen bond	The weak sideways attraction that holds each base pair together. A-T makes two, G-C makes three.
antiparallel	The two DNA strands run in opposite directions, one from 5' to 3' and the other from 3' to 5'.
replication	The copying of DNA before a cell divides, so each daughter cell receives a complete set.
gene	One recipe. A stretch of DNA that carries the instructions to build one protein (or one functional RNA).
chromosome	One printed manual. A single, very long piece of DNA plus its packaging proteins, condensed so it fits in the nucleus.
centromere	The waist of a chromosome. It is where the spindle fibres grip the chromosome during cell division.
telomere	The protective cap at each end of a chromosome.
nucleosome	A spool of protein with DNA wound around it. Many spools make chromatin, and chromatin condenses into a chromosome.
mutation	A typo. A change in the DNA sequence.

4.2 The big picture

Start with the letters. A **nucleotide**, the building block of DNA, has three parts: a **base**, a sugar, and a phosphate group. The sugar and the phosphate join end to end to make a long **sugar-phosphate backbone**: the two rails of a ladder. The bases stick out sideways from the rails, one per rung. DNA uses only four bases: adenine, thymine, guanine and cytosine, written A, T, G and C. RNA, the working copy of a gene, uses uracil, U, in place of thymine.

The rule for the rungs is absolute. A always pairs with T, and G always pairs with C. These are the **base pairs**, and each one is held together by **hydrogen bonds**, weak sideways attractions between a slightly positive hydrogen and a slightly negative oxygen or nitrogen atom. An A-T pair is held by two hydrogen bonds. A G-C pair is held by three. Because the two strands point in opposite directions, one running from 5' to 3' and the other from 3' to 5', the ladder is **antiparallel**. Wind the two strands around each other and you have the **double helix**.

The discovery was an experiment, not a guess. In 1928 **Griffith** worked with the bacterium that causes pneumonia and found that material from dead virulent bacteria could turn live harmless bacteria into virulent ones. He called it the transforming principle and could not say what it was. In 1943 **Avery, MacLeod and McCarty** identified that principle as DNA, using an enzyme test: **DNase**, which cuts up DNA, destroyed the transforming activity, while **protease**, which cuts up protein, did not. The hereditary material was DNA and not protein. **Watson, Crick and Franklin** then built the double-helix model, using Rosalind Franklin's X-ray images and the rule that A and T appear in equal amounts, and so do G and C.

Why does base pairing matter for everything afterwards? Four reasons. First, copying. Each strand is a template, and because pairing is strict, copying is accurate. Second, heat. G-C-rich DNA needs more heat to pull the strands apart, because it has more hydrogen bonds to break, so its melting temperature is higher. Dr. Wong's slide gave the direction, not a number: more GC content means a higher melting temperature. The exam notes also carry a '~40 deg C drop' and a '60 deg C versus 55 deg C' board example, and the two do not reconcile - learn the direction and the reason, not a number. That direction decides the temperature for a PCR primer. Third, repair. A cell can compare a fresh strand against the old one and cut out a letter that does not match, because the pairing rules say what the correct letter should have been. Fourth, damage. When a chemical attaches to a base and changes its shape, the pairing rule breaks, and the next round of copying bakes the mistake into the DNA. That is the route by which cigarette smoke causes cancer.

Now the two big nouns. A **gene** is one recipe: a stretch of DNA on a chromosome that gets copied into RNA and then used to build one protein. A **chromosome** is one printed manual: a single very long piece of DNA, plus the proteins that package it, condensed small enough to fit inside the nucleus. A human body cell has 46 of them, in 23 matched pairs, one from each parent. Each chromosome has a **centromere**, the waist where the cell's division machinery grips it, and a **telomere** at each end, a protective cap.

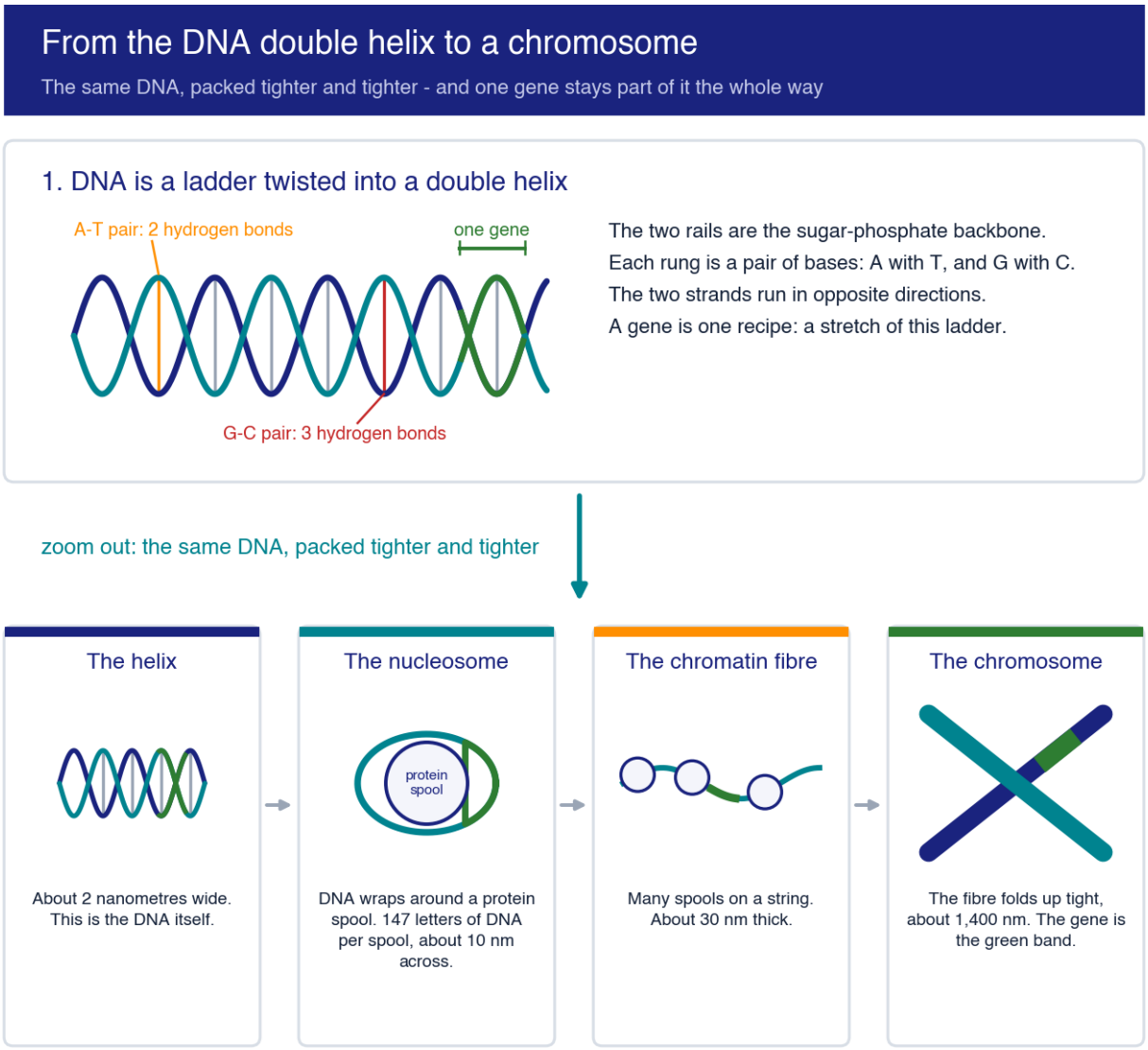
The manual is far too big to fit as it is. A human cell holds roughly a metre of DNA. So DNA is wrapped around protein spools called **histones**, and one spool with DNA wound round it is a **nucleosome**. Nucleosomes on a string form chromatin, the packaged manual, and chromatin folds until it becomes a chromosome. Loose, open chromatin is called **euchromatin** and is readable. Tightly packed chromatin is called **heterochromatin** and is switched off.

Before a cell divides, it copies all of this. **Replication** is **semi-conservative**, which means every new double helix ends up with one old strand and one new one. A team of enzymes does the work: one unwinds the helix, one relieves the twisting strain, one lays a short starting primer, one adds the letters and proofreads as it goes, and one seals the final gaps. Because DNA can only be built in one direction, one strand is built continuously and the other in short pieces that are later joined.

Finally, the **mutation**. A mutation is a typo in the recipe. Some typos change nothing. Some change one amino acid, and occasionally that turns a normal protein into one that drives a tumour. Some shift the reading frame, so everything after the typo is misread and the protein is cut short. Some are much bigger, involving whole genes or chromosomes. A typo that is copied and passed on becomes permanent, which is the basis of both cancer genetics and the inherited diseases in the exam scenarios.

4.3 Walkthrough

Figure F4 - From the DNA double helix to a chromosome



The green mark is one gene - a single recipe. It is the same piece of DNA in every picture.

Base pairing is labelled on the helix, then the same DNA is shown being packed step by step: helix, nucleosome, chromatin fibre, chromosome. A gene is marked on the helix and again on the chromosome, so you can see it is the same piece of DNA the whole way.

4.3.1 Nucleotides: the letters of DNA

- **What it is.** A **nucleotide** is the repeating unit of DNA and RNA. It is made of three parts: a **base** (the letter), a sugar, and one or more **phosphate** groups. Take the phosphate away and what is left, base plus sugar, is called a **nucleoside**. DNA's sugar is **deoxyribose**. RNA's sugar is **ribose**, which has one extra oxygen atom. DNA has four bases: adenine (A), thymine (T), guanine (G) and cytosine (C). RNA has the same three plus uracil (U) in place of thymine.

- **How it works / Why it matters.** The sugars and the phosphates join end to end to make the **sugar-phosphate backbone**, and the bases hang off the backbone and point inward. That arrangement is what allows two strands to pair up and form the helix.

Two small notes finish this off. First, the ends of a strand are numbered. Each sugar has numbered carbons, and the number 5 marks the carbon that carries the phosphate group, while the number 3 marks the carbon on the other side. So every strand has a 5' end and a 3' end. That is what the table in 4.1 is referring to, and it is why two strands can run in opposite directions without being mirror images of one another. Second, the phosphate group has a second job. Phosphates store and release energy, which is why ATP, a molecule with three phosphates, is the cell's spendable energy unit, and why kinases in this course use ATP to attach a phosphate to a protein. So the same small chemical group is doing two jobs in the cell: it is the structural backbone of DNA, and it is the wallet for energy.

- **Exam hook.** Dr. Wong defined nucleoside as base plus sugar and nucleotide as nucleoside plus phosphate in Lecture 1, and he listed the five bases A, T, C, G and U. This is one of the plainest definitional facts in the course, and it is the sort of thing that appears as a short definition in a multiple-choice question. He did not say the definition would be examined.

4.3.2 The double helix, and how it was discovered

- **What it is.** The **double helix** is DNA's final shape: two strands wound around each other like a twisted ladder, with the bases pairing across the middle and the sugar-phosphate backbones on the outside. The two strands run in opposite directions, which is called **antiparallel**, so one is read 5' to 3' and the other 3' to 5'.

- **How it works / Why it matters.** Three discoveries built the picture, and each one answers a different question. The first question was: what carries hereditary information? **Griffith** answered it in 1928 without knowing. He worked with *Streptococcus pneumoniae*, the bacterium that causes pneumonia, and found that something from killed virulent bacteria could convert live harmless bacteria into virulent ones. He named the active substance the transforming principle. The second question was: what is the transforming principle? **Avery, MacLeod and McCarty** answered it in 1943, and they answered it with an enzyme experiment. They showed that **DNase**, which cuts DNA, destroyed the transforming activity, whereas **protease**, which cuts protein, left it intact. The conclusion was that DNA, not protein, is the hereditary material. The third question was: what shape does it take? **Watson, Crick and Franklin** built the double-helix model, using Rosalind Franklin's X-ray diffraction images and the observed rule that the amount of A equals the amount of T, and the amount of G equals the amount of C. The model was immediately useful, because the two strands can be pulled apart by heat, and pulling them apart is exactly what PCR does.

- **Exam hook.** Dr. Wong presented this as the chemistry-and-history section of Lecture 1, with Griffith, Avery-MacLeod-McCarty and Watson-Crick-Franklin named on the slides, and he made one point sharply: the enzyme experiment is the proof. The dates on the slides, 1928 and 1943, were shown as part of the story and he did not flag them as examinable. He also stated that the double helix underpins PCR.

4.3.3 Base pairing: why A-T and G-C matters for everything downstream

- **What it is.** The pairing rule is that adenine pairs with thymine, and guanine pairs with cytosine. Each pair is held by **hydrogen bonds**. A-T pairs are held by **two** hydrogen bonds. G-C pairs are held by **three**.

- **How it works / Why it matters.** Four consequences follow, and all four matter later in the course. First, accurate copying. Because pairing is strict, a template strand tells the copying machinery exactly which letter

belongs opposite each letter. Second, melting temperature, written **Tm** for short. Separating two strands means breaking hydrogen bonds, so a G-C-rich stretch needs more heat than an A-T-rich stretch of the same length. On Dr. Wong's slide, the rule is the direction: more GC content means a higher melting temperature. The exam notes also carry a '~40 deg C drop' and a '60 deg C versus 55 deg C' board example, and the two do not reconcile - learn the direction and the reason, not a number. This is why PCR primer design has to look at GC content, as he stated in Lecture 3. Third, repair. The pairing rules tell the cell what the correct letter should have been, so a mismatch stands out and can be removed. The mismatch repair machinery of Lecture 4 works because of this. Fourth, damage and cancer. A chemical that attaches to a base, for example a tobacco-specific nitrosamine such as NNK, changes the shape of that base. The base then pairs incorrectly, the error is copied, and a mutation is fixed in the DNA. Dr. Wong used this route in Lecture 3 to explain how chemical carcinogenesis produces the KRAS mutations seen in lung cancer. Finally, the pairing rules also set the geometry of the helix itself, which is why a G-C-rich helix is a slightly different shape from an A-T-rich one.

- **Exam hook.** G-C equals 3 hydrogen bonds and A-T equals 2 is on his own list of precise numbers he stated, from Lectures 1 and 3. The melting-temperature **direction** is his fact and he linked it explicitly to PCR primer design: more GC means a higher Tm. **The G-C = 3 / A-T = 2 bond counts are the reliable numbers here.** The temperature example itself is inconsistent between his notes, so do not put a number on it.

4.3.4 What is a gene?

- **What it is.** A **gene** is one recipe. It is a stretch of DNA on a chromosome that carries the instructions for building one product. For most genes that product is a protein. For some genes it is a functional RNA, such as tRNA, rRNA or one of the small regulatory RNAs Dr. Wong covered in Lecture 1.

- **How it works / Why it matters.** The flow of information is the **central dogma**, and it runs in one direction. DNA is copied into RNA by **transcription**. That RNA is then used to build protein by **translation**. The RNA copy of a gene is called **mRNA**, and it is a photocopy of the recipe sent out to the factory floor. A gene is not read all the time. Whether it is on or off depends on how tightly the DNA is packed and on which proteins are present to read it, which is the subject of Lecture 5. That is also why two people with the same gene can be different, and why the same gene can be a driver in one cancer and harmless in another. Knowing the name of a gene is not enough. You also need to know what the gene does, where it is switched on, and what happens when it is broken. The clinical examples in this course all depend on that extra step: *K-RAS* only predicts a failure of cetuximab if you also know where it sits in the signalling cascade.

- **Exam hook.** Dr. Wong used the central dogma, DNA to transcription to mRNA to translation to polypeptide chain, as the spine of Lecture 1, and he returns to it in Lecture 5 to talk about control. The K-Ras story is the worked example of why a gene name alone is never a clinical answer. He did not set out a definition of a gene as a memorised line.

4.3.5 What is a chromosome?

- **What it is.** A **chromosome** is one printed manual: a single, extremely long piece of DNA together with the proteins that package it, condensed into a compact shape. A human body cell contains 46 chromosomes, in 23 matched pairs, one of each pair inherited from the mother and one from the father.

- **How it works / Why it matters.** A chromosome has a recognisable anatomy. The **centromere** is the narrowed waist, and it is the point where the spindle fibres attach during cell division, so it is how the cell pulls one copy of each chromosome to each end. The centromere divides the chromosome into a short **p-arm** and a long **q-arm**. The letter q is the long arm, so the band name q21.1 is read out as q-two-one-point-one, which Dr. Wong was careful to spell out on the Lecture 3 slide. Each end is capped by a **telomere**, which has three parts: a unique sequence right at the tip, then 100 to 300 kilobases of subtelomeric repetitive sequence, and then 3 to 20 kilobases of the repeated TTAGGG motif. Chromosomes are big enough to see. A technique called **karyotyping** stains them with Giemsa dye to produce **G-banding**, a pattern of light and dark bands that lets a trained eye read off large structural changes. Its limit is resolution. G-banding detects changes of roughly 0.5 to 10 megabases, so it finds extra chromosomes and large deletions, but it misses single-letter changes, which need sequencing.

- **Exam hook.** 23 pairs of chromosomes, the p-arm and q-arm with the centromere between them, and the resolution limit of G-banding are all on his list. Two details are worth keeping exactly. First, he was emphatic that q21.1 is q-two-one-point-one and not q-twenty-one-point-one. Second, he noted that the resolution figure appears as 5 to 10 megabases in one part of his teaching and 0.5 to 1 megabase in another, and that single-nucleotide changes always need sequencing.

4.3.6 How DNA packs into a chromosome - the bridge to Lecture 3

- **What it is.** The DNA in one human cell is about a metre long, and it has to fit inside a nucleus that is far smaller than a millimetre. It fits because it is wrapped and folded. The packaging ladder has standard steps.

- **How it works / Why it matters.** Step one is the **nucleosome**. DNA wraps around a core of eight histone proteins, two copies each of H2A, H2B, H3 and H4, with a linker histone H1 at the point where the DNA enters and leaves. One turn of DNA holds 147 base pairs, and the whole spool is about 10 nanometres across. Step two is the fibre. A chain of nucleosomes looks like beads on a string, about 11 nanometres thick, and it is packed into a 30 nanometre **chromatin fibre**, then extended at 300 nanometres, condensed at 700 nanometres, and finally a mitotic chromosome of about 1,400 nanometres, which is 1.4 micrometres. The whole packaged form of DNA with its histones is called **chromatin**. Step three is the switch. Tightly packed chromatin is called **heterochromatin**, and it is switched off, because the packing physically blocks transcription factors and enhancers from reaching the gene. Loose chromatin is called **euchromatin**, and it is being read. The state is not fixed. Nucleosomes slide along the DNA to expose or cover a gene, and two things push them: the intrinsic preference of a DNA sequence for a histone, and ATP-powered **chromatin remodelling complexes** that slide them deliberately. Two facts Dr. Wong added make this concrete. Histones are extremely well conserved, from fruit fly to human, which is why model organisms work. And histones are hard to crystallise because they carry a heavy positive charge, which is why getting the nucleosome crystal structure was a real achievement.

- **Exam hook.** He stated the diameters himself: 2 nanometres for the helix, 11 for beads on a string, 30 for the fibre, and 1,400 for the mitotic chromosome. The nucleosome numbers, 147 base pairs and about 10 nanometres, are on his list of precise figures. He also stated the nucleosome core structure was solved at 2.8 angstrom resolution, with below 2.0 angstrom being "very excellent". The concept he said you must understand is the one he put in those words: euchromatin is open and active, heterochromatin is condensed and silent, and it physically blocks access.

4.3.7 How DNA is copied - the bridge to Lecture 4

- **What it is. Replication** is the copying of DNA before a cell divides. Each daughter cell must receive a complete set of chromosomes, so the entire genome is copied, and it is copied fast.

- **How it works / Why it matters.** First, the rule. Replication is **semi-conservative**. Every new double helix contains one old strand and one newly made strand. Two rival models were on the table, semi-conservative and conservative, in which the old helix would stay whole and a completely new one would be made alongside. A Meselson-Stahl-type experiment with heavy and light isotopes settled it, and Dr. Wong showed the three candidate models on a Lecture 4 slide. Second, the team of enzymes. **Helicase** unwinds the double helix. **Topoisomerase** relieves the twisting strain that builds up ahead of the fork. **Primase** lays down a short RNA primer to give the polymerase a free end to start from. **DNA polymerase** adds the new letters and proofreads as it goes, using a 3' to 5' exonuclease activity to cut out a wrong letter. **Exonuclease** removes the primers afterwards, and **DNA ligase** seals the last gaps with a phosphodiester bond. Third, the direction problem. New DNA can only be built from 5' to 3', but the two templates are antiparallel. So one strand, the **leading strand**, is copied continuously, and the other, the **lagging strand**, is copied in short stretches called **Okazaki fragments**, each of which needs its own primer. Fourth, the scale. A human chromosome is enormous, so copying does not start from one point. It starts at thousands of **origins** at once, each opening a **replication bubble** with a fork running in each direction. Fifth, the ends. Because the last RNA primer on the lagging strand cannot be replaced, the end of a chromosome cannot be finished. This is the **end-replication problem**, and the result is that telomeres shorten by roughly 50 to 200 base pairs with every cell division. When they run out, the cell stops dividing, which is the **Hayflick limit**. **Telomerase**, an unusual enzyme that is a reverse transcriptase with

its own internal RNA template, can rebuild the ends. It is active in embryonic stem cells, in some adult stem cells, in germ cells, and, alarmingly, in about 90 percent of cancers.

- **Exam hook.** In his own words in Lecture 4: *"this is important, you need to understand the elongation happens from 5 prime to 3 prime"*, and *"I think one concept you'll need to understand is there's two different strands. So one is the leading strand, and the other one is the lagging strand."* His summary table gives the pairing: 5' to 3' for elongation, 3' to 5' for proofreading, and ligase sealing a 5' phosphate to a 3' hydroxyl. The thousands of origins and the two forks per bubble are his figures, as is the telomere shortening of roughly 50 to 200 base pairs per division and the telomerase activation in about 90 percent of cancers.

4.3.8 What is a mutation?

- **What it is.** A **mutation** is a change in the DNA sequence. It is a typo in the recipe. The word covers everything from a single letter changed to a whole chromosome rearranged.

- **How it works / Why it matters.** Three things can happen when a typo appears. It can do nothing, because the changed letter still codes for the same amino acid. It can change the protein slightly, which may be harmless, may be useful, or may change the protein's job. Or it can break the protein altogether, for example by creating an early stop signal. A typo only becomes real if it survives. Cells have repair machinery that catches most errors, and any change that is fixed or lost does not matter. A change that is copied into a daughter cell and passed down the family becomes a permanent difference, and that is the raw material of inherited disease and of cancer. The important clinical habit is to separate the two questions. Does the mutation exist? And does it matter? Dr. Wong makes this point repeatedly, most sharply in Lecture 3 with the difference between a **driver** change, which gives the tumour a growth advantage, and a **passenger** change, which is simply carried along.

- **Exam hook.** The driver versus passenger distinction is on his list of classic distractors, and the way he said to tell them apart is to integrate copy number profiles with mRNA expression, so that a copy gain that matches an increase in transcription points to a real driver. His worked example was NPM1 in leukaemia. He did not define "mutation" as an examinable line; the driver versus passenger distinction is the exam-facing version of it.

4.3.9 The kinds of mutation: one letter, or a few letters

- **What it is.** The smallest mutation changes a single letter. The next smallest adds or removes a few letters. Everything bigger is saved for the next block.

- **How it works / Why it matters. (a) A single letter changes.** This is a substitution, and it is called a **point mutation**. It can be **silent**, meaning the protein comes out exactly the same, or **missense**, meaning one amino acid changes, or **nonsense**, meaning the change creates a stop signal early and truncates the protein. The cleanest example in the course is **KRAS G12C**, where a single letter change turns one amino acid and leaves the RAS protein stuck in its active state, so the cell keeps dividing whether or not the receptor upstream is blocked. That is why cetuximab fails in K-Ras-mutant colorectal cancer, and why testing for K-Ras is a companion diagnostic. Dr. Wong also showed the NNK story from Lecture 3: the chemical in cigarette smoke alters a base, the altered base pairs wrongly, and the next round of copying fixes the mistake as a GC to AT change in the K-Ras gene. **(b) Letters are inserted or deleted.** If the number of letters added or removed is a multiple of three, the reading frame is preserved. That is an **in-frame** change, and the protein comes out the right length but with a piece missing; his slide showed an in-frame codon deletion producing MET-PHE-GLY-STOP instead of MET-LYS-PHE-GLY-STOP. If the number is not a multiple of three, the reading frame shifts. That is a **frameshift**, and everything downstream is misread. His slide showed a one-letter deletion in the third codon giving MET-LYS-LEU-ALA and running on, and a one-letter insertion right after the start codon giving an immediate stop. Frameshifts usually produce a short, useless protein. They can also produce **neoantigens**, new peptides that the immune system has never seen, and those are exactly what makes immunotherapy work in some cancers.

- **Exam hook.** The KRAS G12C example and the frameshift cartoons are straight from his Lecture 1 slides. The point he drew out from them is that a mutation matters because of what it does to the protein, and because a frameshift can hand the immune system a completely new protein fragment. He did not ask you to memorise

the codon strings on the slide, and he did not set out the labels silent, missense and nonsense as a memorised list; learn them as the three possible consequences of a one-letter change.

4.3.10 The kinds of mutation: whole genes, whole chromosomes, and chemical damage

- **What it is.** Bigger mutations take out a stretch of DNA, copy it, flip it around, or move it somewhere else. Bigger still change the number of whole chromosomes. And there are chemical injuries that turn into mutations when the cell copies the damaged DNA.

- **How it works / Why it matters. (a) Whole genes and whole chromosomes change. Deletions** remove a stretch, **duplications** copy it, **inversions** flip a stretch around, and **translocations** swap pieces between two chromosomes. Dr. Wong's point about inversions in Lecture 3 is the interesting one: an inversion removes no DNA and changes no gene dosage, yet it can change expression dramatically, because it moves a gene away from its enhancer or brings an enhancer next to the wrong promoter. **(b) The wrong number of chromosomes.** When the number of chromosomes is wrong, the condition is called **aneuploidy**, and it comes from **non-disjunction**, which is the failure of chromosomes to separate properly during meiosis.
  Non-disjunction in meiosis I gives two normal gametes and two abnormal ones, all of the same kind.
  Non-disjunction in meiosis II gives one normal gamete and three abnormal ones, of two different kinds. The results are conditions such as **trisomy 21**, which is Down syndrome, and Turner syndrome, which is 45,X. **(c) Copy number varies.** A **copy number variation** means the same stretch of DNA is present in different numbers of copies in different people. The amylase gene, AMY1, is the textbook example, with higher copy numbers in populations that have historically eaten a lot of starch. Dr. Wong flagged this as an idea rather than a mechanism list, and said so: *"I'm just give you a taste... don't be scared with all these mechanism"*. **(d) Chemical damage becomes a mutation.** Reactive oxygen species from the cell's own metabolism, and tobacco-specific nitrosamines such as NNK, attach themselves to a base and change its shape. The changed base then pairs wrongly, replication copies the error, and the error is fixed in the DNA. This is the chain from cigarette to cancer, and it is how the KRAS mutations in lung cancer arise.

- **Exam hook.** He named the meiosis I versus meiosis II distinction as one of his five exam-type questions, in his own words: *"you might need to distinguish the M1 or the M2 failure"*, and *"I think you need to distinguish these two situations here"*. The table to memorise is that meiosis I fails to separate homologous chromosomes and gives 2 normal plus 2 abnormal gametes, all abnormal ones the same, while meiosis II fails to separate sister chromatids and gives 1 normal plus 3 abnormal gametes, of two different kinds. Recombination happens in meiosis I and not in meiosis II. On inversions he said: *"you need to understand that some of these abnormalities can cause the loss of the genetic materials, but some does not."* He also excluded the copy-number mechanisms from the exam, so learn the amylase example as an idea rather than as a mechanism. The DNA adduct story is his Lecture 3 material and the bridge into the repair material of Lecture 4.

4.4 The twelve things to remember

1. A nucleotide is base plus sugar plus phosphate. Remove the phosphate and it is a nucleoside.
2. DNA has four bases, A, T, G and C. RNA swaps thymine for uracil.
3. The sugars and phosphates form the sugar-phosphate backbone. The bases hang off it and pair inward.
4. A always pairs with T, held by 2 hydrogen bonds. G always pairs with C, held by 3.
5. The two strands are antiparallel, and that is why one is copied continuously and one in pieces.
6. A gene is one recipe. A chromosome is the printed manual that holds thousands of recipes.
7. A human body cell has 46 chromosomes, in 23 pairs. Each has a centromere, a short p-arm, a long q-arm, and telomeres at the ends.
8. Packaging runs from a 2 nanometre helix, through the 147 base pair nucleosome, to the 30 nanometre fibre, to the 1,400 nanometre chromosome.
9. Euchromatin is open and being read. Heterochromatin is packed and silent. It is a switch, not a setting.
10. Replication is semi-conservative. Each new helix has one old strand and one new one.
11. Helicase, topoisomerase, primase, polymerase, exonuclease and ligase do the copying. New DNA is built 5' to 3' and proofread 3' to 5'.
12. A mutation is a typo. It may change nothing, change one amino acid, shift the reading frame, or rearrange a chromosome. What matters clinically is whether it is a driver.

4.5 Check yourself

Q1. Explain what a hydrogen bond does in DNA, state how many hydrogen bonds each base pair makes, and say what that tells you about melting temperature.

A1. First, the mechanism. A hydrogen bond is a weak sideways attraction between a slightly positive hydrogen atom and a slightly negative oxygen or nitrogen atom. In DNA, each base pair is held together by these bonds, so the two strands can be held together or pulled apart. Second, the counting. Adenine pairs with thymine using 2 hydrogen bonds, and guanine pairs with cytosine using 3. Third, the consequence, as a direction rather than a number. Separating the two strands, which is what happens when you heat DNA, means breaking these bonds. So a G-C-rich stretch needs more heat to separate than an A-T-rich stretch of the same length, which is the melting temperature, or Tm: more GC content means a higher Tm. That direction decides the annealing temperature of a PCR primer, which is why GC content matters at the bench. His notes also carry a board example with a specific temperature difference, but it does not reconcile with another figure in the same notes, so do not put a number on it. The same logic also explains why damaged bases are dangerous: if a chemical changes the shape of a base, the pairing rule breaks and a wrong letter gets copied in.

Q2. A patient has a K-Ras mutation and is being considered for cetuximab. Explain, at the level of DNA structure, why the mutation predicts the drug will fail.

A2. Start with the observation: the tumour has a single-letter change in the K-Ras gene, the G12C change, so one amino acid in the RAS protein is different from the normal one. The mechanism is at the protein level, but it is written in the DNA. RAS is a **transducer**. In its normal form it only relays a growth signal while the receptor above it, EGFR, is firing. The G12C change leaves RAS locked in its active state, so it signals whether or not the receptor is blocked. Cetuximab blocks EGFR, the receptor at the top of the cascade, so it removes the trigger but cannot remove a locked-on transducer downstream. The clinical meaning is that this is a **predictive** biomarker, not a prognostic one. The mutation tells you the drug will not work; it does not by itself tell you the patient is sicker. And the limit: you cannot make this call from the gene name alone. You need to know where the gene sits in the pathway, which is exactly the reasoning the exam expects.

Q3. Two people have the same rearrangement in chromosome 8, but one has a normal amount of that region's DNA and one does not. Explain how a rearrangement can change gene expression without changing how much DNA is present.

A3. The rearrangement to consider is an **inversion**, in which a stretch of the chromosome is cut out and put back in the opposite direction. The first point is what an inversion does not do: it removes no DNA and adds none, so the gene dosage, which is simply how many copies of the gene the cell has, is unchanged. The second point is what it does do. A gene does not read itself; it is read with the help of regulatory DNA, such as an **enhancer**, which can sit some distance away. Flipping a stretch can move a gene away from its enhancer, so the gene is no longer boosted. Or it can move an enhancer next to a different gene, so the wrong gene is boosted. The expression of both genes changes even though not one letter has been added or lost. That is Dr. Wong's point in his own words: *"you need to understand that some of these abnormalities can cause the loss of the genetic materials, but some does not."* Clinically, this is why a cytogenetics report showing an inversion is not automatically a finding of no consequence.

Q4. Why do we say replication is semi-conservative, and what would we expect to see if the conservative model were true instead?

A4. A semi-conservative copy means each new double helix contains one original strand and one newly synthesised strand. A conservative copy means the original helix stays completely intact and an entirely new helix is built beside it. The way to tell them apart is to label the original DNA and then follow the label through two rounds of copying. If replication is semi-conservative, the first round gives two hybrids, each with one old and one new strand, and the second round gives two hybrids plus two fully new molecules. If replication is conservative, the first round gives one completely old helix and one completely new one, and that old helix persists as a separate molecule in later rounds. Dr. Wong used the Meselson-Stahl-type isotopic labelling experiment to make exactly this distinction, and showed all three candidate models on a Lecture 4 slide. The reason it matters is that semi-conservative copying is what makes proofreading and repair possible in later rounds. Each new strand has an old strand to be checked against, and that is the basis of mismatch repair, which is a Lecture 4 topic and the basis of the MSI story.

Q5. Explain how a gene can be present and working in a cell but completely silent, and what changes that.

A5. Begin with the packaging, because silence is a packaging problem before it is a genetic one. DNA is wrapped around histone spools to make nucleosomes, and nucleosomes are packed into chromatin. Where the chromatin is tightly packed, the DNA is physically unreachable. Dr. Wong's phrasing was that heterochromatin physically blocks transcription factors and enhancers from reaching the promoter. So the gene is there and its sequence is perfectly intact. Nothing can get to it. Then the switch. The state is dynamic and reversible. Nucleosomes slide along the DNA to expose or cover the gene's promoter region, and two things drive the sliding: the intrinsic preference of a particular DNA sequence for a histone, and ATP-dependent chromatin remodelling complexes that move nucleosomes on purpose. Then the second layer of control. Even when the chromatin is open, the gene still needs the right proteins present to be transcribed, and that is the layer Lecture 5 is about. The clinical consequence is the one Dr. Wong developed in Lecture 3: a tumour can silence a tumour suppressor gene by methylating its promoter, and the brake is off even though not a single letter of the gene has changed.

CHAPTER 5

The cell and its parts

5.0 What this chapter is about

Everything in this course happens inside a cell. A biomarker is a molecule. A molecule is made or used by cells. A drug works on a cell. So before any pathway, mutation or therapy makes sense, you need a picture of what a cell is made of and what each part does. This chapter builds that picture from nothing. It starts with what a cell is, then the two basic designs of cell, then each internal compartment in turn. It ends with the dividing schedule of a cell, because that is the bridge into Lecture 2.

5.1 Words to know before we start

Term	In plain English
Cell	The smallest unit that is alive: a bag of water with a membrane, some DNA, and machinery to copy it.
Organelle	A small compartment inside a cell with one job, like a room in a factory.
Prokaryote	A cell with no nucleus. Its DNA floats loose in the cytoplasm. Bacteria are prokaryotes.
Eukaryote	A cell with a nucleus that holds the DNA. Animal, plant, fungal and yeast cells are eukaryotes.
Plasma membrane	The thin, flexible skin around the cell. It decides what may come in or go out.
Cytoplasm	The watery jelly that fills the cell, where most of the chemistry happens.
Nucleus	The control room. It stores the cell's DNA, the master instruction manuals.
Mitochondrion	The power station. It burns fuel with oxygen to make ATP, the cell's spendable energy.
Ribosome	The machine that builds a protein by joining amino acids in order.
Endoplasmic reticulum (ER)	The workshop beside the nucleus. Ribosomes on it fold and check new proteins.
Golgi apparatus	The packing room. It sorts finished proteins and posts them to their destination.
Lysosome	The recycling and disposal plant. It digests worn-out parts and engulfed germs.
Cytoskeleton	The rope-and-tube frame inside the cell. It gives shape, moves cargo, and splits the cell.
Cell cycle	The repeating schedule a cell follows to copy its DNA and then split into two.

5.2 The big picture

Picture a small factory. It has a boundary wall, a control room, a power station, a workshop, a packing room, a recycling bay and a delivery network. Everything the factory sells, it builds for itself, from its own drawings. That is a cell.

The wall comes first, because a factory needs to hold everything in. The **plasma membrane** is a flexible, oily skin built from **phospholipids** - molecules with a water-loving head and a water-hating tail. Put such molecules in water and they line up tail to tail, heads out, forming a **lipid bilayer** two molecules thick. That is why cell membranes form themselves without anyone assembling them. The membrane is also a filter and a set of doors. Some of those doors are **receptors**: proteins that sit in the membrane, catch one specific molecule from outside, and pass the news inward.

Inside is watery **cytoplasm**, where the chemistry of life happens. The control room is the **nucleus**, a separate compartment holding the cell's DNA. Think of DNA as the master instruction manuals - long printed documents written in only four letters. The manuals stay in the control room, and only authorised copies travel out. This separation is the whole difference between a **eukaryote**, which has a nucleus, and a **prokaryote**, a bacterial cell with no nucleus at all, which keeps its DNA loose in the cytoplasm and copies a recipe and builds the part in the same spot.

The **ribosomes** are the machines on the factory floor. Each one reads a copy of one recipe, an **mRNA** (messenger RNA) - a photocopy sent to the factory floor, sent out of the control room - and strings **amino acids** together to make a **protein**. A protein is a folded machine made from a bead necklace. The folding is what gives it its job.

The **endoplasmic reticulum (ER)** is the workshop next to the control room. Ribosomes sit on its rough surface, and new proteins are folded, checked and sometimes decorated with sugar groups (**glycosylation**) there. The **Golgi apparatus** is the packing room downstream: it sorts finished proteins, puts them into **vesicles**, and posts them either to the cell surface or out of the cell entirely. The **lysosomes** are the recycling and disposal plant - acidic bags full of enzymes that digest worn-out cell parts and engulfed bacteria. The **cytoskeleton** is the frame, the lifting gear and the delivery network of ropes and tubes. It also pins the cell membrane, moves cargo around, and pulls the cell in two.

Then there is the **mitochondrion**, the odd one out. It is the power station, and it is the reason Dr. Wong can talk about the **Warburg effect**, the strange habit cancer cells have of ignoring it. A mitochondrion also keeps its own small circle of DNA, about 17 kilobases carrying 37 genes. That happens because a mitochondrion began life as a free-living bacterium that was swallowed and kept. Because of that history, it is inherited from the mother only.

Finally, the whole factory runs on a schedule. The **cell cycle** goes G1, S, G2, M and then cytokinesis: grow, copy the DNA, check the copy, split the chromosomes, cut the cell in two. Every later topic in this course - cancer, stem cells, immune killing, gene therapy - happens at one of those steps.

5.3 Walkthrough

5.3.1 The smallest unit of life - a cell

WHAT IT IS	A cell is a tiny bag of water wrapped in a membrane, with genetic instructions inside and machines to read them. It takes in raw material, makes its own parts, and does a job. It reproduces by copying itself. Nothing smaller than this counts as alive.
HOW IT WORKS / WHY IT MATTERS	Two things make a cell a cell. First, it has a membrane, so it keeps its contents together and can control what crosses. Second, it has DNA, so its parts can be specified by instruction rather than by chance. Human cells are about 30 micrometres across. A human hair is about 50 to 100 micrometres wide, so two or three human cells would sit across it.
EXAM HOOK	He did not flag "what is a cell" as examinable. He did make it the opening move of Lecture 2, where he says diseases are diagnosed and targeted at the cellular level, and where isolating the right cell population is half the battle.

5.3.2 Two designs of cell - prokaryote and eukaryote

WHAT IT IS	A **prokaryote** has no nucleus. Its DNA sits as a loop in the cytoplasm, in a region called the **nucleoid**. Bacteria and archaea are prokaryotes. A **eukaryote** has a real nucleus, with the DNA inside, plus other membrane-bound rooms such as mitochondria and the Golgi apparatus. Animals, plants, fungi and yeasts are eukaryotes.
HOW IT WORKS / WHY IT MATTERS	The compartment changes the whole workflow. In a prokaryote, copying a recipe and building the part happen in the same place at the same time. In a eukaryote, the recipe is copied in the nucleus, edited, and then shipped out to the cytoplasm for building. That two-stage system is what allows eukaryotes to splice, edit and regulate their messages, and it is why human cells can make many different proteins from one gene. It is also why bacteria are so hard to make into medicine factories: they lack the machinery for folding and decorating a complex human protein.
EXAM HOOK	He used exactly this contrast in Lecture 5, topic 2, when he set up prokaryotic operons against eukaryotic gene control. He did not ask students to memorise the organelle list.

5.3.3 The plasma membrane - the front door and the sensor

WHAT IT IS	The **plasma membrane** is the cell's outer skin: a **lipid bilayer** with proteins floating in it. It is fluid, not rigid. It decides what enters and what leaves.
HOW IT WORKS / WHY IT MATTERS	Two features carry most of the clinical weight. First, it is a barrier, so a cell can hold a high internal concentration of something that would otherwise diffuse away. Second, membrane proteins do the talking. **Receptors** in the membrane bind one specific signal molecule and start a chain of events inside. If a tumour copies a receptor gene many times, the cell ends up with far more of that receptor on its surface - this is exactly what happens to **HER2** in breast and gastric cancer, and it is why an antibody that blocks that receptor works. Membrane proteins are also prime drug targets because a drug can reach them from outside the cell.
EXAM HOOK	He did not flag membrane structure itself. He did make the clinical version exam-relevant: HER2 amplification on the cell surface, and receptor occupancy as the parameter that decides dosing. He stated the occupancy thresholds as 1 % for a typical antibody, 2 % for Herceptin in breast cancer, and 10 % for Herceptin in gastric cancer. Plain meaning: the same drug needs to grab far more of the target in the stomach than in the breast before it does anything.

5.3.4 The nucleus - the control room and the master manuals

WHAT IT IS	The **nucleus** is a membrane-bound room at the centre of the cell. It holds the cell's **DNA** packed into **chromosomes**. Around the DNA is the **nuclear envelope**, pierced by **nuclear pores** so that messages can leave and proteins can enter.
HOW IT WORKS / WHY IT MATTERS	Because DNA is kept separate, the cell can control what gets read and when. The cell decides which genes to open, copies only those, and ships the copies out as mRNA. When something goes wrong in the nucleus, the consequences are large. A **mutation** is a typo in one of the manuals. A **driver** is a typo in an important word that gives the cell a growth advantage. A **passenger** is a typo on page 900 that travels along for free. Drivers cause cancer; passengers are noise. The nucleus is also where a virus such as hepatitis B keeps a permanent reservoir of its own DNA, and where that viral DNA can land next to a human gene and switch it on.
EXAM HOOK	He did not flag nuclear structure. The nuclear stories he did flag are the driver-versus-passenger distinction and the mutant-versus-wild-type comparison for K-Ras, where he told you that K-Ras was historically called undruggable and now has an FDA-approved small-molecule inhibitor. He also stated that p53 mutations are found in about 40 % of cancers. Plain meaning: the cell's own brake fails in roughly two tumours out of every five.

5.3.5 Mitochondria - the power station that keeps its own manual

WHAT IT IS	A **mitochondrion** is a bean-shaped **organelle** with two membranes. The inner membrane is folded into **cristae**, which is where the energy machinery sits. Inside the **matrix** is the mitochondrion's own DNA, a small circle of about 17 kilobases carrying 37 genes.
HOW IT WORKS / WHY IT MATTERS	The mitochondrion runs the **TCA cycle** and **oxidative phosphorylation**, which together make most of the cell's **ATP**. That is why cells with high energy demand - muscle, heart, brain, eye, hearing - suffer first when mitochondria fail. It is also the control centre of **apoptosis**, the cell's own death programme: the outer membrane decides whether the cell lives or dies. Three facts about mitochondria matter clinically, and Dr. Wong treats all three as exam content. First, **endosymbiotic origin**: the mitochondrion is a bacterium that was engulfed and kept. Second, **maternal inheritance**: an affected mother passes it to all her children, an affected father passes it to none. Third, the **threshold effect**: a cell can carry mutant mitochondrial DNA and still be healthy, and disease appears only when the proportion of mutant DNA exceeds what that tissue can tolerate. Tissues with high energy demand have a low threshold, so they fail first.
EXAM HOOK	This is Tier 1, his own words: *"this is you'll have to a 100 % understand, right? Not only the mutation within the mitochondrial DNA can cause the mitochondrial disease. Some mitochondrial disease can actually result from the mutation within the nuclear DNA."* And: *"Just make sure you understand that different mutation can cause the mitochondrial diseases, and all this mutation can cause the different inheritance patterns when you analyze this pedigree tree."* Plain meaning: if a pedigree looks mitochondrial, do not stop there - a nuclear gene can mimic it, and the inheritance pattern will be different.

5.3.6 Ribosomes - the machines on the floor

WHAT IT IS	A **ribosome** is a small machine made of RNA and protein. It reads an mRNA and joins **amino acids** together in the order the message specifies.
HOW IT WORKS / WHY IT MATTERS	The ribosome is where the language of four letters becomes the physical world of folded machines. Read every three letters, and you get one amino acid; that three-letter unit is a **codon**. Start codon AUG means methionine and marks the beginning. The stop codons UAA, UAG and UGA mark the end. The ribosome is also a deliberate target in modern medicine: an mRNA vaccine delivers a message and the patient's own ribosomes build the antigen; an in-vivo CAR-T approach delivers mRNA in **lipid nanoparticles** and the patient's T-cells make the receptor on their own surface.
EXAM HOOK	He asked you to remember the start and stop codons on the Lecture 1 slide. He also excluded the amino-acid table: *"don't memorize this... definitely not include in the exam."* Plain meaning: you need the idea of the genetic code, not the full list of 64 codons.

5.3.7 The endoplasmic reticulum and the Golgi apparatus - workshop and packing room

WHAT IT IS	The **endoplasmic reticulum (ER)** is a network of folded membranes attached to the nucleus. The **rough ER** has ribosomes stuck to it. The **smooth ER** makes lipids. The **Golgi apparatus** is a stack of flattened sacs further downstream.
HOW IT WORKS / WHY IT MATTERS	This is the cell's production line for anything leaving the cell. A protein with an **N-terminal signal peptide** is recognised by the ribosome as it is made, and is fed into the rough ER. There it is folded, checked, and often given **glycosylation** - decorated with sugar groups. A transport vesicle then carries it to the cis face of the Golgi, through the cisternae, out of the trans face, and into a secretory vesicle that fuses with the plasma membrane to release the protein outside. That whole route is the **secretory pathway**. It is the route used for antibodies, for hormones and for many medicines made by cells. He makes the point forcefully: remove the signal peptide and the protein never reaches the ER at all, so it never gets made properly, and your experiment produces nothing. The signal peptide also decides whether a protein ends up on the cell surface, inside a lysosome, or secreted into blood.
EXAM HOOK	Tier 2, his words: *"So remember, signal peptide is really, really important for your study."* He told the story of a colleague who lost four months of work by omitting it. He also warned against confusing it with the **N-terminal domain (NTD)** of a nuclear receptor, which is a completely different thing: that one is a regulatory region on a protein that binds DNA, not a postal address for the ER.

5.3.8 Lysosomes - the recycling and disposal plant

WHAT IT IS	A **lysosome** is a membrane-bound sac with a very acidic inside, packed with **enzymes** that break down large molecules.
HOW IT WORKS / WHY IT MATTERS	Lysosomes are where the cell recycles. They digest worn-out organelles, clear protein clumps, and destroy bacteria the cell has swallowed. They are also the recyclers in **autophagy**, the self-eating process in which a starving cell - for example one in the middle of a solid tumour with no blood supply - digests its own unneeded parts to get nutrients. It is a survival trick, and it is one reason a tumour core can keep growing in a hostile place. The failure mode is straightforward: if the recycling plant breaks, material accumulates, and the cell gets sick.
EXAM HOOK	He taught autophagy in Lecture 2, topic 9, as a survival mechanism in the hypoxic tumour core. He did not flag lysosomal storage diseases as examinable.

5.3.9 The cytoskeleton - the frame, the freight system and the dividing machine

WHAT IT IS	The **cytoskeleton** is a network of protein fibres through the cytoplasm. Three kinds matter: **microtubules** (hollow tubes), actin filaments (thin ropes) and intermediate filaments (rope-like, for strength).
HOW IT WORKS / WHY IT MATTERS	The cytoskeleton does four jobs. It gives the cell its shape and holds the membrane in shape. It acts as tracks and motors to move cargo around the cell. It builds the **mitotic spindle**, the ropes that pull duplicated chromosomes apart during division. And it forms the **contractile ring**, a tightening band of **actin** and **myosin** that pinches the cell in two during **cytokinesis**. The cytoskeleton also carries information: the Hippo pathway in Lecture 1 is switched by F-actin, the cytoskeleton branch, so the cell's shape and its growth-control pathway are physically linked.
EXAM HOOK	The contractile ring is Tier 1, his own words: *"Again, like I - I remember that I asked this question during the midterm in the first year, yeah. I think only half student can answer it correctly... So again this is the type of the question that I would try to include in the midterm."* The question is what you see when you block cytokinesis. See 5.3.11 for the answer.

5.3.10 The cell cycle - how one cell becomes two

WHAT IT IS	The **cell cycle** is the repeating schedule that lets a cell copy itself. It has four stages plus a split. **G1** is the growth phase, where the cell makes proteins and organelles and does its work. **S phase** is where the DNA is copied, so the cell ends up with two identical copies of every chromosome. **G2** is the checkpoint phase, where the copy is checked before division. **M phase** is mitosis, the division of the nucleus, where the duplicated chromosomes are pulled apart. **Cytokinesis** is the division of the cytoplasm, which follows mitosis. Cells that are not dividing rest in a side branch called **G0**.
HOW IT WORKS / WHY IT MATTERS	The schedule is driven by a two-part switch: **cyclins** and **cyclin-dependent kinases (CDKs)**. A CDK is the machine that does the work. A cyclin is the key that switches it on. Each cyclin rises, switches on its matching CDK, and the pair drives the next step. That pairing is the control point drugs attack. **Palbociclib** blocks CDK4/6 and so stops cells leaving G1. You can read the cycle from the bench: stain the DNA with **propidium iodide (PI)** and run the cells on a flow cytometer. Cells in G1 show 2N DNA content, cells in S show anything between, and cells in G2 or M show 4N. To line the cells up at the same starting point, researchers starve them of serum by dropping fetal bovine serum from 10 % to 0.5-1 %.
EXAM HOOK	He excluded the detailed pairings: *"you don't need to memorize all this cyclin or cyclin-dependent kinase, CDK."* He did exclude the detailed signalling cascade on the slide too: *"I don't memorize all this thing, I just highlight here in case some student are really interested in molecular biology."* The concept he wants is the four-stage shape of the cycle and the idea that a drug can freeze the cell at one stage.

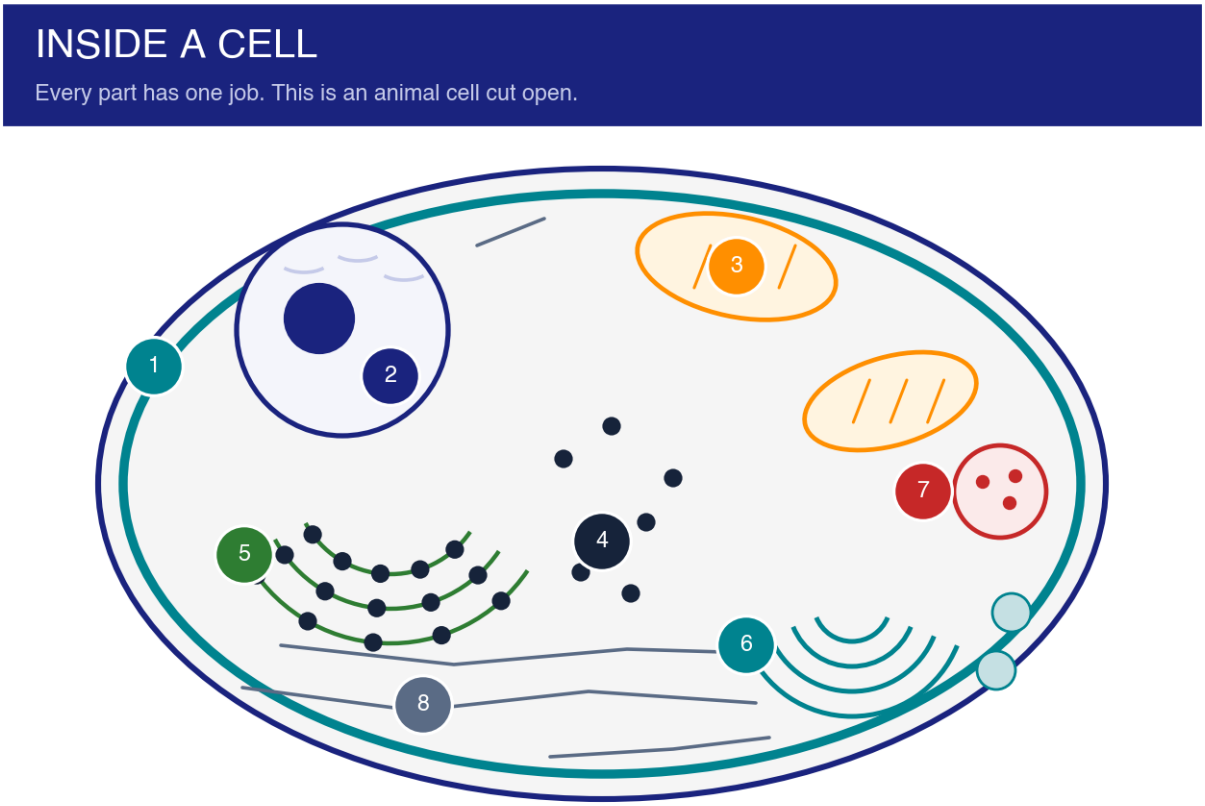
5.3.11 When each part fails - the clinical organelle inventory

WHAT IT IS	Every compartment has a failure mode, and each one has a disease attached to it. The table below is the summary. The first six rows are Dr. Wong's own material. The last three are ordinary background biology included because the chapter scope asks for it.
HOW IT WORKS / WHY IT MATTERS	Read the table as a set of cause-and-effect stories.

Part	What it normally does	What failure looks like	Where it comes from
Plasma membrane	Controls entry and exit; carries receptors	Too many receptors on a tumour cell, or a signal that says "keep growing" when it should say "stop"	Lecture 2, HER2 amplification and PD-L1
Nucleus	Stores DNA; decides which genes are read	Mutations; a driver that gives growth advantage; p53 mutations in about 40 % of cancers	Lectures 2 and 3
Mitochondrion	Makes ATP; controls apoptosis	Mitochondrial disease: energy-hungry tissues fail first, and inheritance follows the mother	Lecture 2, topics 3 and 5
Ribosome	Builds proteins from the message	The cell cannot make the proteins it is told to make	Background biology
Rough ER and Golgi	Fold, decorate, sort and post outgoing proteins	A secreted protein never reaches the cell surface; a folding fault makes the cell sick	Lecture 2, topic 4, on signal peptides
Lysosome	Digests and recycles	Material piles up because the recycling plant is broken	Background biology
Cytoskeleton	Shape, transport, spindle, dividing	The cell cannot pull itself apart; blocked cytokinesis leaves one cell with two nuclei	Lecture 2, topics 7 and 9
Centrosome and spindle	Organises the chromosomes for division	Chromosomes fail to separate properly	Lecture 2, slides 35 and 36
Cell-cycle control	Times the stages; stops when there is damage	Uncontrolled division - the definition of cancer	Lecture 2, topic 6, and p53 in about 40 % of cancers

EXAM HOOK	The contractile-ring question is the one he named first. Here is his own framing, and it is worth learning as a template: *"if you're a genetic consultant"* style questions aside, for this one he says: treat dividing cells with an inhibitor of the contractile ring, or knock out its expression. What do you see? The answer is that **mitosis** and **cytokinesis** are independent processes. Mitosis - the chromosome separation - proceeds normally, because it is done by microtubules, not by the contractile ring. Cytokinesis - the pinching of the cytoplasm - does not happen. So you get a **binucleated cell** - two nuclei and 4N DNA content - and on a flow-cytometry DNA histogram you see a 4N peak instead of separate 2N and 4N peaks. He added a consequence that explains a clinical observation: because organelles and mitochondria are partitioned at random during cytokinesis, two siblings with the same mutation can have very different severity.

Figure 5.1 - Inside a cell



A cut-away animal cell with every compartment named in plain words: the membrane that lets things in and out, the nucleus that keeps the master manuals, the mitochondrion that makes energy, the ribosomes that build protein, the ER workshop, the Golgi packing room, the lysosome recycling plant, and the cytoskeleton frame.

5.4 The 12 things to remember

1. A cell is alive because it has a membrane that holds it together and DNA that specifies its parts.
2. A prokaryote has no nucleus. A eukaryote does. Everything else follows from that one difference.
3. The plasma membrane is a phospholipid bilayer, and it is also a set of receptors and doors.
4. The nucleus is the control room. It stores DNA and decides which recipes get copied.
5. The mitochondrion is the power station, it makes most of the ATP, and it keeps its own DNA because it used to be a bacterium.
6. The ribosome reads mRNA and joins amino acids. Three letters make one amino acid: that is a codon.
7. Rough ER to Golgi to plasma membrane is the secretory pathway, and the signal peptide is the postal address that starts it.
8. Lysosomes are the recycling plant. Autophagy is the cell eating its own spare parts to survive starvation.
9. The cytoskeleton gives shape, moves cargo, builds the spindle and forms the contractile ring.
10. The cell cycle is G1, S, G2, M, then cytokinesis. Cyclins are the keys; CDKs are the machines.
11. Blocking the contractile ring gives one cell with two nuclei and 4N DNA, because mitosis and cytokinesis are separate.
12. Every organelle has a failure disease, and each failure points at a different clinical story.

5.5 Check yourself

Q1. A bacterium has no nucleus. Why does that make it hard to use a bacterium to manufacture a human antibody?

Q2. You add a drug that blocks the contractile ring to a culture of dividing cells. What do you see under the microscope, and what does the DNA-content plot from flow cytometry look like?

Q3. Mitochondria carry their own DNA. Where did that come from, and how does it show up in a family tree?

Q4. Put the compartments a secreted antibody passes through on its way out of the cell in the correct order.

Q5. A patient has a normal gene sequence but the protein never reaches the cell surface. Name two different faults that could cause this, and say which one a deleted signal peptide would cause.

Answers.

A1. Reasoning: a human antibody is a eukaryotic protein, and eukaryotes are separated into compartments. In a prokaryote there is no nucleus and no ER-Golgi secretory pathway, so the machinery that folds, assembles and glycosylates a complex protein is missing. Bacteria are excellent for making simple proteins and bad at making complex glycosylated ones. The clinical version of this is that mammalian cells, such as **CHO cells**, are used when glycosylation matters, and they need **codon optimisation** to read a human gene well.

A2. Reasoning: the contractile ring is actin plus myosin, and its job is **cytokinesis**, the division of the cytoplasm. Mitosis is done by microtubules and is independent of it, so chromosome separation still happens normally and each half gets 2N. But the cell cannot pinch. The result is a **binucleated cell**: one cell containing two nuclei, with 4N DNA content. On a PI-stained flow-cytometry histogram you see a single 4N peak rather than separate 2N and 4N peaks. Note the clinical ripple: mitochondria are partitioned randomly at cytokinesis, so siblings with the same mutation can differ in severity.

A3. Reasoning: the mitochondrion is the **endosymbiotic** case - it began as a free-living bacterium that an ancestral cell engulfed and kept. That is why it has two membranes, its own circular DNA of about 17 kilobases with 37 genes, and its own ribosomes. Because the cell's own machinery destroys paternal mitochondria after fertilisation, inheritance is **maternal**: an affected mother passes the condition to all her children, an affected father passes it to none, and it appears in every generation. The twist he insists on: mitochondrial disease can also be

caused by mutations in **nuclear** DNA, and those follow ordinary Mendelian patterns instead.

A4. Reasoning: the sequence runs cytosol to rough ER to Golgi to secretory vesicle to plasma membrane and out. Step by step: the ribosome on the **rough ER** makes the protein because the **signal peptide** addresses it there; the protein is folded, checked and given **glycosylation** in the ER - the sugar groups are added there; a **transport vesicle** carries it to the **cis face** of the **Golgi**; it moves through the cisternae to the **trans face**; a secretory vesicle buds off and fuses with the plasma membrane, releasing the antibody outside the cell. Note the order matters: glycosylation happens in the ER, sorting happens in the Golgi.

A5. Reasoning: there are two classic routes. First, the protein may be made but never delivered, if the **signal peptide** is missing or broken. That is the answer to the second half of the question: a deleted signal peptide means the protein is never recognised as belonging to the secretory pathway, never enters the rough ER, and is therefore never folded, glycosylated, sorted or exported. Second, the protein may be made and delivered but fail inside the cell, for example a folding fault in the **rough ER** or a sorting fault in the **Golgi**. A signal peptide deletion therefore blocks expression of secreted and membrane proteins entirely - his postdoc lost four months of work that way.

CHAPTER 6

How a cell actually works

6.0 What this chapter is about

Chapter 5 gave you the rooms. This chapter gives you the working day. It starts with **metabolism**, which is simply how a cell turns food into usable energy. Then it follows the information all the way from a gene to a working protein - the **central dogma** - using the factory line picture. After that it explains how a cell receives a message and switches a response on and off, how it decides whether to divide or to die, and how cells talk to each other and to the immune system. Almost every drug and every biomarker in this course is one of those jobs being interrupted.

6.1 Words to know before we start

Term	In plain English
Metabolism	All the chemical reactions in a cell. Breaking things down for energy, and building things up.
Glycolysis	Breaking glucose into smaller pieces without using oxygen. The first step of energy production.
ATP	The cell's spendable energy. Made in the mitochondria and used to power everything else.
Central dogma	The rule that information runs one way: DNA to RNA to protein.
Transcription	Copying one recipe from DNA into a working message (RNA).
Translation	Reading that message and building the protein it describes.
Codon	A three-letter unit on the mRNA. Each one names one amino acid.
Gene expression	The whole process of turning a gene on, making its message, and making its protein.
Enzyme	A protein that makes a chemical reaction go faster.
Kinase	A protein that attaches a phosphate group to another protein. Usually a switch.
Phosphatase	A protein that removes a phosphate group. It is the other half of the switch.
Receptor	A protein, often on the cell surface, that catches one specific signal molecule.
Signal transduction	The relay of a message from the outside of the cell to somewhere inside it.
Apoptosis	Programmed cell death. The cell dismantles itself on purpose.

6.2 The big picture

Start with fuel. A cell takes in sugar and fat and protein, and through **metabolism** it breaks them down to release energy, then rebuilds them into the cell's own parts. It needs raw material as well as energy, and in a cell that is about to divide, raw material matters more. One step in this breakdown is **glycolysis**: glucose is split into smaller molecules without oxygen. When oxygen is available, most cells go further and use the **mitochondrion** to burn the products properly, which yields far more **ATP**, the cell's spendable energy. ATP is the cell's money: it pays for building, moving, copying and signalling. Here is where the Warburg effect comes in. Cancer cells often choose glycolysis even when oxygen is available. On his Lecture 1 slide the numbers are stark: a resting cell with oxygen makes about 36 molecules of ATP per glucose, and without oxygen only 2. A proliferating cancer cell, with or without oxygen, sends about 85 % of its pyruvate to lactate, only about 5 % into mitochondria, and nets roughly 4 molecules of ATP. The inefficient route looks like a mistake. It is not. The half-broken sugar is perfect building material, and a cell about to divide a billion times needs building material far more than fuel.

Now the information half. The rule is the **central dogma**, and it runs one way: DNA to RNA to protein. Think of DNA as the master instruction manuals, locked in the control room. A **gene** is one recipe. Copying a recipe out is **transcription**, and the copy is **mRNA**, a photocopy of one recipe sent to the factory floor. Reading the photocopy and building the machine is **translation**. The machine is a **protein**: a folded machine made from a bead necklace. The ribosomes do the building. The mRNA is read in threes, and each three-letter **codon** names one amino acid.

Then the protein has a life of its own. It folds into a shape, and the shape is the job. It may be decorated with sugar groups in the ER, sorted in the Golgi, and shipped out through the secretory pathway. It may become a receptor, an enzyme, a structural piece, or a signal.

And the cell is listening the whole time. A **receptor** in the membrane catches a specific signal molecule. The message is relayed inward by **signal transduction**, usually through **kinases** that attach a phosphate group and **phosphatases** that remove it. That pair is a light switch. A phosphate group carries charge, so adding one changes a protein's shape and pushes it into an active state; taking it away turns the switch off. The cell can therefore respond in seconds and then recover.

Two huge decisions follow. First, divide or die. A cell with damaged DNA is supposed to stop, or to commit suicide by **apoptosis**. A tumour is a cell that ignored that instruction. Drugs exploit it: CDK4/6 inhibitors freeze the cell cycle, and bispecific antibodies that carry both a tumour marker and a death receptor trigger apoptosis only where the tumour is.

Finally, cells talk. They send **cytokines** to their neighbours, they display fragments of their own proteins on the surface as **MHC** so that immune cells can inspect them, and they receive instructions from the immune system. The most important conversation in modern cancer therapy is the checkpoint: **PD-L1** on the tumour cell binds **PD-1** on the T-cell and switches the T-cell off. An anti-PD-1 antibody blocks that handshake and switches the killer back on. That is why a tumour with many neoantigens, called a **hot tumour**, responds far better than a cold one.

6.3 Walkthrough

6.3.1 Metabolism - how the cell handles its food

WHAT IT IS	**Metabolism** is everything the cell does chemically. Two directions at once: **catabolism**, breaking big molecules down to release energy, and **anabolism**, building big molecules from small ones using that energy.
HOW IT WORKS / WHY IT MATTERS	Sugar is the main starting fuel. It is broken down by **glycolysis** into pyruvate, and then fate depends on oxygen. With oxygen and healthy mitochondria, pyruvate goes into the TCA cycle and oxidative phosphorylation, giving roughly 36 ATP per glucose. Without oxygen, only 2 ATP. But a cell also needs raw material, and the split-up sugar is a better building block than the burned-up version. This is exactly why cancer cells behave oddly, the **Warburg effect**, also called **aerobic glycolysis** because the fermentation happens despite oxygen being present. The tumour gains building material at the cost of efficiency.
EXAM HOOK	He used the Warburg effect as the entry point to Lecture 1 and as the reason metabolomics became a useful way to find cancer biomarkers. He did not ask you to memorise the pathway. He explicitly excluded it: *"Please don't memorize all these chemical structure... They would not be included in my exam."* Plain meaning: understand the trade-off, not the chemistry.

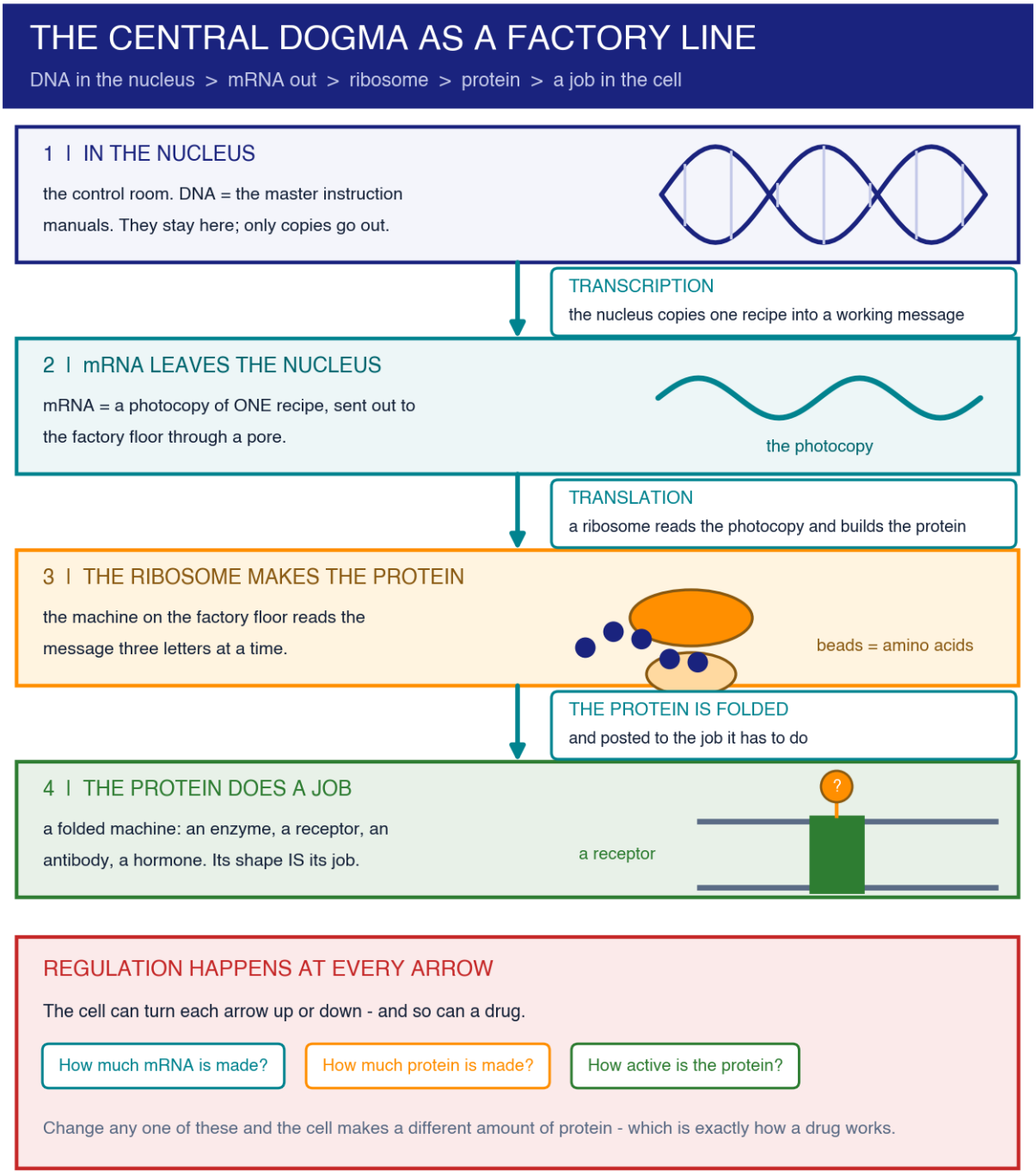
6.3.2 Energy - ATP, the cell's money

WHAT IT IS	**ATP** (adenosine triphosphate) is a small molecule with three phosphate groups in a row. The cell spends energy by breaking off the last phosphate, and it recharges ATP by putting it back.
HOW IT WORKS / WHY IT MATTERS	Every job the cell does is paid for in ATP. Moving molecules, building a protein, copying DNA, opening a channel. A mitochondrion is the biggest ATM: it takes the fuel from sugar and reloads the ATP. This is why the mitochondrion's failure is so dangerous, and why a tumour that switches off oxidative phosphorylation is really switching off its main ATM. It is also why **protein folding** costs the cell energy - the cell has to spend ATP to make the chain fold into the right shape, which is one reason a misfolded protein factory burns fuel and produces nothing.
EXAM HOOK	He stated the ATP yields on the Lecture 1 slide: about 36 molecules with oxygen, 2 without, and about 4 for a cancer cell running aerobic glycolysis. Plain meaning: the cancer cell is running at roughly one-tenth the energy efficiency of a normal aerobic cell, on purpose, because it wants the raw material. He did not ask you to derive these numbers.

6.3.3 The central dogma - a factory line with three arrows

WHAT IT IS	The **central dogma** is the direction rule for genetic information. It runs one way: DNA is copied into RNA, and RNA is read to build protein. Information does not flow backwards from protein to DNA.
HOW IT WORKS / WHY IT MATTERS	There are three arrows, and each one is a step where the cell can turn the output up or down. Arrow one, DNA to RNA, is **transcription**. Arrow two, RNA to protein, is **translation**. Arrow three is what the protein then does in the cell. Because the arrows are separate, each is a separate point of control, and each is a separate opportunity for a drug. This is the single most useful picture in the course, because every later lecture adds a new way of controlling one of these arrows.
EXAM HOOK	He taught the central dogma in Lecture 1, topic 13, and revisited it in Lecture 5, in the promoter and gene-structure topics. He did not list it among the five questions he named as exam-type, but it is the frame that every one of those questions sits inside.

Figure 6.1 - The central dogma as a factory line



DNA stays in the nucleus. A photocopy of one recipe, the mRNA, is carried out to the factory floor. A ribosome reads it and builds a folded protein. The protein then does a job in the cell. Every arrow is a point where the cell can turn the output up or down, and therefore every arrow is a point where a drug can work.

6.3.4 Transcription - making the photocopy

WHAT IT IS	**Transcription** literally means "writing across". An enzyme, **RNA polymerase**, reads along a gene on the DNA and builds a complementary RNA strand. In eukaryotes that first copy is **pre-mRNA**.
HOW IT WORKS / WHY IT MATTERS	The enzyme does not start anywhere. It is recruited to a **promoter**, a specific DNA sequence just upstream of the gene. On the Lecture 5 slide, a complex of general transcription factors assembles at the **TATA box** about 30 bases upstream, and RNA polymerase II is added. Control sequences called **enhancers** and **silencers** can sit thousands of bases away, and DNA-bending proteins loop them onto the promoter. So a gene's output depends on a small DNA word (the promoter) plus distant words (the enhancers) plus the proteins that read them. The pre-mRNA is then edited: **introns** are cut out by the spliceosome and **exons** are joined, which is **splicing**. If the cell chooses different exons, one gene makes several different proteins - **alternative splicing**.
EXAM HOOK	The lac operon is his worked example of transcription control in Lecture 5, topic 5: strong transcription needs lactose present, so that allolactose can bind the **Lac repressor** and lift it off the operator, and glucose low, so that cAMP rises and **CAP-cAMP** can recruit RNA polymerase. Two conditions, not one. He also made the point that *"not all the enhancer are located on the upstream of the sequence"* - some sit downstream of the gene.

6.3.5 Translation - building the machine

WHAT IT IS	**Translation** is the ribosome reading the mRNA and joining amino acids to make a chain. The code is read in threes: each **codon** names one amino acid, and the chain grows as the ribosome moves along the message.
HOW IT WORKS / WHY IT MATTERS	Two details matter for this course. First, the code is not universal in practice: different organisms prefer different codons for the same amino acid, which is called **codon-usage bias**. If you put a human gene into a hamster cell line without **codon optimisation**, the protein comes out badly or not at all. He made this point explicitly: *"So remember that when you use the CHO cell, you need to do the codon optimization."* Second, a mistake in the reading frame is catastrophic. If a mutation adds or removes a number of letters that is not a multiple of three, the reading frame shifts and everything downstream is wrong - that is a **frameshift**, and it usually produces a premature stop and a truncated protein.
EXAM HOOK	Frameshifts are his Lecture 1, topic 15, and they matter because they create **neoantigens** - protein fragments the immune system has never seen, which are ideal targets for immunotherapy. He also stated the start codon AUG (methionine) and the stop codons UAA, UAG and UGA.

6.3.6 What the protein does afterwards - the job

WHAT IT IS	A translated chain is not yet a working protein. It folds into a specific three-dimensional shape, and that shape is what lets it do one job.
HOW IT WORKS / WHY IT MATTERS	The chain is folded in the cytoplasm, or in the rough ER if it carries a signal peptide. **Chaperone** proteins help it fold correctly. Many proteins are then modified - **glycosylation** adds sugar trees, and **phosphorylation** adds phosphate groups. Some proteins are cut in two by enzymes, and that cut is the switch that activates them. The finished protein may end up as an **enzyme** that speeds a reaction, a **receptor** in the membrane, a **cytokine** that is released to signal, an **antibody** that binds a target, or a structural piece such as a part of the cytoskeleton. Its job, its location and its lifetime are all set by the journey.
EXAM HOOK	He excluded the chemistry of folding and the amino-acid R-group table from the exam, but the concept matters for everything else: *"don't memorize this... definitely not include in the exam."* What is examinable is the clinical consequence, which he taught in Lecture 1, topic 7: a single chemical change, such as **deamidation** of asparagine to aspartate at high pH, can alter how well a therapeutic protein binds its target and therefore change its efficacy.

6.3.7 Receiving a message - receptors, kinases and the on/off switch

WHAT IT IS	A **receptor** is a protein that recognises one specific signal molecule and changes shape when it binds. **Signal transduction** is the relay of that news through the cell to the nucleus.
HOW IT WORKS / WHY IT MATTERS	The relay is usually a chain of **kinases**, each one switching on the next by adding a phosphate group, and the switch is reversible because **phosphatases** remove it. The phosphate group carries charge, so adding it changes the protein's shape and its activity. That is the on/off switch. Two examples carry the whole course. First, the **Hippo pathway**: cell-to-cell contact activates MST1, which activates LATS1, which phosphorylates **YAP**; phosphorylated YAP is held in the cytoplasm, so the growth programme stays off. When the pathway breaks, unphosphorylated YAP enters the nucleus and drives growth. Second, **NF-kappaB**: it is normally held in the cytoplasm by **IkappaB**. When lipopolysaccharide (LPS) is detected, IkappaB is degraded, NF-kappaB moves into the nucleus, and the inflammatory genes are switched on. NF-kappaB then makes the inflammatory proteins that keep reinforcing it - a **positive-feedback loop** that makes the response fast. The order of the response is timed: **TNF-alpha** peaks at 4-6 hours, then IL-1beta, then IFN-gamma, then IL-2 and IL-15.
EXAM HOOK	The cascade order LPS to TLR4 to IKK to IkappaB to NF-kappaB to the nucleus is one of his published MCQ shapes. The exact cytokine order is explicitly excluded: *"don't worry about all these cytokines. I'm just want to show you that different responses can be observed in different times."* He did ask you to remember the Hippo phosphorylation logic, and he said the Hippo pathway is a subject of pharmacological targeting, with druggable inhibitors in development.

6.3.8 Deciding to divide, or to die

WHAT IT IS	A cell checks its work before it copies its DNA. If the DNA is damaged, it should stop and repair. If it cannot be repaired, it should die by **apoptosis**, the controlled dismantling programme.
HOW IT WORKS / WHY IT MATTERS	Apoptosis has two routes. The **intrinsic** route starts inside the cell, usually at the mitochondria: stress signals push the balance of the Bcl-2 family, cytochrome c is released, and a cascade of **caspases** is activated. The **extrinsic** route starts outside: a **death receptor** on the cell surface is triggered by its ligand, which starts the caspase cascade directly. Both routes converge on the same executioners, and the cell is taken apart in an orderly way and swallowed by its neighbours. When this fails, the cell keeps dividing and you have cancer. He noted that p53 mutations appear in about 40 % of cancers: the brake is broken. Two drug classes exploit the same logic. **CDK4/6 inhibitors** such as palbociclib freeze the cell in G1 and stop the cycle. A **bispecific antibody** is built to carry two recognition heads at once, one for a tumour-specific antigen and one for a death receptor, so that apoptosis fires only where the tumour is.
EXAM HOOK	He presented the bispecific antibody BI 905711 in Lecture 2 as the clean clinical example, using his own **CDH17** target: apoptosis happens only in tumour tissue, because only tumour tissue has both markers present. He also made apoptosis the connection to the **Warburg effect** and to mitochondrial control of cell death. He did not ask you to memorise the caspase cascade.

6.3.9 Talking to neighbours and to the immune system

WHAT IT IS	Cells communicate by sending molecules and by displaying molecules. They also physically touch, using adhesion molecules to grab each other and to let immune cells slip through tissue.
HOW IT WORKS / WHY IT MATTERS	Three mechanisms carry the clinical weight. First, **cytokines** are released into the space between cells and bind receptors on their targets; the NF-kappaB programme makes a whole cascade of them in a timed sequence. Second, **MHC** molecules on the cell surface display short protein fragments, so that T-cells can inspect what a cell is made of; this is how a mRNA vaccine works - the delivered message is translated, the protein is chopped into peptides, and the peptides are presented for T-cell and B-cell activation. Third, **cell adhesion molecules** control movement through tissue. On his Lecture 2 slide the white-blood-cell journey is in four steps: rolling, using **P-selectin**, **E-selectin** and **integrins**; firm adhesion through **ICAM-1** and **VCAM-1**; squeezing between endothelial cells through **PECAM-1**; and then following a chemical gradient, **MCP-1**. On top of that sit the **immune checkpoints**. **PD-L1** on the tumour cell binds **PD-1** on the T-cell and switches the T-cell off into a state called anergy. The tumour is invisible. An anti-PD-1 antibody blocks that handshake and puts the killer back to work.
EXAM HOOK	He tied the checkpoint story to the **hot tumour** versus **cold tumour** idea: *"So guys, you definitely remember that for the tumor that present many neoantigen and can be treated with the immunotherapy, we call this is a hot tumor, yeah. And not all the solid tumor is a hot tumor."* A tumour with defective mismatch repair accumulates frameshift mutations, produces many neoantigens, becomes hot, and responds to anti-PD-1. He called the dMMR-versus-pMMR survival figure one of the most important figures in modern medicine: *"I can for sure that the presenter will include this figure in his presentation deck."*

6.4 The 10 things to remember

1. Metabolism has two halves: breaking down for energy and building up for new parts.
2. ATP is the cell's money. The mitochondrion is the biggest ATM.
3. The Warburg effect is a strategy, not a mistake: the inefficient route also supplies building material.
4. The central dogma runs one way: DNA to RNA to protein.
5. Transcription copies a recipe out of the nucleus. Translation builds the machine on the factory floor.
6. One gene can make several proteins, because splicing can join exons in different combinations.
7. A signal is relayed inward by kinases and switched off by phosphatases. The phosphate group is the switch.
8. A damaged cell should stop dividing or die by apoptosis. A tumour is a cell that ignored that.
9. Cells talk with cytokines, with MHC display, and with physical adhesion.
10. PD-L1 on the tumour switches the T-cell off; an anti-PD-1 antibody switches it back on. That is why hot tumours respond and cold ones do not.

6.5 Check yourself

Q1. Put the three steps of the central dogma in order and describe each one in factory language.

Q2. A cancer cell has plenty of glucose and plenty of oxygen. Why might it still choose to ferment rather than use its mitochondria?

Q3. What is the difference between a kinase and a phosphatase, and why is that difference so useful to a cell?

Q4. A tumour cell keeps dividing when it should not. Name the two ways a cell normally stops dividing or dies, and say which drug class acts on each.

Q5. A tumour cell has a lot of PD-L1 on its surface. What happens to a T-cell that meets it, and what drug reverses that?

Answers.

A1. Reasoning: the direction rule is DNA, then RNA, then protein. Step one, **transcription**: inside the nucleus, RNA polymerase copies one recipe out of the DNA as mRNA, a photocopy sent to the factory floor. Step two, **translation**: a ribosome on the factory floor reads the photocopy three letters at a time and joins amino acids into a chain, which folds into a working machine. Step three, the job: the finished protein acts - as an enzyme, a receptor, an antibody, a cytokine, or a structural part. Each step is a separate control point, which is why each is a separate drug target.

A2. Reasoning: because efficiency is not the goal. Fermenting glucose to lactate gives only about 4 ATP per glucose instead of about 36, but it leaves the half-broken sugar molecule behind as raw material for nucleotides, lipids and amino acids. A cell that is about to divide many times needs far more building material than fuel, so it switches to the **aerobic glycolysis** of the **Warburg effect** even with oxygen present. This is a deliberate growth strategy, and it is also why metabolomics became a useful way to find cancer biomarkers.

A3. Reasoning: a **kinase** adds a phosphate group, using ATP; a **phosphatase** removes it. The phosphate group carries charge, so adding it changes the protein's shape and pushes it into an active state, and taking it away turns it off again. The usefulness is speed and reversibility: the cell can respond to a signal within seconds and then reset within seconds, which a permanent change could not allow. The **Hippo pathway** is the model: MST1 activates LATS1, which phosphorylates **YAP**, and phosphorylated YAP is trapped in the cytoplasm so the growth programme stays off; phosphatase activity flips that decision.

A4. Reasoning: the two ways are to stop dividing and to die. Stopping is done by the cell-cycle controls: **CDK4/6 inhibitors** such as palbociclib freeze the cell in G1, and this is exactly the drug class he describes in the cell-cycle slide. Dying is done by **apoptosis**, triggered either from inside the cell via the mitochondria or from outside via a

death receptor, and both routes converge on caspases. The **bispecific antibody** acts on the second route: it carries two heads, one for a tumour-specific antigen and one for a death receptor, so apoptosis fires only in tissue that has both - his BI 905711 example built on CDH17.

A5. Reasoning: **PD-L1** on the tumour cell binds **PD-1** on the T-cell, and that handshake switches the T-cell off into anergy. The immune system can see the tumour but will not attack it, which is immune evasion. An **anti-PD-1** antibody blocks the handshake and releases the brake, restoring the T-cell's killing. Whether that works depends on the tumour: a **hot tumour**, loaded with neoantigens because its mismatch repair is broken, responds; a cold one does not. He called the dMMR-versus-pMMR survival figure one of the most important figures in modern medicine.

CHAPTER 7

The plain-English glossary

7.0 Where the glossary is

The complete glossary for this book - every technical term, in one alphabetical list, with a plain-English meaning and the chapter it belongs to - is in **Part 7, section 13.1 (Master glossary)**. It is the only glossary you need. Use this chapter's few lines to get oriented, and Part 7.1 whenever a term stops you mid-sentence.

End of Part 1 Foundations, Chapters 5 to 7. Part 2 begins with Lecture 1, Introduction to Biomolecules.

The complete glossary for the whole book - every technical term, in one alphabetical list, with a plain-English meaning and the chapter it belongs to - is in **Part 7, section 13.1 (Master glossary)**. It is the only glossary you need; you only need to reach for it if you want to revise the foundations vocabulary before you start the lectures.

PART 2

Lecture 1: Introduction to Biomolecules

IN THIS PART

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About this part

Part 2 - Lecture 1 - Introduction to Biomolecules - Dr. Tony Wong - 12 September 2026

Course: GNBF5020 Introduction to Molecular Biology and Genetics - **Instructor:** Dr. Tony Wong

This chapter is Part 2 of the beginner guide. It rewrites the expert shorthand of the existing revision guide so that it can be read by someone who has never studied biology or chemistry. Every number Dr. Wong stated is reproduced exactly and is followed by a plain sentence saying what it means. Where he said nothing about a topic, the Exam hook says so.

CHAPTER 8

Introduction to Biomolecules

8.0 What this lecture is about

Dr. Tony Wong opens the course with the molecules that everything in a cell is built from: sugars, fats, proteins, and DNA and RNA. He then shows what happens when a cell's chemistry goes wrong in cancer, and how a doctor can read that wrongness out of a blood sample. The lecture is really about two jobs - finding a molecule that tells you what is happening inside a patient, and turning that molecule into a treatment decision. By the end of this chapter you should be able to explain, in plain English, the sixteen ideas that the rest of the course keeps reusing.

8.1 Words to know before we start

Term	In plain English
Biomolecule	A molecule that a living cell makes or breaks down. There are four big families: carbohydrates, lipids, proteins, and nucleic acids.
Metabolite	A small molecule the cell makes or consumes while it works. Sugar, lactate and many drugs are metabolites.
Enzyme	A machine that makes a reaction go faster. Usually a protein. Its name usually ends in "-ase".
Antibody	A Y-shaped protein that catches one specific target, the way a hook catches one particular key.
Biomarker	A molecule you can measure in blood to see what the body is doing.
Genotype and phenotype	Genotype = the genes you inherited. Phenotype = the traits those genes actually produced, such as a tumour, an enzyme level, or an outcome.
Gene	One recipe in the DNA. A stretch of four-letter code that makes a protein or a functional RNA.
Protein	A folded machine made from a bead necklace of amino acids. The shape decides what it does.
Membrane and bilayer	The cell's skin. A bilayer is a double layer of lipids that keeps the inside in and the outside out.
Phosphorylation	Adding a small phosphate group to a protein. It is a switch, because the phosphate can be taken off again.
Kinase and phosphatase	The kinase adds the phosphate; the phosphatase takes it away. Together they make a fast on/off switch.
DNA and RNA	DNA is the recipe book kept in the nucleus. RNA is a working copy of one recipe, sent out to the factory floor.
Mutation	A typo in the DNA letters. A typo in an important word can cause cancer.
Survival curve	A graph, called a Kaplan-Meier curve, showing what fraction of a group of patients is still alive over time.

8.2 The big picture

Start every biology course with a factory. A cell is a tiny factory that never closes. It takes in raw material, turns it into building parts, runs itself on chemical energy, copies its own instruction manuals, and ships finished machines out of the door. Everything that happens inside is chemistry. The chemicals involved are called biomolecules - molecules that living things make or break down.

Dr. Wong opens Lecture 1 by sorting these into four families, and the whole course is built on them. Carbohydrates (sugars) are fuel and storage. Lipids (fats) build walls and store energy. Proteins are the machines that do everything. Nucleic acids (DNA and RNA) are the recipe books and the working copies. Part 1 of this guide already taught you the chemistry and the shape of all four. This lecture does something different with them: it asks what happens when the chemistry goes wrong, and how a doctor can read that wrongness out of a drop of blood.

The word you will meet most often is biomarker. A biomarker is a molecule you can measure in blood to see what the body is doing. It can be a sugar, a fat, a protein, or a piece of RNA. What matters is that its level rises or falls

with a disease. Three flavours matter here. A diagnostic biomarker tells you whether the patient has the disease right now. A prognostic biomarker tells you how the disease will go, whatever treatment you give. A predictive biomarker tells you which drug will work for this particular patient.

The lecture's story runs like this. Cancer cells behave oddly with fuel: they break sugar down without oxygen even when oxygen is available. That is the Warburg effect. The reason is not stupidity. The half-broken sugar molecule is more useful to a dividing cell as building material than as energy, so the "inefficient" switch is really a growth strategy. That single observation opens the door to metabolomics, the study of all the small molecules in a sample, and metabolomics is how a great many biomarkers get found.

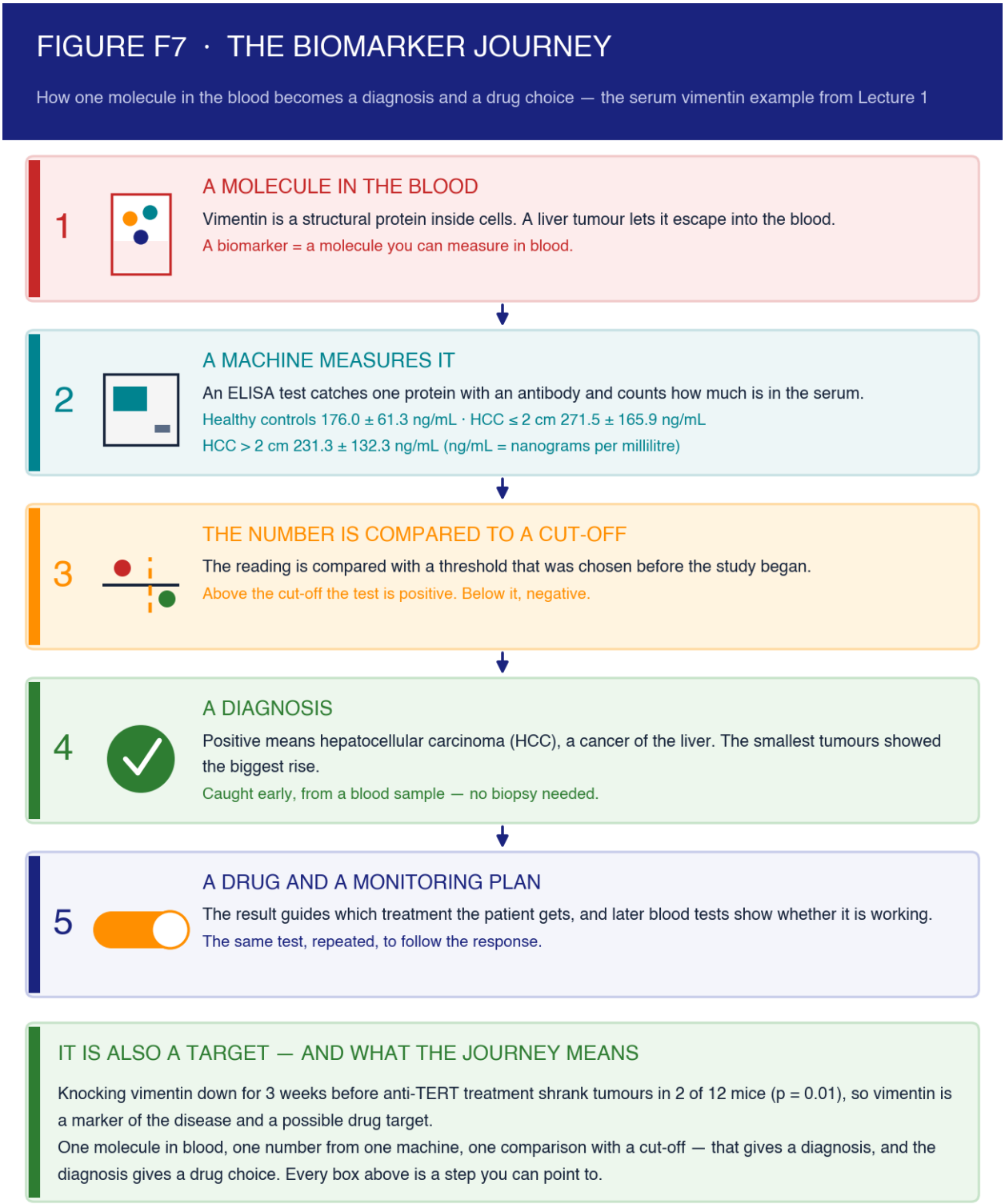
Lipids turn out to do double duty, because they both build membranes and, in the case of bacterial LPS, trigger the inflammation that kills patients in sepsis. Proteins are the busiest family of all. They fold into four levels of shape, they can be switched on and off by gaining or losing a phosphate, and the immune system uses them as guided missiles, because one antibody shape catches exactly one target. That is also why engineers bolt poison onto antibodies to make antibody-drug conjugates.

The second half of the lecture moves to DNA and RNA, where the story turns into inherited typos, into the K-Ras switch that gets stuck on and cannot be switched off, and into small regulatory RNAs that are stable enough in blood to be measured in a routine blood draw.

Two threads run through everything. The first is biomarker discovery: find a molecule that tracks the disease, then validate it in the right samples before you are allowed to call it a test. The second is target nomination: find a molecule the disease depends on, then find a drug that hits it. Most of the numbers in this chapter belong to one thread or the other.

Figure F7 - The biomarker journey

The biomarker journey: a molecule in blood, measured by a machine, compared to a threshold, becomes a diagnosis and then a drug choice.



The biomarker journey, drawn with the serum vimentin example from Lecture 1. Every step below in this chapter is one box in this picture.

8.3 Walkthrough

8.3.1 What the course is about, and what a biomolecule is - Biomolecules

WHAT IT IS	A biomolecule is a molecule that a living system makes or breaks down. Dr. Wong splits them into four families: carbohydrates, lipids, proteins, and nucleic acids. Each family does a different job in the cell. Together they cover everything from fuel and structure to the machinery that reads and copies DNA.
HOW IT WORKS / WHY IT MATTERS	The course goal he states is to translate genomics and molecular biology into clinical and real-world applications. The clinical application he keeps coming back to is the biomarker: a molecule you can measure in blood to see what the body is doing. He separates biomarkers into three kinds and you must keep them apart. A diagnostic biomarker identifies disease now. A prognostic biomarker predicts how the disease will go, independent of treatment. A predictive biomarker predicts the response to a particular therapy. The whole lecture is a series of examples of one or the other. He also frames the course with two running themes, diversity (genetic variation, antibody variability, alternative splicing, meiosis) and stability (DNA repair, epigenetic memory, checkpoints). A third framing point is that the switch from genotype to phenotype is carried by biomolecular processes: **transcription**, **translation**, and their regulation.

The two themes are not decoration. Diversity is everywhere in biology: different individuals differ in their genes, the immune system shuffles antibody genes to make billions of variants, a cell can splice the same message in different ways, and meiosis shuffles chromosome halves at every generation. Stability is the counterweight: a human body copies its DNA tens of billions of times and very rarely makes a mistake that matters, repairs what does break, and passes chemical instructions on to daughter cells so they behave like the parent cell did. Almost every question in this course sits on one side of that balance or the other.

EXAM HOOK	He was explicit about the shape of the assessment and about the language. His words: "multiple choice is for you to pass the exam. But if you want to get a full mark or a better result, you'd better perform well for the short questions, which emphasize on data analysis and also the scenarios questions." He named the two scenario-question families: "if you're a genetic consultant and I give you a medical report, how you would suggest the client whether she or him will get a baby with a certain genetic diseases? Or... I give you some survival data, and then you tell me that how this survival data can really benefit the treatment of the patient." He also set the weights: the midterm is 30 % of the final grade, the final exam is 40 %, and two "super-easy" assignments make up the other 30 %. He said, more than once, that everything is answered in English, and once that he had no choice but to give zero marks to students who answered in Chinese the previous year. The lecture content itself, as opposed to the framing, was not flagged as an examinable topic in its own right.

8.3.2 Sugars: fuel, storage, and structure - Carbohydrates

WHAT IT IS	Carbohydrates are the sugar family of biomolecules. The smallest unit is a **monosaccharide**, such as glucose or fructose. Two of them joined together make a **disaccharide**, such as sucrose. Long chains of them make **polysaccharides**: **glycogen** in animals, **starch** in plants, and **cellulose** in plant cell walls. Part 1 Chapter 2 already gave you their chemistry, so here we only need the job each one does.
HOW IT WORKS / WHY IT MATTERS	Three jobs. Carbohydrates are the primary energy currency: glucose is what cells spend to pay for work. They are the storage form: glycogen is how the liver and muscle bank glucose for later. And they are structural: cellulose is the scaffolding of plant cell walls. Starch is the plant version of the storage idea, and glycogen is the animal version. The disease example he uses shows the job in reverse. **Glucose-6-phosphatase** is the enzyme that releases free glucose from glycogen, and it exists only in the liver and the kidney. If a person lacks it, the liver cannot hand out glucose during a fast, and the body runs into an energy crisis. A **hepatocyte** is simply a liver cell, and the liver is the body's glucose bank: it holds the glycogen, and it is the organ that decides how much glucose to release into the blood when the body needs fuel. The slide maps the route: glycogen is broken down by glycogen phosphorylase to glucose-1-phosphate, phosphoglucomutase converts it to glucose-6-phosphate, and glucose-6-phosphatase finally frees the glucose. The alternative route runs glucose-6-phosphate into glycolysis, then to pyruvate, and pyruvate can go to lactate or on to acetyl-CoA. He also notes that glucose and fructose share the same formula, $C_6H_{12}O_6$, even though their ring shapes differ: glucose sits in a six-membered ring (a pyranose), fructose in a five-membered ring (a furanose). Glycogen's shape is a branched chain, with alpha1,4 **glycosidic bonds** along the chain and alpha1,6 bonds at the branch points, and the branching is functional rather than decorative: every branch end is another place an enzyme can start, so glucose can be released quickly when it is needed instead of only after the whole chain has been worked through.
EXAM HOOK	He said this will not be examined in detail. His words: "Please don't memorize all these chemical structure... They would not be included in my exam." Metabolic pathway details were on the same exclusion list. So do not memorise the pathway, the formulas, or the ring structures. Do understand the three jobs and the logic of the glucose-6-phosphatase example, because that is the idea, not the detail.

8.3.3 Cancer's odd fuel habit - the Warburg effect

WHAT IT IS	The **Warburg effect** is the observation that cancer cells convert pyruvate into **lactate** even when oxygen is plentiful. Breaking sugar down without using oxygen is called **aerobic glycolysis**: aerobic means "with oxygen available", glycolysis means "splitting sugar". Otto Warburg noticed it first, hence the name.
HOW IT WORKS / WHY IT MATTERS	The obvious objection is that this is wasteful, because burning sugar with oxygen makes far more energy. His slide quantifies it. A resting normal cell with oxygen makes about 36 **ATP molecules per glucose** by **oxidative phosphorylation**; without oxygen it manages only 2 **ATP molecules per glucose** by **anaerobic glycolysis**. A proliferating cancer cell, with or without oxygen, sends about 85 % of its pyruvate to lactate and only about 5 % into the mitochondria, because **pyruvate dehydrogenase** is held back by pyruvate dehydrogenase kinases. The result is about 4 ATP molecules per glucose - even worse than the oxygen-starved normal cell. The reason the tumour likes it is that fermentation leaves the half-broken sugar molecule behind, and those leftovers are perfect building material for nucleotides, lipids, and amino acids. A cell that is about to divide many billions of times needs raw material far more than it needs fuel, so the "inefficient" switch is really a growth strategy. This is the entry point to the next topic: a metabolic habit that differs between normal and cancerous tissue is exactly the kind of difference a biomarker can pick up.
EXAM HOOK	He did not ask you to memorise the glycolysis pathway. In fact metabolic pathway details are on his exclusion list, and his words were: "Please don't memorize all these chemical structure... They would not be included in my exam." What he did emphasise is the concept: cancer prefers the pathway that supplies building blocks, and that is why lactate and other metabolites became interesting as biomarkers. The ATP figures on the slide (about 36, 2, and 4 per glucose) were shown for teaching the trade-off, and he did not list them among the numbers to memorise.

8.3.4 Reading every small molecule in a sample - Metabolomics and biomarker discovery

WHAT IT IS	**Metabolomics** is the study of all the small molecules in a sample at once. A metabolite is a small molecule the cell makes or consumes while it works. The instruments that read them are **mass spectrometers**, which separate molecules by their mass-to-charge ratio, and **NMR** (nuclear magnetic resonance), which reads them by their magnetic signature.
HOW IT WORKS / WHY IT MATTERS	The slide shows a seven-step pipeline, and the order matters because most bad biomarker papers fail on steps one to three. Step one, experimental design: a cohort plus animal work. Step two, sample collection: urine, blood and cerebrospinal fluid, or tissue. Step three, sample preparation: homogenise, do liquid-liquid or solid-phase extraction, centrifuge. Step four, instrumental analysis: NMR, **GC-MS** (gas chromatography-mass spectrometry), or **LC-MS** (liquid chromatography-mass spectrometry). Step five, data mining: feature extraction, multivariate statistics, **PCA** (principal component analysis, which finds the axes that best separate the samples), and a heatmap. Step six, biomarker discovery: a volcano plot with cut-offs of $p < 0.05$ and **FDR** < 0.15 (FDR = false discovery rate, the share of the "significant" hits that are actually false alarms), and a Venn diagram that intersects the p-value cut-off with **VIP** > 1.0 (VIP = variable importance in projection, the machine's own score for how well a feature separates the groups) to pick differential metabolites; the slide shows counts of 10, 50, and 25. Step seven, multi-centre validation. The examples he gives are lactate, volatile organic compounds in exhaled breath for **non-invasive** lung or gastric cancer detection, and drug-response biomarkers such as sorafenib response.

The breath work is called **breathomics**, and it is the clearest example of "non-invasive" you will meet. A **volatile organic compound** is simply a small molecule that evaporates. A patient breathes into a Tedlar bag through an MGC preVent II pulmonary function filter, so that outdoor pollution is not mistaken for patient signal. A solid-phase microextraction fibre is then exposed to the collected breath; it sticks to the volatile molecules, and a **GC/MS** reads

them. His slide also lists where those volatile molecules come from inside the body: the cancer cells themselves, the alveolar cells of the lung, T cells, neutrophils, macrophages, fibroblasts, and the cytokines and metabolites they all produce. That list is the real lesson. A volatile molecule is not the product of one cell. It is the sum of everything happening in the tissue, which is why the same idea that finds a biomarker in blood can find one in breath.

The three instruments on his slide are not interchangeable: GC-MS suits volatile, thermally stable molecules; LC-MS suits less volatile and larger ones such as drugs and lipids; NMR suits those present in high concentration. (Standard teaching - he did not state this rule.)

EXAM HOOK	He did not flag the metabolomics pipeline steps as examinable. The numbers on the slide ($p < 0.05$, FDR < 0.15, VIP > 1.0, and the counts 10/50/25) were teaching material for how a discovery study is designed, not items he told you to memorise. The examinable thread is the general one he repeats: a biomarker is a molecule you can measure in blood to see what the body is doing, and a candidate must survive validation before it becomes a test.

8.3.5 Fats that build walls, and fats that carry a warning - Lipids and membranes

WHAT IT IS	Lipids are the water-fearing family of biomolecules. A **fatty acid** is a long carbon chain with a carboxyl group at one end. **Saturated fatty acids** have no double bonds; **unsaturated** ones have one or more. Fatty acids join up in two ways. Three of them plus a glycerol make a **triglyceride**, which is a pure energy store. Two of them plus a phosphate make a **phospholipid**, which is **amphipathic**: it has a water-loving head and a water-fearing tail. That is what lets a cell membrane self-assemble into a bilayer. Part 1 Chapter 2 already built the bilayer picture, so here we move on to what membranes also do.
HOW IT WORKS / WHY IT MATTERS	Membranes are not just walls; they are signal boards. The slide traces one cascade: a membrane lipid, PI, is phosphorylated with ATP to PI(4)P and then PI(4,5)P. When a neurotransmitter binds a G-protein-coupled receptor, **phospholipase C** is switched on and cuts PI(4,5)P into two pieces. **DAG** stays in the membrane and activates **PKC**. The soluble piece, I(1,4,5)P3, travels to the IP3 receptor on the endoplasmic reticulum and releases **calcium ions** from inside it; a phosphatase then switches the signal off. The second half of the topic is the immune alarm. **Lipopolysaccharide**, the **LPS** of Gram-negative bacteria, is carried by LBP, handed to CD14 and MD-2, and presented to **TLR4**, which dimerises. From there are two routes. The MyD88-dependent route (via TIRAP to MyD88) switches on **NF-kappaB** and the pro-inflammatory cytokines. The MyD88-independent route (via TRAM to TRIF) also switches on NF-kappaB and adds type I interferons. LPS is an **endotoxin** and a **pyrogen**, and he reminds you that overwhelming endotoxin in the blood causes **septic shock**, a common cause of death among surgical patients in intensive care.
EXAM HOOK	He said this will not be examined in detail: chemical structures and metabolic pathway details were on his exclusion list ("Please don't memorize all these chemical structure... They would not be included in my exam"). He did not ask you to learn the PI(4,5)P cascade. The one thing worth keeping is the order of the immune arm, LPS -> TLR4 -> NF-kappaB -> inflammatory cytokines -> septic shock, because that chain is the reason NF-kappaB appears again in Lecture 5 and because our exam-focus notes list this cascade order among the likely multiple-choice stems.

8.3.6 A bead necklace that folds into a machine - Amino acids and the protein structure hierarchy

WHAT IT IS	A protein is a chain of amino acids, like a necklace of beads, and it does its job by folding into a specific three-dimensional shape. Each amino acid is the same core with a different side chain, called the **R-group**, sticking out. The R-group decides how stiff or flexible that part of the chain is. **Glycine**, with the smallest R-group of all, is the most flexible. Bulky or charged R-groups such as **tryptophan** and **arginine** make the local chain rigid.
HOW IT WORKS / WHY IT MATTERS	There are four levels of shape, and you must be able to name them in order. The **primary structure** is the sequence itself, just the order of the beads. The **secondary structure** is the local folding that hydrogen bonds create: the **alpha-helix**, a corkscrew, and the **beta-sheet**, where strands lie side by side, held together by C=O---H-N hydrogen bonds. The **tertiary structure** is the whole chain folded into its final three-dimensional shape. The **quaternary structure** is several chains, called **subunits**, assembled together; haemoglobin and antibodies are the examples he gives. A **domain** is a stretch of the chain that folds up and does one job on its own. To read a sequence he points to two methods: **Edman degradation**, the classical way of peeling one amino acid off the end of the chain, and **LC-MS/MS**, the modern mass-spectrometry method. The two are used for different jobs. Edman degradation needs a clean single chain with a free amino group at one end, and it identifies one residue per run, so it is the method for a small, pure peptide. LC-MS/MS chops the sample into small fragments, weighs them, and then breaks each fragment again to read its sequence, so it can identify thousands of proteins in a mixture at once. That difference is why the proteomics workflow later in this lecture uses mass spectrometry and not Edman.

His slide 25 draws the four levels as four pictures, and that slide is worth memorising as a ladder. First the beads on a string. Then the beta-sheet, with the C=O---H-N hydrogen bonds drawn in between the strands, and the alpha-helix beside it. Then a ribbon diagram, with the helices, the sheets and the loops of the folded chain. Last the multi-coloured assembly, where separate subunits come together. Each picture is only reachable by folding what you already have, which is why you cannot skip a level when you explain a protein.

Part 1 Chapter 3 covers all of this in detail, so here the new idea is what the shape hierarchy buys you: the same chemistry, four levels, and each level is a separate thing to go wrong.

EXAM HOOK	He explicitly told you to skip the amino-acid table. His words, in the exclusion list: "don't memorize this... definitely not include in the exam" - that was the table of the 20 standard amino acids with their R-groups on his slide. Chemical structures were excluded too. What you do need is the hierarchy itself (primary -> secondary -> tertiary -> quaternary), the idea of a domain, and the idea that a small R-group means local flexibility. He did not flag the four levels as a named exam question in Lecture 1, but they are the scaffolding for everything about protein shape in the rest of the course.

8.3.7 Why a perfectly good drug protein can go bad in the bottle - Therapeutic protein liabilities

WHAT IT IS	A **therapeutic protein** is a protein made by a factory and given to a patient as a drug. Because it is a drug, the manufacturer has to know every way it can change. Before a trial, they analyse the sequence and predict which parts of the molecule might drift away from the shape they were designed to have.
HOW IT WORKS / WHY IT MATTERS	The example he works through is **deamidation**: an **asparagine** residue, Asn, spontaneously loses an ammonia group and becomes **succinimide**, then takes on water and becomes either **aspartic acid** or **iso-aspartic acid**. The slide gives the masses: Asn is 114.04 Da, the succinimide intermediate Asu is 97.02 Da, and both Asp and isoAsp come out at 115.03 Da. Why does it matter? Because the change alters the local chemistry and can change how well the drug binds its target, and therefore how well it works. Note the two possible end products, Asp and isoAsp. They have the same mass but they are not the same molecule, because the extra acid group ends up in a different place on the chain, so they can fold differently and therefore behave differently. That is the general shape of the problem: a tiny chemical change, a possibly different fold, and a drug that quietly does less than it did. The second topic is the opposite of a liability: **disulfide bonds**, the links between two **cysteine** residues, are the molecular staples that hold the tertiary and quaternary structure together. The practical lesson is that a sequence is not only a recipe for a shape; it is also a list of things that can go wrong later. And because these are real risks, the laboratory never trusts the prediction alone. The same validation tools from the proteomics workflow come back here: **qPCR** (quantitative PCR) for the message, **immunohistochemistry** to see the protein in tissue, **Western blot** for the amount of protein, and **ELISA** for the protein in a fluid such as serum.
EXAM HOOK	He did not flag the deamidation masses, or deamidation itself, as an examinable item. He presented it as an example of the kind of analysis you do before a trial, which is the same kind of "target and liability" thinking that runs through the whole lecture. He did not say anything about how to calculate or predict deamidation.

8.3.8 Billions of antibodies, and antibodies loaded with poison - Antibody diversity and ADCs

WHAT IT IS	An antibody is a Y-shaped protein that catches one specific target, and your immune system makes billions of different ones. The variety comes from **somatic recombination**: in each developing B cell, the gene pieces for the antibody's binding site are shuffled and joined together at random. Every B cell ends up with a slightly different binding site, so the body can recognise a huge range of invaders. An **antibody-drug conjugate**, or **ADC**, is an antibody with a **cytotoxic payload** attached to it. The antibody is the delivery system, and the payload is the poison.
HOW IT WORKS / WHY IT MATTERS	The slide gives the anatomy, and the numbers help you picture the scale. The **antigen-binding site** is at the tip of the Y. Below it sit the **Fv** (fragment variable) piece, about 25 kDa, made of the variable heavy and variable light domains; the **Fab** (antigen-binding fragment), about 50 kDa, which adds the constant heavy chain 1 and the constant light chain; and the **Fc** (crystallisable fragment), also about 50 kDa, which is the constant heavy chain 2 and 3 and is the part that talks to the rest of the immune system. The binding site is a **beta-barrel** made of the variable domains, and the three hypervariable loops inside it are named **CDR1**, **CDR2**, and **CDR3** (complementarity-determining regions). Those three loops are what actually touch the target, and they are the part engineers change. On the drug side, he names **trastuzumab-deruxtecan** as the **HER2** ADC, and he explains why the payload is a DNA-damaging one: **topoisomerase inhibitors** such as **SN-38** and **DXd** are used against HER2 and **Trop2** cancers. **Dual-payload ADCs** use a branched linker to carry two agents, for example a topoisomerase inhibitor plus a **DNA-damage-response** inhibitor, and the aim is to get around resistance that develops against a single payload. One choice you have to make is which kind of antibody to use, and it is not a small one. The antibody class that carries the payload also decides how the immune system treats the target. An **IgG1** (immunoglobulin G1) antibody recruits killer cells to finish off the cell it has bound, a process called **antibody-dependent cellular cytotoxicity**, while an **IgG4** does not, and lasts longer in the blood as a result. He does not settle the question in Lecture 1; he comes back to it in Lecture 2, which is where the antibody isotypes are taught properly.
EXAM HOOK	He did not flag antibody diversity in Lecture 1 as a named exam question. Two related points do matter for the exam, and both come from elsewhere in the course. First, the exclusion list contains ADC structures and dual-payload ADCs, where his words were: "all this would not be included in the midterm. I'm just want you to know more." Second, the V(D)J recombination machinery that generates this diversity is a Tier-1 exam topic in Lecture 4, because a defect in the repair pathway it borrows (NHEJ) causes SCID. So understand the idea of shuffling and of attaching a payload; do not memorise the ADC structures.

8.3.9 A switch that flips in seconds - Reversible phosphorylation and the Hippo pathway

WHAT IT IS	Reversible phosphorylation is the fastest way a cell turns a protein on or off. A kinase adds a phosphate group, spending ATP to do it. A phosphatase takes the phosphate away, using water. The slide draws it as two Pac-Man shapes: the kinase chews up ATP and attaches a negatively charged phosphate, and the mutual repulsion between the new negative charges pries the protein open; the phosphatase uses water and releases inorganic phosphate, which lets the protein close again. Same protein, two shapes, seconds apart.
HOW IT WORKS / WHY IT MATTERS	His worked example is the **Hippo pathway**, which is a brake on cell division. The pathway is "on" when cells touch each other, a signal called **contact inhibition**. A kinase cascade runs: **MST1** (with MST2) activates **LATS1** (with LATS2), and LATS1 phosphorylates **YAP**. Phosphorylated YAP is held outside the nucleus, partly by binding a protein called **14-3-3** and partly because it is marked for destruction. With YAP out of the nucleus, **TEAD**, the transcription factor it would otherwise help, stays inactive, and the cell differentiates or dies instead of dividing. His slide also lists what switches the pathway on: polarity regulators, NF2, AMOT, alpha-catenin, MARKS 1-4, LKB1, SCRIB, and F-actin. When the pathway is "off", a **G-protein receptor** or actin stabilisation inactivates LATS, unphosphorylated YAP enters the nucleus and binds TEAD, and the target genes switch on: **CTGF**, **JAG1**, and **NOTCH2**. The outcome is proliferation, resistance to programmed cell death, stemness, and the **epithelial-to-mesenchymal transition**. The mouse experiment makes the point visually: the slide shows a wild-type mouse liver beside a double Mst1/2 mutant liver, and the mutant liver is massively overgrown. Knocking out MST1/2 removes the brake, unphosphorylated YAP gets into the nucleus, and unchecked organ growth follows, which is cancer. Two broader lessons come out of it: pathways of this kind are conserved across species (**orthology**), which is why mouse models are useful, and animal models of cancer are a standard tool.
EXAM HOOK	He did not flag the Hippo cascade as an examinable item in Lecture 1, and he did not ask you to learn the list of upstream activators. The pathway is revisited in Lecture 5, where Hippo-YAP is one of the named transcriptional switches, so the idea - phosphorylation keeps a co-activator out of the nucleus, and losing the kinase lets it in - is worth having. He did not state any numbers for this topic.

8.3.10 Reading a survival curve - Kaplan-Meier curves

WHAT IT IS	A **Kaplan-Meier curve**, or **KM plot**, is a graph of what fraction of a group of patients is still alive over time. The x-axis is time in months. The y-axis is the probability of survival, and it starts at 1.0 because everybody in the group is alive on day zero. The curve never slides smoothly down. It falls in steps. A flat segment means no events happened during that interval. A downward step means an event - a death or a progression, depending on what is being plotted. The small tick marks along the curve are **censored** patients: people who left the study, or whose follow-up ended, but who were still alive at that moment. They contribute information up to the tick and are then not counted further.
HOW IT WORKS / WHY IT MATTERS	The skill he wants is reading one plot and getting several facts out of it. If a curve drops steeply and early, the gene or marker is associated with early involvement and a poor prognosis. If the curve drops late, the involvement is late. If the two curves separate, the marker separates the two groups of patients, and it is worth measuring. If the curves cross or overlap, the marker does not separate prognosis and measuring it will not help you. His worked example is the one that makes the point sharpest: the same target can be pro-tumour in one cancer and anti-tumour in another. High expression of a **receptor tyrosine kinase** predicts poor survival in pancreatic cancer, but better survival in renal cell carcinoma. The clinical consequence is a rule: a drug target and a prognostic biomarker are always cancer-type specific, and you must never carry a conclusion from one tumour type into another. The parallel example from the **Human Protein Atlas** is **ERBB2**, which is a prognostic marker in renal cancer (favourable), endometrial cancer (unfavourable), and pancreatic cancer (unfavourable), all at $p < 0.001$. In plain English: the same protein, the same amount, three different cancers, three different meanings, and all three comparisons statistically strong.
EXAM HOOK	This is the single highest-confidence exam topic in the whole course, and it is Lecture 1's Tier-1 question. His words: "And I ask this single question every, every time when I talk about the KM plot, yeah. And only a single student give me a good answer last year, yeah." And again: "So again, this is the kind of the question that I would include in the exam, yeah. Just give you a taste about the midterm exam, yeah." And: "But the key is to let you understand that from a single plot, we can really, really find out many, many useful informations for the future study." He also named survival data as one of only two scenario-question families: "I give you some survival data, and then you tell me that how this survival data can really benefit the treatment of the patient." Practise the six-point reading: the curve starts at 1.0; it falls in steps; flat means no events; a step is an event; ticks are censored patients; early and steep means early involvement and poor prognosis. The ERBB2 numbers (renal favourable, endometrial unfavourable, pancreatic unfavourable, all $p < 0.001$) are on the key-numbers list.

The exam-focus notes build this answer shape from what he said he is emphasising - analysing the scientific questions rather than recalling facts. It is not a quote, but it is what a strong answer looks like, and you should memorise its four moves: state the observation, name the mechanism, interpret the clinical meaning, then state the action or the limit. Note their template says "the [log-rank] p-value", which is the test actually used on these curves. The skeleton is: "The curve for [group A] drops from 1.0 to [x] within [n] months, whereas [group B] remains near [y] until [m] months. The [log-rank] p-value is [p], so the two groups are separated. A steeper and earlier drop indicates [early involvement / worse prognosis]. This is consistent with [mechanism]. The same target can behave in opposite directions in different tumour types, so this finding should not be generalised beyond [this tumour type]. The practical implication is [action]." Notice that the last move is not optional. A KM answer without a stated limitation loses the mark that the question is designed to test.

8.3.11 Separating thousands of proteins, then catching one - The proteomics workflow and the Vimentin case

WHAT IT IS	**Proteomics** is the study of all the proteins a cell makes. His worked example is **hepatocellular carcinoma**, **HCC**, which is cancer of the liver. The question was simple: can a protein be found in blood that tells you liver cancer is there, especially early liver cancer? The answer was yes, and the protein was **vimentin**. Vimentin is a structural protein of the cytoskeleton - the internal scaffolding that gives a cell its shape and holds its parts in place. It is the kind of protein you would expect to find inside a cell rather than in a blood tube, so finding it in blood tells you that it has escaped from the cells that made it.
HOW IT WORKS / WHY IT MATTERS	The workflow has to be understood in order. Step one is separation. The slide shows **2D-PAGE**, **two-dimensional gel electrophoresis**: the first dimension is **isoelectric focusing**, which sorts proteins by charge along a strip running from pH 4.0 to 9.0, and the second dimension is **SDS-PAGE**, which sorts them by molecular weight from 220 kDa down to 10 kDa. Together they spread the sample into a grid of spots. Step two is identification: each interesting spot is cut out, digested, run through liquid **chromatography**, ionised, and put into the mass spectrometer, which gives a mass-to-charge plot and then an **MS/MS** (tandem mass spectrometry) spectrum, and the spectrum is matched against a database. Step three is the clinical design: his slide shows the pipeline as IRB-approved tissue, then 2-DE-MS, then clinical data joined to protein identity, then statistics and informatics, and finally validation by qPCR, immunohistochemistry, Western blot, and ELISA. The samples were stratified on purpose: small tumours, large tumours, **cirrhosis**, and healthy transplant controls. His slide shows the result three times over: a 2-DE bar chart of relative percentage volume, and dot plots for Western blot and for qPCR. That repetition is the point. A difference seen by one method is a curiosity; a difference seen by three independent methods is a finding. Vimentin came out as up-regulated in early HCC, and because it can be measured in **serum** by **ELISA**, it is a non-invasive biomarker. ELISA is the right tool for this last step, because it is built for fluid: a known antibody is immobilised on a plate, the serum is washed over it, a second antibody carrying a detectable tag binds, and the signal is proportional to how much vimentin was present. The Human Protein Atlas was the tool used for tissue-expression profiles and for **target nomination**. The therapeutic step belongs to a different lecture: in Lecture 4 he showed an **anti-TERT antisense oligonucleotide** given for 3 weeks before tumour implantation cut the tumour-take rate to **2 of 12 mice**, against **9 of 12** untreated, p = 0.01. In plain English: fewer tumours took hold at all. That is a result about telomerase and TERT, not about vimentin - do not attach it to the vimentin table.
EXAM HOOK	The serum vimentin numbers are exam content and must be exact. In a cohort of n = 152 people: controls (n = 64) 176.0 +/- 61.3 ng/mL, median 163.0 (128.3-212.5); HCC <= 2 cm (n = 44) 271.5 +/- 165.9, median 202.5 (160.0-328.5); HCC > 2 cm (n = 44) 231.3 +/- 132.3, median 179.0 (150.0-264.0). Read this as: the small tumours, which are the hardest to catch early, show the biggest rise above the control group, and both cancer groups sit above the controls. A related key number, from Lecture 4 rather than this one: 3 weeks of anti-TERT pretreatment cut the tumour-take rate to 2 of 12 mice, against 9 of 12 untreated, p = 0.01. He did not give a verbatim "I will ask" line for the vimentin table itself, but he did put these numbers on the same list as the other Tier-3 multiple-choice numbers, so treat them as recall items.

8.3.12 The recipe book, and the two experiments that proved what it was - Nucleic acids: chemistry and history

WHAT IT IS	Nucleic acids are built from three parts. There are five bases: **adenine** (A), **thymine** (T), **cytosine** (C), **guanine** (G), and **uracil** (U). Uracil is the RNA version of thymine. A **base** plus a sugar is a **nucleoside**, and the sugar is **ribose** in RNA or **deoxyribose** in DNA. A nucleoside plus a phosphate is a **nucleotide**, and nucleotides are the actual building blocks of DNA and RNA.
HOW IT WORKS / WHY IT MATTERS	The history is the argument. Griffith worked on pneumonia bacteria in 1928 and noticed that something from dead virulent bacteria could transform live harmless ones into virulent ones. He called it the **transforming principle** but did not know what it was. In 1943 Avery, MacLeod and McCarty repeated the experiment and asked what the transforming principle was made of. They added enzyme after enzyme to the mixture. **Proteases** (enzymes that destroy proteins) did nothing. **DNase** (an enzyme that destroys DNA) killed the transforming activity. So the transforming principle was DNA, and DNA is the hereditary material. Then Watson, Crick and Franklin produced the **double-helix model**, on which the whole of replication and PCR is built. The structural point he puts on the slide is **base pairing**. The two strands of the double helix run **antiparallel**, and the bases pair across: cytosine with guanine through 3 hydrogen bonds, adenine with thymine through 2. Higher GC content means the helix is harder to pull apart, which means a higher **melting temperature**, Tm, and that is the single fact you will be asked to apply to primer design in PCR.

PCR (polymerase chain reaction) is worth a moment of its own, because it is the first practical thing the double helix gave us. You start with a template, add two short pieces of DNA called

primers that bind to the two strands at opposite ends, and add a **polymerase**, the enzyme that builds a new strand. The machine then repeats three steps over and over: **denaturation**, which pulls the two strands apart by heat; **primer annealing**, where the primers bind where they match; and **extension**, where the polymerase copies the strand using free nucleotides. Each round doubles the amount of DNA, so after n cycles you have roughly 2^n copies from one starting molecule. The base-pairing rule is what makes the primer possible: a primer only binds where its sequence is the exact complement of the template.

EXAM HOOK	The G-C = 3 hydrogen bonds and A-T = 2 fact is on the key-numbers list for Lectures 1 and 3. His reason for caring is PCR, where the melting temperature of the primers has to be right. Chemical structures were on his exclusion list, so do not memorise ring diagrams or formulas. The Avery-MacLeod-McCarty logic is worth keeping as a story, because it is one of the two or three experiments he expects a student to be able to reason from.

8.3.13 The factory line - The central dogma and gene-expression flow

WHAT IT IS	The **central dogma** is the one-way flow of information inside a cell. DNA is copied into messenger RNA by transcription. The messenger RNA leaves the nucleus and is read by a **ribosome**, which translates it into a chain of amino acids. The chain folds into a protein, and the protein does a job. You can think of it as: the recipe book stays in the control room, a photocopy of one recipe goes to the factory floor, and the factory floor turns it into a machine.

One more piece of geography, because it matters in every eukaryotic cell. His schematic puts the transcription step inside the nucleus and the translation step outside it. The messenger RNA is made in the nucleus, and it must be exported through the nuclear pores before the ribosomes in the cytoplasm can read it. That separation is why a cell can control the whole flow from one place. Lecture 3 will show that the DNA is packed up and mostly switched off, and Lecture 5 will show that control happens at every arrow in this chain.

HOW IT WORKS / WHY IT MATTERS	His slide walks the flow with a worked example. The **sense strand** and the **anti-sense (template) strand** are read, giving the messenger RNA GUG CAU CUG ACU, which is translated into the chain val-his-leu-thr. He reminds you that translation starts at the **start codon** AUG, which codes for methionine, and stops at one of three **stop codons**: UAA, UAG, and UGA. The applied part of the topic is making a cell produce a protein for you. If you put a human gene into a factory cell, that cell has its own preferred way of writing the same amino acid. The same amino acid can be spelled by several different three-letter **codons**, and a host cell uses some of them much more often than others. This preference is **codon-usage bias**. With **CHO cells**, the hamster cell line used to make human proteins, you therefore have to rewrite the gene - **codon optimisation** - so the cell reads it efficiently. His slide shows the result as a Western blot for two proteins, FALVAC-1 at 38.7 kd and PTP4A3 at 24 kd, against a GAPDH loading control at 39 kd, comparing Original, Genewiz, Thermo, Opt-b, and Opt-a codon-optimised versions across three groups.
EXAM HOOK	He did not flag the codon-optimisation blot in Lecture 1 as a question, but the principle is a Tier-2 item in Lecture 5, where he says: "So remember that when you use the CHO cell, you need to do the codon optimization." He also treats the central dogma as a floor that the rest of the course stands on, and the start codon AUG (methionine) and the three stop codons UAA, UAG, UGA are worth having ready.

8.3.14 The switch that gets stuck on - K-Ras as an oncogene, biomarker, and drug target

WHAT IT IS	**K-Ras** is a small protein that sits in the middle of the signalling chain **EGFR** -> RAS -> RAF -> MEK -> ERK, and on a second branch PI3K -> AKT -> mTOR. In the wild-type protein, K-Ras is switched off. It only becomes active when the upstream receptor EGFR fires, and it then switches itself off again. A mutant K-Ras is different: it is **constitutively active**, meaning it is stuck on and sends growth and survival signals whether or not EGFR fires. A gene like this, one that pushes a cell to keep growing, is an **oncogene**.
HOW IT WORKS / WHY IT MATTERS	Three clinical consequences follow. First, a predictive biomarker. In colorectal cancer, **cetuximab**, an anti-EGFR antibody, works only if K-Ras is wild-type. If K-Ras is mutated, cetuximab fails, because blocking EGFR upstream cannot switch off a signal that is already on downstream. So you must test the tumour before you treat it. Second, the mutation is common enough to matter. His slide gives **KRAS G12C** in 13 % of lung adenocarcinoma and 3 % of colorectal cancer, and explains the mechanism: mutations at positions G12, G13, or Q61 damage the protein's own off-switch, **GAP-mediated GTP hydrolysis**, so Ras stays locked in its active **GTP-bound** form. His slide also stars EGFR, RAS, PI3K, and RAF as the common sites of mutation in cancer. Third, K-Ras used to be called **undruggable**, and then it was not. The summary he gives of the whole field is that previously undruggable targets are becoming tractable through **structure-based design**: you solve the three-dimensional shape of the target, find a pocket that a small molecule can fit into, and design the drug for that pocket. His slide 46 shows what he called a mock FDA news release, dated 26 August 2026: approval of **Rasonque (daraxonrasib)**, a RAS inhibitor, for metastatic pancreatic cancer. The trial had n = 500 adults with previously treated metastatic pancreatic adenocarcinoma, and the **median overall survival** was 13.2 months against 6.7 months for standard chemotherapy. In plain English: patients on the new drug lived roughly twice as long as those on chemotherapy. He also notes the scale of the problem - around 90-95 % of the 67,000 new US pancreatic cancer cases each year are pancreatic adenocarcinoma, which is about 3.2 % of all cancer diagnoses but disproportionately lethal. Two FDA fast-track routes are named: **Breakthrough Therapy Designation**, which skips Phase III, and **Orphan Drug Designation**, which applies to rare diseases under 200,000 people in the US and gives 7 years of market exclusivity. The slide also carries **Priority Review** and the Commissioner's National Priority Voucher pilot.

One point of vocabulary to keep straight here, because our exam-focus notes list it as one of the classic "which one is different" pairs. A predictive biomarker predicts the response to a treatment; a prognostic biomarker predicts how the disease will go regardless of treatment. K-Ras status in colorectal cancer is predictive, because it predicts who will respond to cetuximab. The ERBB2 result in the next-but-one topic is prognostic, because it predicts outcome and no drug is being tested there.

EXAM HOOK	Two numbers are on the key-numbers list: the daraxonrasib median overall survival of 13.2 versus 6.7 months, and the KRAS-mutation survival curve with p = 0.016, where mutated KRAS does worse than non-mutated. The exam-focus notes list "K-Ras mutant -> cetuximab fails" as a likely multiple-choice stem, and the predictive-versus-prognostic biomarker pair is a classic distractor pair. He did not ask you to memorise the chemistry of the G12C mutation or the drug's structure, and he did not flag the FDA designation details as examinable.

8.3.15 When the letters get out of step - Frameshift mutations and neoantigens

WHAT IT IS	A **frameshift** is what happens when letters are inserted into or deleted from DNA by a number that is not a multiple of three. Because the code is read in threes, one missing letter shifts every letter after it out of register. The result is usually a nonsense protein or an early stop.
HOW IT WORKS / WHY IT MATTERS	His slide uses a tiny example. The normal message reads AUG AAG UUU GGC UAA, which becomes the chain MET-LYS-PHE-GLY-STOP. Delete one letter from the third codon and the reading frame shifts, giving MET-LYS-LEU-ALA and everything after is wrong. Insert one letter right after the start codon and you get an immediate STOP. Delete a whole codon - three letters - and the frame is preserved, and you get MET-PHE-GLY-STOP, a shorter but otherwise **in-frame** protein.

Why a frameshift is nastier than a single-letter mistake is worth pausing on. A **point mutation** swaps one amino acid and leaves the rest of the chain exactly as it was. A frameshift shifts the reading position of every codon after the accident, so the tail of the protein is not the same protein at all. That is exactly what his slide shows: the deletion gives MET-LYS-LEU-ALA and then nonsense, whereas the in-frame deletion gives MET-PHE-GLY-STOP, which is simply the original protein with one piece missing. One accident, two completely different consequences, decided by whether the letter count is a multiple of three.

What makes a neoantigen special is also worth being precise about. An **antigen** is anything the immune system can recognise and respond to. The body's own proteins are recognised as self, so the immune system learns to leave them alone; that learned restraint is called **tolerance**. A

neoantigen is a fragment that is genuinely new, so no tolerance has ever been built against it, and to the immune system it looks as foreign as a virus would. That is why it is such a good target for immunotherapy.

So the accident does double damage: it breaks the protein, and it accidentally manufactures the thing the treatment needs to aim at. A tumour with many frameshift-derived neoantigens is therefore a much better immunotherapy candidate than one without them, and that is the connection he picks up in Lecture 5.

EXAM HOOK	He did not flag frameshifts as a named exam question in Lecture 1. The examinable version of the idea arrives in Lecture 5, where he says: "So guys, you definitely remember that for the tumor that present many neoantigen and can be treated with the immunotherapy, we call this is a hot tumor, yeah. And not all the solid tumor is a hot tumor." In plain English: a tumour with many frameshift-derived neoantigens is a hot tumour, and hot tumours respond to checkpoint blockade such as anti-PD-1, while many solid tumours do not. In Lecture 1 he gave the mechanism and left the clinical label for Lecture 5.

8.3.16 The RNAs that are not messages - Diverse RNA species and miRNA biomarkers

WHAT IT IS	Not all RNA is a message. Alongside messenger RNA, transfer RNA, and ribosomal RNA, there are **microRNAs** (**miRNA**), **long non-coding RNAs** (**lncRNA**), and **small interfering RNAs** (**siRNA**). A microRNA is a short regulatory RNA about 22 nucleotides long that stops a target messenger RNA from being used. It is made in stages: a primary miRNA is cut by **Drosha** with its partner **Pasha** into a pre-miRNA hairpin, that hairpin is exported out of the nucleus by **Exportin-5**, **Dicer** cuts it into a short double-stranded miRNA, and that duplex is loaded into **RISC**, the silencing complex. When the miRNA matches its target only partly, it represses translation. When it matches perfectly, it cuts the messenger RNA in pieces.
HOW IT WORKS / WHY IT MATTERS	The biomarker reason is simple and powerful: microRNAs are unusually stable in serum, so they survive the trip from a tumour in the liver to a blood tube in the laboratory. That makes them excellent non-invasive biomarkers. His slide shows the discovery workflow, and the workflow is the lesson. Step one: profile tumour tissue against adjacent non-tumour tissue from n = 96 patients, which gave six candidate microRNAs. Step two: check that the candidate is actually secreted, using cell-line supernatant, which left four. Step three: check it in blood before and after surgery in n = 15 patients, which named miR-15b and miR-130b. The paired pre- and post-operative plots show miR-15b with p = 0.0637 (copies per ng up to 1400) and miR-130b with p = 0.0158 (up to 300). Then a validation cohort of n = 116 people: 30 healthy, 29 with chronic hepatitis B, and 57 with HCC. He attaches one caution that catches people out: tissue next to a tumour is biochemically different from truly normal tissue, so it must not be treated as "healthy" when you are calculating what is abnormal.

How you find the candidate in the first place depends on the tool, and his slide shows three. **RNA sequencing** takes the RNA from the sample, fragments it, converts it to DNA, adds the adapter molecules, and builds a DNA library that a machine reads. The oligonucleotide **microarray** is the older route: a chip carries thousands of known gene sequences, the sample's RNA is converted into biotin-labelled complementary RNA, and the complementary RNA is allowed to stick to whichever spots on the chip it matches. The two-dye microarray puts two samples on the same chip at once, **Cy5** for normal tissue and **Cy3** for cancer tissue, so they compete for the same spots and the ratio of colours at each spot says which genes are up and which are down. All three tools answer the same question, which is exactly why the biomarker claim has to be argued from the data rather than the machine.

EXAM HOOK	The classifier numbers are exam content and must be exact: miR-15b plus miR-130b gives **AUC** (area under the curve, a score from 0 to 1 where 1 is a perfect test) = 0.981, with **sensitivity** 98.25 %, **specificity** 91.53 %, and **accuracy** 94.83 %. Read this as: of 100 people without HCC, the test correctly calls about 91 or 92 of them negative, and of 100 people with HCC it correctly calls about 98 of them positive - close to 95 % of all decisions right. These four numbers are on the key-numbers list for Lecture 1. He did not ask you to memorise the miRNA biogenesis enzymes by name, but he did stress the three-step validation logic and the adjacent-tissue caveat, both of which are the kind of thing a scenario question turns on.

8.4 The ten things to remember

1. A biomarker is a molecule you can measure in blood to see what the body is doing. Diagnostic means "is the disease here now", prognostic means "how will it go", predictive means "which drug will work".
2. Cancer cells ferment sugar even when oxygen is available. The reason is not laziness. The half-broken sugar molecule is more useful to a dividing cell as building material than as energy, so the "inefficient" switch is really a growth strategy.
3. A biomarker is not a measurement. It is a measurement that has been validated in the right samples, against the right controls, before anyone is allowed to call it a test.
4. Proteins have four levels of shape: sequence, local folds, whole fold, and assembly of subunits. A domain is a stretch that folds and works on its own.
5. Phosphorylation is the fastest switch a cell has. A kinase adds the phosphate, a phosphatase takes it off.
6. A Kaplan-Meier curve starts at 1.0 and falls in steps. Flat means no events, a step means an event, a tick means a censored patient. An early steep drop means early involvement and a poor prognosis.
7. The same protein can be pro-tumour in one cancer and anti-tumour in another. ERBB2 is favourable in renal cancer and unfavourable in endometrial and pancreatic cancer, all at $p < 0.001$. Always name the tumour type.
8. DNA is the hereditary material, proved by Avery, MacLeod and McCarty when DNase killed the transforming principle and proteases did nothing. The flow is DNA -> RNA -> protein.
9. A mutant K-Ras is stuck on, so cetuximab fails in K-Ras-mutant colorectal cancer. K-Ras is no longer undruggable: daraxonrasib gave a median overall survival of 13.2 months against 6.7 for chemotherapy.
10. Two weak signals can make one strong test. miR-15b plus miR-130b reached AUC = 0.981 in HCC, with sensitivity 98.25 %, specificity 91.53 %, and accuracy 94.83 %.

8.5 Check yourself

Q1. A Kaplan-Meier plot shows two groups. Curve A falls from 1.0 to about 0.4 within the first six months and then stays flat. Curve B stays near 0.9 until month 30 and then falls. What does the plot tell you, and what would you do next?

A1. The observation comes first: the two curves separate early, so the marker splits the group into two patients with genuinely different outcomes. Curve A falls steeply and early, which by Dr. Wong's rule means early involvement and a poor prognosis for group A; curve B's late drop means late involvement. The mechanism would be that the marker is involved in something that happens before the event in group A. The clinical meaning is that measuring this marker separates patients who need different treatment intensity. The action is to state the limitation: this is a prognostic reading from one tumour type, and the ERBB2 example shows why you must not carry it across - the same marker was favourable in renal cancer and unfavourable in endometrial and pancreatic cancer. So I would look for the mechanism in this tumour type, and confirm the association in an independent cohort before changing treatment.

Q2. A cancer cell makes only about 4 ATP molecules per glucose where a resting normal cell with oxygen makes about 36. Why would a tumour choose the worse option?

A2. This is the Warburg effect, and the naive reading is that the tumour has made a mistake. The mechanism says otherwise. Oxidative phosphorylation makes energy but consumes the sugar intermediate, whereas fermentation leaves the half-broken sugar molecule in the cell as pyruvate. That leftover is the raw material for nucleotides, lipids, and amino acids, which is exactly what a cell about to divide a billion times needs. The clinical meaning is that the "wasteful" metabolism is a growth strategy, and it also leaves a fingerprint: lactate is raised, which is why lactate and other metabolites became candidate biomarkers. The limit on the conclusion is that this is not a claim that the tumour can do without mitochondria; the slide still shows about 5 % of pyruvate going into them.

Q3. Serum vimentin in controls is 176.0 +/- 61.3 ng/mL. A patient with a 1.5 cm liver lesion reads 271.5 +/- 165.9 ng/mL. Does this patient have liver cancer, and what can you not conclude?

A3. The observation is that the value is above the control group: 271.5 +/- 165.9 against 176.0 +/- 61.3 in controls. That figure is in fact the small-tumour group's own mean (HCC <= 2 cm, 271.5 +/- 165.9), so the patient's value *is* the small-tumour group's value, not something above it - and that is the group with the biggest rise of all. The mechanism is that the tumour is releasing vimentin, and the smallest tumours give the largest signal, so this is promising for early detection. The clinical meaning is more modest: serum vimentin is a non-invasive way to look for a signal, and it does not replace imaging - the role of avoiding repeated scans and biopsies belongs to markers such as **CEA**. The action and the limit: you cannot conclude that this patient has HCC from one number. The spread is wide (165.9), the groups overlap, and the figure came from a cohort, not from a diagnostic rule with a cut-off. A single elevated value needs a diagnostic threshold, a second test, and imaging. The related anti-TERT experiment cut the tumour-take rate to 2 of 12 mice against 9 of 12 untreated, p = 0.01 - that is a result about telomerase, not about vimentin.

Q4. A patient with colorectal cancer is found to carry a K-Ras mutation. The oncologist is considering cetuximab. What happens, and why?

A4. The observation is that the tumour is K-Ras-mutant. The mechanism is that cetuximab is an anti-EGFR antibody, so it works by blocking the receptor at the top of the chain, EGFR -> RAS -> RAF -> MEK -> ERK. But wild-type K-Ras is only active when EGFR fires, so blocking EGFR switches the whole chain off. A mutant K-Ras is constitutively active: it is stuck on and sends growth signals whatever EGFR does. So blocking upstream cannot switch off a signal that is already on downstream, and the drug fails. The clinical meaning is that K-Ras status is a predictive biomarker, so you must test before you treat: a K-Ras-mutant patient should not be given cetuximab on the expectation of benefit. The action is to choose a different strategy. The limit is that this is a statement about cetuximab in colorectal cancer; it does not mean K-Ras-mutant cancer is untreatable. K-Ras is no longer undruggable - daraxonrasib, a RAS inhibitor, gave a median overall survival of 13.2 months against 6.7 months for standard chemotherapy in previously treated metastatic pancreatic cancer.

Q5. You find a microRNA raised in tumour tissue compared with adjacent tissue. Why is that not yet a blood test, and what is the "adjacent tissue" trap?

A5. The observation is a difference between two pieces of tissue from the same patient. The mechanism is that the tumour is making and shedding more of that microRNA. The problem is that tumour tissue and adjacent tissue are both from a diseased liver, and the adjacent tissue is already biochemically abnormal, so the comparison does not tell you what "normal" looks like, and it certainly does not tell you what happens in blood. The clinical meaning is that a tissue finding only becomes a test after you have shown the molecule reaches the blood and survives there. The action is the three-step validation Dr. Wong lays out: show it is secreted in cell culture, show it in pre- and post-operative serum, then validate in an independent cohort. His slide runs exactly that route, from n = 96 paired samples down to six candidates, then four after cell-culture testing, then miR-15b and miR-130b in n = 15 patients, and finally a cohort of n = 116 people (30 healthy, 29 with chronic hepatitis B, 57 with HCC). The limit to remember is that microRNAs are unusually stable in serum, which is exactly why they work, and exactly why a promising tissue signal can still fail the trip to the blood.

End of Part 2. The next chapter, Part 3, covers Lecture 2 - Basic Cell Biology. Where this chapter met a topic it only hinted at - the Warburg effect in dividing cells, NF-kappaB, the Hippo-YAP pathway, antibody isotypes, and the central dogma - the next lecture or the ones after it take it further.

PART 3

Lecture 2: Basic Cell Biology

IN THIS PART

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About this part

Source: GNBF5020 Lecture 2, "Basic Cell Biology", 19 September 2026, Dr. Tony Wong. This chapter rewrites the expert shorthand of the existing revision guide so that it can be read by someone who has never studied biology or chemistry. Every number Dr. Wong stated is reproduced exactly and is followed by a plain sentence saying what it means.

CHAPTER 9

Basic Cell Biology

9.0 What this lecture is about

Dr. Tony Wong's second lecture answers a deceptively simple question: how do you look at one cell among billions, and how do you change what that cell does? He starts with the instruments - the machine that counts and sorts cells one at a time, the microscopes that show you where inside a cell two proteins are sitting, and the probes that count how many copies of a gene a tumour has. Then he walks through the cell itself: what each compartment does, why the mitochondrion is the odd one out, how a cell divides, and why one inherited disease follows the mother and not the father. In the second half he turns all of that into medicine: stem cells, genome editing, antibodies, the drug pipeline, CAR-T, liquid biopsy, immune checkpoints, and the viral DNA that drives liver cancer. By the end of this chapter you should be able to explain, in plain English, all eighteen ideas that the rest of the course keeps reusing.

9.1 Words to know before we start

Term	In plain English
Cell	A tiny factory. It takes in raw material, copies its own instruction manuals, builds machines, and ships finished machines out of the door.
Antibody	A Y-shaped protein that catches one specific target, the way a hook catches one particular key.
Isotype	The class of an antibody, such as IgG1 or IgG4. The class decides how long it stays in the blood and how the immune system treats whatever it has caught.
Flow cytometry	Counting cells one at a time with a laser. Fluorescently tagged antibodies act as name tags, so the machine can say which cells are present.
FACS	Flow cytometry with a sorting arm. It physically separates the cells you want from the ones you do not, and keeps them alive in a tube.
Gating	Drawing a boundary on a flow-cytometry plot to keep only one population and throw the rest away.
Chromosome	A printed manual. One long DNA molecule with its protein packaging, carrying many genes.
Cell cycle	The schedule one cell follows to become two: grow, copy the DNA, check the copy, separate the chromosomes, cut the cell in half.
Mitosis	Dividing the contents. The duplicated chromosomes are pulled apart so that each new nucleus gets a complete set.
Cytokinesis	Cutting the cell in two. A tightening ring of protein pinches the cytoplasm into two separate cells.
mtDNA	The mitochondrion's own small circle of DNA - about 17 kilobases carrying 37 genes - and it comes from the mother only.
iPSC	An induced pluripotent stem cell: an ordinary body cell pushed back into being able to turn into almost any cell type.
CAR	A chimeric antigen receptor: a synthetic receptor bolted onto a T cell so the T cell can recognise a tumour cell it would otherwise walk past.
Neoantigen	A brand-new molecular fragment that the immune system has never seen before, and therefore has no tolerance for.

9.2 The big picture

Every doctor who treats cancer faces the same awkward problem. The tumour is somewhere inside the body, it is made of billions of cells, and those cells are not all alike. The treatment you want to give may work on one population and be useless on another. So before you can measure anything, diagnose anything, or treat anything, you have to be able to pick the right cells out of a mixed soup and hold onto them without killing them. Dr. Wong's second lecture is built around that single idea.

The first half of the lecture is the toolbox. Flow cytometry is the core instrument: a laser interrogates cells one at a time, and fluorescently tagged antibodies act as name tags, so a machine can tell you how big each cell is, how cluttered its interior is, and which marker proteins it is wearing. Sorting that stream of cells - FACS - lets you collect a pure population alive, ready to grow or to assay. When the problem is the tissue rather than the suspension, laser capture microdissection cuts a clean piece out of a slide, and FISH reads gene copy number straight from the DNA using a probe instead of an antibody. Immunofluorescence and confocal microscopy then show you where inside the cell two proteins sit, which is how you discover that a pathway component has moved from the cytoplasm into the nucleus and switched something on.

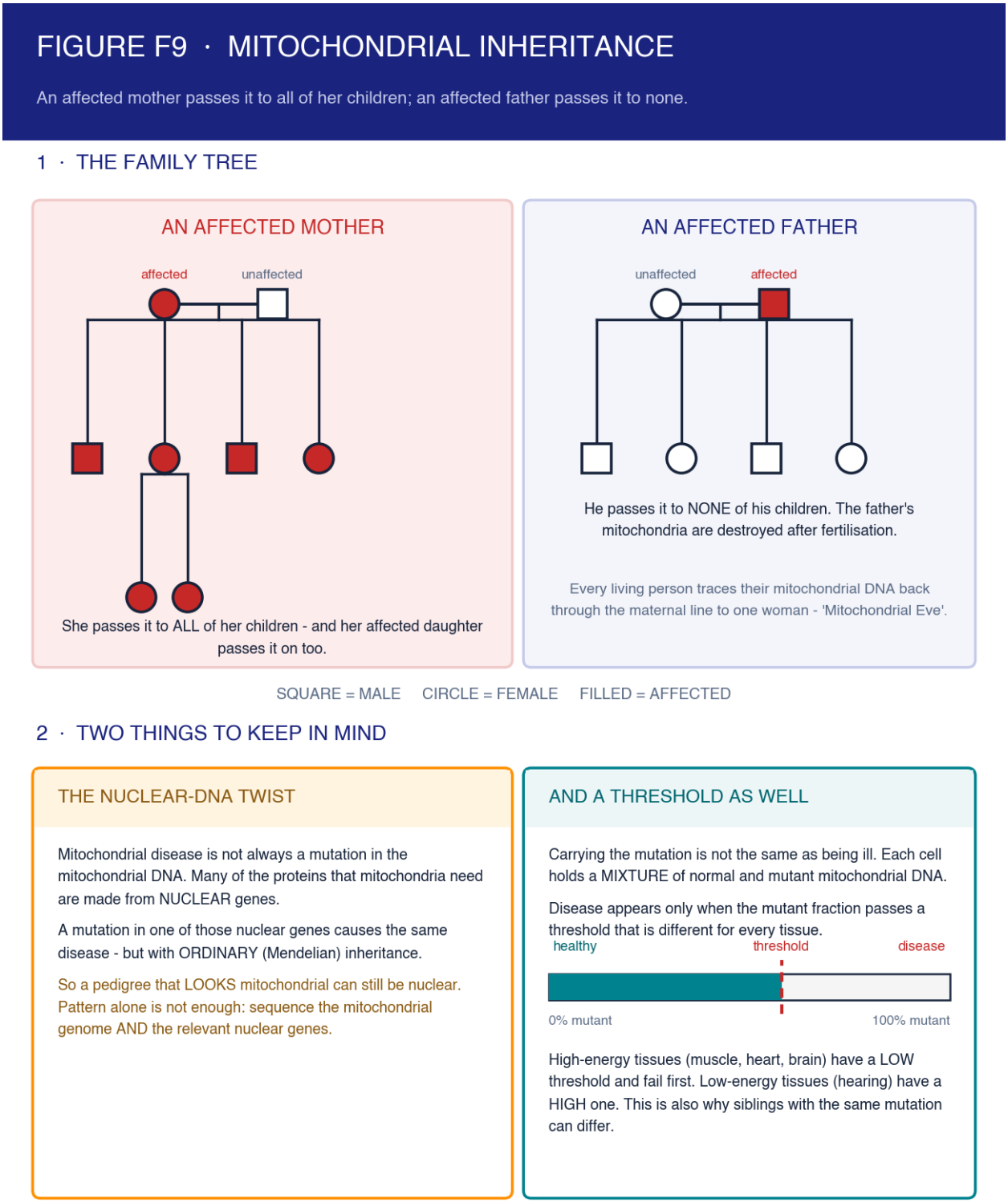
Then the lecture steps back and looks at the cell itself. Which room does which job? The nucleus stores the master manuals. The endoplasmic reticulum is the workshop where new proteins are folded and checked; the Golgi apparatus is the packing room; the cytoskeleton is the frame, the freight system, and the machine that pulls the cell in two. The mitochondrion is the power station, and it turns out to be the odd one out: it keeps its own small circle of DNA, because it used to be a free-living bacterium that was swallowed and kept. That single historical accident explains a whole family of inherited diseases, and it is why the pedigree you will be asked about in the exam is a pedigree of mothers.

The middle of the lecture is about how one cell becomes two. There is a schedule, the cell cycle: G1 to grow, S to copy the DNA, G2 to check the copy, M to separate the chromosomes, and then cytokinesis to cut the cell in two. The schedule is run by two parts that fit together like a key and a lock - cyclins and cyclin-dependent kinases - and a drug can jam the lock. And here is the distinction the whole lecture is built around. Mitosis is dividing the contents. Cytokinesis is cutting the cell in two. They are separate jobs, done by different machines, and blocking one leaves the other running.

The last third of the lecture turns that biology into medicine. Stem cells can be reprogrammed back into being able to become almost anything, which gives us disease models in a dish and a way to test a drug's toxicity before a human ever takes it. CRISPR gives us a pair of molecular scissors. Antibodies come in versions: IgG1 recruits killer cells, IgG4 does not but stays in the blood longer. CAR-T takes a patient's own T cells and bolts a synthetic receptor onto them. Liquid biopsy reads a blood sample instead of cutting into the tumour. And the tumour microenvironment explains why some tumours let the immune system in and others hide. Every one of these is a version of the same move: characterise the right cell population accurately enough that you can turn the characterisation into a biomarker or a therapy.

Figure F9 - Mitochondrial inheritance

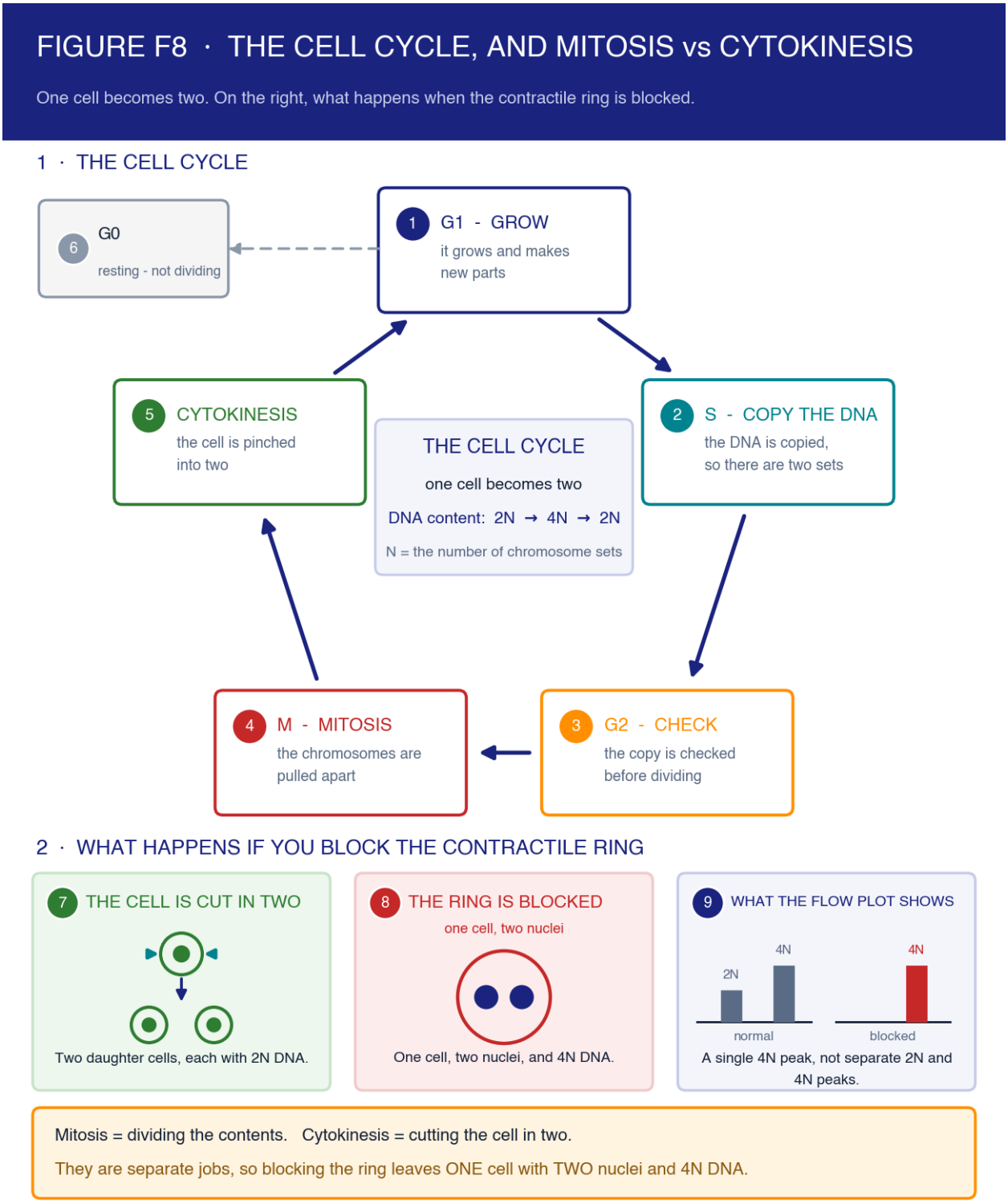
A family tree showing a mitochondrial disease: an affected mother passes it to all of her children, an affected father passes it to none of them, and a caveat box explains that a nuclear gene can imitate the same pattern.



Mitochondrial inheritance. Squares are males, circles are females, and a filled symbol means the person is affected. Read the three panels as a single argument.

Figure F8 - The cell cycle, and mitosis versus cytokinesis

The cell cycle drawn as a ring of phases, with a second panel showing what happens when the contractile ring is blocked: one cell with two nuclei and 4N DNA instead of two cells.



The cell cycle and the single most important exam idea in this lecture: mitosis and cytokinesis are separate.

9.3 Walkthrough

9.3.1 Picking cells out of a soup - Flow cytometry and FACS

WHAT IT IS	**Flow cytometry** is a machine that counts and characterises cells one at a time. Cells are stained with antibodies carrying a fluorescent dye, so each antibody acts as a name tag for one particular protein on the cell surface. As each cell passes a laser, the machine records how much light scatters forwards and sideways, and how much fluorescence comes back. Two scatter channels do most of the work: **forward scatter (FSC)** is a rough measure of **cell size**, and **side scatter (SSC)** is a rough measure of **granularity**, meaning how cluttered the inside of the cell is. **FACS** - fluorescence-activated cell sorting - is the same instrument with a sorting arm: instead of only counting, it physically deflects the cells you have chosen into a tube, alive and sterile, ready to be grown or assayed.
HOW IT WORKS / WHY IT MATTERS	The physical trick is **hydrodynamic focusing**. A stream of **sheath fluid** squeezes the sample so that the cells pass the laser in single file, one at a time, which is what makes the counts accurate. The laser is then interrogated by several detectors at once: one for forward scatter, one for side scatter, and a set of filters and mirrors that split the emitted fluorescence by colour onto separate **photomultipliers**, one per fluorescent dye.

You then draw boundaries on the resulting plots, and that is called **gating**. Dr. Wong's slide shows the first gate on an FSC-versus-SSC plot: **granulocytes** sit high on side scatter because they are full of granules, **monocytes** are in the middle, and **lymphocytes** are low on both because they are small and comparatively empty. Only after that gate is drawn does he look at the fluorescence channels. His second plot is **CD3-positive** against **CD4-positive** - the markers on a **helper T cell** - split into four quadrants, and the example shows the double-positive population at **41.6 %** in one condition against **11.3 %** in the other. In plain English: in one sample, four out of every ten of the gated cells were helper T cells; in the other, barely one in ten was.

Three more tools belong in the same drawer. **Laser capture microdissection (LCM)** is the opposite of a suspension experiment: under the microscope, a laser cuts a precise region off a tissue section, so that when you analyse a tumour biopsy you are not also measuring the fibroblasts and the immune cells that were sitting next to the tumour. **FISH** (fluorescence *in situ* hybridisation) counts genes rather than proteins: a labelled DNA probe sticks to the gene you care about, and you count the dots. That is how **HER2 amplification** is measured, and therefore how you decide whether a patient is a candidate for Herceptin. Finally, the two ways of staining tissue are easy to confuse: **immunohistochemistry (IHC)** uses an enzyme that deposits a brown pigment you can see down a microscope, whereas **immunofluorescence (IF)** uses a glowing dye you read on a fluorescent microscope. His own slide uses **FITC-anti-CD3** and

Alexa Fluor 647-anti-CD4 together, which is the simplest possible demonstration of why fluorescence is more flexible than IHC: you can read several markers at once and ask which cells carry which combination.

EXAM HOOK	He made the core reading of a flow plot a Tier-2 item: **FSC = size, SSC = granularity**, and **gating** isolates the target population while **FACS** physically sorts viable cells into tubes or wells. The exam-focus notes list a Flow-plot archetype from their case bank: "Flow plot - FSC vs SSC; which gate; which population; what the contaminant is." He named only two scenario families outright: a genetic-counselling report and survival data. What he explicitly put out of scope is the FISH protocol detail, with the words: *"Definitely not include in the exam. I will not ask you about what's the denaturations, what's the temperature."* In plain English: do not memorise the temperatures and the denaturation steps of a FISH probe. Do understand that FISH uses DNA probes rather than antibodies, and that it counts gene copies. He did not flag the FACS gating percentages as a named question, but they are on the slide, so treat them as recall items.

9.3.2 Seeing where proteins sit - Immunofluorescence and confocal microscopy

WHAT IT IS	**Immunofluorescence** is the fluorescent version of antibody staining. A primary antibody binds your protein of interest, a second antibody carrying a **fluorescent dye** binds the first, and you look under a fluorescent microscope. **Multiplex immunofluorescence** does several of these at once on the same cell, using different dyes for different proteins, so you can ask not only "is this protein here?" but "is it here *together with* that one?". The standard nuclear stain is **DAPI**, which glows blue and marks the nucleus in every panel.
HOW IT WORKS / WHY IT MATTERS	**Co-localisation** means two proteins are found in the same place inside the same cell. That is worth knowing because a pathway's whole logic is often spatial: a protein sitting in the cytoplasm means the signal is off, and the same protein sitting in the nucleus means the signal is on. Dr. Wong's example is the **Hippo pathway** - the components **MST1**, **LATS** and **YAP** - which Lecture 1 taught as the switch that decides whether a cell divides or differentiates. Staining all three in the same cell shows you at a glance whether YAP is being held outside the nucleus.

Confocal microscopy is the upgrade that solves one specific problem. In a wide-field microscope, light from above and below the focal plane also reaches the detector, so a whole thick cell lights up at once and everything is blurred together. A confocal microscope uses a

pinhole in front of the detector to block out-of-focus light, so it can take an optical slice through the cell at one depth at a time. Stack the slices and you get a **3D reconstruction** of the subcellular architecture. His slide shows the ray path - excitation laser, dichroic mirror, objective, focal plane - beside a side-by-side comparison of wide-field and confocal images of the same specimen, which is the clearest way to see what the pinhole buys you.

EXAM HOOK	He did not flag microscopy technique as examinable in its own right. What he did do is put the *cascade behind the picture* out of scope: *"I don't memorize all this thing, I just highlight here in case some student are really interested in molecular biology."* In plain English: you are not expected to memorise the signalling cascade drawn on the slide, but you should know that IF can show you several proteins in one cell, that co-localisation is the question it answers, and that confocal microscopy resolves depth rather than colour.

9.3.3 The rooms inside a cell - Eukaryotic organelles and their clinical context

WHAT IT IS	A **eukaryotic** cell is a set of rooms, each with one job. The **nucleus** stores the genomic DNA. The **endoplasmic reticulum (ER)** is the workshop: the **rough ER** carries ribosomes and makes and folds proteins, and the **smooth ER** makes lipids such as phospholipids. The **Golgi apparatus** is the packing room. The **cytoskeleton** gives the cell its shape and moves cargo. The **mitochondrion** is the power station. Part 1 Chapter 5 already taught you this inventory with the factory picture, so this is the lecture's clinical pass over the same list.
HOW IT WORKS / WHY IT MATTERS	The slide sets the cartoon cell next to a real **TEM** (transmission electron microscope) picture of a mouse L-cell printed at 15,000 times magnification, with the same parts labelled on both. The point of putting them side by side is that every "room" is a real structure, not a drawing convention.

The ER-to-Golgi route is the one to know in detail, because it is the production line for anything leaving the cell. On the rough ER, ribosomes build the protein and **chaperones** fold it and check it, and **glycosylation** decorates it with sugar groups. A transport **vesicle** then carries it to the **cis face** of the Golgi, moves through the stacked **cisternae**, leaves from the **trans face**, and enters a secretory vesicle that fuses with the plasma membrane to release the protein outside. That route has a name - the **secretory pathway** - and it is where the next topic's signal peptide acts.

The mitochondrion carries two clinical weights at once. It runs the **TCA cycle** and

oxidative phosphorylation, which together make most of the cell's **ATP**, and it is also the control centre of **apoptosis**, the cell's own death programme - which is why a cell that cannot keep its mitochondria healthy also cannot decide properly when to die. And here the lecture loops back to Lecture 1: cancer cells frequently shift from oxidative phosphorylation to **aerobic glycolysis**, the **Warburg effect**. You now know which organelle they are refusing to use, and why refusing it can be an advantage.

EXAM HOOK	He did not flag the organelle inventory as a memorised list, and Part 1 Chapter 5 already carries it. He did treat the Warburg effect as a recurring thread: it is where Lecture 1 opened, and this is where it gets attached to an organelle. He did not flag the TEM field labels as examinable.

9.3.4 The postal address on a protein - Signal peptides and protein targeting

WHAT IT IS	A **signal peptide** is a short address label at the **N-terminal** end of a protein - the very first stretch of amino acids, which is the end that emerges from the ribosome first. If that address says "rough ER", the ribosome hands the growing chain over to the ER membrane as it is being made, and the protein enters the **secretory pathway**: rough ER, then Golgi, then either the plasma membrane or out of the cell as a secretion.
HOW IT WORKS / WHY IT MATTERS	The clinical consequence is enormous and entirely practical. A protein that is going to sit on the cell surface - a **receptor** such as HER2 or PD-L1 - or that is going to be secreted into the blood - an **antibody**, a **hormone**, a **cytokine** - must travel this route. And if you put the gene for such a protein into an experimental system and delete the signal peptide, the protein is made but never reaches the ER. It never folds properly, never gets glycosylated, never gets to the surface, and your experiment produces nothing at all. Dr. Wong's point is not that this is a subtlety; it is that it is a total failure that looks like nothing whatsoever, so you can waste months on it.

The address also decides the destination, and that is how a cell can hold the same protein in different compartments. With one kind of address the protein ends up on the plasma membrane; with another it is sent to a lysosome instead; with the secretion address it is released into the blood. This is why the signal peptide is a favourite thing for a medicinal chemist to engineer when designing a therapeutic protein.

EXAM HOOK	This is a Tier-2 item he marked explicitly, with the words: *"So remember, signal peptide is really, really important for your study, yeah."* The story behind it is that his postdoc lost **4 months** of work by omitting the signal peptide. In plain English: four months of his postdoc's life, because a fifteen-amino-acid address was missing. He also warns you not to confuse the signal peptide with the **N-terminal domain (NTD)** of a nuclear receptor, which Lecture 5 covers: that one is a regulatory region on a DNA-binding protein, not a postal address for the ER. The signal peptide itself is on the distractor-pair list, paired with NTD.

9.3.5 The power station that came from a bacterium - Mitochondria: origin, inheritance and disease

WHAT IT IS	A **mitochondrion** is a bean-shaped **organelle** with two membranes. The inner membrane is folded into **cristae**, where the energy machinery sits; the space inside is the **matrix**. Crucially, a mitochondrion carries its own DNA: **mitochondrial DNA (mtDNA)** is a small double-stranded circle of about **17 kilobases** carrying **37 genes**, including the two ribosomal RNAs, the transfer RNAs, and the thirteen proteins of the respiratory chain. The circular gene map on his slide names them: the D-loop, 12S RNA, 16S RNA, the ND genes, the COX genes, ATP 8 and ATP 6, CYTB, and the tRNAs.
HOW IT WORKS / WHY IT MATTERS	The reason a mitochondrion has its own DNA is evolutionary. The **endosymbiotic** theory says that long ago an ancestral cell swallowed a bacterium and, instead of digesting it, kept it. The two then became one organism. Every mitochondrion in every one of your cells is the descendant of that one ancient bacterium, which is why it has a circular genome like bacteria do, its own ribosomes, and double membranes instead of one.

That history creates the inheritance pattern, and it is the reason this topic is a Tier-1 exam question. After fertilisation, the **paternal** mitochondria are tagged for destruction, so the embryo's mitochondria come from the egg and therefore from the mother. The consequence runs through the whole population: all living humans trace their mitochondrial DNA back through the maternal line to a single woman, the so-called **Mitochondrial Eve**. Her pedigree has three hallmarks. An affected **mother** passes the condition to **all** of her children. An affected

father passes it to **none** of them. And the condition appears in **every generation**. His slide 28 shows a three-generation pedigree where an affected mother has five affected children - three sons and two daughters - the affected daughters have affected children of their own, and the affected sons have none.

But carrying the mutation is not the same as being ill, and this is where most people stop too early. A cell can hold a mixture of normal and mutant mtDNA, and that mixture is called

heteroplasmy; when every copy is normal you have **homoplasmy**. Disease appears only when the proportion of mutant mtDNA crosses a **threshold** that differs from tissue to tissue. Tissues with a high energy demand - contracting **skeletal muscle** is the example he uses - have a **low** threshold and are the first to fail; tissues with a low demand, such as **hearing**, have a **high** threshold and can carry a lot of mutant DNA for years without symptoms. His slide 30 draws exactly this: a cell goes from all-normal, through a mixed state, across a dashed threshold line, to all-mutant, and the health of the cell flips at that line.

The body map on slide 27 shows how wide the consequences are, because mitochondrial disease is never one disease: neurodegeneration, strokes, epilepsy, retinitis pigmentosa, hearing loss, cardiomyopathy, liver disease, kidney dysfunction, diabetes, infertility, anaemia, and muscle weakness. Two further slides round it off. Slide 29 contrasts mitochondrial inheritance with

X-linked dominant and **X-linked recessive** pedigrees, because those are the patterns you must be able to tell apart on paper. And slide 27 also shows **mitochondrial replacement therapy**, a six-step procedure that moves the mother's mitochondria into the egg so that a child does not inherit the disease.

Finally, mitochondria are not fixed furniture. Exercise changes them. Dr. Wong's scatter plot puts mitochondrial content measured by TEM against training volume and gets **R2 = 0.82; P = 0.003** - in plain English, the more someone trains, the more mitochondrial material they have, and the relationship is tight. That build-up of mitochondria is called **mitochondrial biogenesis**, and it is the good-news end of the same organelle.

EXAM HOOK	This is Tier 1, in his own words: *"So, this is you'll have to a 100% understand, right? Not only the mutation within the mitochondrial DNA can cause the mitochondrial disease. Some mitochondrial disease can actually result from the mutation within the nuclear DNA."* And again: *"Just make sure you understand that different mutation can cause the mitochondrial diseases, and all this mutation can cause the different inheritance patterns when you analyze this pedigree tree, yeah."*

Read that carefully, because the second sentence is the twist that catches people. The trap is assuming that any pedigree which looks mitochondrial must be mtDNA. It need not be. Many of the proteins that make mitochondria work are encoded in the **nuclear** genome, not the mitochondrial one, and a mutation in one of those nuclear genes produces the same clinical picture with a completely different inheritance pattern - ordinary Mendelian inheritance, which can look like dominant or recessive. So the exam answer has two halves: give the three hallmarks of maternal inheritance, and then state that a nuclear gene can imitate the pattern, so the pedigree cannot be assigned to mtDNA on pattern alone.

The threshold logic comes with it, because it explains two things at once: why two siblings with the same mutation can have very different severity, and why the pedigree alone cannot predict an individual child's risk. The number **17 kilobases and 37 genes** is on his slides but he did not name it as a memorised figure. He did not ask you to memorise the mtDNA gene map or the list of clinical manifestations.

9.3.6 The schedule that turns one cell into two - The cell cycle and its molecular control

WHAT IT IS	The **cell cycle** is the repeating schedule a cell follows to copy itself. It has four stages and a split. **G1** is the growth phase, where the cell makes proteins and organelles and does its work. **S phase** is where the DNA is copied, so the cell ends up with two identical copies of every chromosome. **G2** is the checkpoint phase, where the copy is checked before division. **M phase** is mitosis, the division of the nucleus, where the duplicated chromosomes are pulled apart. **Cytokinesis** is the division of the cytoplasm, which follows mitosis. Cells that are not dividing rest in a side branch called **G0**.
HOW IT WORKS / WHY IT MATTERS	The schedule is driven by a two-part switch. A **cyclin-dependent kinase (CDK)** is the machine that does the work; a **cyclin** is the key that switches it on. Each cyclin rises, fits its matching CDK, and the pair drives the next step. The slide adds the brakes: the **INK4** inhibitors (p16, p14ARF, p15, p18, p19) and the **Cip/Kip** inhibitors (p21, p27, p57). And it adds the release mechanism: mitogen signal, cyclin D with CDK4/6, phosphorylation of **pRb**, release of the transcription factor **E2F**, and then the S-phase genes switch on, including cyclin E, which starts a positive feedback loop. **Palbociclib** blocks CDK4/6, which stops cells leaving G1 - and since **p53** mutations are found in about **40 %** of cancers, the cell's own emergency brake is also broken in roughly two tumours out of every five.

You can read all of this from the bench. Stain the DNA with **propidium iodide (PI)** and run the cells on the flow cytometer: cells in G1 show a **2N** DNA peak, cells in S show anything between 2N and 4N, and cells in G2 or M show a **4N** peak. To line the cells up at the same starting point, researchers **arrest them by serum starvation**, dropping **fetal bovine serum (FBS)** from 10 % down to **0.5-1 %**. In plain English: they remove almost all the growth factors the medium was providing, so the cells fall out of the cycle and sit at the G1 gate waiting for the same signal, and then all restart together.

EXAM HOOK	He explicitly excluded the detail: *"you don't need to memorize all this cyclin or cyclin-dependent kinase, CDK."* The concept he wants is the shape of the cycle and the fact that a drug can freeze a cell at one stage. One number is on his key-numbers list and you should keep it exactly: **serum starvation down to 0.5-1 % FBS** for G1 arrest and synchronisation. The **~40 % p53-mutation** figure is on his slide 41 but is not on the key-numbers list, so treat it as context rather than as a memorise item. The cyclin-CDK pairings on the slide - cyclin D with CDK4/6 in G1, cyclin E with CDK2 at the G1/S boundary, and so on - are the material he told you to skip.

9.3.7 Two different jobs that people confuse - Mitosis vs cytokinesis

WHAT IT IS	These are two separate jobs with two separate machines. **Mitosis** is dividing the contents: the duplicated chromosomes are lined up on the **mitotic spindle** (microtubules) and pulled apart, so that each new nucleus receives a complete set of chromosomes. **Cytokinesis** is cutting the cell in two: a tightening band of **actin** and **myosin** - the **contractile ring** - pinches the cytoplasm in the middle until the cell splits.
HOW IT WORKS / WHY IT MATTERS	Because the two jobs are done by different machines, you can break one and leave the other running. Block the contractile ring - with an inhibitor, or by knocking out its expression - and mitosis still happens perfectly: the DNA copies, the chromosomes separate, two nuclei form. What does not happen is the pinching. So you are left with **one cell containing two nuclei**: a **binucleated cell**, and if the block is longer or the cell divides again, a **multinucleated** one. On a flow-cytometry DNA histogram you would see a single **4N** peak where you would normally see separate 2N and 4N peaks, because the cell has completed its division of contents and never gave any of it away.

There is a second consequence that explains a clinical observation, and it is the one Dr. Wong connects to the previous topic. During cytokinesis, organelles and mitochondria are **partitioned at random**. Two siblings who inherited the same mutant mitochondrial DNA therefore do not inherit the same proportion of mutant mitochondria in each of their cells. One sibling may sit well below the threshold in every tissue and be asymptomatic; the other may sit above it in muscle and heart and be severely affected. Same mutation, same family, very different disease. That is the threshold effect of the previous topic, and random partitioning is the mechanism.

EXAM HOOK	This is the first Tier-1 question he named in this lecture, in his own words: *"Again, like I - I remember that I asked this question during the midterm in the first year, yeah. I think only half student can answer it correctly, yeah... Actually quite a surprise to me, yeah. So again this is the type of the question that I would try to include in the midterm, yeah. So when you do the revision, try to think in this way, yeah."*

In plain English: this question was asked in the first-year midterm, only about half the class got it right, and he is telling you to expect it again. The answer must contain four moves. First, the observation: one cell with two nuclei, and a 4N DNA peak on the flow plot instead of separate 2N and 4N peaks. Second, the mechanism: the contractile ring of actin and myosin drives cytokinesis, and it is a different machine from the microtubule spindle that drives mitosis, so blocking the ring leaves mitosis running normally. Third, the clinical meaning: the cell has divided its contents but cannot cleave, and at the level of the whole body, random partitioning of mitochondria at cytokinesis is why siblings carrying the same mitochondrial mutation can have very different severity. Fourth, the limitation: **mitosis and cytokinesis are separate processes**, and that distinction is the entire point of the question. He did not flag the act-versus-myosin detail as examinable, but the ring, the two nuclei and the 4N peak are all named.

Figure F8, above, shows the whole thing in one picture: the ring of phases on the left, and on the right the blocked cell with its two nuclei and its 4N DNA content.

9.3.8 Making four different gametes - Meiosis and recombination

WHAT IT IS	**Meiosis** is the specialised division that makes gametes - sperm and eggs - and it is two divisions in a row. In **meiosis I**, **homologous chromosomes** separate: the chromosome you got from your mother goes one way and the one you got from your father goes the other. In **meiosis II**, the **sister chromatids** separate, exactly as in mitosis. The end result is **four haploid** gametes - each carrying one set of chromosomes instead of two - and each one different from the others.
HOW IT WORKS / WHY IT MATTERS	The source of that difference is **crossing over**, also called **recombination**. While the homologous chromosomes are paired in meiosis I, they swap segments with each other. His slide 40 draws a parent with two homologous pairs and alleles labelled A, B, C on one and a, b, c on the other. After a crossover, the products are A-B-c and a-b-C - **recombinants** - alongside the unchanged A-B-C and a-b-c. In plain English: the information that came from your mother and the information that came from your father are physically spliced together, so a single egg or sperm carries a combination of genes that neither parent ever had. That is the **diversity** half of Dr. Wong's two running themes, and it is why meiosis is the raw material for everything from natural variation to the antibody diversity of Lecture 1 and the neoantigens of Lecture 4.

The comparison on the same slide is worth having straight. Mitosis from a **diploid** parent with two homologous pairs gives **two diploid daughters that are identical to the parent**. Meiosis from the same parent gives first two non-identical haploids, then **four haploids**. The difference is not just the number of divisions; it is that meiosis shuffles.

EXAM HOOK	He did not flag meiosis itself as a named question in Lecture 2; he used it as the diversity example. The reason to know it precisely is that it comes back in Lecture 3 as a Tier-1 question, and he says there: *"And you might need to distinguish the M1 or the M2 failure, because if there's a M1 failure, there's a formation of the germ cell without a losing copy of the chromosome after the M1..."* and *"And I think you need to distinguish these two situations here, yeah."* So learn which chromosomes separate in each division, and that recombination happens in meiosis I and not in meiosis II. The non-disjunction detail itself belongs to Lecture 3.

9.3.9 Two ways for a cell to die - Apoptosis and autophagy

WHAT IT IS	**Apoptosis** is programmed cell death: the cell deliberately dismantles itself because it is damaged, infected, or no longer needed. **Autophagy** is the opposite gesture - the cell digests its own unneeded parts to get nutrients, rather than dying.
HOW IT WORKS / WHY IT MATTERS	Apoptosis has two routes, and his slide 42 draws them side by side. The **intrinsic** route starts with stress inside the cell: the mitochondria release **cytochrome c**, which joins **APAF-1** and procaspase-9 into an **apoptosome**, which activates caspase-9. The **extrinsic** route starts with a signal from outside: the ligand **TRAIL** binds a **death receptor** on the cell surface, which switches on caspase-8, which activates caspase-3 and also cuts **bid** into **tbid** to feed the signal into the mitochondria. Either way the cell ends up executing itself, in an orderly way, without spilling its contents and causing inflammation. When this goes wrong in the wrong direction, the result is cancer: a cell that should have died survives and keeps dividing. That is why deregulation of apoptosis is a hallmark of cancer.

Two therapeutic ideas follow. First, **CDK4/6 inhibitors** such as palbociclib work the other way round - they do not kill the cell, they arrest it, so that the cell cannot keep dividing. Second, and more elegant, the **bispecific antibody** approach: an antibody with two binding heads, one aimed at a death receptor and one at a tumour-specific antigen, so that it only triggers death where both are present. Dr. Wong's worked example is **BI 905711**, a **tetravalent bispecific antibody** targeting **TRAILR2** and **CDH17** (cadherin-17), the protein his own twenty-year research is built on. His slide shows the three-tissue logic in three columns: normal liver has no CDH17 and is highly TRAIL-sensitive, so nothing happens; normal gut has CDH17 but is only weakly TRAIL-sensitive, so again

nothing happens; colorectal cancer has both, so the tumour apoptosis fires and the normal tissue is spared. Below that, patient-derived xenograft curves show

BI 905711 given every seven days suppressing tumour volume over 0-25 days against vehicle.

Autophagy is the survival mechanism. His example is the **hypoxic** core of a solid tumour, where there is no blood supply and therefore no oxygen and no nutrient. Cells there engulf and digest non-essential organelles to keep going. In plain English: the tumour eats its own furniture to stay alive in a neighbourhood that cannot supply it.

EXAM HOOK	He did not flag the caspase cascade for memorisation; the detail on the slide is teaching material, and the same exclusion he applied to signalling cascades applies here - *"I don't memorize all this thing, I just highlight here in case some student are really interested in molecular biology."* The caspase detail itself is not on any exclusion list, so this is an inference, not something he said about apoptosis. The concept he does want is the two-route logic, the link between apoptosis failure and cancer, and the selectivity idea - a drug that kills tumour cells and spares normal tissue by requiring two things to be present at once. The BI 905711 three-tissue comparison is the clearest clinical illustration of that idea in the whole lecture.

9.3.10 Turning a skin cell back into a stem cell - Stem cells and iPSCs

WHAT IT IS	A **stem cell** is a cell that can copy itself and can also turn into more specialised cells. There are three classes, and his slide 46 arranges them by how many things they can become. **Embryonic stem cells (ESC)** come from a blastocyst, are **pluripotent** - able to become almost any cell type - and can renew themselves indefinitely. **Adult stem cells** are found in the body, are **multipotent** - they can become a limited family of related cell types - and have a finite capacity. The **haematopoietic stem cell (HSC)** is the example, and his slide 48 traces the whole blood-cell family tree from long-term HSC down through short-term HSC and the multipotent progenitor to the common lymphoid and common myeloid branches, ending at granulocytes, macrophages, megakaryocytes, red cells and T cells. **Induced pluripotent stem cells (iPSC)** are the third class, and they are the revolution: they are ordinary body cells pushed back into a stem-cell-like state.
HOW IT WORKS / WHY IT MATTERS	The reprogramming was done in 2006 by Takahashi and Yamanaka, and the recipe is four factors, the **Yamanaka factors**: **Oct3/4**, **Sox2**, **c-Myc** and **Klf4** (his slide also lists Nanog and LIN-28). Feed those four into an ordinary **fibroblast**, a skin-connective-tissue cell, and within a few weeks it stops behaving like a fibroblast and starts behaving like an embryonic stem cell: it grows in tight, round colonies, and it can be pushed into making nerve cells, muscle cells, fat cells, pancreatic beta cells or blood cells.

But claiming a cell is an iPSC is easy and proving it is hard, so there is a validation ladder. Step one is **morphology**: does the cell look like an embryonic stem cell? Step two is markers: does it express the right surface markers and the right genes, by PCR and flow cytometry? Step three is the **gold standard**, the **teratoma assay**: inject the candidate cells into an immunodeficient mouse and check what they grow into. If you find tissue from all three **germ layers** - **ectoderm** (skin and nerve), **mesoderm** (muscle, cartilage, blood) and **endoderm** (gut lining and internal organs) - then the cells really are pluripotent. His slide 51 is the photograph that makes the point unforgettable: a mature cystic teratoma with hair, oily sebaceous material and an actual tooth in it.

Why this matters clinically is that iPSCs solve the ethical problem and add two new capabilities. Because the cells come from an ordinary skin biopsy rather than an embryo, the embryo question largely disappears. And because you can make a cell from *this patient*, you can grow that patient's own disease in a dish. His slide 52 splits into two branches. The drug branch: differentiate the cells into **organoids**, run drug screens in multi-well plates, read out activity, and validate a target - which is how you spot **cardiotoxicity** on iPSC-derived cardiomyocytes before a human ever takes the drug. The therapy branch: correct the faulty gene in the iPSC, differentiate it into the healthy specialised cell, and give it back as an autologous treatment.

EXAM HOOK	He flagged the quality-control logic rather than the marker list, with the words: *"unique expression pattern and timing, and also the negative staining to exclude the contaminated cell. Don't forget it."* In plain English: an iPSC is not proved by looking positive for one marker; you need the right combination of markers, at the right time, and you need the negative stains that show no contaminating cell is present. He did not ask you to memorise the full haematopoietic marker panel or the complete marker list from the 2006 paper. He did treat the teratoma assay as the gold standard, and the four Yamanaka factors are worth knowing by name.

9.3.11 Correcting a faulty gene - Gene therapy and genome editing

WHAT IT IS	**Gene therapy** means treating disease by changing the genetic material inside a patient's cells. In **ex vivo autologous** gene therapy, you take the patient's own cells out, fix the faulty gene in the laboratory, grow the corrected cells, and put them back into the same patient. Nothing is given to anyone who is not the donor, so there is no risk of an immune reaction against foreign material.
HOW IT WORKS / WHY IT MATTERS	The tool that changed this field is **CRISPR/Cas9**. **Cas9** is an enzyme; **gRNA** (guide RNA) is the address label that tells it where to cut. The enzyme finds a short DNA sequence next to a **PAM** (protospacer adjacent motif) site, cuts the DNA there using its two nuclease domains, **RuvC** and **HNH**, and leaves a double-strand break. What happens next is the choice you make on purpose. If the cell repairs the break by **non-homologous end joining (NHEJ)**, the broken ends are simply joined back together with small insertions or deletions - an **indel** - and the gene is knocked out. If the cell repairs it by **homology-directed repair** using a supplied template, a whole new piece of sequence can be written in, and the gene is knocked in. Cas9 is short enough to fit inside a **lipid nanoparticle** or an **adenovirus**, which is how the instructions get delivered into cells in the first place. Doudna and Charpentier shared the 2020 Nobel Prize in Chemistry for it.

Dr. Wong makes one comparison point that matters practically: **CRISPR/Cas9 is now standard for gene knock-out and knock-in, and the older siRNA and shRNA knockdown techniques are largely superseded.** In plain English: for ten years the field used small RNA molecules that quietly dimmed a gene's output; CRISPR now lets you switch the gene off outright, or rewrite it, and it does the job faster and more precisely.

He also spends a moment on the business side, because the science does not exist in a vacuum. The CRISPR patent was contested between the Doudna-Charpentier group at UC Berkeley and the Zhang group at the Broad Institute, and he uses it to teach a concept a biotech scientist needs:

Freedom to Operate. That is the question of whether you may build a commercial product even if you did not invent the tool, and it is a legal question that is decided before the first experiment. His slide 54 also carries a *Science* news article from 10 January 2023 with the headline "FDA no longer needs to require animal tests before human drug trials", with the pipeline running Target Validation -> Lead Identification and Optimization -> Preclinical MOA -> Toxicology studies -> Phase 1 Clinical Trial.

EXAM HOOK	The comparison between CRISPR and the older knockdown methods is on his Tier-2 list, in his words: *"CRISPR/Cas9 is now standard for gene knock-out/knock-in; older siRNA/shRNA knockdown techniques are largely superseded."* He also put the CR ISPR-Cas9 repair-pathway logic on the exam-focus list in Lecture 4, because knock-out goes through NHEJ and knock-in goes through HR. He did not ask you to memorise the patent dispute or the vector-delivery details, and he did not flag the patent story as a named question.

9.3.12 Two versions of the same antibody - Therapeutic antibody isotypes

WHAT IT IS	An antibody has two ends. The **Fab** end is the target-recognition business end - the hook that catches one specific target. The **Fc** end is the business end - the part that tells the rest of the immune system what to do. The **isotype** is the class of the Fc end, and it is the clinical dial. The two classes that matter here are **IgG1** and **IgG4**.
HOW IT WORKS / WHY IT MATTERS	**IgG1** is the fighter. Its Fc region binds **FcgammaRIII** receptors on **natural killer (NK) cells**, and that recruits them to kill the cell the antibody is stuck to. The mechanism has a name: **antibody-dependent cellular cytotoxicity (ADCC)** - the antibody does the grabbing, and the body's own killer cell does the killing. **IgG4** is the pacifist. Its Fc region binds those Fc receptors poorly, so it does **not** trigger ADCC - but that same property makes it much more stable in the blood, so it has a **longer serum half-life**, meaning it stays in the circulation longer before it is cleared.

His worked example is **trastuzumab** (Herceptin), an IgG1 against **HER2**. And his slide 20 shows the four-step consequence of **HER2 amplification**: more gene copies, then more mRNA, then more receptor on the cell surface, then more of the receptor's outer piece shed into the blood. Trastuzumab works in two ways at once: it blocks HER2 from pairing with HER3, which switches off the downstream **PI3K** signal (subunits p85 and p110), and its Fc end recruits NK cells to the tumour membrane through ADCC.

The antibody machinery also comes back in the exam from a completely different direction. Dr. Wong's Tier-1 question in Lecture 4 asks what happens if the **NHEJ** repair pathway is defective, and his answer is: *"Therefore, if I ask you, if the people showing a defective in this non-homologous end joining, what's the outcome? Definitely the outcome is the severe immune deficiency, right? Because he cannot generate a sufficient amount of the antibody pool to recognize the thousand copies of the foreign antigen. Yeah, that's the clinical outcome of this genetic defect."* The chain you must be able to draw: NHEJ is borrowed by **V(D)J recombination** to assemble the V, D and J gene segments into a working antibody or T-cell receptor, and the segment copy numbers he gives are **V about 39 copies, D = 27 copies, J = 6 copies**. Break NHEJ and you cannot build the repertoire, and the result is **SCID**, severe combined immunodeficiency.

EXAM HOOK	**IgG1 vs IgG4 is on the exam-focus list as a named distractor pair**, with the discriminating fact spelled out: IgG1 gives you ADCC, IgG4 gives you no ADCC but a longer half-life. He said in Lecture 1 that he would come back to the isotypes properly in Lecture 2, and this is that lecture. He did not ask you to memorise the full immunoglobulin class list. The NHEJ-to-SCID question itself is a Lecture 4 item, but the antibody half of it is here.

9.3.13 From a target to a patient - The drug-development pipeline

WHAT IT IS	Getting a drug to a patient is a long, expensive, strictly ordered pipeline. It starts with **target identification** - finding a molecule the disease genuinely depends on. Then **medicinal chemistry** - turning that target into an actual compound with the right shape, solubility and potency. Then the **preclinical** stage: **MOA** (mechanism of action, the evidence that the compound does what you think it does) and **toxicology** (the evidence about what it does to a whole animal). Then **IND-enabling studies** - the formal package that licenses the company to go to the regulator, the **IND**, an Investigational New Drug application. Then the human trials, which are the part you will meet on the exam.
HOW IT WORKS / WHY IT MATTERS	The human trials are split into stages with different jobs. **Phase 1A** is **dose escalation**: small groups of patients, starting with a tiny dose and increasing it until either the drug works or the toxicity appears. **Phase 1B** is the **expansion** stage: more patients, at the dose that Phase 1A settled on, to find the right dose and to look for response. **Phase 2** and **Phase 3** then test **efficacy** - Phase 2 in patients, Phase 3 in large randomised trials comparing the new drug against the current standard of care.

Before any of that, companies hold **pre-IND** and **pre-pre-IND** meetings with the FDA. The purpose is to agree on the animal model, and this is a genuine scientific argument: a

pharmacologically relevant non-human primate may model the human disease far better than a

humanised mouse, but a primate study is vastly more expensive and ethically harder, while a humanised mouse is fast and cheap but may not reproduce the human phenotype. Choosing the model badly at this meeting wastes the whole programme.

And the cost is why the pipeline exists in the shape it does. Dr. Wong's key number for the clinical stage is **~US$200,000 per patient**. In plain English: enrolling one person in a clinical trial costs on the order of two hundred thousand US dollars, so a trial's design - how many patients, which arms, which endpoints - is a business decision as much as a scientific one.

EXAM HOOK	Two of his key numbers sit directly on this topic: the **~US$200,000 per patient** cost of a trial, and the **Phase 1A (dose escalation) -> Phase 1B (expansion) -> Phase 2/3 (efficacy)** structure. He did not ask you to memorise the regulatory meeting paperwork or the full IND checklist. What he wants is that you can say what each phase is *for*: Phase 1A finds the dose, Phase 1B expands the population, and Phase 2 and 3 test whether it works.

9.3.14 Bolting a receptor onto a T cell - CAR-T cell therapy

WHAT IT IS	A **T cell** is one of the immune system's own killer cells. A **CAR** (chimeric antigen receptor) is a synthetic receptor built by joining together an antibody's recognition head with the signalling machinery of a T cell. The patient gets their own cells back with this new receptor on the surface, and those cells can now recognise a tumour they previously ignored.
HOW IT WORKS / WHY IT MATTERS	The anatomy is the whole story. The extracellular end is an **scFv** (single-chain variable fragment), which is the **VH** and **VL** variable domains of an antibody joined by a **linker** - the same binding machinery as a normal antibody, so the CAR recognises exactly what the antibody would. Below that comes the **hinge**, then the **transmembrane domain** that anchors the whole thing in the T-cell membrane, then the **co-stimulatory domains** - **CD28** and **4-1BB** are the standard choices - and finally the intracellular signalling part, built from **CD3-zeta** and **ZAP70**. When the CAR's scFv latches onto its antigen, the intracellular domains fire, the T cell switches on, and it kills: it secretes **perforin**, which punches a pore in the tumour cell, and **granzymes**, which enter through the pore and trigger the tumour cell's own death programme.

His slide 55 contrasts this with full T-cell activation, which needs two separate signals - the T-cell receptor meeting its antigen on MHC, plus a co-stimulatory signal from a dendritic cell in the lymph node. The CAR collapses both into one receptor. That is why CAR-T works: it does not need the lymph node.

His slide 59 gives the **manufacturing cycle** as five numbered steps: extract the T cells from the patient's blood; engineer them with the CAR; grow millions of them; infuse them back into the patient; and then they bind the cancer cells and kill them. In practice the activation step uses anti-CD3 and anti-CD28 antibody-coated beads plus the cytokines **IL-2** and **IL-15**, and before the infusion the patient receives **lymphodepletion** chemotherapy, which clears some of their own lymphocytes so the CAR-T cells have room to expand.

There are three ways to get the CAR onto a T cell, and this is one of the named distractor triples. **Autologous** CAR-T uses the patient's own cells: no rejection risk, but the cells have been through chemotherapy and may be in poor condition - poor **T-cell fitness** - and manufacturing and shipping is slow and expensive. **Allogeneic** ("off-the-shelf") CAR-T is made from healthy donors, or from iPSCs, and is manufactured in advance as a standard product; his slide 67 shows iPSC-derived cells being gene-edited at the **TRAC** locus with a **ZFN** or CRISPR/Cas9 so that the cells will not attack the recipient. **In vivo** CAR-T skips the manufacturing step entirely: an mRNA encoding the CAR is delivered inside a **lipid nanoparticle (LNP)** targeted to the patient's own T cells, which then make the receptor on themselves. His slide 68 traces that route precisely - an anti-CD5-decorated LNP, CD5-targeted endocytosis, escape from the endosome, degradation of the LNP to release the mRNA,

translation by the ribosome, and the **FAP** CAR appearing on the T-cell surface, where it binds **FAP** on an activated fibroblast.

Two commercial facts frame all of this. Slide 63 lists the **solid-tumour barriers** that stop CAR-T working outside blood cancers: the inhibitory checkpoints **CTLA4**, **PD1**, **LAG3**,

TIM3 and **VISTA** on the effector cells, with **PDL1** and **PDL2** on the target cells; missing chemokine receptors **CCR2b** and **CCR4**; the suppressing cells **Tregs**, **MDSCs** and **TAMs**; the signals **TGFbeta** and **IL-10**; a stroma-rich **ECM** with **HSPGs**; and a hostile physical environment of hypoxia, low pH, **CD47**, **CD73**, adenosine and **IDO**. The same slide puts the price on it: **USD 370,000 to 475,000** per CAR-T treatment. Slide 65 adds the cryo-logistics cycle - cryotransport, cryostorage in liquid-nitrogen dewars, lymphodepleting chemotherapy, infusion, and monitoring - with a world map showing shipping between centres. And slide 69 carries a press release from 14 September 2022: "Capstan Therapeutics Launches with $165 Million to Deliver on the Clinical Promise of Precise In Vivo Cell Engineering", made up of a

$102 M Series A and a **$63 M** seed round.

EXAM HOOK	Two pairs are on the exam-focus distractor list, and you must keep them apart: **autologous vs allogeneic** CAR-T, patient-derived versus donor- or iPSC-derived; and **in vivo CAR-T**, which is mRNA in lipid nanoparticles with the cells engineered inside the body. **Receptor occupancy** is not one of those pairs - it is a key number rather than a pair. The occupancy figures he states are exact: Herceptin in **breast** cancer needs **2 %** and Herceptin in **gastric** cancer needs **10 %**; the **1 %** that a typical antibody needs is a Lecture 5 figure rather than a Lecture 2 one. In plain English: the same drug has to grab far more of its target in the stomach than in the breast before it does anything, which is why the same antibody needs different dosing in different tissues. He did not ask you to memorise the full CAR domain architecture or the solid-tumour barrier list; he flagged the three flavours of CAR-T and the occupancy numbers.

9.3.15 Reading the tumour from a blood sample - Liquid biopsy and circulating tumour cells

WHAT IT IS	A **circulating tumour cell (CTC)** is a cancer cell that has broken away from a solid tumour and entered the bloodstream. **Liquid biopsy** means using that blood sample as your specimen, instead of cutting a piece out of the tumour with a needle.
HOW IT WORKS / WHY IT MATTERS	The isolation is a flow-cytometry problem, and it uses two markers that pull in opposite directions. **EpCAM** is a protein on the surface of epithelial cells, and most solid tumours are epithelial, so you keep the cells that are **EpCAM-positive**. **CD45** is the classic marker of white blood cells, and you want to exclude them, so you keep the cells that are **CD45-negative**. Selection is therefore **EpCAM-positive / CD45-negative**. The clinical payoff is that a blood draw can be repeated as often as you like, which lets a doctor monitor a metastatic tumour over time without repeatedly operating on the patient.

The catch, and it is a real one, is **EMT** - the epithelial-to-mesenchymal transition. A cell that has left the tumour and travelled in the blood often changes shape and changes its surface markers, losing the epithelial proteins it started with. So the marker profile of a CTC is not automatically the marker profile of the primary tumour. In plain English: the sample in the blood tube is a biased and partly transformed picture of the tumour, and reading it as if it were the tumour itself will mislead you.

His slide 73 shows why CTC counting is worth doing. The **Kaplan-Meier** curve for overall survival in breast cancer, split by CTC count, has three lines: patients with **CTC0** have the best survival, at about 67 % alive at 100 weeks; patients with **CTC1-4** are intermediate, at about 50 % alive at around 80 weeks; and patients with **CTC >= 5** do worst, at under 20 % alive after 80 weeks. Read that with his Kaplan-Meier rule from Lecture 1: the curve that drops steeply and early is the bad group, and here the number of CTCs separates the patients into groups with genuinely different outcomes.

EXAM HOOK	He did not flag the CTC count itself as a named question. The line *"the kind of the question that I would include in the exam"* was said about the Kaplan-Meier plot question, not about this curve - but the CTC-stratified survival curve is case-bank material, and it reads like a Kaplan-Meier plot like any other: curve starts at 1.0, falls in steps, flat means no events, a step is an event, ticks are censored patients, and a curve that drops steeply and early means early involvement and a poor prognosis. The **EpCAM-positive / CD45-negative** selection rule and the **EMT** caveat are the two things to hold on to, because a scenario question about a liquid biopsy will turn on the contamination and the marker-mismatch problems.

9.3.16 Why some tumours hide from the immune system - The tumour microenvironment and immune checkpoints

WHAT IT IS	A tumour is not a lump of cancer cells. It is an **ecosystem**, the **tumour microenvironment (TME)**: cancer cells, plus immune cells that are trying to kill them, plus **cancer-associated fibroblasts (CAFs)** which build the scaffolding, plus the **ECM**, the extracellular matrix that holds it all together. The **immune checkpoint** system is the part of the immune system that prevents over-reaction, and tumours abuse it.
HOW IT WORKS / WHY IT MATTERS	The mechanism is a two-molecule handshake. **PD-L1** is a protein on the surface of tumour cells. **PD-1** is a protein on the surface of T cells. When PD-L1 binds PD-1, the T cell is switched off into a state called **anergy** - it is still there and still attached, but it has been told to stand down. That is **immune evasion**, and it is why an anti-PD-1 antibody works: it binds PD-1 and stops the handshake, and the brake is released.

Not every tumour manages this. Dr. Wong's contrast is between a **hot tumour** and a cold one. A tumour with **microsatellite instability high (MSI-H)** status - which means its DNA mismatch repair machinery (**MMR**) is defective, so copying errors go uncorrected - accumulates many

frameshift mutations, and frameshifts manufacture **neoantigens**, brand-new protein fragments the immune system has never seen. A high neoantigen load means the tumour is visible and the immune system can get in: that is a hot tumour, and hot tumours respond to anti-PD-1 or anti-PD-L1. His key number is that **15 %** of colorectal cancers are MSI-H - in plain English, about one colon cancer in seven. Cold tumours, where neoantigen load is low, are the ones that do not respond.

His slide 16 puts the two groups side by side on a **progression-free survival** Kaplan-Meier curve in colorectal cancer (Le DT and colleagues, *New England Journal of Medicine*, 2015), with the MMR-deficient cohort clearly above the MMR-proficient cohort and **P < 0.001 by log-rank test**. He later calls that dMMR-versus-pMMR figure *"one of the most important figures in modern medicine"*.

One more piece of the microenvironment story, and it is the bridge to why CAR-T struggles in solid tumours: getting an immune cell to the tumour is a four-step journey, drawn on slide 19. First

rolling along the blood vessel wall, using **P-selectin**, **E-selectin** and integrins. Second **adhesion and activation** through **ICAM-1** and **VCAM-1**, some of which are soluble and shed into the blood. Third **transmigration** through the vessel wall via **PECAM-1**. Fourth

chemotaxis towards the tumour, following a chemical gradient such as **MCP-1**. Every step is a checkpoint a therapy can be blocked at, and a tumour that is surrounded by dense stroma makes all four steps harder.

EXAM HOOK	The hot-versus-cold distinction is his own words, in Lecture 5: *"So guys, you definitely remember that for the tumor that present many neoantigen and can be treated with the immunotherapy, we call this is a hot tumor, yeah. And not all the solid tumor is a hot tumor."* The dMMR-versus-pMMR survival curve is also his: *"I would say this figure is one of the most important figures in modern medicine, because when you attend any seminars about the immune checkpoint inhibitors or immunotherapies or colorectal cancer, I can for sure that the presenter will include this figure in his presentation deck."* The **15 % MSI-H** figure is on his key-numbers list, and "MSI-H -> anti-PD-1" is on his likely multiple-choice stems: *"A patient with X is most likely to respond to which drug?"* He did not ask you to memorise the four-step extravasation molecules.

9.3.17 A virus that switches on a human gene - HBV integration in hepatocellular carcinoma

WHAT IT IS	**HBV** is hepatitis B virus. **Hepatocellular carcinoma (HCC)** is cancer of the liver, the same tumour Dr. Wong used for the serum vimentin biomarker in Lecture 1. HBV **integration** means pieces of the viral DNA have become permanently inserted into the human genome inside the tumour cell.
HOW IT WORKS / WHY IT MATTERS	The virus has a lifelong hiding place. Its life cycle, drawn on slide 17, goes: the virus binds a receptor and enters; the capsid is uncoated; the relaxed circular DNA (**rcDNA**) is carried into the nucleus; and there it is converted into **cccDNA**, a small circular molecule that sits in the nucleus as an **episome** - a piece of DNA that replicates independently of the host chromosome and is not physically part of it. That reservoir is why HBV is so hard to eradicate: even when you clear the virus from the blood, the cccDNA keeps sitting there. From it, the virus transcribes its own pgRNA and mRNA, translates its proteins, packages them with reverse transcription back to DNA, buds into the endoplasmic reticulum and is secreted, together with the HBeAg and HBsAg antigens.

The clinically decisive event is where the viral DNA lands. When HBV DNA integrates into the host genome it does so at places that switch human genes on inappropriately. His slide 18 shows the actual breakpoints, read straight off **Sanger chromatograms**: **TERT** on chromosome 5 (HCC #65T, human position 1,298,958 -> HBV position 1,825), **CCNE1** on chromosome 19 (HCC

#200T, position 30,315,004 -> HBV 1,799), and **MLL4** on chromosome 19 (HCC #70T, position 36,212,309 -> HBV 128). TERT, CCNE1 and MLL4 are all **oncogenes** - genes that push a cell to keep growing. Landing next to one and switching it on is how a chronic viral infection turns into a cancer.

And the more integration sites there are, the worse the outcome. His slide 18 also carries the survival analysis: patients with **fewer than 3** integration sites (n = 51) versus patients with

3 or more (n = 32), **P = 0.037**, with the >= 3 group significantly worse. In plain English: the more places the virus has inserted itself, the shorter the patient's survival.

EXAM HOOK	This is on his key-numbers list as **3 or more HBV integration sites -> significantly poorer survival**, and it is a genuine prognostic finding rather than a mechanism: it tells you how the disease will go. He did not ask you to memorise the Sanger chromatograms or the chromosome positions, and he did not ask you to memorise the whole viral life cycle; the two things to keep are that **cccDNA is a persistent nuclear reservoir that is hard to eradicate**, and that **integration near TERT, CCNE1 and MLL4 drives hepatocarcinogenesis**.

9.3.18 The hardest proteins to measure - Membrane proteomics

WHAT IT IS	**Membrane proteins** are the ones built into the lipid bilayer - receptors, channels, transporters. **Membrane proteomics** is the attempt to study them all at once with **mass spectrometry**, the same technique that read the vimentin proteome in Lecture 1.
HOW IT WORKS / WHY IT MATTERS	The technical problem is real and specific. The parts of a membrane protein that sit in the middle of the bilayer are **transmembrane segments**, and they are covered in amino acids that are water-hating. Ordinary mass spectrometry works on molecules dissolved in water; a protein with several water-hating segments will not dissolve and will fall out of solution before it ever reaches the instrument. You need **detergents** or lipids to hold it together, and then the detergent gets in the way of the measurement. So the method that reads thousands of soluble proteins at once is exactly the method that struggles with the ones that matter most.

And yet these are prime drug targets: **GPCRs** (G-protein-coupled receptors), **receptor tyrosine kinases** and **ion channels** all sit in the membrane, and a drug can reach them from outside the cell without having to get inside. They are also the best cell-surface biomarkers, because a cell-surface protein is the one a blood test can actually see. That is the direct connection back to the earlier topics in this chapter: **HER2** and **PD-L1** and **EpCAM** are all membrane proteins, all are measurable, and all are drug targets. When one of them is amplified, you need a **companion diagnostic** - a test that tells you whether this patient is going to benefit from this drug - which is exactly what FISH and IHC provide.

EXAM HOOK	He did not flag membrane proteomics as a named exam question. The concept to keep is the paradox: the proteins that are hardest to measure are the ones that are most druggable and most useful as biomarkers. He did not ask you to memorise detergent chemistry or any specific mass-spectrometry method.

9.4 The ten things to remember

1. Flow cytometry reads cells one at a time: **FSC = size**, **SSC = granularity**, and **gating** is how you pick the population you want out of the plot.
2. FACS physically sorts live cells into tubes. LCM cuts a clean region out of a tissue section. FISH counts gene copies with a DNA probe - that is how HER2 amplification is found.
3. The mitochondrion is a bacterium that was swallowed and kept. That is why it has its own DNA, why it is inherited from the mother only, and why an affected father passes it to none.
4. Mitochondrial disease needs a **threshold**: disease appears only when the mutant fraction of mtDNA exceeds what that tissue can tolerate. High-energy tissues fail first. A **nuclear** gene can imitate the whole pattern.
5. The cell cycle is G1, S, G2, M, then cytokinesis. Cyclins are the keys and CDKs are the machines. Serum starvation down to **0.5-1 % FBS** arrests cells at G1 so they can be synchronised.
6. **Mitosis is dividing the contents. Cytokinesis is cutting the cell in two.** Block the contractile ring and you get one cell with **two nuclei and 4N DNA**, because mitosis is unaffected. This is a named exam question.
7. Cytokinesis partitions mitochondria at random, which is why siblings with the same mutation can have very different severity.
8. Meiosis shuffles: crossing over between homologous chromosomes produces four genetically distinct haploid gametes. That is the raw material for diversity.
9. IgG1 recruits NK cells through Fc receptors and triggers ADCC. IgG4 does not, but it stays in the blood longer. The same antibody can need 2 % receptor occupancy in breast and 10 % in stomach.
10. A tumour is an ecosystem, not a lump: cancer cells, immune cells, fibroblasts and matrix. PD-L1 shuts the T cells down; MSI-high tumours have too many neoantigens to hide, so they respond to anti-PD-1. HBV integration near TERT, CCNE1 and MLL4 drives liver cancer, and **3 or more integration sites** means significantly worse survival.

9.5 Check yourself

Q1. You treat a culture of dividing cells with a drug that blocks the contractile ring. What do you see under the microscope, and what does the DNA-content plot from flow cytometry look like?

A1. The observation comes first: you see a single cell with two nuclei, and on the DNA-content histogram a single 4N peak where you would normally expect separate 2N and 4N peaks. The mechanism is that the contractile ring of actin and myosin drives cytokinesis, while mitosis is done by a completely different machine, the microtubule mitotic spindle. Because the two jobs are independent, blocking the ring stops the cytoplasmic division while the chromosome separation proceeds perfectly - the DNA copies, the chromosomes separate, two nuclei form, and the cell simply never cleaves. The clinical meaning is twofold: at the cell level you have a binucleated cell with 4N DNA content; at the level of the whole body, the same independence explains why siblings carrying the same mitochondrial mutation can have wildly different severity, because organelles and mitochondria are partitioned at random during cytokinesis. The action and the limit are in the distinction itself: this is the question Dr. Wong says he asked in the first-year midterm, where only about half the class got it right, and the point he is testing is that mitosis and cytokinesis are separate processes. You cannot answer it as "the cell stops dividing" - the cell does divide, it just cannot cut itself in two.

Q2. A mother's blood test shows a mitochondrial mutation. She has four children: a son and a daughter are affected, and the other two are not. Does this fit mitochondrial inheritance, and what would change your mind?

A2. The observation is the pattern of transmission: an affected mother with some affected and some unaffected children. The mechanism is that paternal mitochondria are tagged for destruction after fertilisation, so a child gets its mitochondria from the egg, and therefore from the mother. The classic hallmarks of mitochondrial inheritance are that an affected mother passes the condition to all her children, an affected father passes it to none, and the condition appears in every generation. The critical move is that this pedigree does **not** fit that classic pattern,

because an affected mother has unaffected children, and under strict maternal inheritance that should not happen. The clinical meaning is that you should suspect a **nuclear** gene instead: many of the proteins that make mitochondria work are encoded in the nuclear genome, and a mutation in one of those produces the same clinical picture with ordinary Mendelian inheritance. The action is to sequence both the mitochondrial genome and the relevant nuclear genes rather than assuming from the shape of the tree. The limit you must state is the **threshold effect**: even with a confirmed mtDNA mutation, disease appears only when the mutant fraction exceeds what that tissue tolerates, so a mutation is a probability, not a certainty, and the two unaffected siblings are not lucky - they may simply sit below threshold.

Q3. A colleague has been trying for months to express a therapeutic antibody in cells. The gene is correct, the protein sequence is correct, and nothing works. What is the first thing to check, and why?

A3. The observation is that the protein is not appearing where it should. The mechanism is that the gene is missing its **N-terminal signal peptide** - the short address label that tells the ribosome to hand the growing protein over to the rough ER. Without it the protein is made but never enters the ER, so it is never folded by chaperones, never glycosylated, never routed through the Golgi, and never reaches the plasma membrane or the outside of the cell. For a secreted antibody that means no product at all, and the failure is invisible, because the cell looks alive and the machinery is working. The clinical meaning is that this is the single most common way to waste months on a therapeutic protein that cannot possibly work. The action is to check the signal peptide first, before anything else, and to remember that it is also the address that decides the destination - surface, secretion, or lysosome. The limit: a signal peptide is not the same thing as the **N-terminal domain (NTD)** of a nuclear receptor, which is a regulatory region on a DNA-binding protein and has nothing to do with ER targeting. Dr. Wong's own postdoc lost

4 months of work this way.

Q4. Two patients have the same HER2-positive breast cancer and the same drug. One responds, one does not. The drug is trastuzumab, an IgG1 antibody. Give the mechanism, and then say what changes if you move the same drug to gastric cancer.

A4. The observation is that the same antibody behaves differently in two patients and in two tissues. The mechanism has two halves, and they come from the two ends of the antibody. The

Fab end binds HER2 and blocks it from pairing with HER3, which switches off the downstream PI3K signal. The **Fc** end is the business end: because trastuzumab is an **IgG1**, its Fc region binds **FcgammaRIII** on **natural killer cells**, which then kill the tumour cell - this is

antibody-dependent cellular cytotoxicity (ADCC). The clinical meaning is that the choice of

isotype is a clinical decision, not a manufacturing detail. If the drug were an **IgG4** instead, it would not trigger ADCC at all, because IgG4 binds those Fc receptors poorly - but it would have a longer serum half-life and stay in the blood longer. And the second half of the answer is the **receptor occupancy**: Herceptin needs about **2 %** receptor occupancy in breast cancer and about **10 %** in gastric cancer, with a typical antibody needing around **1 %** - a figure that comes from Lecture 5 rather than from this lecture. In plain English: the same drug has to grab far more of its target in the stomach than in the breast before it does anything. The action is that dosing has to be tissue-specific and that the isotype and the occupancy threshold both have to be in the dosing decision. The limit: this is a HER2 story; the same marker behaves in opposite directions in different tumour types, as the ERBB2 data in Lecture 1 showed, so you cannot carry the conclusion from one tissue to another.

Q5. A liver cancer patient has been infected with hepatitis B virus for twenty years. The biopsy shows the viral DNA inserted into the tumour genome at four separate places. What is happening, and what does the count tell you?

A5. The observation is that HBV DNA has integrated at four sites in the tumour genome. The mechanism is that the virus keeps a permanent reservoir in the nucleus as **cccDNA**, an episome that replicates independently of the host chromosomes, which is why the infection is so hard to eradicate. When the viral DNA integrates, it does so near human **oncogenes** - genes that push a cell to keep growing. His slide reads the actual breakpoints off Sanger chromatograms: **TERT** on chromosome 5, **CCNE1** on chromosome 19 and **MLL4** on chromosome 19. Sitting next to one of these and switching it on is how twenty years of chronic viral infection becomes a liver cancer. The clinical meaning is that the number of integration sites is a **prognostic** biomarker, not just a mechanistic detail: his slide compares patients with fewer than 3 integration sites (n = 51) against those with 3 or

more (n = 32), with **P = 0.037**, and the >= 3 group survives significantly less well. In plain English: the more places the virus has inserted itself, the shorter the survival. The action is that the integration count belongs in the prognostic assessment alongside the staging. The limit: it is a prognostic marker, not a predictive one - it tells you how the disease will go, not that a particular drug will work - and it does not tell you which sites are the functional ones without doing the breakpoint analysis.

End of Part 3. The next chapter, Part 4, covers Lecture 3 - DNA and Chromatin Structure, where the cell-cycle and checkpoint material from this lecture meets the way DNA is packed and switched on and off.

PART 4

Lecture 3: DNA and Chromatin Structure

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About this part

Source: GNBF5020 Lecture 3, "DNA and Chromatin Structure", 3 October 2026, Dr. Tony Wong. This chapter rewrites the expert shorthand of the existing revision guide so that it can be read by someone who has never studied biology or chemistry. Every number Dr. Wong stated is reproduced exactly and is followed by a plain sentence saying what it means.

CHAPTER 10

DNA and Chromatin Structure

10.0 What this lecture is about

Dr. Tony Wong opens Lecture 3 with a simple question: what happens to DNA when you put it inside a cell? The answer is that it gets folded, and the way it is folded decides which genes can be read. He then shows how a cell marks its own DNA with chemical tags that switch genes on or off without changing a single letter, and how doctors can read those tags in a patient's blood or stool to decide a treatment. The last movement of the lecture is about chromosomes themselves: what happens when the number of chromosomes goes wrong, when a whole chunk of the genome is copied twice or deleted, and how you tell a typo that causes cancer from a typo that just came along for the ride. By the end of this chapter you should be able to explain, in plain English, the seventeen ideas that this lecture builds on.

10.1 Words to know before we start

Term	In plain English
Chromatin	DNA together with the proteins that package it, inside the nucleus. The manual and its packaging.
Histone	A small protein that DNA is wound around. Strongly positively charged, and very similar in every species.
Nucleosome	A DNA double helix wrapped once around a bundle of eight histone proteins - 147 base pairs in all. A spool the recipe is wound onto.
Euchromatin	Loosely packed chromatin. The genes are exposed and can be read. "Open".
Heterochromatin	Tightly packed chromatin. The genes are buried and cannot be read. "Closed".
Methylation	Adding a small carbon-and-hydrogen tag (a methyl group) to a base. It is a chemical label, not a change of letter.
CpG island	A short stretch of DNA rich in C-next-to-G pairs, usually in front of a gene. Often the switch that turns a gene on.
Epigenetics	Turning a gene on or off without changing the DNA letters themselves. A label on the same recipe.
Histone code	The system in which chemical tags on the histone proteins record "open" or "closed".
Karyotype	A photograph of a person's chromosomes, cut out and arranged in matching pairs.
Centromere	The pinched middle of a chromosome where the spindle attaches to pull it apart.
Telomere	The repetitive DNA cap at the end of a chromosome. It acts as a shrinking timer.
Aneuploidy	Having the wrong number of chromosomes, because one pair failed to separate properly.
Copy-number variation (CNV)	A stretch of DNA that some people carry in several copies and other people carry in one.

10.2 The big picture

Start with the cell as a factory. The nucleus is the control room, and it holds the master instruction manuals. Those manuals are written on a very long molecule, DNA. But there is a problem: a cell has to carry two metres of DNA inside a nucleus you could cover with a thumbnail. So DNA is not loose in the nucleus. It is folded, tightly and in layers, like a printed manual bound into a book.

That folding is the whole subject of this lecture, and it is not just tidiness. A gene is a recipe. If a recipe is buried inside a tightly bound book, nobody can read it. If it lies on an open page, the cell can copy it and make the protein. So the folding state of the DNA is the first gate of control, before the later gates Lecture 5 will teach.

The folding has levels. A bare DNA helix is about 2 nanometres across. It wraps around eight histone proteins to make a **nucleosome**, a bead about 10 nanometres across, and the chain of beads is the "beads on a string" picture. The string coils into a 30 nm fibre, that fibre loops and thickens, and by the time the cell is ready to divide,

the whole thing has condensed into the familiar X-shaped chromosome, about 1,400 nanometres wide. Every level of folding is reversible: a cell must be able to open a gene now and close it in an hour.

Chromatin comes in two flavours. **Heterochromatin** is tightly packed and silent. **Euchromatin** is loosely packed and active. His slide 8 labels them "Silent" and "Active". These are not two different substances. The same DNA, the same histones, folded differently. And the folding is moved on purpose. Nucleosomes slide along the DNA to expose a promoter or bury it, for two reasons: because the DNA sequence prefers certain positions, and because ATP-powered

chromatin remodelling complexes push them around.

Then the lecture turns to chemistry that switches genes off: a cell can attach a methyl group to the cytosine base at particular spots. This is **DNA methylation**, the clearest example of

epigenetics: the letters are unchanged, but a tag has been added that decides whether the page can be read. Methylation is established by one family of enzymes and copied onto the daughter DNA by another, so a cell can pass the tag on to its children. When the wrong gene gets tagged, disease follows. A methylated tumour-suppressor promoter silences the gene that brakes cell division, and that alone can push a normal cell towards cancer. Because methylation leaves a chemical fingerprint, doctors can measure it. That is the point of **methylation-based biomarkers**, and the numbers he labours over here - odds ratios, confidence intervals, areas under a curve - are the evidence that these marks predict something real.

The same idea applies to the histones themselves. **Histone modifications** are chemical tags on their tails, and together they form the **histone code**. Acetylation tends to open chromatin; methylation can open or close it depending on which amino acid is tagged. The tags also talk to each other, and they talk to DNA methylation, in **positive feedback loops**. That is what makes

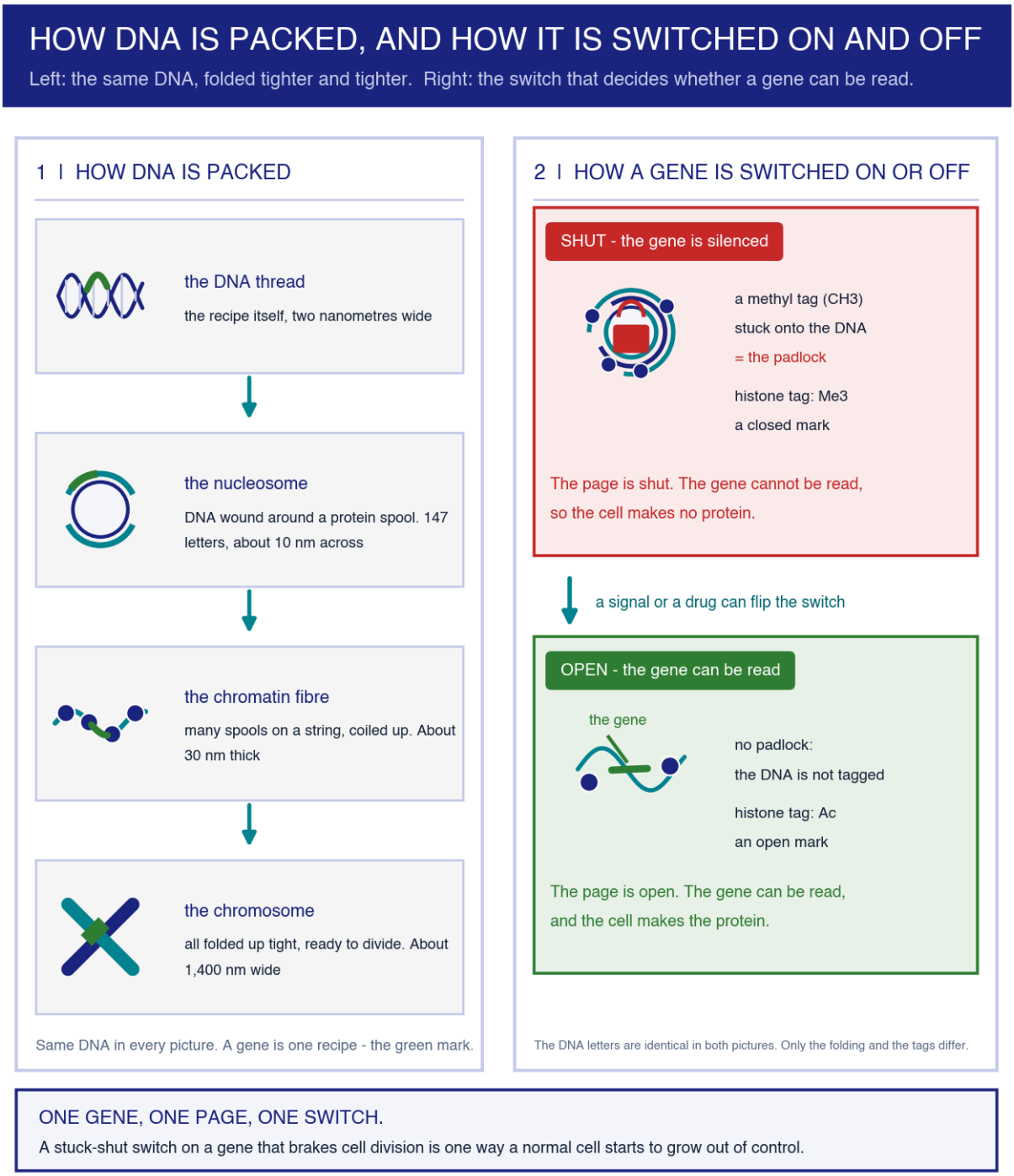
cellular memory possible: a liver cell can divide for years and still know it is a liver cell, because the marks reinforce each other and are copied at every division.

Around the middle of the lecture the scale jumps out to the whole chromosome. A **karyotype** is a photograph of the chromosomes, and **cytogenetics** is the study of what such a photograph can and cannot show. A chromosome has a short arm and a long arm joined at a **centromere**, and at each end sits a **telomere**, which shortens a little with every cell division until the cell hits the Hayflick limit and stops dividing.

The last third is about what goes wrong at the chromosomal level. **Aneuploidy** happens when one pair of chromosomes fails to separate properly during meiosis, and that single slip produces Down syndrome or Turner syndrome. **Copy-number variation** is the milder version: a stretch of DNA that some people carry in seven copies and other people carry in five. When the whole genome is read at once, the hard part is telling a **driver** - a typo in an important word - from a **passenger** - a typo on page 900. And finally, DNA **adducts**: chemicals from cigarette smoke and from the cell's own reactive oxygen species stick to the bases and change what gets copied - how most lung cancers start.

Figure F10 - Chromatin packing, and the marks that open or close it

Left: DNA wrapped into nucleosomes, then into a fibre, then into a chromosome. Right: closed chromatin with a methylation padlock and a histone tag, beside open chromatin that can be read.



Left: the same DNA, folded tighter and tighter, from a 2 nm thread to a chromosome. Right: the switch. The two right-hand panels are the same DNA with different labels on it - one shut, one open.

10.3 Walkthrough

10.3.1 The recipe book, one more time - DNA basics revisited

WHAT IT IS	DNA is a long molecule written in only four letters: A, T, C and G. Two strands run alongside each other, and each letter has exactly one partner. Adenine pairs with thymine. Cytosine pairs with guanine. This pairing is what lets one strand act as a template for the other, so the cell can copy its manuals before it divides.
HOW IT WORKS / WHY IT MATTERS	The pairs are held together by **hydrogen bonds**, which are weak, reversible attractions between atoms. A G-C pair has **3 hydrogen bonds** and an A-T pair has **2 hydrogen bonds**. So a GC-rich stretch of DNA is genuinely harder to pull apart than an AT-rich stretch of the same length. That hardness has a name: the **melting temperature**, or **Tm**, the temperature at which the two strands come apart. What his slide gives you is the direction, not a number: more GC content means a higher melting temperature, and that is what matters when you design a primer for **PCR** - a GC-rich primer needs noticeably more heat before it lets go, so you cannot ignore how much G and C you put in it. The exam notes also carry a '~40 deg C drop' and a '60 deg C versus 55 deg C' board example, and those two do not reconcile with each other - so learn the direction and the reason, and do not put a number on it.

Denaturation, also called **melting**, is the act of unwinding the double helix into two single strands. It is a normal everyday event inside a cell, and it is the first thing that happens in the PCR machine in Lecture 4. The practical point of the Tm value is that it tells you how hot you have to make the machine to separate the strands without cooking the primer.

The second half of the topic is **histones**, the proteins DNA is wound around. He makes two points about them. First, they are **highly conserved across species**, from Drosophila to human: the histone in a fruit fly is recognisably the same protein as the histone in you. That is why mouse and fly experiments can tell you something about human disease. Second, they are

difficult to crystallise, because they carry a heavy positive charge. Charged, floppy proteins do not sit neatly in a crystal lattice, and the crystal structure of the nucleosome was a hard-won prize.

EXAM HOOK	He did not flag "DNA basics" as a named exam question, but he did put the base-pairing numbers on his key-numbers list: **G-C = 3 hydrogen bonds; A-T = 2**, tagged to Lectures 1 and 3. The **~40 deg C drop in Tm** is on the exam notes' list too, but those notes disagree with themselves about that figure, so learn the direction rather than the number. The exam hook is downstream: he links GC content to **PCR primer design**, and PCR is examined in Lecture 4, so this is groundwork rather than a question in itself. Histone conservation he used as the reason model organisms work, and the crystallisation difficulty he mentioned as the reason the 2.8 A structure (below) took real effort. He did not ask you to memorise histone sequences.

10.3.2 From a thread to a printed manual - The chromatin hierarchy

WHAT IT IS	**Chromatin** is the DNA plus the proteins that package it. Inside the nucleus it is folded in layers, like a printed manual bound into a book. The nucleosome is the first level of that binding: eight histone proteins form a core, and about 147 letters of DNA wrap around it once, like a thread wound onto a spool.
HOW IT WORKS / WHY IT MATTERS	His slide 4 gives the diameters, and they are worth learning as a ladder because they show how extreme the compression is: **2 nm** for the bare helix, **11 nm** for the "beads on a string" of nucleosomes, **30 nm** for the chromatin fibre, **300 nm** for the extended form, **700 nm** for the condensed form, and **1,400 nm** for the mitotic chromosome. Read that as: the same DNA, folded up about 700-fold from the 2 nm thread to the 1,400 nm chromosome.

The nucleosome itself has a precise composition. The core is an **octamer**, which means eight copies: two of each of **H2A**, **H2B**, **H3** and **H4**. Around it sits 147 base pairs of DNA double helix. A fifth histone, **H1**, is called the linker histone and sits where the DNA enters and leaves the core, like the strap on a spool. The whole core particle is about **10 nm** across (his slide also gives it as 5.7 nm tall by 11 nm across). His slide 9 carries the caption

"Crystal structure of nucleosome core particle at 2.8 A". The unit there is the

angstrom (A), one ten-billionth of a metre, which is about the size of an atom. A structure solved at 2.8 A means the atoms can be located almost individually. He marked **under 2.0 A** as "very excellent".

Then the two names that matter most. **Heterochromatin** is the tightly packed form: the DNA is squeezed shut so that transcription factors and enhancers physically cannot reach the promoter. **Euchromatin** is the loose, open form: the DNA is exposed and the gene can be read. Slide 8 labels them "Silent" and "Active". And the state is **dynamic and reversible**: it is a setting, not a label. The binding that holds histones to DNA is **non-covalent** - hydrogen bonds plus the electrostatic attraction between positively charged histones and the negatively charged phosphate backbone - and non-covalent bonds are exactly the kind that can be made and broken on demand.

EXAM HOOK	He put **chromatin accessibility** in the Tier 2 list - the group of concepts the exam notes describe as the ones he says you must understand and remember. His wording for the pair: **Euchromatin = open, transcriptionally active; Heterochromatin = condensed, silent** - and it physically blocks transcription factors and enhancers from reaching the promoter. The **~10 nm** nucleosome diameter and the **2.8 A** structure (with **< 2.0 A** = "very excellent") are on his key-numbers list. The **23 pairs** of chromosomes in the karyogram on the same slide are also on the list. What he did **not** want you to do is memorise the pictures: his exclusion list contains **electron-microscopy chromatin images**, with his words *"don't memorize all these figures, yeah. It's not necessary."* Know the hierarchy, the two names, and the fact that it is reversible. Do not learn the electron micrographs.

10.3.3 Sliding the spools along the string - Nucleosome translocation

WHAT IT IS	A **nucleosome** can slide along the DNA it is wrapped around. When it slides aside, the promoter of a gene becomes exposed and the gene can be switched on. When it slides back, the promoter is buried and the gene goes quiet.
HOW IT WORKS / WHY IT MATTERS	Slide 10 shows what happens when this goes right. There are **DNase-I hypersensitive sites** - regions of DNA that an enzyme called DNase I cuts easily, because they are not covered by a nucleosome and are therefore exposed. The slide names three kinds: a **nucleosome-free enhancer region**, a **nucleosome-free promoter region**, and a **nucleosome reposition region**, where the nucleosome is present but has moved to a new spot.

Two forces drive the sliding. The first is **intrinsic DNA-histone affinity**: the DNA sequence itself has a built-in preference for certain shapes, and will sit more comfortably on some stretches than on others, because certain sequences favour bending and certain histone tails fit particular grooves. The second is **ATP-dependent chromatin remodelling complexes** - molecular machines that burn ATP to push, slide or eject nucleosomes. He names four families on slide 12:

SWI/SNF, **INO80**, **ISWI** and **CHD**, and tabulates the members in yeast, fly, human and mouse.

The clinical reason this matters is the same reason the whole hierarchy matters. If you cannot open the page, you cannot read the recipe. A drug that blocks a remodeler, or a mutation that stops a remodeler working, is a drug or a disease that changes gene expression without touching a single letter of DNA.

EXAM HOOK	He listed chromatin accessibility as Tier 2, and within it the two drivers: **intrinsic DNA-histone affinity** and **ATP-dependent chromatin remodelling complexes**. He did **not** ask you to memorise the comparative tables of the four remodeler families across four species - that is reference material on the slide, not an exam item. The same goes for the DNase-I hypersensitivity picture. The examinable idea is the gate: **a gene you cannot open is a gene you cannot read.**

10.3.4 Photographing the chromosomes - Karyotyping and cytogenetics

WHAT IT IS	A **karyotype** is a photograph of someone's chromosomes. Cells are grown, their division is stopped at the stage where the chromosomes are at their clearest, the chromosomes are stained, and then each one is cut out of the photograph and arranged in matching pairs.
HOW IT WORKS / WHY IT MATTERS	The stain he uses is **G-banding**, which stands for Giemsa banding. Giemsa is a blue-purple dye mix. The banding works because the dark bands are the **AT-rich** regions and the light bands are the **GC-rich** regions, and because the two types of chromatin condense differently through the cell cycle. This is not decoration: the banding pattern is a barcode. Every chromosome has a recognisable band pattern, which is how a laboratory knows that chromosome 8 has lost a piece of its long arm.

The crucial limitation is **resolution**. G-banding reliably detects gross structural change at roughly **0.5-10 Mb**, where an **Mb** is a megabase, one million letters of DNA. He quotes

5-10 Mb in one place and **0.5-1 Mb** in another, and both figures are in his material. Read that as: the useful window runs from about half a million letters up to ten million. Below that, the change is too small to see. So a karyotype finds trisomies and large deletions, and it

misses single-nucleotide changes entirely. For a one-letter mutation you need **sequencing**, whether Sanger or next-generation. His slide 3 shows a human G-banded karyogram with **23 pairs** of chromosomes plus the X and Y.

This is exactly the kind of tool that has a defined job and a defined limit, and knowing the limit is what a doctor needs. A normal karyotype does not mean the genome is normal. It means nothing large is wrong.

EXAM HOOK	Karyotyping resolution is Tier 2, and he flags the discrimination directly: **point mutations and single-nucleotide variants require sequencing**, because karyotyping cannot see them. The **23 pairs** of chromosomes is on his key-numbers list. The resolution figure itself is not on that list, but it is the fact his Tier 2 entry is built on, so be ready to say "G-banding detects structural change of roughly 0.5-10 Mb and misses point mutations, which need sequencing." He did not ask you to memorise the Giemsa recipe or the staining protocol.

10.3.5 One chromosome, named - Chromosome anatomy

WHAT IT IS	Every chromosome has two arms joined at a narrow waist. The short arm is called the **p-arm**, and the long arm is called the **q-arm**. The waist is the **centromere**.
HOW IT WORKS / WHY IT MATTERS	The **centromere** is the part that matters mechanically. It is where the **spindle microtubules** attach during metaphase, so that when the cell divides, the two halves are pulled to opposite poles. It is also where the two sister copies are held together until the moment they are told to separate.

The arms are not named by size. The short arm is **p** for petite, the French for small, and the long arm is **q** because **q** comes after **p** in the alphabet. His slide 13 is a full

ideogram for chromosome 8, a labelled diagram of one chromosome with every band named: the p-arm bands p23.3, p23.2, p23.1, p22, p21.3, p21.2, p21.1, p12, p11.2, p11.1, then the centromere, then the q-arm bands q11.1, q11.21, q11.22, q11.23, q12, q13, q21.1, q21.2, q21.3, q22.1, q22.2, q22.3, q23, q24.1, q24.2, q24.3.

And he leaves a note on that slide about how to say the names, because it is a classic trap. His words: **"q21.1 means q-two-one-point-one NOT q-twenty-one-point-one"**. It is read digit by digit, and that is how cytogeneticists say it out loud.

EXAM HOOK	He put the **q21.1 pronunciation** in the Tier 2 list next to the karyotyping resolution limit, so it is examinable as a name you must be able to say and locate. He did not ask you to memorise the whole chromosome 8 band list - that is a reference diagram for reading cytogenetic reports, not a list to recite.

10.3.6 The shrinking timer at the end of each chromosome - Telomeres and the mitotic clock

WHAT IT IS	The two ends of a chromosome are capped with **telomeres**: long stretches of repeated DNA. The telomere is made of three parts. At the very tip is a unique sequence. Next to it lies a block of **100-300 kb of subtelomeric repetitive sequence** ("subtelomeric" means "near the telomere"). And at the very end there is a stretch of **3-20 kb of TTAGGG repeats**, which is the same six-letter sequence over and over.
HOW IT WORKS / WHY IT MATTERS	The problem is called the **end-replication problem**. When a cell copies its DNA before dividing, the copying machinery cannot quite finish the very last stretch at the end of a chromosome. His slide 14 puts the loss at roughly **50-200 bp at each somatic cell division**. So every time an ordinary body cell divides, its telomeres get a little shorter. They are a **mitotic clock**: a counter that ticks down with every division.

When the telomeres reach a critical threshold, the cell stops dividing. That limit is the

Hayflick limit, and the cell enters **senescence** - a state where it is alive and working but permanently retired from dividing. This is the molecular reason our organs age.

Some cells cheat the clock. **Telomerase** is an enzyme that adds telomeric repeats back. It is a

reverse transcriptase, meaning it makes DNA from an RNA template, and - unusually - it carries its own template inside itself, so it never needs an external script. Telomerase is active in

embryonic stem cells and **germ cells**, and he notes that

~90 % of cancers upregulate telomerase. That is a very large fraction, and it is the same idea as Lecture 1's K-Ras: a cancer cell needs to keep dividing, so it keeps a switch on that normal cells turn off.

The liver connection ties this lecture back to Lecture 2. Telomere shortening underlies

hepatocyte aging in chronic liver disease, whether that is hepatitis B or fatty liver, so a liver that has been inflamed for years ages faster, cell by cell.

EXAM HOOK	The two telomere lengths are on his key-numbers list: **telomere 3-20 kb TTAGGG** and **100-300 kb subtelomeric**. Read them as: the very tip carries 3 to 20 thousand copies of the six-letter TTAGGG repeat, and just behind it sits 100 to 300 thousand letters of other repetitive DNA. There is one more telomere finding you should know, because it is counterintuitive: the **top 10 % longest telomeres predicted worse recurrence-free survival** in hepatocellular carcinoma. In plain English: for this patient group, the tumours with the longest telomeres came back sooner. The exam-focus document files that number under Lecture 4, where telomere biology returns, but it is a telomere result and it is on his key-numbers table. The **Hayflick limit**, the **end-replication problem** and the fact that telomerase is a reverse transcriptase with an internal template are all things he said explicitly.

10.3.7 Putting a chemical tag on the recipe - Epigenetics and DNA methylation

WHAT IT IS	**Epigenetics** is the study of how genes are switched on and off without changing the DNA letters. The mechanism he teaches is **DNA methylation**: the addition of a **methyl group** (-CH3, one carbon with three hydrogens) to carbon 5 of the **cytosine** base. The letter is still C. The recipe is unchanged. But a tag has been added.
HOW IT WORKS / WHY IT MATTERS	The chemistry first. The methyl group comes from **S-adenosyl methionine (SAMe)**, the cell's universal methyl donor. The site matters: on DNA it happens at a **CpG site**, which is a **C-phosphate-G in the 5'->3' direction** - a cytosine followed by a phosphate followed by a guanine. Around the promoters of many genes sit clusters of these sites, called **CpG islands**.

There are two kinds of methylation enzyme, and the difference between them is a classic exam pair. **De novo methyltransferases** - **DNMT3A** and **DNMT3B** - establish new marks where there were none. **Maintenance methyltransferases** - **DNMT1** - copy existing marks onto the new daughter strand after replication. That is how the tag is inherited: one enzyme writes the label, the other copies it. His slide 19 shows the timing during development. Methylation is at its

minimum at the morula and blastocyst stage, rises rapidly after the blastocyst in differentiated tissues, and - importantly - the **germ cells in the gonad undergo demethylation**, so the next generation starts from a fresh slate. That is how an embryo can become any kind of cell.

And this is where the disease comes in. Slide 20 draws **two pathways to cancer**, and they pull in opposite directions. The first is the inside of a **promoter CpG island** of a

tumour-suppressor gene (TSG): normally unmethylated, so the brake gene is expressed; when it becomes **hypermethylated**, the gene is transcriptionally repressed and the brake is gone. The second is **pericentromeric heterochromatin**, the repetitive DNA near the centromere: normally heavily methylated; when it becomes **hypomethylated**, the cell loses the repression of those repeats, which allows **mitotic recombination** and **genomic instability**. So the same chemistry can cause cancer by silencing a brake and by unleashing a chaos region.

EXAM HOOK	This is Tier 2, and he states the core idea in his own words: *"The most important aspect of the DNA methylation is it has a very significant functions on these gene expressions, because DNA methylation is usually involved in the suppression of the gene expressions."* He also flags the pair directly: **de novo versus maintenance methylation** - establish a new mark versus copy a mark onto the daughter strand. The consequence he emphasises is that **promoter hypermethylation of tumour-suppressor genes is a common cancer driver**, and his summary of the whole lecture says it in plain words: perturbing the chromatin state is enough to drive cancer **sometimes without changing a single nucleotide**. He did not ask you to memorise the embryogenesis methylation curve, and he did not ask for the full lists of stage- specific methylated genes.

10.3.8 Reading methylation in a patient's sample - Methylation-based clinical biomarkers

WHAT IT IS	A **methylation-based biomarker** is a measurement of methyl tags that tells you something clinical about a patient. Because the tag is a chemical mark on the DNA, it can be detected in a sample of tumour, in blood, or in stool.
HOW IT WORKS / WHY IT MATTERS	The headline clinical idea on slide 21 is the **CIMP** - the **CpG Island Methylator Phenotype**. A subset of colorectal cancer patients have many promoter CpG islands methylated at once, and this group has a poorer outcome. Read that as: the methylation is not a random side-effect in these tumours; it is a recognisable subtype with its own behaviour.

And the location of the mark changes the answer. Slide 22 measures **CDKN2A** (also known as

p16, a tumour-suppressor) in two exons. The average methylation is about

17 % in adjacent normal tissue versus ~28 % in tumour for exon 1, and about

16 % adjacent versus ~41 % tumour for exon 2. The ROC curves then give the diagnostic value:

Exon 1 AUC = 0.847, **Exon 2 AUC = 0.924**, and **Exon 1 + Exon 2 combined AUC = 0.938**. The

AUC, the area under the receiver operating characteristic curve, is a single number from 0 to 1 for how good a test is; 1.0 is a perfect test. So exon 2 is the better region on its own, and measuring both regions together is better still. That is the point of the whole slide:

where you look changes what you can conclude.

Slide 23 turns the same data into a **multivariate logistic regression**, which asks what predicts methylation and what predicts the gene being lost. The **odds ratio (OR)** is the measure of how strong that association is: an OR of 12.7 means the odds of the outcome are about twelve times higher in the group with the risk factor than in the group without. The

95 % confidence interval (CI) is the range the true value most likely sits in, given the size of the study; if that range is narrow, the estimate is solid. The two headline numbers are:

- **CIMP-high (versus CIMP-0) OR 39.6, 95 % CI 20.6-76.1, p < 0.0001** as the top predictor of CDKN2A methylation. In plain English: a patient in the CIMP-high group is roughly forty times more likely to have methylated CDKN2A than a patient with none of the CIMP pattern, and the result is highly statistically significant. - **CDKN2A methylation OR 12.7 (7.84-20.7), p < 0.0001** as the top predictor of **CDKN2A (p16) loss**. In plain English: when the CDKN2A promoter is methylated, the chance that the tumour has also lost the p16 protein is about twelve times higher. Methylation is a marker of the gene being knocked out.

The same slide shows the survival curves:

CDKN2A methylation (-) versus (+) log-rank p = 0.0035, and

p16 expressed versus lost log-rank p = 0.026. The slide also lists other predictors with their odds ratios - CIMP-low OR 5.30 (3.52-8.00), PTGS2 OR 0.45 (0.28-0.72), BRAF OR 2.62 (1.41-4.90), high tumour grade OR 2.50, age OR 1.38, TP53 OR 1.70.

The most important clinical biomarker of the lecture is **MGMT promoter methylation**. **MGMT** encodes an enzyme that repairs DNA damage. If the promoter is methylated, the enzyme is not made, so the damage the drug causes is never repaired. Slide 25 draws the two patients: patient A, methylated MGMT promoter, treated with **TMZ** (temozolomide, an alkylating chemotherapy drug), comes out as **Sensitive**; patient B, unmethylated, repairs the damage and comes out as

Resistant. Slide 26 gives the original numbers from the New England Journal of Medicine paper by Hegi and colleagues:

MGMT promoter methylation (versus no methylation) prognostic HR = 0.41 (0.29-0.57), p < 0.001, and **predictive HR = 0.49 (0.32-0.76), p = 0.001**, on **114 unmethylated versus 92 methylated** patients, **P < 0.001**. The plain reading: methylated patients do markedly better, and they do better specifically because of the treatment, not just because of luck.

Finally, the sampling. Methylation can be read without an operation. **Stool tests** -

Cologuard and **Coloclear** - and blood-based **cell-free DNA (cfDNA)**, which is DNA floating free in the plasma. Slide 24 maps the whole panel across four boxes: tissue-based prognostic (CDKN2A, CHFR and IGF2 methylation, H3K27me2, H3K56ac, H4K16ac, miR-31), tissue-based predictive (MGMT, TFAP2A, miR-106, miR-19a), stool-based diagnostic for early disease (ITG4, SFRP2, Cologuard, miR-21, miR-92a, miR-135b) and for late disease (SFRP2, VIM, TFPI, miR-21, miR-92a, miR-223), and blood-based early (SEPT9, SFRP2, miR-21, miR-92a, miR-29a) and late (SDC2, SEPT9, H3K27me, miR-21, miR-92a, miR-221).

EXAM HOOK	This is the most number-dense part of the lecture and the part where he laboured longest, so the numbers must be exact. **OR 39.6 (95 % CI 20.6-76.1, p < 0.0001)** for CIMP-high versus CIMP-0 predicting CDKN2A methylation, and **OR 12.7 (7.84-20.7, p < 0.0001)** for CDKN2A methylation predicting p16 loss - both are on his key-numbers list as Tier 3, ready-made MCQ stems. He also named **MGMT promoter methylation** as a Tier 2 concept, and it is one of only two methylation items he placed there. The **AUC** values and the MGMT hazard ratios are on his slides and are fair game for a scenario answer, because they are the numbers the clinical claim rests on. Two things he did **not** ask you to memorise: the long lists of stage-specific methylated genes on slide 21 (SLC5A8, MINT1, MINT31, SFRP1, SFRP2, CDH13, CRBP1, RUNX3, then p14ARF, HLTF, ITGA4, CDKN2A, CDH1, ESR1, then TIMP3, CXCL12, ID4, IRF8), and the complete anatomical biomarker map on slide 24. Keep the ideas: the CRC multistep model, and the fact that a single site is not the whole story.

10.3.9 How the lab actually finds the methylated bases - Methods for methylation profiling

WHAT IT IS	**Bisulfite sequencing** is the method that reads methylation directly. There is also a cheaper alternative: **immunohistochemistry (IHC)** with antibodies against modified histones.
HOW IT WORKS / WHY IT MATTERS	The chemistry of bisulfite is elegant. **Sodium bisulfite** is a reagent that reacts with cytosine, but only when the cytosine is **unmethylated**. It converts unmethylated cytosine into **uracil**, which reads as thymine when the DNA is sequenced. Methylated cytosine is left alone. So you run the chemistry, then sequence, and every cytosine that now reads as a T was methylated, and every cytosine that still reads as a C was not. The result is a **methylation map at single-base resolution** - you get the answer letter by letter, not just "this region is methylated".

The alternative is **immunohistochemistry**, which uses an antibody raised against a modified histone, such as H3K27me3, and a colour-producing enzyme to show where the antibody stuck. He notes the honest limitation: IHC for modified histones is **clinically less reproducible** than sequencing, because antibody staining depends on the lab and the reader. It is nevertheless used in **glioblastoma prognosis**, because the answer it gives is faster and cheaper.

EXAM HOOK	He put **bisulfite sequencing = gold standard** in his Tier 2 notes, with the mechanism stated as unmethylated C -> U; methylated C preserved. So the mechanism is examinable, not just the name. The slide-level detail of how a sequencing library is prepared he did not flag, and the antibody names used for IHC are on the slide as reference only.

10.3.10 Tags on the spool - Histone modifications and the histone code

WHAT IT IS	**Histone modifications** are small chemical groups added to the histone proteins after they are made. They are called **post-translational modifications**, and they happen on the flexible **tails** of the histones, which stick out from the core. The four families he names are **acetylation**, **methylation**, **ubiquitination** and **phosphorylation**.
HOW IT WORKS / WHY IT MATTERS	The reason the tail is the right place is charge. The histone tails carry lots of positively charged **lysine** residues, and those positive charges are what grip the negatively charged DNA backbone. So adding a group to a lysine changes how tightly that histone holds on. **Acetylation** puts a small neutral group onto the lysine and cancels its positive charge. The histone holds the DNA less tightly, the chromatin loosens, and the gene becomes readable. **Deacetylation** puts the charge back and the chromatin closes. Methylation does not simply add charge, which is why it can push either way: it depends on which residue carries it.

The **histone code** is the idea that the combination of marks on the histones records a decision. And the marks are written in a code you must be able to read. The standard example is **H3K9me3**, which decodes as: **Histone 3**

(H3), **Lysine** (K) at **position 9**, with **trimethylation** (me3). His slide 28 spells out a second one in full, **H3K4me3**: **Histone 3 (H3)** -

Lysine (K) - **position 4 from the N-terminal end** - modification type (**me**, methylation) - **tri-methylated (3)**.

Clinically, the pattern itself carries information. Slide 33 shows

histone modification pattern as a prognostic marker in glioma, with two survival curves over 70 to 80 months and patients grouped into ten clusters by their modification pattern. And slide 31 shows the genetics behind it: a **chromatin regulator gene** that is either inherited-mutant or newly-mutated can leave a healthy cell with gene A active and gene B repressed, and in disease those two are swapped. Stress and environment can drive the transition.

EXAM HOOK	The **histone code nomenclature** sits inside his Tier 2 methylation and epigenetics item: H3K9me3 = Histone 3, Lysine at position 9, trimethylation. So being able to read and write that code is examinable. He did not ask you to memorise every mark on slide 28 - the full list of H3 and H4 sites is a reference table. He also did not put the glioma clustering plot on any list; the idea that a modification pattern can stratify patients is what the slide is for.

10.3.11 The two labels talking to each other - Epigenetic cross-talk and cellular memory

WHAT IT IS	**Epigenetic cross-talk** means the different marks influence each other. And **cellular memory** is the result: a cell that has settled into one identity keeps that identity across many cell divisions.
HOW IT WORKS / WHY IT MATTERS	Slide 28 gives the clearest examples of cross-talk. **H2B K123ub** - ubiquitination of histone H2B at lysine 123 - pushes on **H3 K79me** and **H3 K4me**. On the other side, **H4 K8ac** and **H4 K12ac** cross-talk with **H2A K119ub**. One mark sets up the next, so the marks are not independent switches but a network.

The DNA-methylation side of the conversation is on slides 27 and 30. **Suv39h** and **HP1** cooperate to spread K9 methylation along the chromatin. **Boundary elements** act as fences, keeping active and inactive regions apart so a mark cannot spread everywhere. The **TET** enzymes remove a methyl group from DNA - they are the eraser. And

H3K4me3 inhibits DNMT binding to CpG, which means a region already carrying the active H3K4me3 mark is a permissive environment: it keeps the methylation enzymes away. So the marks and the DNA methylation push each other in the same direction.

That is what a **positive feedback loop** means. Once a region is closed and methylated, the histone marks keep the DNA methylated, the DNA methylation keeps the histone marks closed, and the next round of cell division copies the whole arrangement. Slide 27 labels the outcome: replication of parental methylation gives **epigenetic inheritance**, which is cellular memory. This is why a liver cell does not turn into a nerve cell just because it has the same DNA as one. Note the contrast with a mutation: a mutation changes the letters and is inherited that way; an epigenetic mark does not change the letters at all, and is inherited because it is copied.

EXAM HOOK	Cross-talk and cellular memory are inside his Tier 2 item, in exactly this form: *"DNA methylation <-> histone marks reinforce each other in positive feedback loops -> cellular memory."* This is one of the shorthand lines that made the old revision guide unreadable, so make sure you can explain the loop in plain words: each mark holds the other in place, and both are copied when the cell divides. He did not ask you to memorise the individual cross-talk arrows on slide 28.

10.3.12 Drugs that undo the labels - Epigenetic drugs

WHAT IT IS	**HDAC inhibitors** are drugs that block the enzymes that remove acetyl groups from histones. If you keep the acetyl on, the chromatin stays open and silenced genes can be read again.
HOW IT WORKS / WHY IT MATTERS	Slide 34 makes the case. Cancers show a **global loss of H4K16 acetylation**, and **low H3K18ac predicts poor survival** in pancreatic, breast, prostate and lung cancers. The **HDACs are over-expressed in cancers**, which means they are taking the acetyl off faster than normal, which means tumour suppressors stay switched off and cancer-promoting pathways stay switched on. The **HDAC inhibitors** restore the acetylation and re-express the silenced tumour suppressors. He also notes that HDACs do not only work on histones: their **non-histone substrates** include **p53**, **HSP90**, **NF-kappaB** and **Tubulin**, so blocking them has effects beyond chromatin.

The same slide draws a three-stage carcinogenesis pathway: **normal cell -> initiation** (cell cycle up, apoptosis down, autophagy down) **-> promotion** (autophagy up, angiogenesis up, metastasis up, immune homeostasis down) **-> progression** to a vascularised tumour.

And then there is the clever twist, which is the best idea in the lecture. HDAC inhibitors can

induce PD-L1 on the tumour cells' surface. PD-L1 is the "don't attack me" sign that the immune system reads, and it is exactly what the checkpoint-blocking antibodies **anti-PD-1** and

anti-PD-L1 neutralise. So a drug that opens the chromatin and switches on a gene the tumour had switched off can make the tumour visible to immunotherapy. He calls the state

gene addiction: the tumour has become dependent on the drug to keep PD-L1 up, and when you stop the drug the tumour is exposed. Slide 50 gives the working examples: **RGFP 966**, an HDAC3 inhibitor, works through **BRD4** to raise PD-L1 and increase sensitivity to PD-L1 blockade; **HDAC6** raises **STAT3**, which raises PD-L1; **Nexturastat A** blocks melanoma progression; **MPT0G612** blocks IFN-gamma-induced PD-L1 and pushes the cell towards apoptosis. Earlier panels name the checkpoint landscape itself: **CTLA-4** with CD80/CD86 (**Ipilimumab** against CTLA-4), and

PD-1/PD-L1 (**Pembrolizumab**, **Nivolumab** against PD-1; **Atezolizumab**, **Durvalumab**,

Avelumab against PD-L1).

EXAM HOOK	He did not ask you to memorise the drug names on slide 50 - that is a dense four-panel reference figure. The examinable idea is the chain **HDAC inhibitor -> histone acetylation restored -> silenced tumour suppressor re-expressed**, and the clinical twist **HDAC inhibitor -> PD-L1 induced -> "gene addiction" -> vulnerability to anti-PD-1/PD-L1 checkpoint blockade**. In the exam-focus notes this appears as the mechanism behind **cold tumours can be re-programmed hot**, and the MCQ stem list includes "A patient with promoter hypermethylation -> which drug?" with **HDAC inhibitor** as the answer. Note that he places "cold versus hot tumour" in Lecture 5, but the mechanism itself is taught here.

10.3.13 CAR-T is not the only way - Adoptive cell therapies revisited

WHAT IT IS	**Adoptive cell therapy** means taking cells out of a patient, engineering them in the laboratory, and putting them back to attack the tumour. **CAR-T** cells carry a **chimeric antigen receptor**, a synthetic receptor that lets the T cell recognise a target on the cancer.
HOW IT WORKS / WHY IT MATTERS	The problem he names is that **CAR-T struggles to penetrate solid tumours**. It works beautifully in blood cancers, where the cells can meet their target in the blood. But a solid tumour is a physical mass, with a stroma and a hostile micro- environment, and the CAR-T cells often cannot get in or cannot survive there.

His answer is to change the effector cell. **CAR-NK** cells use a natural killer cell as the chassis instead of a T cell. **CAR-macrophages (CAR-M)** use a macrophage, which is a cell whose normal job is to eat things, so it is naturally good at getting into tissue and at handling a hostile environment. Both have better infiltration profiles than CAR-T. He does not present this as a finished answer - it is a direction of travel.

EXAM HOOK	He did not flag this as examinable in Lecture 3. The CAR-T material itself is Lecture 2, the antibody isotypes that decide how the immune system handles the CAR-T cell are Lecture 2, and CAR-T versus CAR-NK versus CAR-M is a landscape slide, not a question. Know the one-sentence reason: CAR-T cannot get into solid tumours well, so engineers borrowed cell types that can.

10.3.14 When the chromosome count goes wrong - Chromosomal abnormalities

WHAT IT IS	**Aneuploidy** is having the wrong number of chromosomes. It happens because of **non-disjunction**, which is simply a failure of a chromosome pair to separate properly when a cell divides. **Structural variants** are the other family, where the structure of a chromosome is rearranged.
HOW IT WORKS / WHY IT MATTERS	Start with the division that makes gametes, which is **meiosis**. There are two rounds, meiosis I and meiosis II, and they can fail in two different ways. That difference is the single most important thing in this topic.

In **meiosis I**, what fails to separate is the **homologous** chromosomes - the one copy you got from your mother and the one you got from your father. In **meiosis II**, what fails to separate is the **sister chromatids** - the two identical copies of a single chromosome made during replication. The consequences differ, and his slides 38 and 39 draw both. A meiosis I failure produces **2 normal and 2 abnormal gametes**, and the two abnormal ones are the same type; they carry **non-identical homologues**, because the mother's and father's copies are not the same. A meiosis II failure produces **1 normal and 3 abnormal gametes** - two of one type and one of another - and the abnormal ones carry **identical sister chromatids**. One more difference:

recombination, the crossover, occurs between homologues in meiosis I and does not occur in meiosis II.

Fertilisation then does the arithmetic. A gamete with one chromosome too few produces a

monosomy (2n - 1); a gamete with one too many produces a **trisomy (2n + 1)**. The famous examples on slide 37: **Trisomy 21** is **Down syndrome**, with three chromosome 21s circled; **Turner syndrome (45,X)** is a single X chromosome with no second sex chromosome.

The structural variants come next, and his slide 40 lists five, illustrated with chromosome 4 and chromosome 20. **Deletion** - a segment is lost. **Duplication** - a segment is copied again. **Inversion** - a segment is cut out and put back the other way round, flipped 180 deg . **Substitution** - a segment of chromosome 4 is inserted into chromosome 20. **Translocation** - chromosomes 4 and 20 exchange segments reciprocally.

And then the inversion gets a paragraph of its own, because it is the elegant one. An inversion changes **no genetic material and no gene dosage**. Nothing is gained and nothing is lost. Yet it can drastically change how much a gene is expressed, because it moves the gene away from its

enhancer, or puts an enhancer next to a different promoter. The expression changes without the DNA amount changing at all.

EXAM HOOK	This is the **Tier 1** item for Lecture 3 - the question he explicitly flagged as exam-type. His words: *"And you might need to distinguish the M1 or the M2 failure, because if there's a M1 failure, there's a formation of the germ cell without a losing copy of the chromosome after the M1..." and "And I think you need to distinguish these two situations here, yeah."* The discriminating facts you must be able to state: **M1 separates homologues, M2 separates sister chromatids; M1 gives 2 normal + 2 abnormal gametes, M2 gives 1 normal + 3 abnormal; recombination happens in M1, not M2.** On the inversion he said: *"you need to understand that some of these abnormalities can cause the loss of the genetic materials, but some does not."* **23 pairs** of chromosomes, **Trisomy 21** and **Turner 45,X** are on his key-numbers list. **M1 versus M2 non-disjunction** is also one of his named distractor pairs. The scenario-question archetype that follows directly from this is pedigree plus aneuploidy: work out M1 or M2, the resulting gametes, and the syndrome.

10.3.15 When a whole chunk of the manual is photocopied - Copy-number variation

WHAT IT IS	**Copy-number variation (CNV)** is a stretch of DNA that exists in a different number of copies in different people. Some of the genome is repetitive, and the number of those repeats varies between individuals and between populations.
HOW IT WORKS / WHY IT MATTERS	The classic example is the **amylase gene (AMY1)**. Amylase is the enzyme in saliva that starts digesting starch. Populations that have historically eaten a lot of starch carry more copies of AMY1, which makes more amylase, which makes starch digestion more efficient. This is **adaptive CNV** - the copy number is not a defect, it is an adaptation to diet. Slide 41 gives the numbers: **high-starch populations n = 133, median 7, mean 6.72 +/- 2.35**, against **low-starch populations n = 93, median 5, mean 5.44 +/- 2.04**. The populations he names as high-starch are **Japanese, Hadza and European American**; the low-starch groups are **Biaka, Mbuti, Datog and Yakut**. Slide 42 puts this in an evolutionary frame with a mammalian phylogenetic tree: H. sapiens 7 copies, P. troglodytes 3, Cercopithecidae 3, Ceboidea 2.

How do new copies appear? Slide 43 lists four **CNV-generation mechanisms**. **NAHR** -

non-allelic homologous recombination - is unequal crossover between misaligned repeats, which produces a duplication in one place and a deletion in the other. **FoSTeS** - fork stalling and template switching - is a replication-based route. **NHEJ** - non-homologous end joining - is DNA repair joining two ends that do not perfectly match. **MEI** - mobile element insertion - is RNA copying itself into a break in another chromosome. In the main teaching points, the two he names are **NHEJ during DNA repair** and **NAHR**, unequal crossover at repetitive sequences during meiosis.

Slide 44 then turns CNV into a cancer tool: bar charts of gene loss and gain across chromosomes 1 to 22, stratified by size (1-10 kb, 10-50 kb, 50-100 kb, above 100 kb), a heatmap of differential expression in tumour versus non-tumour, and a protein-protein interaction network with hub proteins including **NPM1, CDK4, HSP90AB1, TUBB2A, TBCE, PABPC1, RPL8 and EIF3H**.

EXAM HOOK	He put **CNV molecular mechanisms in depth** on his exclusion list, with his words: "I'm just give you a taste... don't be scared with all these mechanism." So know that NAHR and NHEJ are the two mechanisms named in the teaching points, know that NAHR is unequal crossover at repeats, and do not try to memorise FoSTeS and MEI. The AMY1 example he did labour over, and the numbers - high-starch median 7 versus low-starch median 5 - are on slide 41 for exactly that reason. The phylogenetic tree and the CNV network figure are reference material. He did not ask you to memorise the hub protein list.

10.3.16 Which typo matters? - Cancer genomics: driver vs passenger

WHAT IT IS	When you read the genome of a tumour, you find many changes. Almost all of them are **passengers** - bystander alterations that hitched a ride because the region was unstable or nearby. A few are **drivers**: alterations that actually give the tumour its growth advantage.
HOW IT WORKS / WHY IT MATTERS	The plainest way to say it: a **driver is a typo in an important word**, and a **passenger is a typo on page 900**. Both are real changes. Only one of them changes what the cell does. A chromosome that is unstable tends to lose and gain big chunks all at once, and most of the genes in those chunks are innocent bystanders. If you count every change and call them all drivers, you will drown.

His method for finding the real ones is to **integrate the CNV profile with the mRNA expression**. The logic is that a genuine driver should show two things together: the gene is present in extra copies, and it is actually being transcribed more. Copy gain plus transcriptional up-regulation, in the same gene, is much stronger evidence than either alone. The worked example is **NPM1** in leukaemia, and the slide gives the survival result: NPM1-high versus NPM1-low gives

overall survival P = 0.0257 and **cumulative recurrence P = 0.0115**. Read that as: the tumours with high NPM1 both survived less well and recurred more often, and both comparisons are statistically significant.

This is a bioinformatics skill, and it is the same skill the whole course is about: turn a pile of measurements into a clinical claim, and state the limit of the claim.

EXAM HOOK	**Driver versus passenger** is Tier 2, and he stated the discriminator exactly as the glossary does: a driver **confers a growth advantage**; a passenger is carried along by chromosomal proximity or instability and does **not** functionally contribute. The method he named is **integrating CNV profiles with mRNA expression** to find concordant copy gain and transcriptional up-regulation, with **NPM1** in leukaemia as the worked example. Driver versus passenger is also on his list of the classic distractor pairs. The exact NPM1 survival p-values are on slide 44; he did not put them on the key-numbers list, so treat them as supporting detail rather than a memorised stem.

10.3.17 When a chemical sticks to the recipe - DNA adducts and chemical carcinogenesis

WHAT IT IS	A **DNA adduct** is a chemical that has attached itself to a base in the DNA and changed it. The base is still there, but it now carries a lump of foreign chemistry.
HOW IT WORKS / WHY IT MATTERS	Two sources feed into this. The cell's own **reactive oxygen species (ROS)** - the unstable, chemically aggressive molecules that cell respiration produces - damage DNA constantly. And **tobacco-specific nitrosamines**, of which **NNK** and **NNN** are the names he uses, arrive from cigarette smoke. Your liver's **P450** enzymes (the cytochrome P450 family, which chemically modifies drugs and poisons) activate them into forms that react with DNA, producing **POB-DNA** and **PHB-DNA** adducts, and methyl adducts. Slide 16 shows the four kinds of reaction on the bases: **arylamination**, **alkylation**, **oxidation** and **two-site alkylation**.

Slide 17 then runs a full carcinogenesis chain with four named carcinogens -

4-aminobiphenyl (4-ABP), **aristolochic acid I (AA-I)**, **benzo(a)pyrene (B(a)P)** and

PhIP. The chain reads like this. The carcinogen reacts with intact DNA at the sequence

3'-GAAGGA-5' / 5'-CTTCCT-3', forming a **DNA adduct (dA-AL-I)**. If the cell does

not repair it, the next round of replication makes an error, and the adduct persists. A second round of replication then **fixes the mutation** as **3'-GATGGA-5' / 5'-CTACCT-3'**. And that fixed mutation can become cancer. The teaching point of the whole slide is the word **fixed**: the error is not a wobble, it is written into the genome permanently.

The specific consequence he names is the one that links back to Lecture 1:

NNK induces a GC->AT mutation at the K-ras gene in mice. That is the same K-Ras gene whose mutant form makes a lung cancer cell stuck on and unresponsive to the drug that works when K-Ras is normal. A chemical in cigarette smoke is doing, step by step, what that mutation then does for the patient.

EXAM HOOK	He did not ask you to memorise the exact base sequences on slide 17, and the individual carcinogen names are reference material. The examinable chain is: **carcinogen -> metabolic activation by P450 -> DNA adduct -> unrepaired -> replication error -> fixed mutation -> cancer**, and the named example **NNK induces a GC->AT mutation at K-ras in mice**, which is the bridge to the K-Ras story from Lecture 1. ROS as an endogenous source of damage is the point that connects this topic to Lecture 4's DNA repair pathways: the cell is damaged continuously, and repair is what stands between that damage and a fixed mutation.

10.4 The ten things to remember

1. Chromatin is DNA plus its packaging. It folds in levels - 2 nm helix, 11 nm nucleosome, 30 nm fibre, 1,400 nm chromosome - and the folding decides which genes can be read.
2. A nucleosome is eight histones plus 147 base pairs of DNA wound around them, about 10 nm across. Its crystal structure was solved at 2.8 A; under 2.0 A counts as very excellent.
3. Heterochromatin is condensed and silent; euchromatin is open and active. The switch is reversible, because histone-DNA binding is non-covalent.
4. Nucleosomes slide to expose or bury promoters. Two drivers: intrinsic DNA-histone affinity and ATP-dependent chromatin remodelling complexes. A gene you cannot open is a gene you cannot read.
5. G-banding finds structural change of roughly 0.5-10 Mb and misses point mutations, which need sequencing. The p arm is short, the q arm is long, the centromere is the spindle attachment point. "q21.1" is said "q-two-one-point-one".
6. Telomeres lose about 50-200 bp at each somatic cell division. That is the mitotic clock, and hitting its threshold is the Hayflick limit. Telomerase is a reverse transcriptase with its own internal template, and ~90 % of cancers switch it on.
7. Methylation is a methyl tag on cytosine, not a change of letters. De novo enzymes write it, maintenance enzymes copy it at every division. Promoter hypermethylation silences a tumour suppressor, and that is enough to drive cancer on its own.
8. MGMT promoter methylation is the clearest predictive biomarker: the enzyme is silenced, the alkylating drug's damage is never repaired, and the patient becomes sensitive. Prognostic HR = 0.41 (0.29-0.57), predictive HR = 0.49 (0.32-0.76).
9. M1 and M2 are different failures. M1 separates homologues and gives 2 normal + 2 abnormal gametes; M2 separates sister chromatids and gives 1 normal + 3 abnormal. An inversion changes no DNA and no dosage, yet still changes expression.
10. A driver is a typo in an important word; a passenger is a typo on page 900. To tell them apart, integrate the CNV profile with mRNA expression - the NPM1 example. And the physical cause of many cancers is a chemical that stuck to a base and was never repaired.

10.5 Check yourself

Q1. A pathologist reports a karyotype as normal for a patient who has a classic inherited syndrome. Explain how that is possible, and say what test would be used instead.

A1. The observation to start from is that the karyotype is the wrong instrument for this patient's question. A **karyotype** only sees large-scale structure: G-banding detects gross structural change of roughly **0.5-10 Mb**, so it will find a trisomy or a large deletion, and it will find nothing at all if the change is small. The mechanism behind the miss is simple: a single-letter change is far below the resolution of a microscope. Many inherited syndromes are

caused by exactly that, a single changed letter or a very small change inside one gene, so a normal karyotype is entirely consistent with the diagnosis. The clinical meaning is that a normal karyotype does not mean the genome is normal - it means nothing large is wrong, and the diagnostic question is still open. The action is to go to **sequencing**, whether Sanger for a single gene or next-generation for a panel, and to test the genes that the syndrome implies. The limit to state is the direction of the error: karyotyping has excellent sensitivity for large abnormalities and no useful sensitivity at all for point mutations, so it is a screening tool for a specific question, not a general test for genetic disease.

Q2. A patient with colorectal cancer is found to have a methylated **CDKN2A** promoter. Name the two other findings that go with it, and say why the location of the methylation matters.

A2. The observation is the methylated promoter, and the two accompanying findings come straight off Dr. Wong's slide 23. The first is the **CIMP** status:

CIMP-high versus CIMP-0 gives OR 39.6 (95 % CI 20.6-76.1), p < 0.0001 as the top predictor of CDKN2A methylation. In plain terms, a CIMP-high patient is roughly forty times more likely to have methylated CDKN2A than a patient with none of the CIMP pattern, and the narrow confidence interval means the estimate is solid. The second is the loss of the protein itself:

CDKN2A methylation gives OR 12.7 (7.84-20.7), p < 0.0001 as the top predictor of

CDKN2A (p16) loss. Read that as: when the promoter is methylated, the tumour has also lost the p16 protein, so the methylation is a marker of the gene being knocked out. The survival curves on the same slide say the same thing from the other side, with log-rank p = 0.0035 for methylation-positive versus negative and p = 0.026 for p16 expressed versus lost.

The mechanism is the one from Lecture 3's centre: methylation at a promoter CpG island suppresses transcription, so the tumour-suppressor gene is silenced at the level of being read, and silencing the brake is a common way a normal cell becomes a cancer cell. The clinical meaning is that the methylation is not just a passenger of the tumour - it is functionally connected to the loss of the brake.

Now the location, which is the part most candidates get wrong. On slide 22, **Exon 1 AUC = 0.847**,

Exon 2 AUC = 0.924, and **Exon 1 + Exon 2 combined AUC = 0.938**. So exon 2 is the more informative region on its own, and measuring both regions together is better than either. The average methylation is about 17 % adjacent normal versus ~28 % tumour in exon 1, and 16 % adjacent versus ~41 % tumour in exon 2. The action is therefore to measure the right region, and preferably to combine two. The limit: a single number is a group-level statement from a cohort, and about 17-41 % methylation is a signal to interpret, not a yes/no answer for an individual patient.

Q3. A patient has an inversion inside one chromosome. Nothing is gained and nothing is lost. How can the amount of a protein change?

A3. The observation is that the inversion is structurally neutral: no deletion, no duplication, so no change in gene dosage and no change in how much DNA is present. The mechanism is

positional, not quantitative. A gene's expression depends not only on its own sequence but on where it sits relative to its **enhancer**, which is a regulatory stretch of DNA that boosts transcription of a gene, sometimes quite far away. An inversion flips a segment, and in doing so it can move a gene away from its enhancer, or park an enhancer next to a different promoter. The effect is on **transcription**, not on copy number.

The clinical meaning is that a patient can have a genuinely abnormal expression profile with a completely normal-looking structural analysis of the genome, which is why sequencing that looks only for gains and losses will miss it. The action, if you are the geneticist, is to look for inversions as well as for changes in amount. The limit is worth stating because Dr. Wong says it directly: *"you need to understand that some of these abnormalities can cause the loss of the genetic materials, but some does not."* An inversion does not cause loss of material, but it can still cause disease. The related point is that the same principle applies to a **translocation**, where a segment moves to another chromosome entirely and lands next to a different enhancer.

Q4. Explain why a tumour genome contains many more mutations than are responsible for its behaviour, and describe the method Dr. Wong uses to pick the ones that matter.

A4. The observation is that a tumour genome is full of changes, and only a small number of them actually drove the tumour. The mechanism is that a cancer genome is usually **genetically unstable**, and instability causes big chunks of chromosome to be lost and gained all at once. Any gene that happens to sit inside one of those chunks is carried along without being selected for. That is the definition of a **passenger** - a typo on page 900 - as against a **driver**, which is a typo in an important word that gives the cell a growth advantage.

The method is to **integrate the CNV profile with the mRNA expression**. The reasoning is that a genuine driver should leave two traces at once: the gene is present in more copies, and it is actually being transcribed more. A gene whose copy number rose but whose expression did not is probably a passenger that happened to fall inside a gained segment. A gene whose expression rose in step with its copy number is a much better candidate. The worked example is **NPM1** in leukaemia, and the outcome on slide 44 is **overall survival P = 0.0257** and

cumulative recurrence P = 0.0115 for NPM1-high versus NPM1-low.

The clinical meaning is that the whole pipeline of personalised cancer genomics depends on making this distinction correctly. The action is to combine the two data types and then test the candidate functionally. The limit is that even a concordant copy gain and expression rise is evidence, not proof, and the reason he asks for the integration is precisely that single-layer analysis produces false drivers.

Q5. A patient with glioblastoma is being considered for temozolomide plus radiotherapy. A test on the tumour reports that the MGMT promoter is methylated. What does that mean, and why does it change the treatment decision?

A5. The observation is that the tumour's **MGMT promoter is methylated**. The mechanism is a chain with a twist in it. **MGMT** encodes an enzyme that repairs a particular kind of DNA damage - the alkylation damage caused by the chemotherapy drug **TMZ**, temozolomide. Methylating the promoter switches the gene off, so the repair enzyme is not made. Now the drug's damage cannot be repaired, the tumour cell accumulates damage, and it dies. So the promoter methylation does not make the drug stronger; it removes the tumour's ability to defend itself against the drug.

That is why the methylation is a **predictive** biomarker rather than only a prognostic one. Slide 25 draws the two patients: patient A with a methylated MGMT promoter comes out **Sensitive**, and patient B with an unmethylated promoter comes out **Resistant**. And the original *New England Journal of Medicine* numbers, from Hegi and colleagues, are

prognostic HR = 0.41 (0.29-0.57), p < 0.001 and **predictive HR = 0.49 (0.32-0.76), p = 0.001**, on **114 unmethylated versus 92 methylated** patients, **P < 0.001**. In plain English: methylated patients do markedly better overall, and they do better because of the treatment, not merely because their disease was milder.

The clinical meaning is that the test changes what the patient receives. Temozolomide plus radiotherapy is the standard of care, and this result is the reason a pathologist is asked to do it at all: it identifies the patients in whom the treatment is most likely to work, and it avoids wasting a toxic regimen on patients whose tumour can simply repair the damage.

The action is therefore to test before you treat, and to report the result as predictive rather than prognostic. The limit is the one the four-move answer template is built around: this is a statement about MGMT promoter methylation predicting response to alkylating chemotherapy and radiotherapy in glioma, and it should not be carried into other tumour types or other drugs. The related point, and it is Dr. Wong's own framing of the whole lecture, is that this is a case where a tumour has been driven by a change that is **not** a single-letter mutation at all - it is a chemical tag on the DNA that switched a gene off.

End of Part 4. The next chapter, Part 5, covers Lecture 4 - DNA Replication, Repair and Recombination. Where this chapter left threads - how the DNA is copied, how damage is repaired, how methylated cytosines are read out and how editing tools find their targets - the next lecture takes them further.

PART 5

Lecture 4: DNA Replication, Repair and Recombination

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About this part

Part 5 - Lecture 4 - DNA Replication, Repair and Recombination - Dr. Tony Wong - 10 October 2026

Course: GNBF5020 Introduction to Molecular Biology and Genetics - **Instructor:** Dr. Tony Wong

This chapter is Part 5 of the beginner guide. It rewrites the expert shorthand of the existing revision guide so that it can be read by someone who has never studied biology or chemistry. Every number Dr. Wong stated is reproduced exactly and is followed by a plain sentence saying what it means. Where he said nothing about a topic, the Exam hook says so.

CHAPTER 11

DNA Replication, Repair and Recombination

11.0 What this lecture is about

Dr. Tony Wong starts this lecture with a plain question: how does a cell copy its DNA without destroying it? The answer turns out to be a small factory with five or six machines, working in opposite directions at the same time. He then asks the darker question: what happens when the copying goes wrong? DNA is damaged every second of every day - by radiation, by chemicals, and by the cell's own chemistry - and the cell keeps a whole repair shop for each kind of damage. Finally he shows that the same machinery we use to mend DNA can be borrowed to make a drug, and that a bacterium's own immune system, CRISPR, can be turned into a scalpel for editing genes. By the end of this chapter you should be able to describe, in plain English, all twenty-seven ideas this lecture builds on.

11.1 Words to know before we start

Term	In plain English
DNA replication	Making a second copy of a cell's DNA before the cell divides. The copy is a recipe book duplicated exactly.
Semi-conservative replication	Each new DNA molecule contains one old strand and one brand-new strand. Like copying a printed page: the copy keeps the original page on one side.
Replication fork	The Y-shaped opening where the double helix is being unzipped and copied. The two arms of the Y are the two template strands.
Primer	A short piece of RNA laid down on a template so the copying machine has something to extend. A polymerase cannot start from nothing.
Leading strand, lagging strand and Okazaki fragment	Leading = one long new strand made in one run. Lagging = many short new strands, each started by its own primer. An Okazaki fragment is one of those short pieces.
DNA ligase	The machine that seals a nick in the DNA backbone. It is the glue that joins Okazaki fragments into one long strand.
Topoisomerase	The machine that relieves the strain ahead of the fork by cutting the DNA, letting it swivel, and re-joining it.
Telomere and telomerase	Telomere = the repeated DNA tail at the end of a chromosome. Telomerase = the enzyme that rebuilds that tail.
DNA Damage Response (DDR)	The cell's alarm system for DNA damage. It detects the damage, then either repairs it or kills the cell.
Homologous recombination (HR)	Repairing a broken double strand by copying from a matching template, the sister copy of the same chromosome. Exact, but only possible while the cell is copying its DNA.
Non-homologous end joining (NHEJ)	Repairing a broken double strand by simply joining the loose ends. Fast, needs no template, and often introduces small errors.
Mismatch repair (MMR)	The system that finds and corrects letters that were mismatched during copying, and slips in repeated sequences.
Microsatellite instability (MSI)	The fingerprint-like repeating DNA in the genome changing length because MMR is broken. MSI-H means "highly unstable".
Synthetic lethality	Two faults that the cell can survive one at a time, but not together. A drug can exploit this to kill tumour cells only.

11.2 The big picture

Every cell in your body is a factory that is constantly copying itself. Before a cell divides, it duplicates all of its DNA - roughly three billion letters - so that each daughter cell receives a complete copy of the instruction manual. That copying has to be almost perfect, because a typo in an important word can turn a healthy cell into a cancer cell.

The lecture follows the life of a DNA molecule from the moment it is copied, through the damage that will eventually happen to it, through the repair shops that mend it, and finally to the ways doctors and drug companies exploit that machinery.

Copying is a mechanical problem. The double helix is a zip. To copy it you must unzip it, and you can only unzip it by breaking the hydrogen bonds that hold the two halves together. That unwinding creates a mess of strain ahead of the unzipping point, so a machine called topoisomerase has to cut the DNA ahead of the fork, let it swivel, and re-join it. Then you need a starting point, because a polymerase cannot begin a strand from nothing. A short piece of RNA called a primer has to be laid down first. And because the two strands of the helix point in opposite directions, one of them can only be copied in one continuous run, while the other has to be built backwards in short pieces called Okazaki fragments. Your body does this billions of times a day.

The end of a chromosome is a special problem. The last piece of the lagging strand cannot be finished, because there is no primer far enough along. So the ends of chromosomes carry a tail of repeated DNA, the telomere, and an enzyme called telomerase can rebuild it. When the telomeres run out, the cell stops dividing. When a cancer cell switches telomerase back on, it can divide more or less forever.

Then the story turns from copying to damage. DNA is attacked all day, by the cell's own chemistry and by the world outside. Sunlight makes bulges in the helix. X-rays cut it. Chemicals in cigarette smoke glue things onto it. And the copying machinery itself makes mistakes. Each kind of damage needs a different repair job, and the cell keeps a separate shop for each one: one for a single damaged letter, one for a bulky scar, one for a mismatched letter, and two for a broken rope.

The most striking part of the lecture is that these repair jobs are not only repairs. Cells deliberately break and re-join their own DNA to build the millions of different antibodies your immune system can make. If the repair machine they borrow is broken, the child has no working antibodies at all, and develops severe combined immunodeficiency, or SCID. Take the same idea and put it into a tumour, and it becomes a drug. If a tumour has already lost one repair gene, you can block the second one with a drug and kill the tumour while leaving normal cells alone. That is synthetic lethality.

Finally, the lecture shows that we can now aim at DNA on purpose. CRISPR-Cas9 is a bacterial immune system that engineers have turned into molecular scissors with an address label. And it closes on a reminder that science runs on money and law as well as on molecules. Two universities fought over who owned CRISPR, and that fight is still working its way through the patent offices.

11.3 Walkthrough

11.3.1 Setting up the experiment - Experimental set-up for studying replication

WHAT IT IS	Before you can study how DNA is copied, you have to make every cell start at the same point. Two tools do this. **Flow cytometry** is a machine that measures cells one at a time as they stream past a laser. If you first stain the DNA with **propidium iodide**, a dye that binds to DNA, the machine reports how much DNA each cell contains. **Cell-cycle synchronisation** means putting all the cells into the same phase of the cell cycle at once, so that all their clocks are set to the same time.
HOW IT WORKS / WHY IT MATTERS	His slide shows the flow-cytometry histogram with two peaks. The tall peak at DNA content 1 is the G1 population, the cells that have one copy of their genome. The valley between the peaks is S phase, the cells that are part-way through copying. The second peak at DNA content 2 is G2 and M, the cells that have finished copying and are about to divide. The second graph explains why: the red DNA-content line stays flat at 1 through G1, climbs steadily from 1 to 2 across S phase, and stays at 2 through G2 and M. The black RNA and protein line climbs the whole time, because the cell keeps making protein whatever the DNA is doing.

The synchronisation trick he teaches is **serum starvation**. **FBS**, fetal bovine serum, is the liquid fraction of cow blood, and culture media use it to feed cells. Take the FBS down to **0.5-1 %** and the cells stop dividing and sit quietly in G1. Put the serum back and every cell leaves G1 at the same moment. That gives you a clean starting gun for a replication-fork experiment. At time zero every cell is in the same place, so anything you see later was

caused by replication and not by the cells having started at random stages.

EXAM HOOK	He did not flag the histogram or the DNA-content graph as an examinable item in its own right. The number is what matters: **serum starvation down to 0.5-1 % FBS** is on the key-numbers list for both Lecture 2 and Lecture 4, because it is the standard way to hold cells at G1 so that every cell starts copying at the same moment. Flow cytometry itself is taught in Lecture 2, where he gave the detail: **FSC = size**, **SSC = granularity**, and propidium iodide staining gives you G1, S and G2-M. Learn the number and the one-line purpose, and you have the exam content.

11.3.2 One old strand plus one new strand - Semi-conservative replication

WHAT IT IS	**Semi-conservative replication** means each new DNA molecule is made of one old strand and one brand-new strand. Imagine copying a printed page: the copy has the original page on one side and a freshly printed one on the other. Two rival ideas existed. The **conservative** model says the old double helix stays together intact and a completely new one is built alongside it. The **dispersive** model says the old strands get chopped up and scattered into the new ones.
HOW IT WORKS / WHY IT MATTERS	The experiment that settled it is a **Meselson-Stahl-type isotopic labelling experiment**. The trick is a heavy isotope. DNA built with a heavy form of nitrogen does not sit in the middle of the test tube when you spin it, so you can tell the two apart. Grow bacteria in heavy nitrogen for one generation so all their DNA is heavy. Move them into light nitrogen and let them divide. After one round of division, if semi-conservative replication is right, every molecule is half heavy and half light, and the tube shows one band in the middle. If conservative replication is right, you get a heavy band and a light band. After two rounds, semi-conservative predicts a half-and-half band plus an all-light band. The experiment matched semi-conservative replication.

Why it matters beyond history: this is why information can be inherited unbroken. Every strand of DNA you carry today has one continuous line of ancestors running back through your parents.

EXAM HOOK	He did not flag the Meselson-Stahl experiment or the three models as a named exam question in Lecture 4. The slide, with all three models drawn side by side, was teaching material. The examinable core is the definition itself: each daughter helix has **one old strand and one newly synthesised strand**. Always answer "semi-conservative", and never "conservative". If you are asked to distinguish the three models, the discriminating facts are these: semi-conservative gives hybrid molecules after the first cycle; conservative gives one untouched parental molecule plus one new one; dispersive gives molecules with old and new segments mixed in both strands.

11.3.3 The five machines that copy DNA - Core replication enzymology

WHAT IT IS	Five enzymes do the copying, and each has one job. **Helicase** unwinds the double helix. **Topoisomerase** relieves the strain that unwinding creates ahead of the fork. **Primase** lays down a short starting stub. **DNA polymerase** adds the letters. **DNA ligase** seals the gaps that are left behind.
HOW IT WORKS / WHY IT MATTERS	Start at the fork. Helicase uses energy from ATP to break the hydrogen bonds between A and T, and between C and G, so the two strands separate. Ahead of the fork, topoisomerase binds to the double helix and relieves the strain, cutting and re-joining the DNA so it can swivel. Behind the fork, **single-strand binding proteins (SSBs)** stick to the separated strands so they do not simply snap back together.

Then primase, which is really a small RNA polymerase, makes a short RNA **primer** on the template. The primer matters because DNA polymerase cannot start a strand from nothing. It needs a free

3'-OH group to add the next letter onto, and the primer supplies it. The primer is about **10 nucleotides** long. DNA polymerase then adds a **deoxyribonucleoside triphosphate (dNTP)** to the 3' end of the primer, and releases **pyrophosphate** as it does so. His slide draws the polymerase as a right hand with **fingers, palm and thumb**, gripping the template and the growing strand.

At the end, the primers are removed by an **exonuclease**, an enzyme that cuts nucleotides off the end of a strand. His slide names **RNase H** as the enzyme that erases the RNA primers. The gaps left behind are filled in by DNA polymerase, and the last nick in the backbone is sealed by **DNA ligase**, which forms a **phosphodiester bond** between a 5' phosphate and a 3' hydroxyl.

EXAM HOOK	He said, verbatim: "So this is important, you need to understand the elongation happens from 5 prime to 3 prime." The **5'->3' versus 3'->5'** pairing is on the distractor-pair list, and the discriminating fact is exactly this: **nucleotide addition (elongation) goes 5'->3'**, while **proofreading and excision go 3'->5'**, and **ligase seals a 5' phosphate to a 3' hydroxyl**. He did not ask you to memorise the polymerase's three domains or the finger-palm-thumb picture. Learn the five enzymes, what each one does, and the direction rule.

11.3.4 One strand continuous, one strand in pieces - Leading vs lagging strands

WHAT IT IS	The two strands of the double helix point in opposite directions. They are called **antiparallel**. Because DNA polymerase can only add letters to the 3' end of a growing strand, one of the two templates can be copied in one continuous run and the other can only be copied backwards, in short pieces. The continuous one is the **leading strand**. The piece-by-piece one is the **lagging strand**, and its short pieces are called **Okazaki fragments**.
HOW IT WORKS / WHY IT MATTERS	Think of two people pulling a rope in opposite directions and both trying to write on it with a pen that only writes left to right. The person on the left can write smoothly. The person on the right has to write in small pieces, starting a new piece every time, and then join the pieces up afterwards.

The lagging strand therefore needs a **new primer for every fragment**. His slide 10 gives the numbers: the primers are made at intervals of about **200 nucleotides**, and each primer is about **10 nucleotides** long. The processing sequence on that slide is the one to remember. Primase makes a new RNA primer. DNA polymerase extends it to make a new Okazaki fragment. The old RNA primer is erased and replaced with DNA. What is left is a **nick** - a break in one strand - and DNA ligase seals it, joining the new Okazaki fragment to the growing chain. The polymerase that makes the leading strand needs only **one** primer for the whole strand.

EXAM HOOK	He said, verbatim: "I think one concept you'll need to understand is there's two different strands. So one is the leading strand, and the other one is the lagging strand." On top of that, the 5'->3' rule: "So this is important, you need to understand the elongation happens from 5 prime to 3 prime." He did not ask you to memorise the enzyme names for primer removal. The facts worth having are these: **leading = continuous, one primer**; **lagging = discontinuous, Okazaki fragments, each with its own RNA primer**; **all synthesis runs 5'->3'**; **ligase joins the fragments**.

11.3.5 Poisoning the machine that untangles DNA - Topoisomerase inhibitors

WHAT IT IS	A **topoisomerase inhibitor** is a drug that blocks topoisomerase, the enzyme that relieves strain ahead of the replication fork. Topoisomerase works by cutting the DNA, letting it swivel or pass through, then re-joining the cut. An inhibitor freezes the enzyme onto the DNA with the cut still open. The strand is left broken and cannot be re-joined, so the cell cannot finish copying.
HOW IT WORKS / WHY IT MATTERS	This is where the lecture turns molecular biology into a drug class. The idea he uses is the one from Lecture 1: an **antibody-drug conjugate (ADC)** is an antibody with a poison attached, and the antibody is the delivery system that only finds its target. The payloads he names here are topoisomerase inhibitors: **SN-38** and **DXd**, the payload in **deruxtecan** drugs. The targets in the clinic are **HER2** and **TROP2**, with **trastuzumab-deruxtecan** and **datopotamab-deruxtecan** as the named examples.

The clinical logic is worth stating plainly, because it is exactly the kind of reasoning the exam likes. A topoisomerase inhibitor kills any dividing cell, which is useful against cancer but useless as a plain drug, because it would kill healthy dividing tissue too. Put the same poison on an antibody that only binds a receptor on the tumour, and the drug is carried to the tumour and released there. The antibody is the postcode; the poison is the letter.

EXAM HOOK	He did not flag the ADC structures or the payload chemistry as examinable, and the exclusion list covers it: "all this would not be included in the midterm. I'm just want you to know more". What he did emphasise is the reasoning, and the exam-focus notes list "Why is a topoisomerase inhibitor in the antibody?" as one of the likely short-question scenarios. So know the answer to that: selective delivery. A topoisomerase inhibitor alone kills every dividing cell, so attaching it to a tumour-targeting antibody concentrates it at the tumour and spares normal tissue. Know the two targets (HER2 and TROP2) and the two named drugs. Do not memorise the linker chemistry.

11.3.6 Carrying two poisons at once - Dual-payload ADCs and synthetic lethality

WHAT IT IS	A **dual-payload ADC** uses a **branched linker** to carry two different drugs at the same time. One is usually a topoisomerase inhibitor. The other is a **DNA Damage Response (DDR)** inhibitor, for example a blocker of **ATM**, one of the two sensor kinases the cell uses to spot damage.
HOW IT WORKS / WHY IT MATTERS	The reason is resistance. A tumour that survives a single payload has usually found a backup route. If the first poison is a topoisomerase inhibitor, the cell may cope by leaning on its DNA damage signalling and repair machinery. Cutting that off as well removes the escape. You are not adding more poison. You are removing the tumour's second line of defence.

This is the same idea as synthetic lethality in a later topic, just applied inside one drug molecule instead of to two genes. The lecture uses it to make one point: a single-agent ADC eventually runs into resistance, and the fix is to attack two repair routes at once.

EXAM HOOK	**He said this will not be examined.** The exclusion list carries the ADC structures and dual-payload ADCs, with his words: "all this would not be included in the midterm. I'm just want you to know more". Read that carefully, because it is narrower than it sounds. He excluded the **structures and the chemistry** - how the branched linker is built and what each half does. He did not exclude the **concept**. Know that dual-payload ADCs exist, know that the two payloads are usually a topoisomerase inhibitor plus a DDR inhibitor such as an ATM blocker, and know the reason: to defeat compensatory resistance. Do not learn the linker structures.

11.3.7 Many forks on one huge chromosome - Eukaryotic replication bubbles

WHAT IT IS	An **origin of replication** is a marked site on the DNA where copying starts. In a eukaryotic cell - a cell with a nucleus - there are **thousands of origins**, and they all fire at roughly the same time. Each one opens a **replication bubble**, and each bubble has **two replication forks** moving away from each other in opposite directions.
HOW IT WORKS / WHY IT MATTERS	The reason is sheer size. One human chromosome carries billions of bases. If only one origin fired, copying would take far too long, and the cell could not divide. His slide puts the numbers plainly: **several hundred sites in yeast, and tens of thousands in the human genome**. Where the origins sit is not random; their locations are influenced by local DNA topology and by the chromatin environment, because the DNA has to be accessible enough to be opened.

His electron micrograph of replicating eukaryotic DNA shows multiple bubbles along the fibre, with a scale bar of **5 kb**. The three-step schematic next to it shows bubbles opening at multiple origins, expanding towards each other until they fuse, and leaving two daughter DNA molecules. When two forks meet, the polymerases simply stop when they encounter another strand. The whole arrangement is called multiple **replicons**.

EXAM HOOK	He did not name this as a Tier-1 question, but the fact sits on the Tier-2 sheet for Lecture 4: **thousands of origins, each with two bidirectional forks**. That is the number to carry. The reason it matters clinically is that most of the anticancer drugs that damage DNA, and most of the mutagens we are exposed to, are trying to break something that is being copied thousands of times over at once. Do not memorise the micrograph.

11.3.8 The ends of the line - Telomeres, telomerase and the Hayflick limit

WHAT IT IS	A **telomere** is a long tail of repeated DNA at the end of a chromosome. In humans the repeat is **TTAGGG**. A **telomerase** is the enzyme that rebuilds that tail, and it is unusual: it is a **reverse transcriptase**, an enzyme that builds DNA from an RNA template, and it carries its own template inside itself. The **Hayflick limit** is the number of times a normal human cell can divide before it stops and becomes senescent.
HOW IT WORKS / WHY IT MATTERS	The problem the telomere solves is the end-replication problem. The lagging strand is built backwards in pieces, and at the very end of a chromosome there is no room for the last primer to sit, so a gap is left behind and every round of copying shortens the chromosome a little. His slide 11 shows the four-step telomerase mechanism: the enzyme carries an associated RNA template, **CAAUCCCAAUC**, which pairs with the telomere's 3' overhang, **5'-CCATGCATTGGTTAG-3'**; the enzyme then extends the overhang with **GGTTAG** repeats; the enzyme shifts along and repeats the process; finally primase and DNA polymerase complete the complementary strand, which contains **AATCC**. The telomerase complex itself has two named parts: **TERC**, the RNA template, and **TERT**, the catalytic enzyme. **TZAP** is a protein that trims telomeres back down by cutting off a t-loop.

The liver cancer story on his slides 13 and 14 is the clinical payoff. The sequence runs: fetal liver has long telomeres and high TERT expression; the healthy adult liver has short telomeres and TERT is switched off; chronic liver injury drives hepatocyte turnover and telomere shortening; cirrhosis reaches the **Hayflick limit**, which activates the DNA damage response and apoptosis; then a **TERT genetic alteration** raises TERT messenger RNA, raises telomerase activity, restores telomere length, and lets the tumour keep dividing forever. In the numbers on slide 14, the overall series is

n = 1,148. TERT promoter substitutions were found in **57.8 % (n = 664)**, made up of **-124 C>T at 55.7 % (n = 638)**, **-146 C>T at 2.1 % (n = 24)** and **-57 A>C at 1 % (n = 12)**, plus a 5'UTR deletion in **0.5 % (n = 6)**. In **R0-resected hepatocellular carcinoma (n = 578)**, the patients were split by tumour telomere length, lower 90 % against top 10 %, and the top 10 % did worse:

early-recurrence-free survival P = 0.006 and **disease-specific survival P = 0.0093**. The multivariate forest plot agrees: **tumour telomere length in the top 10 % gave a hazard ratio of 1.66 (1.05-2.61), p = 0.029**, against cirrhosis at HR 1.64 (1.17-2.28), p = 0.004.

Read that as: the tumours that had managed to keep their telomeres long were the ones that came back early. Keeping telomeres long is not a good thing for the patient.

Finally the therapeutic version, slide 15. An **antisense oligonucleotide (ASO)** against TERT is given to immunodeficient nude mice carrying flank xenografts. Tumour take rate falls as the pretreatment gets longer: **9/12 with no pretreatment**, **4/12 after 2 weeks** of anti-TERT pretreatment (Fisher's p = 0.1), and **2/12 after 3 weeks** (Fisher's p = 0.01). The in vivo messenger-RNA measurement was **100 mcg ASO over 48 h intratumoural**, **p = 0.0022**, and the tumour volume curves gave **p = 0.01 at week 5** for the scrambled control (n = 6) against the anti-TERT ASO (n = 8).

EXAM HOOK	Two things are exam content. First, the key-numbers list carries: **the top 10 % of longest telomeres gives worse recurrence-free survival**. Notice the direction - longest telomeres, worst outcome. Second, the mechanism is examinable as a chain: chronic injury, hepatocyte turnover, telomere attrition, the Hayflick limit, senescence, then TERT reactivation and telomere re-lengthening in the tumour. He did not ask you to memorise the forest plot, and he did not ask you to learn the TERT promoter mutation percentages for the exam. But he did present telomere length as a real prognostic biomarker, so understand why it points the way it does.

11.3.9 A driver is not automatically a drug target - Druggability vs disease relevance

WHAT IT IS	Two different questions that get confused. **Disease relevance** asks: does this molecule matter to the disease? **Druggability** asks: can we make a drug that hits this molecule and this molecule only, without wrecking the rest of the cell?
HOW IT WORKS / WHY IT MATTERS	His worked contrast is the cytoskeleton. Cytoskeletal components are vital for metastasis - a tumour cell crawling through tissue needs to push and pull - but they resist selective small-molecule attack. Blocking them would hurt every cell in the body, because every cell has a cytoskeleton. So a molecule can be completely central to a disease and still be a bad drug target.

Telomerase is the opposite example, and this is why the topic sits next to the telomeres. Telomerase is genuinely druggable, because antisense oligonucleotides and small molecules against it are in trials. The reason is that telomerase is a switch-like target: most normal adult cells barely use it, so blocking it does little harm, while a tumour that has switched it on depends on it.

EXAM HOOK	**He said this will not be examined.** The exclusion list says of the pharma target nomination and druggability workflow: "it's not included in exam, but I think it's really more interesting". Note the nuance, because it is the same nuance as everywhere else in this lecture. He excluded the **workflow** - the company process, the criteria, the slide-by-slide commentary. He did not exclude the **idea**. So know that a driver is not automatically a druggable target, know the cytoskeleton example, and know that telomerase is the good example of a target that is both relevant and attackable. Do not memorise the workflow.

11.3.10 How a company chooses what to attack - Industrial target nomination

WHAT IT IS	**Target nomination** is the decision a pharmaceutical company makes when it picks a molecule to build a drug against. Dr. Wong gives three conditions, and a candidate has to pass all three.
HOW IT WORKS / WHY IT MATTERS	Condition one is **disease-specific overexpression in patient data**: the molecule has to be measurably raised in the patients' own samples, not just in a dish. Condition two is **structural tractability**: there has to be a way to attack it. That may be a pocket a small molecule can fit into, an antibody that can bind it, or a nucleic-acid therapy that can switch it off. Condition three is a **non-crowded competitive landscape**: if six companies are already doing this target, the commercial reason to do it a seventh time is weak, however good the science is.

Read the three together and you get the shape of the whole drug-development story in Lecture 1: find a molecule that is changed in the disease, check it is something you can actually make a drug against, then check that making that drug is still worth doing.

EXAM HOOK	**He said this will not be examined**, with the words: "it's not included in exam, but I think it's really more interesting". As with the other exclusions, the exclusion is of the **workflow**, not the concept. Understand that a target must be relevant, attackable and commercially sensible, and you will have understood something that also explains why he spends time on druggability at all.

11.3.11 The quality check inside the copier - Polymerase proofreading

WHAT IT IS	**Proofreading** is the error check built into the copying machine. DNA polymerase has a **3'->5' exonuclease** activity, which means it can cut a wrongly-paired letter off the end of the strand it has just made and put the right one back. A **mismatch** is a base paired with the wrong partner - for example A opposite G.
HOW IT WORKS / WHY IT MATTERS	The sequence is: the polymerase adds a wrong base; the 3' end wobbles out and the polymerase stalls; the exonuclease activity cuts the wrong base off; the polymerase adds the correct base and carries on. His slide 17 draws the fork with **Pol epsilon** on the leading strand and **Pol delta** on the lagging strand, each carrying a **POL** domain and an **EXO** domain, and shows the mismatch repair heterodimer **MSH2/MSH6** with **MLH1/PMS2** standing by as a second safety net.

Then the twist, which is the clinical hook. **Nucleoside analogues** are drugs designed to look like normal nucleotides. They get incorporated into DNA and stop chain growth, which is how they kill a dividing cell. But if the 3'->5' exonuclease activity removes the drug instead, the polymerase simply puts a normal base back and the cell carries on living. That is **drug resistance**, and it develops in a portion of patients. His slide 19 shows the fork in two: if p53's exonuclease excises the nucleoside analogue, the result is **drug resistance**; if instead further DNA synthesis is blocked, the result is

cell death signalling. He notes that the exonuclease activity is modulated by the local concentration of p53, by the nature of the analogue, and by the local DNA composition.

EXAM HOOK	The direction pair is the exam content: **elongation 5'->3', proofreading and excision 3'->5'**. It is one of the listed distractor pairs, and the discriminating fact is exactly that. He did not flag the A-versus-G mismatch example as examinable in its own right, and he did not ask you to remember the Pol epsilon and Pol delta domain names. What you should be able to say is this: proofreading catches a wrong base and removes it, and the same machinery can remove a drug, which is one way tumours and viruses become resistant.

11.3.12 The cell's own quality-control team - p53 and other DNA-integrity factors

WHAT IT IS	**p53** is the protein Dr. Wong introduced in Lecture 1 as a tumour suppressor whose mutation causes cancer, and given its cell-cycle-brake role in Lecture 2. Here he gives it a second job. Beyond the polymerase's own proofreading, p53 has its own **3'->5' exonuclease** activity and acts as an external proofreader.
HOW IT WORKS / WHY IT MATTERS	His slide 17 gives the four-step mechanism, adapted from Preston and colleagues (2011) and Rahav and colleagues (2013). Step one: **misincorporation** - the polymerase adds an incorrect base. Step two: **polymerase dissociation and partition to p53** - the polymerase lets go and p53 is recruited to the mismatch. Step three: **mispair excision by the p53 exonuclease** - p53 cuts the wrong base out. Step four: **p53 dissociation, partition back to the polymerase and DNA elongation** - the polymerase comes back and copying resumes.

The mouse experiments on slide 18 show what happens when you remove these safety nets. He compares wild-type mice with mice carrying proofreading-defective **Pole** and **Pold1** and with **Mlh1** knockout mice, and with the **Pole e/e Pold1 e/e** double mutant, which has the shortest survival of all. The tumour-incidence table reads: **thymic lymphoma, 0 % in wild-type against 82 % in the double mutant**; **intestinal adenoma or adenocarcinoma, 6 % in wild-type against 45 % in Pole e/e**; and

histiocytic sarcoma, 22 % in wild-type against 36 % in Pole e/e. Read that as: take away the cell's proofreading and repair machinery and the animals get cancer by themselves, without anybody giving them a carcinogen.

EXAM HOOK	He did not flag the mouse table or the four-step cartoon as a named question. The examinable idea is the general one from the source: **loss of function in proofreading polymerases or in mismatch repair gives a high burden of spontaneous tumours**. p53 also matters elsewhere in the course: it is one of the two **effectors** in the DNA Damage Response cascade, and the topic on neoantigens shows how a proofreading defect can end up helping the patient rather than harming them.

11.3.13 Wrong proteins are the fuel - Neoantigens as immunotherapy fuel

WHAT IT IS	A **neoantigen** is a piece of protein built from a mutated gene. It is new. The immune system has never seen it, so no tolerance has ever been built against it, and to the immune system it looks as foreign as a virus would.
HOW IT WORKS / WHY IT MATTERS	The chain is the whole point. If the cell's repair machinery is defective, it accumulates a very high **mutation burden**. A high mutation burden means a large number of mutant peptides, and a large number of mutant peptides means many neoantigens. Those neoantigens are presented on the cell surface by **MHC class I** molecules, which are the display cases that T cells inspect. A **cytotoxic T lymphocyte (CTL)**, a T cell that kills, recognises the neoantigen through its **T-cell receptor (TCR)** and kills the cell. That is the rational basis for **checkpoint inhibition**: the neoantigen is the fuel, and the checkpoint antibody is the match.

The beautiful counter-example is on slide 20. A **proofreading mutation in Pol epsilon** turns the cell into an **ultramutator**: neoantigens go up, **tumour-infiltrating lymphocytes** go up, and the outcome is a

good prognosis. The Kaplan-Meier curve for POLE-mutant patients sits above the curve for POLE-wild-type patients (Xing and colleagues, 2022). In plain English: a broken proofreading machine makes a tumour that is easier to see and easier to kill. The same fault is bad news in one context and good news in another.

EXAM HOOK	The examinable version of the idea, and the figure he singles out. His words about the slide showing mismatch-repair-deficient versus mismatch-repair-proficient colorectal cancer on anti-PD-1: "I would say this figure is one of the most important figures in modern medicine, because when you attend any seminars about the immune checkpoint inhibitors or immunotherapies or colorectal cancer, I can for sure that the presenter will include this figure in their presentation deck." He did not ask you to memorise the POLE Kaplan-Meier curve. But the chain is examinable, and it comes back in Lecture 5 with the phrase you should use: a tumour with many neoantigens is a **hot tumour**, and hot tumours respond to checkpoint blockade, while many solid tumours do not.

11.3.14 Making millions of copies of one piece of DNA - PCR

WHAT IT IS	**PCR**, the polymerase chain reaction, makes millions of copies of one chosen stretch of DNA. You need four ingredients. The **template** is the DNA you want to copy. The **primer pair** is two short pieces of DNA that say where to start, one for each strand. The **polymerase** is the copying machine. The **dNTPs** are the free building blocks.
HOW IT WORKS / WHY IT MATTERS	The machine repeats three steps over and over. **Denaturation**: heat pulls the two strands apart by breaking the hydrogen bonds. **Annealing**: the temperature comes down and the primers stick to the places where their sequence matches the template. **Extension**: the temperature comes back up and the polymerase copies along the template, adding a nucleotide to the 3' end each time. Each round of the three steps doubles the amount of DNA, so after n cycles you have roughly 2 to the power n copies from one starting molecule. After 30 cycles, that is about a billion copies.

His slide 21 also shows **hot-start PCR**, which is the version used in real laboratories. The polymerase is supplied with an antibody bound to it; the heat of the first denaturation knocks the antibody off, and only then does the enzyme become active. Without this, the enzyme would start nibbling at the template before the first cycle has properly begun, and the result is a mess.

The base-pairing fact from Lecture 1 is what makes the primer possible at all: a primer only binds where its sequence is the exact complement of the template, and because G pairs with C through three hydrogen bonds while A pairs with T through two, a GC-rich primer needs a higher **melting temperature** than an AT-rich one.

EXAM HOOK	He did not flag PCR as a named Lecture 4 exam question, because he introduced it in Lecture 1. What is examinable is the principle behind it: **G-C = 3 hydrogen bonds and A-T = 2**, on the key-numbers list for Lectures 1 and 3, and the reason it matters is primer design, because GC content sets the melting temperature. For Lecture 4, PCR is the tool the next two topics depend on, and the exponential-amplification idea is what lets us find a single mutated cell in a sample.

11.3.15 Cutting and pasting DNA - Molecular cloning

WHAT IT IS	**Molecular cloning** means putting a chosen piece of DNA into a small circular DNA molecule called a **plasmid**, which acts as a **vector**: a vehicle that carries the DNA into a cell. The cell then acts as a photocopier, making millions of copies for you.
HOW IT WORKS / WHY IT MATTERS	Two tools do the work. **Restriction enzymes** are enzymes that cut DNA at a specific short sequence, and they leave a short single-stranded overhang - a **sticky end**. His slide 21 shows EcoRI: the red DNA backbone ends with a G and a CTTAA overhang, and the blue insert carries the complementary AATT and G overhang. Complementary sticky ends base-pair with each other, so the two pieces stick together by themselves, and **DNA ligase** then seals the backbone.

His slide 21 also shows the full workflow. The **donor plasmid** carries an **Amp** marker, a **NotI** site and the **insert**, and is amplified by PCR and then digested with **EcoRI** and **NotI**. The

recipient plasmid carries Amp and the restriction sites XhoI, EcoRI, HindIII and NotI, and is digested with EcoRI and NotI to linearise it. The two are ligated together, producing a circular

recombinant plasmid carrying the insert. The **Amp** marker means ampicillin resistance, and its job is to let you select, in a plate of bacteria, only the cells that actually took up the plasmid.

A plasmid is useful for three different reasons: **propagation**, making more copies of the DNA; **expression**, making the cell produce the protein the gene codes for; and **sequencing**, reading out the letters.

EXAM HOOK	He did not flag the cloning workflow as an examinable item, and the details of any one restriction enzyme are slide-reading rather than exam content. What is worth holding on to is the concept in one sentence: cut with an enzyme that leaves sticky ends, glue with ligase, and grow the plasmid in bacteria. That sentence is enough to answer "how do you make a recombinant plasmid?".

11.3.16 The catalogue of damage - Classification of DNA damage

WHAT IT IS	DNA damage comes in sizes and it comes from two directions. The size question runs from a single base with a chemical tag on it, all the way to both strands of the helix cut in the same place. The source question is: did it come from inside the cell or from outside it?
HOW IT WORKS / WHY IT MATTERS	The **endogenous** sources are the cell's own doing. **Replication errors** - the polymerase simply got it wrong. **DNA strand breaks from abortive topoisomerase** - the enzyme cut the DNA and could not re-join it. And **metabolic by-products**, above all the attack of **reactive oxygen species (ROS)** on the telomere, which is the one place where the cell's own chemistry and the telomere story meet.

The **exogenous** sources come from outside. **Ultraviolet light**, which makes **pyrimidine dimers** - two adjacent bases chemically stuck together. **Radiation**, such as x-rays and gamma rays, which make double-strand breaks. **Mutagenic chemicals**, such as DNA intercalating agents, which slide between bases and distort the helix. And **viruses**, which can integrate into the genome, as HBV does in liver cancer, which you met in Lecture 2.

The lesion types on his slide 22 are: single-strand break, double-strand break, bulky adducts, mismatch, insertion and deletion, and base modifications such as alkylation. These are not random labels. Each one is the job description of one of the repair shops in the next few topics.

EXAM HOOK	He did not flag the taxonomy as an examinable item in its own right. But the mapping from damage type to repair pathway is examinable, because it is the classic "which enzyme or pathway is responsible for X" multiple-choice stem. Learn the pairing: **one damaged base goes to base-excision repair, a bulky multi-nucleotide lesion goes to nucleotide-excision repair, a copying mistake goes to mismatch repair, and a broken double strand goes to homologous recombination or non-homologous end joining**.

11.3.17 The alarm system - The DNA Damage Response (DDR)

WHAT IT IS	A **DNA Damage Response (DDR)** is the cell's alarm system for DNA damage. It has three jobs: detect the damage, pass the signal on to the right people, and then act.
HOW IT WORKS / WHY IT MATTERS	His slide 23 draws the cascade, adapted from Zhou and colleagues in Nature (2000). **Signals** are the things that hurt, such as DNA damage itself and replication stress. **Sensors** sit on the damaged DNA and notice. **Transducers** pass the message along. **Effectors** then carry it out - and his slide names **p53** and **BRCA** here. From the effectors come four possible outcomes: **stop the cell cycle**, **turn on transcription**, **carry out DNA repair**, and **trigger apoptosis**, the controlled suicide of the cell. The slide marks the cell-cycle stop with a road sign and apoptosis with a tombstone, which tells you the two outcomes he is pointing at.

The two key signalling components are **ATM** and **ATR**, both **protein kinases**, which is a machine that switches other machines on by adding a phosphate. One of the first things the cell does is inhibit **cyclin-dependent kinases**, so that the cell cycle slows down and the cell has more time to repair itself. If the damage is beyond repair, **caspase cascades** and signals from the mitochondria trigger apoptosis.

And here is the cancer connection that ties the whole lecture together: **cancer cells often disable this kill-switch**. A cell that has lost the ability to commit suicide when its DNA is broken will keep dividing and keep accumulating damage.

EXAM HOOK	He did not flag the DDR cascade as a named question, and the details of the pathway names are slide-reading. Two things are worth carrying. First, the logic of the cascade: sensors, effectors, then four possible responses, of which repair and death are the two he emphasises. Second, **ATM** is the enzyme that appears as a payload partner in dual-payload ADCs, so it is not only a pathway name but a drug target.

11.3.18 The five repair shops - The major repair pathways

WHAT IT IS	The cell keeps separate repair shops, each built for one kind of damage. **Direct reversal** is the simplest of all. **Base-excision repair (BER)** handles a single damaged base. **Nucleotide-excision repair (NER)** handles a bulky lesion. **Mismatch repair (MMR)** handles copying mistakes. **Homologous recombination (HR)** and **non-homologous end joining (NHEJ)** handle broken double strands.
HOW IT WORKS / WHY IT MATTERS	**Direct reversal** needs no nucleotide removal, no resynthesis and no ligation. His example is **MGMT**, which removes the alkyl group from O6-methylguanine and puts the DNA back exactly as it was.

BER deals with small but highly mutagenic lesions: a base modified by a metabolic event such as oxidation, or by ionising radiation. A **DNA glycosylase** removes the damaged base, leaving an

abasic site, an **AP site**, where the sugar is still there but the letter has gone. The gap is then excised and a new base put in. In detail on his slide 25, the long-patch route replaces 2 to 13 nucleotides using Pol delta or Pol epsilon with **PCNA, FEN1 and Lig1**, and the short-patch route replaces a single nucleotide using **Pol beta, XRCC1 and Lig3**.

NER deals with bulky lesions that distort the helix and block copying - thymine dimers from UV, bulky adducts, and intra-strand cross-links. It is a large system, with more than 30 proteins. There are two flavours. **GG-NER**, global genome repair, spots lesions anywhere using **UV-DDB (DDB1 and DDB2/XPE)** and **XPC**. **TC-NER**, transcription-coupled repair, notices lesions that stall the

RNA polymerase II that is reading that gene, using **CSA and CSB**. Both routes converge on

TFIIH (XPB and XPD), then **XPA**, **RPA**, and the two endonucleases **XPF and XPG**, which cut out a piece of **22-32 nucleotides**. The gap is then filled in by polymerase and sealed by ligase.

MMR fixes the mistakes polymerase makes: a **mismatch** such as a G opposite a T, and small

insertions or deletions in repetitive sequence. His slide 26 gives the two recognition pairs. A single-base mismatch is spotted by **MSH2/MSH6 (MutSalpha)**, and a repeat mismatch by **MSH2/MSH3 (MutSbeta)**; both then call in **PMS2/MLH1 (MutLalpha)**. An **exonuclease** cuts out the wrong piece, and

DNA polymerase plus DNA ligase put it back right.

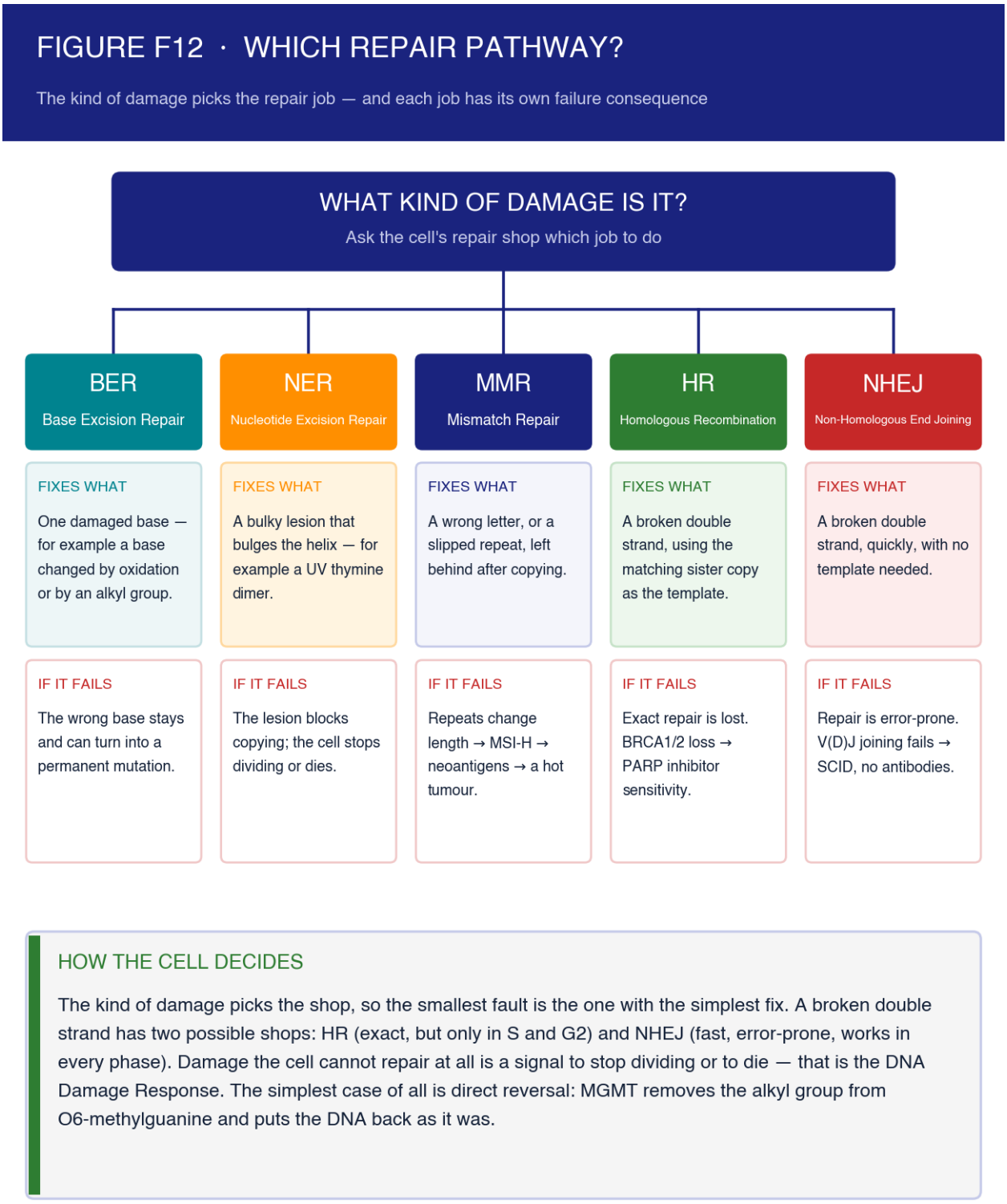
EXAM HOOK	The **BER versus NER** pair is a listed distractor pair, and the discriminating fact is the one to memorise: **BER fixes a single damaged base; NER fixes a bulky lesion that spans several nucleotides**, such as a UV dimer. The same logic gives you the MMR versus HR pair: **MMR fixes replication mismatches and small indels; HR is the high-fidelity repair of a broken double strand**.

He also excluded one part of this topic. His words: "I can ensure that they will not be included in the coming midterm", which was about **DNA glycosylase and abasic (AP) site chemistry**. Read the exclusion narrowly, as

the exam-focus document itself says it should be read: do not learn the glycosylase and AP-site chemistry. You still have to know that BER removes a single damaged base and that NER removes a stretch of nucleotides around a bulky lesion.

Figure F12 - The repair decision map

The repair decision map: the kind of damage picks the repair pathway, and each pathway carries its own failure consequence.



Which repair pathway does which job, and what happens when that job is not done. This is the decision tree behind the last four topics: the kind of damage picks the shop, and a broken double strand has two shops to choose between.

11.3.19 When the repair shop stays shut - MMR deficiency and MSI

WHAT IT IS	**Microsatellites** are short tandem repeats scattered throughout the genome, in coding and non-coding regions alike: single nucleotides, or two or three repeated over and over. His slide 28 gives two examples, **TATATATATATA** as a dinucleotide microsatellite and **GTCGTCGTCGTC** as a trinucleotide one. The length of these repeats varies from person to person, and that variation is the basis of the **DNA fingerprint**.

Microsatellite instability (MSI) is what happens when the mismatch repair shop is broken and the lengths of these repeats start changing. **MSI-H** means the instability is high; **MSS** means the microsatellites are stable.

HOW IT WORKS / WHY IT MATTERS	His slide 27 shows the mechanism with a repeat template **3'-GTGTGTGTGTGT-5'**. When the polymerase copies a repeated sequence it can **slip**, leaving an unpaired **CA loop** in the new strand, **5'-CACAC[CA]ACACA-3'**. If mismatch repair is **intact**, the loop is spotted and corrected, and the result is a stable microsatellite, **MSS**. If mismatch repair is **deficient**, the slippage is not corrected, the repeat gets longer or shorter, and the result is **MSI-H** or **dMMR**, deficient mismatch repair.

The consequences run all the way to the clinic. Because the slipped repeats are often in genes, the cell accumulates **frameshift mutations**, which produce truncated proteins and, importantly, lots of

mutation-associated neoantigens (MANA). His slide 30 shows MANA presented on the tumour cell by an MHC complex to a T-cell receptor, and the tumour trying to escape by putting **PD-L1** on its surface to bind **PD-1** on the T cell and switch the T cell off. That is **T-cell anergy**. An **anti-PD-1** antibody blocks the interaction, the T cell wakes up, and it kills the tumour cell. So the chain is: MMR deficiency, then MSI-H, then frameshifts, then neoantigens, then a **hot tumour**, then a response to immune checkpoint inhibitors.

His slide 29 gives the colorectal cancer numbers: **MSI results from defective DNA mismatch repair in 15 % of colorectal cancer**. The slide also shows the adenomatous pathway, with **APC** early (loss of control of beta-catenin signalling), **KRAS** intermediate (a constitutive growth signal), **p53** late (loss of tumour suppression) and increasing MSI as the polyp becomes an adenocarcinoma, plus the serrated pathway with **BRAF** early. And the cause of MMR loss in many of these tumours is

hypermethylation of the hMLH1 promoter. This is where Lecture 3 meets Lecture 4 - the switch that silences MMR is itself a methylation mark. (That link is the guide's own; he does not make it in the lecture.)

EXAM HOOK	**15 % of colorectal cancers are MSI-H**, and that number is on the key-numbers list. The chain matters more than the number: **MMR deficiency leads to MSI-H, which leads to frameshift mutations, which leads to neoantigens, which makes the tumour hot, which predicts a response to anti-PD-1**. This is the mechanism behind the figure he called one of the most important in modern medicine, and it is one of the likely "which drug would this patient respond to" stems: an MSI-H colorectal cancer is an anti-PD-1 patient.

11.3.20 A fetal protein that comes back - CEA as a solid-tumour biomarker

WHAT IT IS	**Carcinoembryonic antigen (CEA)** is a protein that is made normally only while a baby is still developing. Many cancer cells switch it back on. So an adult with CEA in the blood is telling you something.
HOW IT WORKS / WHY IT MATTERS	CEA is used as a **circulating surrogate marker** for tumour burden in colorectal cancer. A surrogate marker is a cheaper stand-in for the thing you really want to measure. The thing you really want to measure is tumour size on a CT or MRI scan, but scanning a patient repeatedly is expensive and inconvenient. A blood test is cheap, quick and repeatable, so CEA in the blood is used to follow how much tumour is left.

His slide 31 is the evidence that the marker works, and it is the same anti-PD-1 story. The graph plots the **change in tumour marker level (%)** against **days, 0 to 400**, with a dashed line at **0 % (no change)**. Three groups are

shown. **Mismatch repair-proficient colorectal cancer** (red) mostly rises above 0 %, some patients going up to more than **+200 %**, meaning the tumour was getting bigger and the marker was tracking it. **Mismatch repair-deficient colorectal cancer** (blue) drops sharply towards

-100 %. **Mismatch repair-deficient non-colorectal cancer** (black dashed) also drops towards -100 %. The citation is Le and colleagues, New England Journal of Medicine, 2015.

Read that as: the tumours that shrank were exactly the ones whose mismatch repair was broken, and the CEA level fell as they did. The biomarker is measuring the response, and the response is what the MMR status predicts.

EXAM HOOK	He did not flag CEA itself as a Tier-1 question, but he did present the slide behind it as one of the most important figures in modern medicine, using the words: "I would say this figure is one of the most important figures in modern medicine, because when you attend any seminars about the immune checkpoint inhibitors or immunotherapies or colorectal cancer, I can for sure that the presenter will include this figure in their presentation deck." So be ready to combine two facts: **CEA is a circulating surrogate marker for tumour burden that replaces repeated scanning**, and **the direction of the CEA change after anti-PD-1 differs between MMR-proficient and MMR-deficient tumours**.

11.3.21 How a drug gets tested in people - How clinical trials are structured

WHAT IT IS	Before a drug can be given to patients it has to pass through a ladder of clinical trials, and each rung asks a different question.
HOW IT WORKS / WHY IT MATTERS	**Phase 1A** is **dose escalation**. Small numbers of patients who already have the disease are given increasing doses, and the question is safety: what dose is safe, and what is the maximum dose the body will tolerate. **Phase 1B** is **expansion cohorts**: more patients, at the chosen dose, to see the side effects properly and to look for early signs of benefit. **Phase 2 and Phase 3** are about **efficacy**: does it work, and does it work better than the treatment already being used?

The reason this topic sits between the biomarker and the repair pathways is cost and measurement. Trials are expensive: about **US$200,000 per patient**. With a cohort of hundreds of patients, that is tens of millions of dollars. So a trial needs an outcome that can be measured cheaply and often, and that is exactly what a circulating biomarker gives you. **CEA** and similar blood markers are used as

surrogate endpoints for tumour size, which lets the trial read out a response without scanning everyone every few weeks.

EXAM HOOK	The numbers are on the key-numbers list for both Lecture 2 and Lecture 4: **about US$200,000 per patient**, and the phase structure **Phase 1A dose escalation, Phase 1B expansion, Phase 2 and 3 efficacy**. He did not name trial design as a standalone exam question in Lecture 4, but these are ready-made multiple-choice numbers, and the surrogate-endpoint idea is what links the biomarker topic to the therapy topics.

11.3.22 Fixing a rope cut in two - Double-strand-break repair

WHAT IT IS	A **double-strand break (DSB)** is the worst structural damage there is. Both strands of the helix are cut in the same place, and the chromosome falls apart into two pieces. The cell has two routes to fix it, and they are built on completely different principles.
HOW IT WORKS / WHY IT MATTERS	**Homologous recombination (HR)** repairs the break by copying the information from a **homologous template**: the **sister chromatid**, the identical copy of the same chromosome made when the cell copied its DNA before dividing. Because the template is identical, the repair is exact. The price is that the sister chromatid only exists during S and G2, so HR is **restricted to S/G2**. His slide 33 gives the steps: the broken ends are processed by **Rad50, Mre11 and Nbs1** to give 3' overhangs; then **Rad51, Rad52 and RPA** carry out **strand invasion** into the matching duplex; then repair synthesis and **branch migration** by Rad54; then **resolution**.

Non-homologous end joining (NHEJ) needs no template at all. It simply grabs the two loose ends and joins them. His slide 37 shows **Ku** binding the broken ends, then a polymerase, a nuclease and a ligase working on the ends, and then the possible outcomes: a **deletion**, or a **deletion and addition**. That is the error-prone part. NHEJ works in **every phase of the cell cycle**, including G1 and in cells that will never divide again, such as a terminally differentiated neuron.

His slide 42 shows the cell's decision. In G1, S and G2 the factors **53BP1, RIF1 and Shieldin** push the cell down the NHEJ route, with **KU and DNA-PKcs** protecting the ends and ligase joining them with minimal processing. In S and G2, long-range resection opens the ends, **RPA** coats the single-stranded DNA, **RAD51** coats it, and **BRCA1 and BRCA2** direct strand invasion into the sister chromatid. Two other routes sit in between: **alternative end joining (a-EJ)**, which needs only a few base pairs of homology, and **single-strand annealing (SSA)**, which needs more than about 20 base pairs. He also lists the factors that push the cell one way or the other: the cell cycle, the state of differentiation, how complicated the break is, the tumour suppressor status, and the level of the repair proteins themselves.

EXAM HOOK	This is the pair he calls the single most likely multiple-choice distractor. Keep the five facts apart in your head. **HR uses a template and is exact, but only in S and G2; NHEJ uses no template, works in all phases, and is error-prone.** HR is used by meiotic recombination, NHEJ is used by **V(D)J recombination** in the immune system, and that link is the Tier-1 question in the next topic. In CRISPR, HR gives **knock-in** and NHEJ gives **knock-out**. He also stressed that HR is tightly controlled by cyclin and CDK, and that BRCA1 is needed for efficient HR.

11.3.23 Shuffling the deck between generations - HR in meiosis

WHAT IT IS	**Crossing over** is the deliberate exchange of DNA between the maternal and paternal copies of a chromosome during meiosis I, the first division that makes eggs and sperm. It is the same machine as homologous recombination, used deliberately.
HOW IT WORKS / WHY IT MATTERS	His slide 34 shows it. A **diploid parent cell (2n = 4)** copies its DNA, then the **homologous chromosomes** line up. Chromatids that are not sisters - one from the mother, one from the father - swap segments. The slide uses alleles A, B, C on the blue chromosome and a, b, c on the red one, and after the crossover you get **recombinant chromatids** carrying A-B-c and a-b-C, alongside **non-recombinant chromosomes** carrying A-B-C and a-b-c.

Then meiosis finishes its two divisions and produces haploid daughter cells that are not identical to the parent cell or to each other. Crossing over is only one of three sources of variety: the shuffling of segments, **random segregation** of the chromosomes into the daughter cells, and **random fertilisation**. Together they give an almost limitless number of possible children from the same two parents. The slide makes the contrast with mitosis explicit: mitosis gives two diploid daughter cells identical to the parent and to each other; meiosis gives four haploid daughter cells, none of them identical.

EXAM HOOK	He did not flag meiotic crossover as a Lecture 4 exam question in its own right. The M1 versus M2 non-disjunction question is a Lecture 3 item, and it is already on the Tier-1 list. What Lecture 4 adds is the connection: **the same repair machinery that fixes a broken chromosome is the machinery that deliberately shuffles your genes between generations**. That link appears in the HR versus NHEJ table, where HR is listed as the pathway used by meiotic recombination.

11.3.24 Killing only the cells that are already broken - Synthetic lethality in cancer

WHAT IT IS	**Synthetic lethality** describes two faults that a cell can survive on its own but not together. A cell missing one repair gene may be perfectly healthy. A cell missing both dies.
HOW IT WORKS / WHY IT MATTERS	His slide 35 draws the logic in one picture. The **normal cell** has both **Gene A** and **Gene B** working. A **lightning bolt** labelled "Gene A mutation" turns it into a **cancer cell** with only Gene B working. Now an **inhibitor of B** is applied to both. The normal cell survives, because Gene A compensates, and it is drawn with a smiling sun and the word **Viable**. The cancer cell dies, because both its workarounds are gone at once, and it is drawn with a skull and crossbones and the word **Lethal**.

So the drug is not aimed at the tumour by magic. The tumour has already damaged itself, and the drug removes the only thing that was keeping it alive. That is what gives the treatment its

therapeutic window: the margin between the smallest dose that works and the dose that becomes toxic. Synthetic lethality widens that window, because normal cells have a working copy of the first gene and survive the same dose that kills the tumour.

EXAM HOOK	He did not attach a number to this topic, and he did not name it as a standalone question. The definition and the picture are the exam content: **losing one gene alone is survivable, losing both is lethal; drug the second gene in a tumour that has already lost the first; normal cells with the wild-type partner are unaffected**. This is the idea that the next topic applies to the immune system, and that the PARP inhibitor case study turns into a treatment.

11.3.25 Why your antibodies are all different - NHEJ in adaptive immunity (V(D)J recombination)

WHAT IT IS	During the development of a B cell or a T cell, the cell deliberately cuts and re-joins its own antibody and T-cell-receptor genes. This is **V(D)J recombination**, and it is how your immune system ends up with billions of different antibodies. **V** stands for variable, **D** for diversity, **J** for joining. **NHEJ** is the repair machinery borrowed to do the joining.
HOW IT WORKS / WHY IT MATTERS	His slide 40 lays it out. The **immunoglobulin heavy chain germline DNA** carries **Variable (V) 39**, **Diversity (D) 27**, **Joining (J) 6** and **Constant (C) 6** segments. The cell performs **somatic recombination** in two rounds: first **D to J**, then **V to D-J**. The result is transcribed and spliced into a messenger RNA and translated into the heavy chain. The **light chain** is simpler: **V 36**, **J 5** and **C (1 kappa, 4 lambda)** segments, recombined **V to J**.

Two details make the mechanism work. Each segment is flanked by **recombination-signal sequences**, which are recognised by the **RAG1/RAG2 protein complex**; RAG makes the cut and the machinery that processes and ligates the ends is **NHEJ**. And because the joining is random, the possible combinations are enormous. Slide 41 does the same for the T cell: the **alphabeta T-cell receptor**, with an alpha chain made by V-J recombination and a beta chain made by V-D-J recombination, forming an **alphabeta heterodimer** on the cell surface next to **CD3epsilon**, **CD3delta** and the **zeta chain**, on a CD4 T cell reading an antigen presented by **MHC class II**.

Now the failure. If NHEJ is defective, the cell cannot assemble the segments, so it cannot build the repertoire. A child born with such a defect has **severe combined immunodeficiency (SCID)**. He puts it plainly: the child cannot generate enough of the antibody pool to recognise the many copies of a foreign antigen.

EXAM HOOK	**This is Lecture 4's Tier-1 exam question.** His words, verbatim: "Therefore, if I ask you, if the people showing a defective in this non-homologous end joining, what's the outcome? Definitely the outcome is the severe immune deficiency, right? Because he cannot generate a sufficient amount of the antibody pool to recognize the thousand copies of the foreign antigen. Yeah, that's the clinical outcome of this genetic defect."

The chain you must be able to draw is this: **NHEJ is co-opted by V(D)J recombination to join V, D and J segments, which generates antibody and TCR diversity, so a defective NHEJ means the repertoire cannot be built, which means SCID**. The segment copy numbers he gave are **V ~ 39 copies, D = 27 copies, J = 6 copies**, and these are on the key-numbers list. If you are asked why a normal cell can use NHEJ for general repair but not for V(D)J, the answer is that the immune cell is deliberately making permanent changes to its own genome, so an error-prone pathway is exactly what it needs.

11.3.26 A bacterial scalpel with an address label - CRISPR/Cas9

WHAT IT IS	**CRISPR** - clustered regularly interspaced short palindromic repeats - is a bacterial immune system, and **Cas9** is the enzyme it uses. You design a **guide RNA (gRNA)** that matches the DNA you want to cut, and Cas9 cuts both strands at that spot.
HOW IT WORKS / WHY IT MATTERS	His slide 43 shows the mechanism. The Cas9 protein and the gRNA find their target next to a **PAM**, the short DNA sequence the enzyme requires before it will cut. The cut is made by two nuclease domains, **RuvC** and **HNH**, and the result is a **double-strand break**. Now the cell's own repair machinery takes over, and this is exactly where the last two topics come together.

If the cell repairs the break by **NHEJ**, it makes small **insertions or deletions**, and because the coding sequence is read in threes, those small changes usually destroy the gene. That is a **gene knock-out**, and it works in every phase of the cell cycle. If the cell repairs by **homology-directed repair (HR)**, and a **donor template** is supplied, the new sequence is copied in accurately. That is a

gene knock-in, and it only works in **S and G2**, when a sister chromatid is available. The difference between the two choices is therefore not a preference; it is a limitation of the cell biology, and it is exactly the HR versus NHEJ difference from the previous topics.

His slide also shows the therapeutic pipeline: take somatic cells from a **patient**, reprogramme them into **induced pluripotent stem cells (iPSCs)**, edit them with CRISPR-Cas9 to make **isogenic iPSC controls**, differentiate them into different cell types, characterise them and do quality control for safety, purity and activity, and transplant the patient-matched healthy cells back. He notes that CRISPR-Cas9 won the **Nobel Prize in Chemistry in 2020**, and his bullet list says it plainly:

powerful and simple.

EXAM HOOK	He did not name CRISPR as a Tier-1 question, but it is on the Tier-2 sheet, and the cross-lecture note says: "CRISPR/Cas9 is now standard for gene knock-out/knock-in; older siRNA/shRNA knockdown techniques are largely superseded." The examinable content is the choice and its reason: **NHEJ gives a knock-out and works in every phase; HR with a donor template gives a precise knock-in and is restricted to S and G2**. He did not ask you to memorise the RuvC and HNH domain names or the iPSC pipeline.

11.3.27 Who owns the tool? - The CRISPR IP landscape

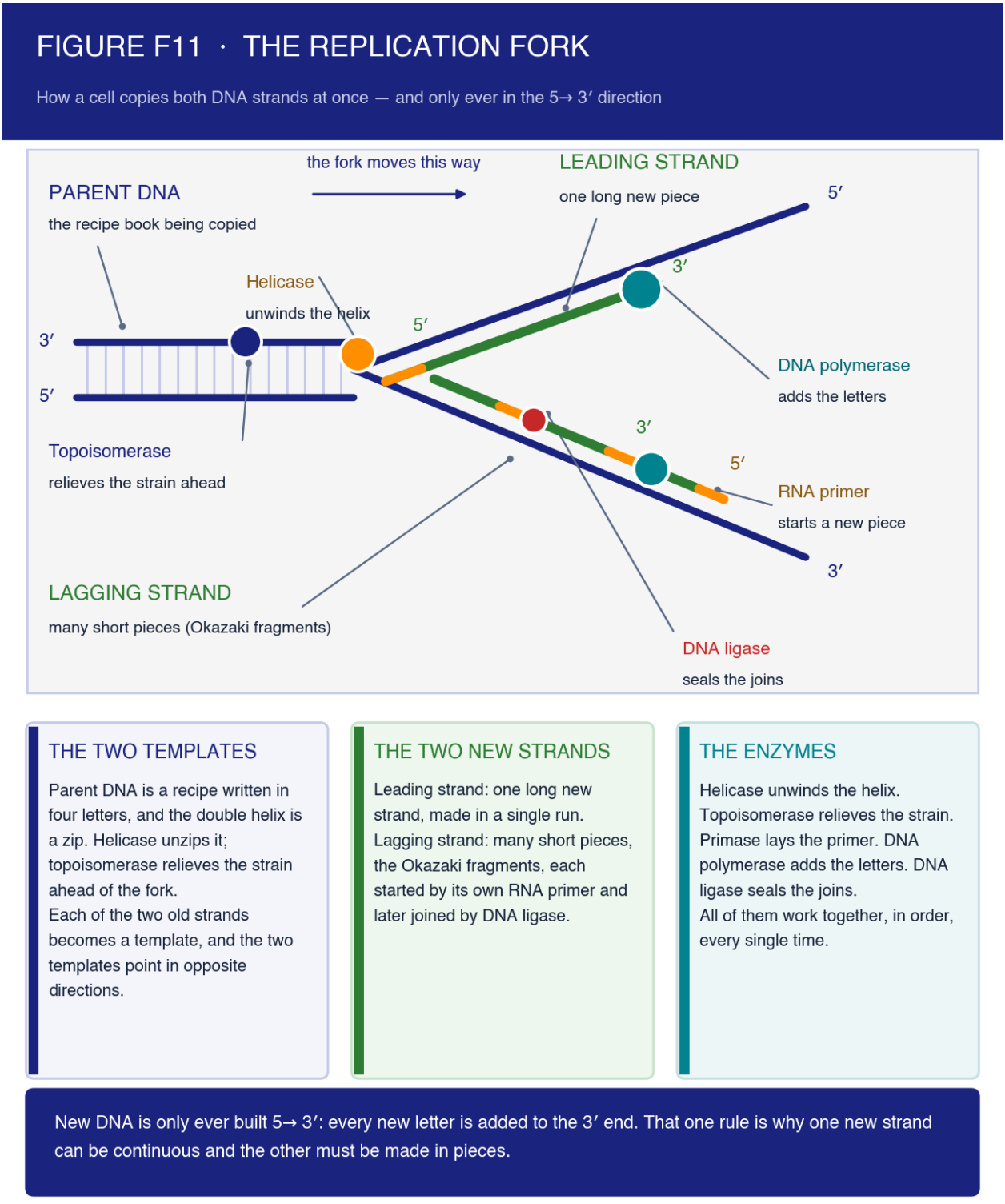
WHAT IT IS	**IP** means intellectual property. It is the legal question of who owns the right to use a piece of technology, and for CRISPR it is a real and ongoing fight. Two tracks run through it. The **UC Berkeley** track, held by **Jennifer Doudna** and **Emmanuelle Charpentier**, covers the invention of CRISPR-Cas9 itself. The **Broad Institute** track, held by **Feng Zhang**, covers the use of CRISPR technology in eukaryotic cell engineering.
HOW IT WORKS / WHY IT MATTERS	His slide 44 is a two-track timeline. On the Berkeley track: first patent application **25 May 2012**, first scientific article in bacteria **28 June 2012**, the eLife paper showing CRISPR editing human cells **29 January 2013**, the interference proceedings announced **11 January 2016**, the European Patent Office granting the first EU CRISPR-Cas9 patent on eukaryotic uses **10 May 2017**, the appeal to the Federal Circuit **25 July 2017**, and a second interference beginning **June 2019**. On the Broad track: first USPTO application **12 December 2012**, the Science paper on mouse and human cells **13 January 2013**, the first CRISPR-Cas9 patent for plant and animal cells **15 April 2014** under the **first-to-invent** principle rather than first-to-file, the PTAB finding no interference **15 February 2017**, the Federal Circuit ruling for Broad **10 September 2018**, and the PTAB ruling that the Berkeley invention was not patentably distinct **10 September 2020**. His slide 45 is a Broad Institute statement updated **26 March 2026**, saying the Broad patents are "patentably distinct" and that a commercial user might require a licence from both sides.

Three ideas sit underneath the fight, and they are the transferable ones. **Inventions versus specific applications**: you can own the tool, or you can own a particular use of the tool, and the law does not settle that neatly. **Non-obviousness**: an invention has to be more than an obvious extension of what was already known. And **Freedom to Operate (FTO)**, a review carried out before commercialisation to ask the practical question: if I build this product, will I be sued? And a **patent pool**, in which several owners license their patents to each other on agreed terms, which is what Broad has been pressing for.

EXAM HOOK	**He did not flag this as examinable.** The exam-focus document's exclusion list does not include the CRISPR patent topic, so treat the timeline and the dates as reference material rather than exam content. Keep the three transferable ideas - inventions versus specific applications, non-obviousness, and Freedom to Operate - and leave the dates. The concept is still worth keeping, because it explains why a discovery does not automatically become a product: someone has to own it, and the commercial users may have to pay two organisations. That is a real constraint on medicine, and it is the same kind of constraint as the cost of a clinical trial.

Figure F11 - The replication fork

The replication fork: the parental double helix opening, the leading strand made in one continuous run, the lagging strand made as Okazaki fragments, each with its own RNA primer, and DNA ligase sealing the joins.



The replication fork, drawn the way Dr. Wong's slide 7 and slide 10 draw it. Read it from the top down: the parental helix opens at the fork, the leading strand is made in one continuous run with its 3' end at the polymerase, and the lagging strand is made in short Okazaki fragments, each with its own orange RNA primer, that DNA ligase later joins.

11.4 The ten things to remember

1. New DNA is only ever built in one direction, **5'->3'**. Proofreading and cutting go the other way, **3'->5'**. Ligase seals a 5' phosphate to a 3' hydroxyl.
2. Replication is **semi-conservative**: every new double helix has one old strand and one new one.
3. The five copying machines are **helicase** (unwinds), **topoisomerase** (relieves strain), **primase** (lays the primer), **polymerase** (adds the letters) and **ligase** (seals the gaps).
4. Because the two templates are **antiparallel**, the **leading strand** is continuous and the **lagging strand** is made in **Okazaki fragments**, each with its own RNA primer, about 200 nucleotides apart.
5. **Telomerase** rebuilds the telomere at the end of a chromosome, because the lagging strand cannot finish there. When telomeres run out, the cell stops dividing: the **Hayflick limit**.
6. Each kind of damage has its own shop: **BER** for one damaged base, **NER** for a bulky lesion, **MMR** for a copying mistake, **HR** and **NHEJ** for a broken double strand.
7. **MMR deficiency** produces **MSI-H**, which produces frameshifts, which produce neoantigens, which makes a **hot tumour** that responds to **anti-PD-1**. **15 % of colorectal cancers are MSI-H**.
8. **HR uses a template and is exact but only works in S and G2. NHEJ uses no template, works in every phase, and is error-prone.** That single difference explains knockout versus knock-in in CRISPR.
9. **NHEJ is borrowed by V(D)J recombination** to build antibody and TCR diversity, so a defective NHEJ gives **SCID**. The segment counts are **V ~ 39, D = 27, J = 6**.
10. **Synthetic lethality** lets you kill only the tumour: drug the second gene in a cell that has already lost the first. And **CRISPR-Cas9** is the engineered version of the same story, cutting the DNA and letting the cell's own repair machinery decide the outcome.

11.5 Check yourself

Q1. A cell is treated with a drug that stops DNA ligase working. What exactly does that block, and would you expect the cell to copy its DNA fine but fail to divide?

A1. The observation first: ligase is needed to seal a **nick in the DNA backbone**, and the only nicks left after ordinary copying are the joins between **Okazaki fragments** on the lagging strand, plus the small gaps left where the RNA primers were removed. So the drug does not block the copying itself; it blocks the **finishing**. The mechanism is that ligase forms a **phosphodiester bond** between a 5' phosphate and a 3' hydroxyl, and without it the lagging strand stays as a chain of separate fragments. The clinical meaning is that the cell accumulates nicks and unligated fragments, which are exactly the kind of damage that feeds the DNA damage response and, if the cell cannot cope, apoptosis. The action is to think of this as a drug that leaves the replication machinery half-finished. The limit is that the leading strand is copied continuously and needs no joining, so the damage is concentrated on one strand and on the lagging side, and it takes a replication round for the problem to appear at all.

Q2. A tumour is shown to have lost both copies of BRCA1. What treatment follows from that fact, and why is it less toxic than it sounds?

A2. The observation is that the tumour has lost **BRCA1**, and BRCA1 is needed for efficient

homologous recombination. The mechanism is that HR is the high-fidelity way to fix a double-strand break, so an HR-deficient cell has to fall back on error-prone NHEJ, and its genome gets more and more broken. The clinical meaning is **synthetic lethality**: a **PARP inhibitor** blocks PARP, which is trapped on single-strand breaks and causes the replication forks to stall into double-strand breaks. A cell with working HR can repair those breaks and survives. A cell without BRCA1 cannot, and dies. So the drug kills the tumour while normal cells, which still have working HR, are spared. The action is to test the tumour for BRCA status before treating. The limit is that this is a statement about HR-deficient tumours and PARP inhibitors, and the same fact will not transfer to every tumour or every drug.

Q3. A child has no functional antibodies and no functional T cells. Your colleague says the cause is "a DNA repair problem". Explain which repair problem, and what the molecular chain looks like.

A3. The observation is a **SCID** phenotype, severe combined immunodeficiency, which is exactly the Tier-1 chain Dr. Wong named for Lecture 4. His words: "Therefore, if I ask you, if the people showing a defective in this non-homologous end joining, what's the outcome? Definitely the outcome is the severe immune deficiency, right? Because he cannot generate a sufficient amount of the antibody pool to recognize the thousand copies of the foreign antigen." The mechanism: during B-cell and T-cell development the cell deliberately breaks and re-joins its own antibody and T-cell-receptor genes, joining **V**, **D** and **J** segments to build the repertoire. The breaks are made by **RAG1/RAG2** and the ends are processed and ligated by **NHEJ**. If NHEJ is defective, the segments cannot be joined, the repertoire cannot be built, and the child has no usable antibodies or T cells. The numbers he gave for the heavy chain are **V ~ 39 copies, D = 27 copies, J = 6 copies**. The clinical meaning is that this is a genetic disease of the repair machinery rather than of the immune system itself. The limit is that many other genetic causes of SCID exist; the chain he wants is specifically the NHEJ one.

Q4. A colorectal cancer is reported as MSI-H. A colleague asks whether that means the tumour is "more mutated" and whether you should now use chemotherapy. What is the correct answer?

A4. The observation is the **MSI-H** label, which means the microsatellites in the tumour genome are changing length, which means the **mismatch repair** system is defective, which means the copying errors that MMR normally corrects are being kept. The mechanism is **polymerase slippage** in repeated sequences: the polymerase slips, a small loop is left in the new strand, and if MMR is intact the loop is corrected and the microsatellite stays stable, whereas if MMR is deficient the repeat length changes. Because microsatellites sit inside genes, this produces **frameshift mutations** and a high load of

neoantigens, and that is what makes it a **hot tumour**. The clinical meaning is that MSI-H predicts a response to **immune checkpoint blockade such as anti-PD-1**, because the neoantigens are the fuel and the checkpoint antibody unblocks the T cells that recognise them. So the answer to the colleague is not chemotherapy, it is immunotherapy. And the number to quote is that **15 % of colorectal cancers are MSI-H**. The limit is that MSI-H predicts a response on average; individual patients can still relapse, and the same tumour's behaviour depends on other things as well.

Q5. You want to insert a corrected copy of a gene into a cell's genome rather than simply destroy the gene. Which repair pathway will do the job, and what limits it?

A5. The observation is that the task is a **knock-in**, a precise insertion, not a knock-out. The mechanism is that **CRISPR-Cas9** cuts both strands at a chosen address, using a guide RNA matched to the target next to a **PAM**, and the cut is made by the RuvC and HNH nuclease domains. The cell then repairs the break by one of two routes, and the route decides the outcome. If the cell uses **NHEJ**, it makes small insertions and deletions, which usually destroys the gene, and that is the **knock-out** route. If the cell uses **homology-directed repair**, and a **donor template** carrying the corrected sequence is supplied, the new sequence is copied in accurately, and that is the **knock-in** route. The clinical meaning is that the editing strategy is not a free choice. The limit is the cell biology: HR-based editing is **restricted to S and G2**, because it needs a sister chromatid to copy from, whereas NHEJ works in every phase of the cell cycle. So a knock-out can be made in any dividing or non-dividing cell, while a knock-in generally has to be done while the cell is copying its DNA.

End of Part 5. The next chapter, Part 6, covers Lecture 5 - Control of Gene Expression. Where this chapter opened threads - the p53 brake, the checkpoint inhibitors, the hot-tumour idea, the code optimisation needed to make a protein in a factory cell, and the CRISPR tools - the next lecture takes them further.

PART 6

Lecture 5: Control of Gene Expression

IN THIS PART

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12.4 The ten things to remember

12.5 Check yourself

About this part

Part 6 - Lecture 5 - Control of Gene Expression - Dr. Tony Wong - 17 October 2026

Part 7 - The Exam Pack - the whole course

Course: GNBF5020 Introduction to Molecular Biology and Genetics - **Instructor:** Dr. Tony Wong

This file is the last part of the beginner guide. It contains the fifth and final lecture, and then the exam pack for the whole course. The lecture rewrites the expert shorthand of the existing revision guide so that it can be read by someone who has never studied biology or chemistry. Every number Dr. Wong stated is reproduced exactly and is followed by a plain sentence saying what it means. Where he said nothing about a topic, the Exam hook says so. The exam pack gathers, in one place, the five questions he named as exam-type, the numbers, the things he said will not be asked, the pairs he uses as distractors, and a twenty-question self-test.

CHAPTER 12

Control of Gene Expression

12.0 What this lecture is about

Dr. Tony Wong's fifth lecture is where the whole course comes together. He starts with a simple question - why does a cell switch some genes on and others off - and then works through every level at which that decision is made, from the way DNA is packed right through to how long the finished message survives. This is the bridge between the chemistry of Lectures 1 and 2 and the clinical genetics of Lectures 3 and 4. Its payoff is that almost every drug and every biomarker he has shown you works by changing gene expression somewhere along that chain. By the end of this chapter you should be able to explain all twenty-four ideas in plain English, and say which of them he has told you he will test.

12.1 Words to know before we start

Term	In plain English
Gene expression	The whole process of turning a gene on, copying the message, and building the protein.
Promoter	The DNA word that tells the copying machinery where a gene starts.
Operator	The DNA word that a repressor sits on to block a bacterial operon.
Enhancer	A switch that turns a gene up. It can be far from the gene it controls.
Silencer	A switch that turns a gene down, wherever it happens to sit.
Transcription factor	A protein that binds a switch to help or hinder copying.
Operon	A group of neighbouring genes switched on and off together by one shared control region.
Exon	A stretch of a gene that stays in the mature message.
Intron	A stretch of a gene that is cut out and thrown away.
Alternative splicing	Joining the same exons in different ways, so that one gene makes several different proteins.
Reporter assay	The experiment that measures how strong a promoter is, by reading out a gene that makes light.
Receptor occupancy	The fraction of receptors that must be filled by drug before there is a clinical effect.
MicroRNA	A short RNA, about 22 nucleotides long, that switches a target message off.
Chromatin	DNA plus the proteins that package it. Open chromatin can be read; closed chromatin cannot.

12.2 The big picture

Picture a factory that has one set of master manuals but makes hundreds of different products. The manuals never change. What changes is which page is copied, how many copies are made, how long each copy is used for, and which finished part is shipped where. Biology calls the manual DNA, the page a gene, the copy the messenger RNA, and the choice of which page to copy gene expression. Dr. Wong's whole fifth lecture is the story of how that choice is made, and of what happens when it is made wrongly.

The lecture has three big movements. The first is the prokaryotic classic: the lac operon in bacteria. It is a beautiful, simple switch, and it teaches every idea you need - a promoter, an operator, a repressor, and the rule that a gene is only made when two separate conditions are met at once. He then shows the mirror image, the tryptophan operon, which is switched off by its own end product. From there he moves to the eukaryotic version, which is far more complicated. Your cells keep DNA in a nucleus, so copying happens there and building happens out in the cytoplasm, and in between the message is edited. A human gene is not one recipe but a set of recipes with optional pieces, so the cell can splice the same message several ways and make several different proteins from one gene. That is why a nerve cell and a liver cell, carrying the same DNA, behave completely differently.

The third movement is the pharmacology. Dr. Wong shows you that the switch itself is a drug target. The estrogen receptor is a protein with three working parts, and a drug can grab one part and change what the whole protein

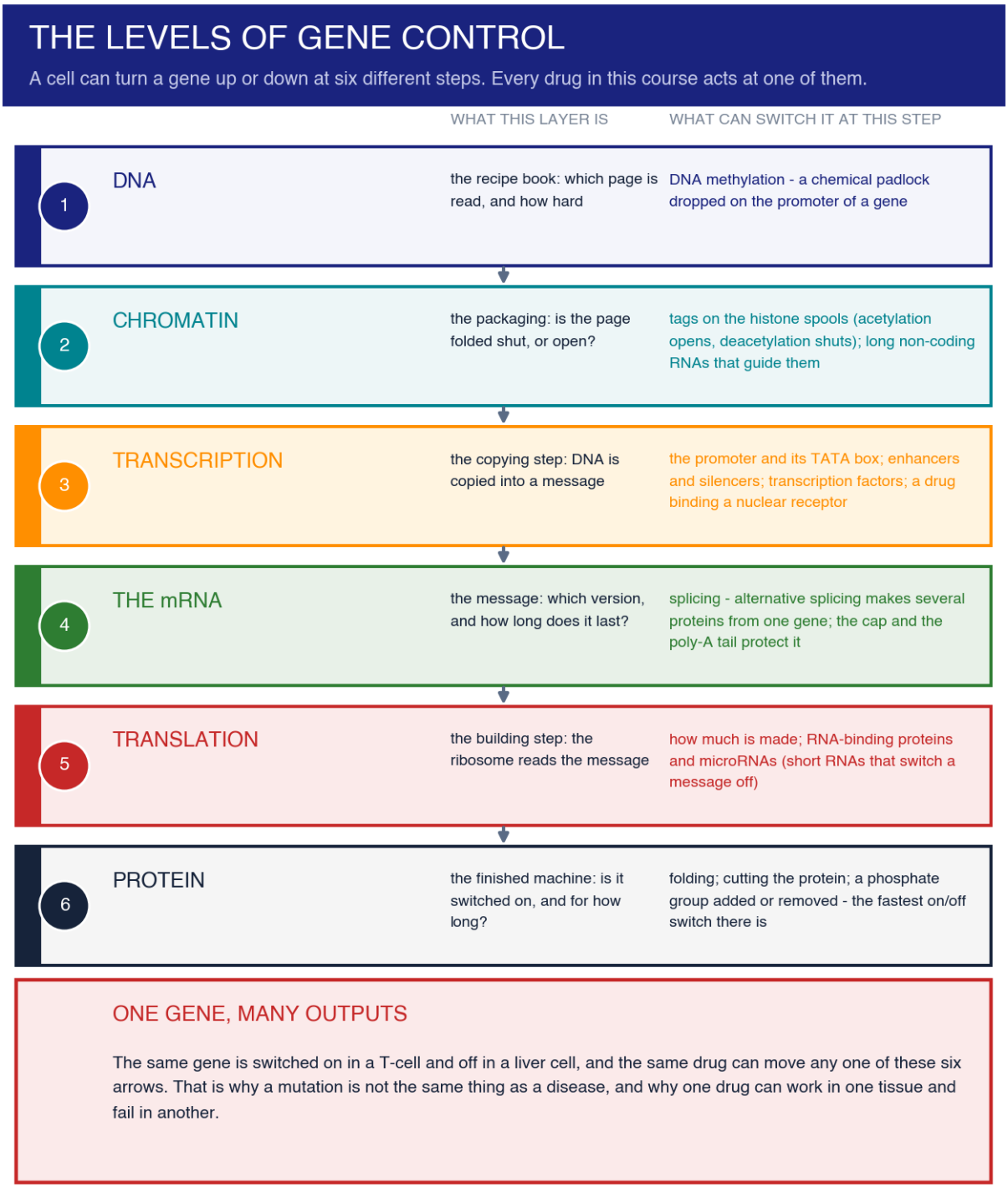
does. The same drug can be an antagonist in one tissue and a partial agonist in another, which is exactly what tamoxifen does in breast tissue versus the endometrium. The same theme runs through the dimerisation switch, the receptor occupancy numbers, the NF-kappaB inflammatory cascade with its strict time course, the oxygen sensor HIF-1alpha, the Hippo pathway that switches growth off when cells touch, and Notch, which lets one cell talk to its neighbour. He finishes with the newest layer: long non-coding RNAs that guide the chromatin machinery without making any protein at all, microRNAs that switch messages off, and mRNA therapies that turn the cell's own copying machinery into a drug delivery system.

Two ideas run through everything and you should hold on to them. First, regulation happens at several layers at once, and the sum of those layers is what turns a genotype into a phenotype. That is why carrying a BRCA mutation does not guarantee disease, and why a drug can turn a cold tumour into a hot one. Second, almost every clinical biomarker in this course - diagnostic, prognostic and predictive - works through gene expression. Once you understand how the cell decides what to read, you can read any patient's transcriptome.

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Figure F13 - The levels of gene control

The levels of gene control: DNA, chromatin, transcription, mRNA, translation and protein, each with the control mechanism that acts there.



The six levels at which a cell decides how much of a gene to make. Every drug and every biomarker in this course acts at one of these six arrows.

12.3 Walkthrough

12.3.1 Why bother controlling genes at all - Multiple layers of regulation

WHAT IT IS	Every cell in your body carries the same DNA, and almost every cell carries the same set of genes. A liver cell and a nerve cell do completely different jobs because they read different recipes. Gene expression control is the whole set of decisions that decides which recipes get read, how much of each, and when. Your **genotype** is the recipes you inherited; your **phenotype** is what those recipes actually produced.
HOW IT WORKS / WHY IT MATTERS	Dr. Wong's framing is that the flow of genetic information can be regulated at multiple layers, and that the sum of those layers is what turns a genotype into a phenotype. The clinical consequence is immediate. Mis-regulation explains why the same mutation, such as a BRCA mutation, does not guarantee disease: two people with the same typo in the same manual can end up with completely different disease, because the other layers are different. The same logic explains why some cold tumours, which carry few neoantigens and respond poorly to immunotherapy, can be re-programmed by drugs into hot tumours. He also says this lecture sits at the centre of the course because most clinical biomarkers act through gene-expression regulation. Once you understand the mechanism, you can interpret any patient's transcriptome.
EXAM HOOK	He did not flag this as examinable in its own right. But the "genotype is not phenotype" idea is embedded in the material he did flag, and he says it plainly when he puts expression in the context of clinical genetics.

12.3.2 Two cell designs, two control systems - Eukaryotic versus prokaryotic gene regulation

WHAT IT IS	A **prokaryote** is a cell with no nucleus, such as a bacterium. A **eukaryote** is a cell with a nucleus, such as an animal, plant, fungus or yeast cell. They solve the problem of controlling gene expression in different ways.
HOW IT WORKS / WHY IT MATTERS	Because a prokaryote has no nucleus, transcription and translation happen in the same compartment. A ribosome can start reading a message while it is still being copied. A eukaryote compartmentalises: transcription happens in the nucleus, the message is edited there, it is exported through a pore, and translation happens on the rough endoplasmic reticulum and in the cytosol, before the finished protein is sorted to its destination. Those compartments are the reason eukaryotes need splicing, and the reason a eukaryotic protein can carry complex sugar decorations that a bacterial one cannot. That difference is exactly why Dr. Wong chooses different host cells for different recombinant proteins in the next topic.
EXAM HOOK	He did not flag this as examinable in his own words. He used it as the setup for the two systems he does examine.

12.3.3 Switches on the same page, and switches made elsewhere - Cis versus trans regulation

WHAT IT IS	A **cis-regulatory element** is a piece of DNA that regulates the gene on the same DNA molecule it sits on. The examples are the promoter, the enhancer, the silencer and the insulator. A **trans-acting factor** is a protein, such as a transcription factor or a co-activator, that is encoded somewhere else and floats freely through the cell until it binds a cis element.
HOW IT WORKS / WHY IT MATTERS	The picture is a factory with a control panel and a hand-held remote. Cis elements are the panels wired to one particular machine. Trans factors are the remotes, and one remote can switch on many different machines, even machines in a completely different room. Dr. Wong's example is a repressor encoded on chromosome 1 that acts on a target gene on chromosome 13. Because trans factors are diffusible, trans-regulation is the more common kind in eukaryotic systems. He also warns you not to picture the enhancer only as something upstream. A regulatory sequence can sit at the 3 prime end of the target gene just as well.
EXAM HOOK	He put this in his Tier 2 list, the concepts he says you must understand and remember. His exact words about enhancers: "Not all the enhancer are located on the upstream of the sequence. Some of the regulatory sequences can be located in the downstream sequence, so that means in the 3' ends of your target genes. This is one of the concepts that you need to understand." And about trans-regulation: he notes it is "much more common at least in eukaryotic system". The chromosome 1 repressor acting on a chromosome 13 target is one of his named numbers for the exam.

12.3.4 How long the message lasts - mRNA stability as a regulatory layer

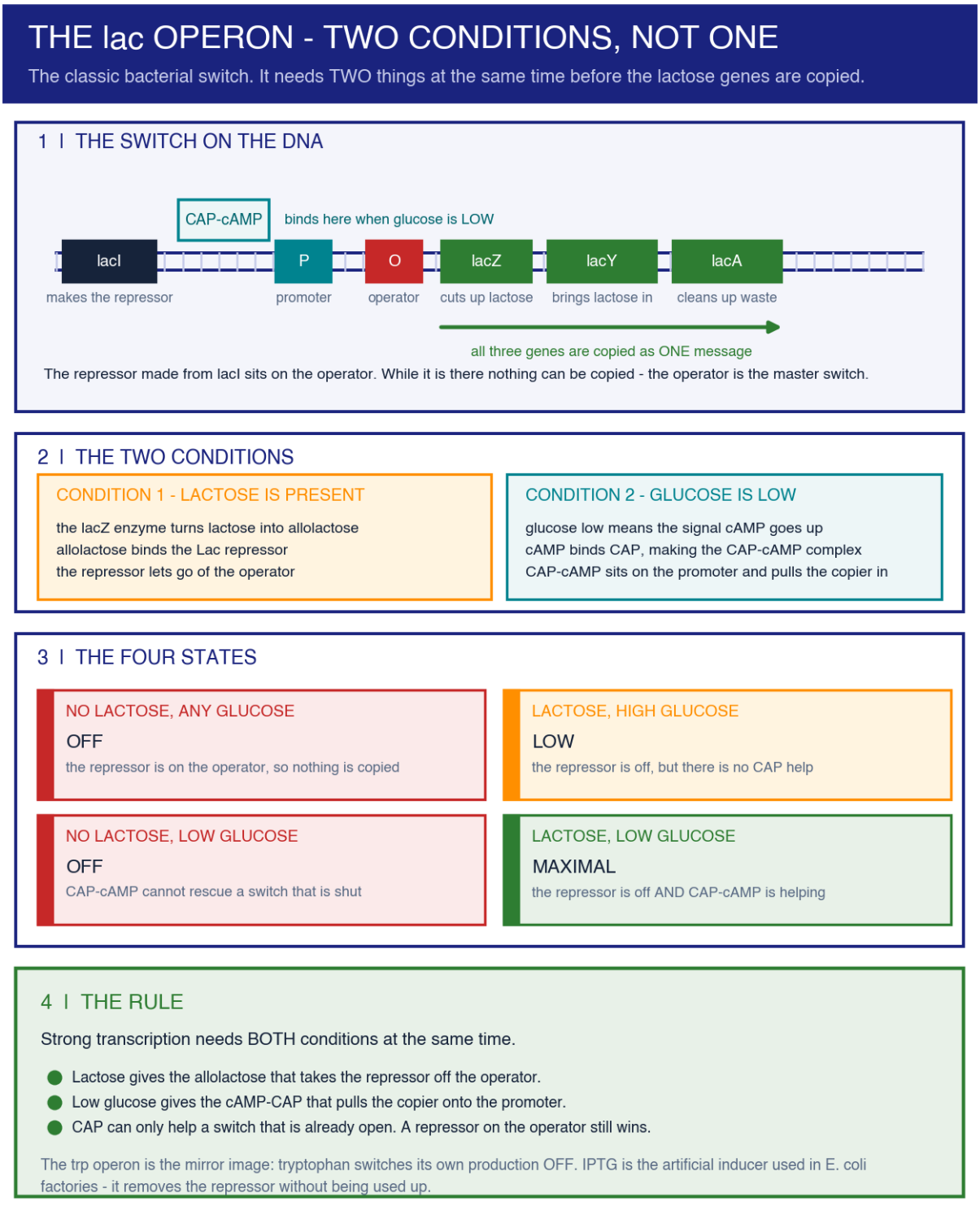
WHAT IT IS	Once the message has left the nucleus, it is not automatically read. How long it survives, called its **half-life**, is itself a control point. An mRNA that is destroyed quickly produces very little protein. An mRNA that hangs around for hours produces a lot.
HOW IT WORKS / WHY IT MATTERS	A eukaryotic mRNA carries three features that set its lifetime. There is a **5 prime cap** on the front, which protects the message and tells the ribosome to pick it up. There is a **poly-A tail** on the back, and the rule he gives is simple: a longer poly-A tail means more translation, while a shorter one means the message decays rapidly. And there is the **3 prime untranslated region**, the part of the message that is not translated into protein but carries binding sites for RNA-binding proteins and for microRNAs. Those proteins act as the cell's delivery system for turning a message off. The clinical relevance is the same as everywhere else in this course: how much of a protein a cell makes, and for how long, is often what a biomarker is measuring.
EXAM HOOK	He did not flag this as examinable. The direction of the poly-A tail rule is worth knowing because it reappears in the alternative polyadenylation material later in the lecture.

12.3.5 The classic bacterial switch - The lac operon

WHAT IT IS	The **lac operon** is the best-understood control system in biology, and it is the one Dr. Wong uses to teach the whole idea. In the bacterium E. coli, three genes for using lactose sit side by side on the DNA: lacZ, which makes the enzyme that breaks lactose down, lacY, which brings lactose into the cell, and lacA. Ahead of them sit two short DNA words: the **promoter**, where the copying machinery lands, and the **operator**, which is where a blocking protein can sit. A separate **regulatory gene** makes that blocking protein, the **Lac repressor**. All three lactose genes are copied together as one long message, called a **polycistronic mRNA**.
HOW IT WORKS / WHY IT MATTERS	Here is the rule, and the rule has two conditions, not one. First, **lactose** must be present. The enzyme made by lacZ converts lactose into **allolactose**, and it is allolactose, not lactose itself, that binds the Lac repressor and knocks it off the operator. Second, **glucose** must be low. When glucose is low, the small second messenger **cAMP** rises, and cAMP binds the protein **CAP**. The cAMP-CAP complex then sits on the promoter and pulls the copying machinery in, which turns transcription from slow to strong. Put the two together and you get his four states. With no lactose, the repressor is on the operator and nothing is copied. With lactose but no cAMP-CAP, the repressor is off but transcription is only slow basal. With cAMP-CAP present, transcription is strong. And with cAMP-CAP present but the repressor still sitting on the operator, it is still blocked, because the repressor is the master switch and CAP can only add to a system that is already open. The **trp operon** is the mirror image of this. Tryptophan is the end product, and when it is plentiful it binds the Trp repressor as a **corepressor**, which enables the repressor to sit on the operator and shut down tryptophan biosynthesis. The cell does not waste energy making a substance it already has. The five structural genes are trpE, trpD, trpC, trpB and trpA.
EXAM HOOK	This is one of the four Lecture 5 topics he put in his Tier 2 list - the tier of concepts he says you must understand and remember. The exam-focus document heads the entry "The lac operon - two conditions, not one", and it is scenario case 16 in the case-study bank, under the title "Reading the lac operon logic correctly", and the marker looks for exactly four things: the two conditions, allolactose, cAMP-CAP and IPTG. The chromosome 1 and chromosome 13 example belongs to this topic family as well. He also emphasises the word "two" - the commonest wrong answer to this question is to name only the lactose condition.

Figure F14 - The lac operon

The lac operon: promoter, operator and genes, the lactose and glucose states, allolactose and cAMP-CAP labelled, with the two-condition rule.



The lac operon and the two-condition rule. If you can redraw this from memory, you can answer the question.

12.3.6 Choosing a factory to make the protein - Recombinant protein expression in CHO cells and E. coli

WHAT IT IS	A **recombinant protein** is a protein made by a host cell that has been given a human gene. Making one in bulk is a manufacturing decision, and the choice of host cell is the first decision.
HOW IT WORKS / WHY IT MATTERS	Mammalian cells, and the **CHO cell** line in particular, are used when the protein needs to be glycosylated, that is, decorated with the right sugar groups, or when the protein is too complex for a bacterium to handle. But a human gene does not necessarily read well in a hamster cell, because different species prefer different spellings of the same amino acid. That is why the gene has to be rewritten, which is called **codon optimisation**. Bacterial systems, usually E. coli, are cheaper and faster, and they often use the lac system with **IPTG** as an artificial inducer. IPTG is a molecule that looks enough like allolactose to switch off the Lac repressor, but the cell cannot metabolise it, so you can add it and take it away again and the switch stays where you put it. His slide shows a pET-style plasmid carrying a T7 promoter, a LacO site, the open reading frame of the target gene, the lacI gene, an origin of replication and an antibiotic resistance marker. Timing matters as much as the inducer. If you induce too early, the cell is busy dividing rather than producing, and the protein may be made in the wrong amount or fold wrongly into hard, insoluble clumps called inclusion bodies. Waiting until the culture is in its exponential growth phase gives a far higher yield of correctly folded protein. Cost-effectiveness is what finally decides the platform.
EXAM HOOK	He said the codon-optimisation point out loud: "So remember that when you use the CHO cell, you need to do the codon optimization." His slide for the IPTG experiment shows integrated peak area against IPTG concentration from 0 to 1000 micromolar, with the optimum at 100 micromolar. He did not otherwise flag this topic as a named exam question.

12.3.7 How a gene is chopped up - Eukaryotic gene structure and splicing

WHAT IT IS	A eukaryotic gene is not one continuous recipe. It is a short stretch of **exons**, which survive into the mature message, separated by long stretches of **introns**, which do not. The whole lot is copied into a first draft called **pre-mRNA**, and then the introns are cut out and the exons joined together by the **spliceosome**.
HOW IT WORKS / WHY IT MATTERS	His slide uses the mouse beta-globin gene as the worked example: Exon 1, Intron 1, Exon 2, Intron 2, Exon 3, and the mature message with a 5 prime untranslated region, exons 1 to 3 and a 3 prime untranslated region, which then becomes a protein with amino acids at positions 1, 30, 104 and 146. The spliceosome is built from **snRNPs**, small nuclear RNAs with their proteins. His second slide shows the three-step mechanism: five snRNPs bind the intron, the intron loops out so that the exons are brought together, and then the intron is released and the snRNPs recycle for the next message. The reason this matters clinically is **alternative splicing**, covered in the next topic.
EXAM HOOK	He did not flag this as examinable in his own words. It is background that you need in order to understand the next three topics.

12.3.8 When the splicing goes wrong in cancer - Alternative splicing in cancer

WHAT IT IS	**Alternative splicing** is the cell joining the same exons in different combinations, so that one gene makes several different protein versions, called **isoforms**. In a cancer cell, the splicing machinery often makes different, wrong choices.
HOW IT WORKS / WHY IT MATTERS	There are three consequences, and all three are clinical. First, aberrant or cancer-specific splicing can create **neoantigens**, protein fragments the immune system has never seen, which is exactly what makes immunotherapy work in some cancers. Second, splicing out the part of a protein that a drug binds to produces **drug resistance**, because the drug no longer has anything to grip. Third, a single letter can create a new splice site. His worked example is a liver-cancer gene in which a single-nucleotide substitution creates a novel splice site and a shorter protein isoform. In patients who carry the variant, overall survival is significantly worse; and in mouse models, over-expressing the variant drives more aggressive tumours. His slides carry the detail. The agarose gel shows two band sizes, 425 base pairs for the full-length exon 5-6-7-8 form and 250 base pairs for the exon 7-skipped form, with the intronic sequence and the polymorphic base marked. The survival comparison is 23 patients without alternative splicing against 19 with it, P less than 0.001 on the log-rank test. A further slide lists six cancer isoforms whose two versions do opposite things: HIF3alpha with intron retention giving a tumour suppressor against a hypoxia regulator; MCL1 giving a short pro-apoptotic form against a long anti-apoptotic form; PKM giving the adult form against the embryonic and tumoural form; VEGF165 giving an angiogenic form against an anti-angiogenic one; the androgen receptor giving AR-V7, which is constitutively active and resists enzalutamide, against the normal receptor; and Bcl-x giving a long anti-apoptotic form against a short pro-apoptotic one.
EXAM HOOK	He did not flag this as examinable on its own, but it is the mechanism behind the hot-versus-cold tumour point he does flag, and behind the neoantigen story that runs through Lecture 4. His words: "So guys, you definitely remember that for the tumor that present many neoantigen and can be treated with the immunotherapy, we call this is a hot tumor, yeah. And not all the solid tumor is a hot tumor." His seven-type survival panel gives the combined figure as P less than 2.2 times 10 to the minus 16, and an area under the curve of 0.993 for the combination.

12.3.9 The engine room of a gene - The core promoter and the TATA box

WHAT IT IS	The **core promoter** is the stretch of DNA right at the start of a gene where the transcription machinery assembles. The **TATA box** is a six-letter DNA word inside it, sitting about 30 bases upstream of the point where the new RNA copy begins.
HOW IT WORKS / WHY IT MATTERS	The machinery is **RNA polymerase II**, the enzyme that copies protein-coding genes, plus a set of **general transcription factors** that are needed at any gene rather than at one particular gene. His slide shows the assembly order: TFIID, which contains the TATA-box-binding protein, arrives first and sits on the TATAAA sequence at minus 30; then TFIIA and TFIIB; then RNA polymerase II arrives with TFIIE, TFIIF and TFIIH, where TFIIH opens the DNA and starts the polymerase moving. The plus 1 position is the transcription start site. Enhancers and silencers can sit thousands of base pairs away, and **DNA-bending proteins** loop the DNA so that they can touch the promoter. Repression works in three ways on his slide: a repressor simply outcompeting an activator at the enhancer, or quenching it by binding the activator's DNA-binding domain, or by binding its activation domain. A separate slide lists seven ways a transcription factor can be switched on: simply being made, binding a ligand, being phosphorylated, gaining a second subunit, being unmasked from an inhibitory partner, being released to enter the nucleus, or being cut loose from a membrane by proteolytic cleavage.
EXAM HOOK	He did not flag this as examinable; neither the TATA box nor the assembly order is named. He did say that the machinery which loops a distant enhancer onto its promoter is one of the key combinatorial levers of the lecture.

12.3.10 Finding the working switches in a promoter - Promoter mapping by reporter assay

WHAT IT IS	This is the experimental method the whole field uses to find out which parts of a promoter actually do the work. A **reporter assay** measures how active a piece of DNA is by putting it in charge of a gene that gives an easy signal.
HOW IT WORKS / WHY IT MATTERS	You take a candidate promoter and put it upstream of a **reporter gene**, usually **luciferase**, the gene from fireflies that makes light. You then make a series of **serial deletions**, cutting the promoter back piece by piece, and you measure the light output each time. A promoter fragment that still drives the reporter is functional; the moment you delete the piece and the light drops, that piece was doing the work. The classic use of the method is to find out which version of a TATA-box variant is actually functional, and the method is the reason we know where promoters begin at all. The logic is worth spelling out, because it is the logic of half of this course: change one thing, measure one number, and the difference tells you what that thing was doing.
EXAM HOOK	He did not flag this as examinable in his own words. He presented it as the standard experimental logic for locating promoter elements.

12.3.11 Why a liver cell is not a nerve cell - Cell-type and developmental specificity

WHAT IT IS	Every cell in your body has the same DNA, but a T-cell and a dendritic cell run completely different programmes. They express different sets of transcription factors, and those factors switch on different target genes. That difference is called **cell-type specificity**.
HOW IT WORKS / WHY IT MATTERS	The second half of the idea is time. A gene can be on in an embryo and off in an adult, or on in one stage of development and off in the next. His slide shows a single-cell map, the t-SNE plot from Schelker and colleagues in 2017, in which single cells from peripheral blood, melanoma and ascites fluid separate into clear clusters: macrophages and monocytes, ovarian carcinoma cells, melanoma cells, natural killer cells, dendritic cells, cancer-associated fibroblasts, T-cells, endothelial cells and B-cells. A second slide plots expression across developmental time points, from a one-month embryo to a thirty-month adult, for three named transcripts and their average. How a cell knows which programme to run is the job of **signal transduction**: an external signal starts a phosphorylation cascade or a proteolytic cleavage, which moves a transcription factor into the nucleus, which changes which target genes are read. The clinical payoff is enormous. It is why single-cell sequencing can tell you which cells are in a tumour, and why the same drug can work on one cell type and fail on another.
EXAM HOOK	This is one of the points he says he always emphasises. His words: "the expression is usually cell-specific and at the same time, it is time-specific... This is the concept that I always emphasize." He puts it in his Tier 2 list, the concepts you must understand and remember.

12.3.12 Where the protein ends up - Signal peptide and localisation, revisited

WHAT IT IS	A **signal peptide** is a short stretch of amino acids at the very start of a protein that acts as a postal address. It tells the cell which post office to use.
HOW IT WORKS / WHY IT MATTERS	A protein carrying an N-terminal signal peptide is taken into the endoplasmic reticulum, then to the Golgi apparatus, and from there it can be sent to the plasma membrane, to a lysosome, or straight out of the cell as a secreted protein. Removing the signal peptide blocks expression of secreted and membrane proteins entirely; the protein simply never arrives anywhere. Dr. Wong tells this as a story rather than a fact. He had a postdoc who lost four months of work because the signal peptide had been left off the construct. It is a reminder that in this field the detail that looks trivial is often the detail that costs you four months.
EXAM HOOK	He put it in his Tier 2 list. His words: "So remember, signal peptide is really, really important for your study, yeah." He also warns you not to confuse it with the N-terminal domain of a nuclear receptor, which is a completely different thing, and that confusion is one of his named distractor pairs.

12.3.13 Three boxes in one protein - Nuclear-receptor architecture and its functional domains

WHAT IT IS	A **nuclear receptor** is a protein that sits inside the cell, waits for a hormone or a drug to arrive, and then goes into the nucleus and switches genes on or off. Its architecture is always the same three-part plan.
HOW IT WORKS / WHY IT MATTERS	The **N-terminal domain**, the NTD, is the variable and regulatory end. It is the part that differs between receptors of the same family, and it carries the first activation function, AF-1. The **DNA-binding domain**, the DBD, is the highly conserved middle. It contains **zinc fingers**, small folded shapes held together by a zinc ion, and it recognises the receptor's own DNA word, called a response element. The **ligand-binding domain**, the LBD, is the end that catches the hormone or the drug. When the ligand lands, the LBD changes shape, and that change is what lets the receptor recruit co-activators or co-repressors, which is how the receptor decides whether the gene will be switched on or held down. His domain map shows a flexible hinge between the DBD and the LBD, and a variable C-terminal region. The activation route also differs between receptor types. For a Type I receptor such as the estrogen receptor, the receptor sits in the cytoplasm held by a chaperone called HSP90 until estrogen arrives; the chaperone lets go, the receptor pairs with itself, the pair moves into the nucleus, co-activators are recruited, and the pair binds the estrogen response element. For a Type II receptor such as the thyroid hormone receptor, the receptor is already sitting on its DNA word with RXR and with co-repressors, and the arrival of the ligand simply swaps the co-repressors for co-activators.
EXAM HOOK	He put the three domains in his Tier 2 list, and the DBD-versus-LBD distinction is one of his named distractor pairs. The fact you must be able to state is the NTD versus LBD distinction: the NTD is variable and regulatory, the LBD is where the hormone or the drug binds and where the shape change happens.

12.3.14 Two are better than one - Dimerisation as a regulatory switch

WHAT IT IS	Many transcription factors work as pairs. A **homodimer** is two of the same protein joined up. A **heterodimer** is two different ones.
HOW IT WORKS / WHY IT MATTERS	Monomers usually bind DNA poorly. Dimerisation dramatically increases the DNA-binding affinity, because two DNA-binding domains can grip the DNA word together and each can help the other fold correctly. His worked example is **HER2**, the receptor whose gene is amplified in some breast and gastric cancers. HER2 cannot homodimerise efficiently, so it pairs with HER1, HER3 or HER4 instead. That pairing is what drives the growth signal, and that is exactly what **Herceptin**, the drug **trastuzumab**, blocks. So a drug that does nothing at all to the receptor's activity still works, because it stops the receptor from pairing up. It is one of the cleanest examples in the course of a drug blocking a physical interaction rather than a chemical reaction.
EXAM HOOK	He put dimerisation in his Tier 2 list, and the dimerisation-versus-monomer pair is one of his named distractors. The discriminating fact to remember is short: monomers bind DNA poorly, dimers bind strongly.

12.3.15 The same drug, two different tissues - Tissue-selective pharmacology and Tamoxifen

WHAT IT IS	**Tamoxifen** is a **Selective Estrogen Receptor Modulator**, a SERM. It acts on the estrogen receptor, but it does not always produce the same effect.
HOW IT WORKS / WHY IT MATTERS	In breast tissue, tamoxifen occupies the receptor and blocks estrogen from binding, so the co-activator cannot be recruited and the cell does not proliferate. It is an **antagonist** there, and that is why it is an anti-tumour drug. In the endometrium, the tamoxifen-receptor complex does recruit a co-activator, so the cell proliferates. It is a **partial agonist** there, and that is why long-term tamoxifen carries a small increased risk of endometrial cancer. His slide shows three panels: breast with estrogen driving proliferation, breast with tamoxifen blocking it, and endometrium with tamoxifen driving proliferation. The lesson is that the receptor is the same protein and the drug is the same molecule, but the tissue supplies different co-activators and co-repressors, so the outcome differs. Context is not a detail; context is the mechanism.
EXAM HOOK	He said this out loud: "This is the concept that you have to bear in mind." The exam hook is the tissue pairing itself. Tamoxifen is an antagonist in breast and a partial agonist in endometrium, with a small increased risk of endometrial cancer.

12.3.16 How much drug is enough - Receptor occupancy as a clinical parameter

WHAT IT IS	**Receptor occupancy**, the RO, is the fraction of receptors that have to be bound by the drug before there is a clinical effect.
HOW IT WORKS / WHY IT MATTERS	A drug does not have to fill every receptor to work. If a signal is amplified downstream, a small fraction of occupied receptors is enough to produce the full effect. The number therefore becomes a dosing number, and it is tissue-specific, because the number of receptors and the strength of the downstream signal differ from tissue to tissue. Dr. Wong's numbers are the ones you must know. For a typical antibody, the threshold is 1 %. For Herceptin in breast cancer it is 2 %. For Herceptin in gastric cancer it is 10 %. In plain English: in breast cancer the drug only has to occupy two receptors in every hundred, whereas in gastric cancer it has to occupy ten in every hundred. Same drug, same target, different dosing strategy, because the required occupancy is different in the two tissues.
EXAM HOOK	He put the receptor occupancy thresholds in his Tier 2 list and in his list of precise numbers: 1 % for a typical antibody, 2 % for Herceptin in breast cancer and 10 % for Herceptin in gastric cancer. The discriminating idea is that RO is tissue-specific, which is why the same antibody is dosed differently in the two diseases.

12.3.17 The inflammation clock - NF-kappaB and temporal cytokine cascades

WHAT IT IS	**NF-kappaB** is a transcription factor that sits in the cytoplasm doing nothing until danger arrives. It is held there by **I-kappa-B**, or IkappaB, which wraps around it and hides its DNA-binding part.
HOW IT WORKS / WHY IT MATTERS	The signal for bacterial infection is **LPS**, the endotoxin of Gram-negative bacteria, which is detected by the cell-surface receptor TLR4. LPS triggers the IkappaB kinases, which add a phosphate to IkappaB-alpha inside the p50, p65, IkappaB-alpha complex. The phosphorylated IkappaB is then destroyed, which releases NF-kappaB, which moves into the nucleus, binds the promoters of inflammatory genes, and starts making inflammatory proteins. The part Dr. Wong emphasises is the timing. The response comes in a strict order. **TNF-alpha comes first, peaking at 4 to 6 hours.** Then comes IL-1beta, then IFN-gamma, then IL-2 and IL-15. And the loop closes on itself: the cytokines that NF-kappaB causes to be secreted, in particular TNF-alpha and IL-1beta, feed back and push NF-kappaB harder. That **positive-feedback loop** is why the response is so fast, and it is also why it has to be shut down carefully. Later, anti-inflammatory mediators resolve the inflammation. This is the same cascade that Lecture 1 described from the other end, when sepsis was shown to be a runaway of the body's own inflammation.
EXAM HOOK	He gave the numbers exactly: TNF-alpha peaks at 4 to 6 hours, followed by IL-1beta, then IFN-gamma, then IL-2 and IL-15. And he then did something unusual, which you must notice. He put the exact cytokine sequence on his not-examined list, with the words "don't worry about all these cytokines. I'm just want to show you that different responses can be observed in different times." So the sequence is not to be memorised as a list. What you must be able to explain is the shape: a defined time course, and a positive feedback loop that makes the response rapid.

12.3.18 How a cell feels the oxygen - Environmental sensing and HIF-1alpha

WHAT IT IS	**HIF-1alpha** is the cell's oxygen sensor. It is made all the time, and in normal oxygen it is destroyed almost as fast as it is made. When oxygen runs low, it survives.
HOW IT WORKS / WHY IT MATTERS	The destruction machinery is worth understanding, because it is what makes the switch. In normal oxygen, the enzymes PHD and FIH add an oxygen group to HIF-1alpha. That mark is recognised by the protein VHL, which tags HIF-1alpha with ubiquitin and hands it to the proteasome for destruction. In **hypoxia**, that marking step cannot happen, so HIF-1alpha survives, moves into the nucleus and pairs with HIF-1beta. It then recruits p300 and CBP, binds the **hypoxia-responsive element**, a specific DNA word, and switches on its target genes: **VEGF**, which grows new blood vessels, **GLUT1**, the glucose transporter that lets a cell keep working without oxygen, and **EPO**, the red-blood-cell growth factor. The result in a solid tumour is angiogenesis, invasion, metastasis, cancer stemness and metabolic reprogramming. Dr. Wong then shows how synthetic biologists have taken this switch and used it as a delivery system. Put the expression of a CAR under the control of a hypoxia-responsive element, and the T-cell only arms itself in the oxygen-starved middle of a solid tumour, sparing the healthy tissue around it. His slide notes that about half of the solid tumours studied contain hypoxic tissue, and that when the hypoxia is removed the HIF-controlled CAR falls to less than 0.2 normalised signal within two to three hours, while the always-on CAR stays between 0.8 and 1.0.
EXAM HOOK	He did not flag this as examinable in his own words. It is scenario case 14 in the case-study bank, "A compound that switches itself off in the right cells", where the marker looks for the HRE-controlled CAR, hypoxia, and a single gate.

12.3.19 When cells touch, growth stops - Hippo and YAP revisited

WHAT IT IS	The **Hippo pathway** is the mechanism by which a cell that touches its neighbours stops dividing. **YAP** is the protein at the end of it, and the switch is a phosphate group.
HOW IT WORKS / WHY IT MATTERS	When the Hippo pathway is on, the kinase MST1 activates LATS1, and LATS1 phosphorylates YAP. The phosphorylated YAP is caught in the cytoplasm by the protein 14-3-3 and then degraded, so YAP never reaches the nucleus and growth stays off. When the pathway is off, unphosphorylated YAP enters the nucleus, binds the protein TEAD on the DNA, and drives the transcription of growth genes. His slide fills in the upstream detail. At the cell junction, the proteins KIBRA, MER and WILLIN recruit and trigger MST phosphorylation; SAV then carries the signal from MST to LATS. The clinical consequence is that losing the pathway is dangerous. Loss-of-function mutations or knockouts leave YAP in the nucleus, and the result is tissue overgrowth and spontaneous tumours. His slide shows Drosophila tissue with massive folded overgrowth, mouse livers from animals lacking both MST1 and MST2, and mouse livers with discrete tumour nodules developing into an advanced multi-nodular mass covering the whole organ. He notes that druggable inhibitors are in development.
EXAM HOOK	He did not flag this as examinable in his own words. He did present it as the sharpest example in the course of what happens when a control system fails, and he returns to the pathway in Lecture 1 as the cleanest illustration of a reversible phosphorylation switch.

12.3.20 One cell telling another - Notch and synNotch

WHAT IT IS	**Notch** is a receptor that lets one cell change the behaviour of its neighbour. A **synNotch** is an engineered version of it.
HOW IT WORKS / WHY IT MATTERS	In normal Notch signalling, a **delta ligand** on the neighbouring cell touches the Notch receptor, whose extracellular part is a stack of EGF repeats. That triggers **proteolytic cleavage**, which cuts loose the intracellular part of Notch, the NICD, and the NICD moves into the nucleus and acts as a transcription factor. The whole system is direct cell-to-cell contact, not a secreted hormone. Synthetic biologists have rebuilt this so that it can be used as a safety switch. In a synNotch CAR, binding to cancer antigen 1 causes the cleavage, and the intracellular transcription factor then drives the conditional expression of the CAR, which goes on to target cancer antigen 2. The CAR only fires when both antigens are present, which is an **AND-gate**, and that is the point: it raises specificity enormously, because a normal cell expressing one of the two antigens on its own cannot trigger the kill.
EXAM HOOK	He did not flag this as examinable in his own words. The principle to carry away is the AND-gate: two inputs are required, so the false-alarm rate falls.

12.3.21 Switching genes off without changing the letters - Epigenetic regulation

WHAT IT IS	Epigenetic regulation changes which genes are read without changing a single letter of the DNA. It is the difference between the same manual and a different reading of it.
HOW IT WORKS / WHY IT MATTERS	Two mechanisms carry the weight. First, **DNA methylation** of the CpG islands in a promoter silences the gene next to it, and when the gene is a tumour suppressor, silencing it removes the brake. In cancers such as colorectal cancer this is associated with poor survival. Second, **histone modifications** open or close the chromatin around a gene: acetylation loosens the nucleosome packing so that DNA becomes accessible, while deacetylation tightens it. His slide also shows the cross-talk, which is the important part: hypomethylation of the promoter of HDAC, the enzyme that removes acetyl groups, raises HDAC, which lowers histone acetylation, which lowers transcription. The two systems are not separate; they reinforce each other. Lecture 3 covered the same chemistry from the methylation-biomarker side, including the drugs that block histone deacetylases and the methylation-based tests used to choose treatment.
EXAM HOOK	He did not flag this as examinable in Lecture 5, because it belongs to Lecture 3. In his Tier 2 list he treats DNA methylation as a concept you must understand: "The most important aspect of the DNA methylation is it has a very significant functions on these gene expressions, because DNA methylation is usually involved in the suppression of the gene expressions."

12.3.22 RNA that makes no protein - Long non-coding RNAs

WHAT IT IS	A **long non-coding RNA**, an lncRNA, is an RNA longer than 200 nucleotides that is not translated into a protein. It is a message that is never used as a message.
HOW IT WORKS / WHY IT MATTERS	Instead, lncRNAs work as a **scaffold** and a **guide**. A scaffold holds two or more proteins together at the right spot. A guide carries a protein to the DNA word it should act on. His slide lists four archetypes. **APOLO** forms an RNA-DNA hybrid to regulate a gene. **PNCTR** uses its YYUCUY motifs to bind PTBP1 and control splicing. **XIST**, the classic, carries repeat elements that bind SPEN and PRC2 to silence one whole X chromosome. **TINCR** base-pairs with an mRNA and binds Staufen 1 to control how stable that message is. His knockout slide shows how many different jobs these molecules do: Xist, Airn, Kcnq1ot1, H19 and Meg3 for dosage compensation and growth; Pnky and linc-Brn1b for brain development; Tug1 for infertility; Maenli for limb development; Norad for genome stability and ageing; lincRNA-Cox2, lincRNA-Eps, Gaplinc, lnc-Epav, lnc-Lsm3b and Firre for the immune system; and Fendrr, Peril, Mdgt, Upperhand, Handsdown, Charme and Chaserr for plain viability. The clinical interest is that they are emerging drug-discovery targets and biomarkers.
EXAM HOOK	He did not flag this as examinable. The number worth keeping is the definition: longer than 200 nucleotides, and no protein made.

12.3.23 Features written on the message itself - Post-transcriptional control, mRNA features and microRNAs

WHAT IT IS	After the message has been copied and edited, there is still more regulation to be done, and it happens on the message itself. The three features that carry it are the **5 prime cap**, the methylated guanosine at the front; the **poly-A tail** at the back; and the **3 prime untranslated region**, the untranslated stretch in between.
HOW IT WORKS / WHY IT MATTERS	How long the message lives, how well it is translated, how far it travels inside the cell and where the protein ends up are all decided by what is bound to those three features. Choosing where to put the poly-A tail is itself a regulatory step, and it is called **alternative polyadenylation**. His slide contrasts a proximal polyadenylation site, which gives a short 3 prime untranslated region, with a distal site, which gives a long one that can bind RNA-binding proteins, microRNAs and lncRNAs, and thereby controls stability, translation, nuclear export, where the message goes and where the protein goes. The BDNF example is his illustration: the short isoform stays near the cell body, while the long isoform is trafficked into the dendrites, so that the protein is made where it is needed. Then there are the **microRNAs**. A microRNA is about 22 nucleotides long. It finds its target by base-pairing with a partially complementary **seed sequence**, positions 2 to 8, in the 3 prime untranslated region of a message. The pairing either destroys the message or stops it being translated. The biogenesis is worth knowing as a chain: RNA polymerase II makes a primary microRNA transcript, the microprocessor complex, made of DGCR8 and DROSHA, cuts it into a hairpin, the hairpin is exported by exportin-5 with RanGTP, DICER1 with TRBP cuts it into a duplex, and the duplex is loaded onto AGO1 to AGO4 to form the silencing complex with GW182 in the P-body, where the target message decays. The clinical payoff is that microRNAs are unusually stable in serum, which is why they are popular non-invasive biomarkers, the same argument that made the serum microRNA classifier in Lecture 1 work.
EXAM HOOK	He did not flag this as examinable in his own words, but the numbers on his slides are worth knowing as background. His liver-cancer slide analysed 220 microRNAs in 97 paired tumour and adjacent non-tumour tissues, and the 14q32.2 cluster was over-expressed in tumours, correlating with log10 of AFP at P equal to 4.23 times 10 to the minus 10 and with survival at P equal to 0.009 and Chi squared equal to 6.81 over zero to 100 months. The tools he names for target prediction are TargetScan and miRTargetLink 2.0.

12.3.24 Using the same gene twice - Alternative splicing revisited, and mRNA therapeutics

WHAT IT IS	This topic brings three threads together: the fact that one gene can make several proteins, the fact that a good biomarker can be a soluble version of a membrane protein, and the fact that you can now treat a patient by giving them the message itself.
HOW IT WORKS / WHY IT MATTERS	First, the biomarker trick. A single gene can be spliced into a secreted, soluble isoform that has no transmembrane domain. That isoform can be measured in blood, which turns a membrane protein into a practical test. Second, the immuno-targeting thread. Aberrant splicing produces neoantigens, and neoantigens are immuno-targetable. And single-nucleotide mutations can create new splice sites, which creates entirely new isoforms. Third, the therapeutic thread. An **mRNA vaccine** delivers a transcript encoding an antigen. The patient's own cells take it up, their own ribosomes translate it, the antigen is presented on **MHC**, and T-cells and B-cells are activated. His slide follows the whole route: intramuscular injection, mRNA loaded into lipid nanoparticles entering what he calls the vaccinated cell, the ribosome translating the protein, the proteasome cleaving it into peptides, the endoplasmic reticulum and endosome pathway, MHC presentation, recognition by the T-cell receptor, and then the immunostimulatory signals that activate T- and B-cells.
EXAM HOOK	He did not flag this as examinable in his own words. The principle worth carrying is that mRNA therapy uses the patient's own copying machinery, so the only thing you have to get right is the sequence you deliver.

12.4 The ten things to remember

1. Regulation happens at several layers at once, and the sum of those layers is what turns a genotype into a phenotype. The same BRCA mutation does not guarantee disease, and a drug can turn a cold tumour into a hot one.
2. A cis element regulates the gene on the same DNA molecule it sits on. A trans factor is a diffusible protein that can act on a different chromosome altogether. Dr. Wong's example: a repressor on chromosome 1 acting on a target gene on chromosome 13.
3. The lac operon needs two conditions, not one: lactose present, so that allolactose removes the Lac repressor from the operator, and glucose low, so that cAMP-CAP recruits the copying machinery onto the promoter. CAP can only add to a system the repressor has already opened.
4. The trp operon is the mirror image. The end product, tryptophan, acts as a corepressor and shuts down its own synthesis.
5. Eukaryotic genes are chopped up. Exons survive, introns are cut out by the spliceosome, and alternative splicing lets one gene make several different proteins. That is why your liver cell and your nerve cell, with the same DNA, behave differently.
6. Nuclear receptors have three working parts: the variable N-terminal domain for regulation, the highly conserved DNA-binding domain that recognises the receptor's own DNA word, and the ligand-binding domain, where the hormone or drug lands and changes the protein's shape so it can recruit co-activators or co-repressors.
7. Dimerisation is a switch in its own right. Monomers bind DNA poorly; dimers bind strongly. HER2 cannot pair with itself, so it pairs with HER1, HER3 or HER4, and Herceptin blocks that pairing.
8. The same drug can do opposite things in different tissues. Tamoxifen is an antagonist in breast tissue and a partial agonist in the endometrium, which is why it carries a small increased risk of endometrial cancer. "This is the concept that you have to bear in mind."
9. The numbers you must be exact about: receptor occupancy is 1 % for a typical antibody, 2 % for Herceptin in breast cancer and 10 % for Herceptin in gastric cancer. TNF-alpha peaks at 4 to 6 hours, then IL-1beta, then IFN-gamma, then IL-2 and IL-15.
10. Control systems can fail. Lose the Hippo pathway and YAP stays in the nucleus, and the tissue grows out of control. That is the same lesson as a driver mutation, seen from the regulatory side.

12.5 Check yourself

Q1. A culture of E. coli is growing on lactose but not on glucose. The lac genes are being transcribed, but only slowly. Another culture has glucose and no lactose, and the lac genes are not being transcribed at all. Explain both observations with the two-condition rule, and say what the second culture would do if you added lactose.

A1. Start with the observation. In the first culture, lactose is present but glucose is also present, so transcription is happening at only the basal level. In the second culture, neither condition is met. The mechanism is the two-condition rule. Lactose is converted by the lacZ enzyme into allolactose, and allolactose is what binds the Lac repressor and knocks it off the operator. So if lactose is absent, the repressor stays on the operator and no transcription happens at all, whatever the glucose level. Glucose is the second, independent condition. When glucose is low, cAMP rises, cAMP binds CAP, and the cAMP-CAP complex sits on the promoter and recruits the copying machinery, which turns slow basal transcription into strong transcription. The first culture has condition one but not condition two, so it is slow. The second culture has neither, so it is off. If you now add lactose to the second culture, allolactose removes the repressor and the genes turn on at the slow basal level, because glucose is still high and so there is still no cAMP-CAP. The limit on this answer is that CAP can only add to a system the repressor has already opened: even with cAMP-CAP present, the repressor on the operator means the genes stay blocked.

Q2. A tumour is removed from a patient and the pathologist reports that the tumour-suppressor gene is present in the DNA, at normal copy number, but the gene is not making any RNA. The patient also has a normal number of

the protein in surrounding normal tissue. Explain what has happened and what kind of drug might help.

A2. The observation is that the gene is there and is switched off, only in the tumour, and that the neighbouring normal tissue is making the protein normally. The mechanism is epigenetic silencing: the promoter CpG island of the gene has been methylated, which is what Lecture 3 described, and the chromatin around it is closed, so the transcription machinery cannot reach the promoter. No letter of the DNA has changed, which is why the copy number is normal and why the same gene works in normal tissue. The clinical meaning is that this is a targetable state rather than a missing gene. The action is to use a drug that opens the chromatin back up, such as an inhibitor of histone deacetylase, which stops the removal of the acetyl groups and lets the chromatin loosen. The limit is that opening chromatin globally is not specific, so these drugs are less selective than a targeted therapy. This is also the mechanism behind the cross-talk on his slide, where the methylation state of the HDAC promoter itself pushes the whole region towards silencing.

Q3. A colleague gives you a plasmid for bacterial expression with a T7 promoter, a LacO site, the target gene, and the lacI gene, and asks you to add IPTG at the start of the culture. You think that is a mistake. Explain why, and give the two things that go wrong.

A3. Take the observation first: the construct is a standard pET-style inducible system, and the colleague proposes to induce at the moment of inoculation. The mechanism is that IPTG removes the Lac repressor from the LacO site so that the T7 machinery can read the target gene, but the timing matters as much as the inducer. At the start of the culture the bacteria are adapting, not yet dividing, and the machinery for high-volume protein production is not yet at full strength. So the protein is made too early and too slowly. The first thing that goes wrong is **toxic pre-expression**: the protein is produced before the cells are ready, which can damage or kill the culture. The second is **misfolding**: a protein made incompletely or too fast folds badly and clumps into an **inclusion body**, which is unrecoverable. The clinical and manufacturing meaning is yield and purity, because an inclusion body is not a product. The action is to wait until the culture is in its exponential growth phase and then add IPTG. His slide also shows that the dose has an optimum, at 100 micromolar, over a tested range of 0 to 1000 micromolar. The limit is that this is a bacterial system: if the protein needs mammalian sugar decorations, the host cell has to be a CHO cell instead, and then the gene needs codon optimisation.

Q4. A patient with HER2-positive breast cancer is treated with trastuzumab. A patient with HER2-positive gastric cancer is treated with the same drug but at a different schedule. Using receptor occupancy and dimerisation, explain why the same drug behaves differently, and what tamoxifen teaches you about the same idea.

A4. The observation is the same target, the same drug, and two different diseases. The mechanism has two halves. The first is dimerisation. HER2 cannot pair efficiently with itself; it pairs with HER1, HER3 or HER4, and that pairing is what drives the growth signal. Trastuzumab blocks the pairing, which is why it works even though it is not an enzyme inhibitor. The second half is receptor occupancy, the fraction of receptors that must be filled by drug before there is a clinical effect. A typical antibody needs 1 % occupancy. Trastuzumab needs 2 % occupancy in breast cancer but 10 % in gastric cancer, so the two diseases need different dosing strategies from the same molecule. The clinical meaning is that occupancy is a tissue-specific number, not a property of the drug, so a drug's schedule has to be worked out per disease. The second half of the question is the general lesson, and tamoxifen is the sharpest example. The same molecule occupies the same receptor in breast tissue and in the endometrium, and produces opposite outcomes, because the two tissues supply different co-activators and co-repressors. Tamoxifen is an antagonist in breast, and therefore an anti-tumour drug there, and a partial agonist in the endometrium, and therefore carries a small increased risk of endometrial cancer. The limit is that neither of these numbers tells you about toxicity, because the therapeutic window, the gap between the effective dose and the toxic dose, is a separate question.

Q5. A pathologist reports that a liver tumour carries a single-nucleotide substitution that creates a new splice site, and that patients with it survive significantly less well than patients without it. Give the mechanism, the immune consequence, and the drug consequence.

A5. The observation is that a single letter has created a new splice site, and that patients with the variant do worse. The mechanism is that the new splice site makes the splicing machinery build a shorter isoform, one that excludes an exon. His slide shows the two band sizes, 425 base pairs for the full-length form and 250 base pairs for the exon 7-skipped form, and the survival comparison of 23 patients without alternative splicing against 19 with it at P less than 0.001. The immune consequence is that a protein the body has never made becomes a

neoantigen, a novel fragment the immune system has not seen. That is what makes a tumour "hot" in his terms: "for the tumor that present many neoantigen and can be treated with the immunotherapy, we call this is a hot tumor, yeah. And not all the solid tumor is a hot tumor." The drug consequence has two halves. The first is immunotherapy: a neoantigen is something the immune system can be pointed at, which is the whole logic of checkpoint blockade in neoantigen-rich disease. The second is the opposite of benefit: because splicing can delete the very part of a protein that a drug binds to, an aberrant isoform can cause

drug resistance, and that resistance is then inherited by every daughter cell. The limit is that these are associations and mechanisms, not a treatment rule. You cannot treat a patient on the strength of a splice variant alone; you need the drug, the tumour type and the evidence that the particular isoform in this particular tumour carries the variant.

PART 7

The Exam Pack

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CHAPTER 13

The Exam Pack

This chapter is the end of the book and the beginning of your revision. It gathers everything that matters for the midterm in one place, and it separates three different things very carefully: what Dr. Wong said he would test, what he said he would not test, and the background that explains the science behind both. Read section 13.4 as well as section 13.2. Knowing what to skip saves you hours.

13.1 Master glossary

This is the book's only glossary. It merges the translation table that used to sit in Part 1 with every technical term introduced in the five lecture chapters, so it is complete and self-sufficient. Read it in two ways. As a dictionary, when a term stops you in the middle of a chapter. As a revision list, by covering the middle column and saying the meaning out loud before you look. The third column tells you where the term does its work: a chapter of this guide, or one of Dr. Wong's five lectures. A term marked **Background** does no work in the lectures at all - it is here for context only.

The rows are alphabetical, ignoring capital letters, brackets and punctuation, so ADP sits next to adenosine and RNA sits next to ribosome. Where a term has more than one common spelling, the usual one is used and the alternative is given in brackets. Terms that appear in more than one place are listed once, with every location where they do real work.

> *This section is a one-line quick-reference table. The **detailed explanation** of each term - what it is, how it works, why it is examined - is in 《GNBF5020 術語詳解》 **/ GNBF5020 Terminology Deep Dive**, the file GNBF5020_Terminology_Deep_Dive.pdf, a separate volume that ships alongside this guide as its own PDF. The Traditional Chinese edition of the same book is* GNBF5020_術語詳解_繁體中文.pdf.

Term (technical, English)	Plain-English meaning	Where it appears
14-3-3	A family of scaffold proteins, each with a pocket that recognises a phosphate group. Phosphorylated YAP is held by it and the complex is then degraded. It is a general-purpose "phosphorylated-protein holding pen" that many other pathways use as well.	Lect 1, Lect 5
3 prime untranslated region (3 prime UTR)	The stretch of an mRNA after the stop codon that is not translated into protein but carries binding sites for proteins and microRNAs.	Lect 5
4-1BB	One of the two co-stimulatory signals built into a CAR, alongside CD28.	Lect 2
Acetylation	Adding an acetyl group to a histone, which loosens the chromatin and lets the gene be read.	Ch 3, Lect 3, Lect 5
Actin	One of the two proteins of the contractile ring that pinches a cell in two. Also the track that cargo moves along.	Ch 5, Ch 6
Activation energy	The energy hill every reaction has to climb before it can happen. An enzyme lowers the hill.	Ch 3
Activator	A protein that increases transcription by binding an enhancer.	Lect 5
Active site	The pocket on an enzyme where the substrate fits and the chemistry happens.	Ch 3
Adenosine	The sugar-and-ring part of ATP, without the phosphate tails.	Ch 6
Adoptive cell therapy	Taking cells out of the patient, treating them, and putting them back to fight the tumour.	Lect 2, Lect 3
ADP (adenosine diphosphate)	ATP after it has spent one phosphate. Getting the phosphate back is how the cell recharges it.	Ch 6
Aerobic glycolysis	Breaking sugar down to lactate even when oxygen is available. The cancer cell's preferred route.	Ch 6, Lect 1

Term (technical, English)	Plain-English meaning	Where it appears
AF-1 and AF-2	The two activation functions of a nuclear receptor, which are the parts that recruit co-activators.	Lect 5
AGO	The protein of the silencing complex that the microRNA sits in.	Lect 5
Alkylation	The chemical damage in which a small group is stuck onto a DNA base, which can lead to a mispair.	Lect 3
Allele	One version of a gene. A person carries two, one from each parent.	Ch 4, Lect 3
Allogeneic	Taken from a donor rather than from the patient. An off-the-shelf cell therapy.	Lect 2
Allolactose	The isomer of lactose made by the lacZ-encoded beta-galactosidase. It is allolactose, not lactose itself, that knocks the Lac repressor off the operator.	Lect 5
Allostery	A molecule changes shape and so changes its activity because something else binds at a different site from the main working one.	Ch 3
Alpha helix (alpha-helix)	A corkscrew shape that a protein chain often takes, held in place by hydrogen bonds between neighbouring turns.	Ch 3
Alternative polyadenylation	Choosing a different place to put the poly-A tail, which changes the length of the 3' untranslated region.	Lect 5
Alternative splicing	Joining exons in different combinations, so that one gene makes several different proteins.	Lect 5
Amino acid	One of the twenty small molecules that are joined together to make a protein. Each has a different side chain that decides how it folds.	Ch 3, Ch 6
AMP (adenosine monophosphate)	ATP after it has spent two phosphates.	Ch 6
Anabolism	The building-up half of metabolism. It spends energy to make the cell's own large molecules.	Ch 6
Anaphase	The stage of mitosis in which the two copies of each chromosome are pulled apart and travel to opposite ends.	Ch 5, Lect 3
AND-gate	A switch that needs two inputs at once before it fires. It is how synNotch CARs avoid false alarms.	Lect 5
Anergy	The switched-off state of a T-cell after a checkpoint such as PD-1 has fired.	Lect 2
Aneuploidy	A cell with the wrong number of chromosomes, because a chromosome failed to separate properly.	Ch 5, Lect 3
Angstrom	The unit used for atomic distances. 2.8 angstrom is a good crystal structure; below 2.0 is excellent.	Lect 3
Antibody	A Y-shaped protein made by B-cells that sticks to one specific antigen and either blocks it or marks it for destruction.	Ch 3, Lect 1, Lect 2
Antibody-dependent cellular cytotoxicity (ADCC)	When an antibody's tail grabs a killer cell, such as an NK cell, and tells it to kill the cell the antibody is stuck to.	Lect 1, Lect 2
Antibody-drug conjugate (ADC)	An antibody used as a delivery vehicle for a toxic drug, so the poison goes only where the antibody binds.	Lect 1, Lect 4
Antigen	Anything the immune system recognises as foreign and tries to attack.	Lect 1, Lect 2, Ch 3
Antigen-presenting cell	A cell, such as a dendritic cell, that displays protein fragments on its surface so that T-cells can inspect them.	Lect 2, Lect 4
Anti-inflammatory mediator	A molecule that switches the inflammation off once the threat has passed.	Lect 5
Antiparallel	The arrangement of the two DNA strands, which run in opposite directions, one 5' to 3' and the other 3' to 5'.	Ch 4

Term (technical, English)	Plain-English meaning	Where it appears
Anti-PD-1 antibody	A drug antibody that blocks the PD-1 checkpoint and puts the T-cell back to work. It is the immunotherapy for hot tumours.	Ch 6, Lect 2
Anti-PD-L1 antibody	A drug antibody that blocks PD-L1 on the tumour cell, with the same effect of releasing the T-cell.	Lect 2
Anti-TERT	An antisense oligonucleotide (ASO) that knocks down TERT messenger RNA, so the cell cannot keep rebuilding its chromosome ends.	Lect 4
APOLO	A long non-coding RNA that forms an RNA-DNA hybrid in order to regulate a gene.	Lect 5
Apoptosis	Programmed cell death. The cell dismantles itself neatly instead of bursting.	Ch 6, Lect 2
Apoptosome	The wheel-shaped structure that forms when cytochrome c is released and sets caspase 9 going.	Lect 2
Apo-repressor	A repressor protein with no ligand attached, which therefore cannot bind the operator.	Lect 5
AR-V7	The shortened androgen receptor made by aberrant splicing. It is constitutively active and resists enzalutamide.	Lect 5
Arylamination	The step in which an activated tobacco precursor (such as an oxidation intermediate of NNK) adds an aromatic amine group onto a DNA base, forming a DNA adduct. If the adduct is not repaired, the next round of copying fixes it as a mutation (NNK causes a GC->AT change at K-ras in mice).	Lect 3
Atezolizumab	An anti-PD-L1 checkpoint inhibitor.	Lect 3
Atom	The smallest unit of an element. Molecules are built from atoms joined together.	Ch 1
Atomic nucleus	The dense centre of an atom, holding the protons and neutrons.	Ch 1
ATP (adenosine triphosphate)	The cell's spendable energy. Breaking off the last phosphate releases the energy that powers everything else.	Ch 1, Ch 6
AUC (area under the curve)	A single number between 0 and 1 that summarises how well a test separates two groups. Higher is better. He quotes 0.981 for the miRNA classifier.	Lect 1
Autologous	Taken from the patient and given back to that same patient.	Lect 2
Autophagy	The cell eating its own spare parts to get nutrients when food is short.	Ch 5, Lect 2
Avery, MacLeod and McCarty	The three who showed in 1943 that the transforming principle was DNA, using DNase and protease.	Ch 4, Lect 1
Basal transcription	The low, always-on level of transcription that happens before any activator arrives.	Lect 5
Base	One of the four letters of DNA: A, T, G or C. In RNA, U takes the place of T.	Ch 2, Ch 4
Base excision repair (BER)	The repair pathway that fixes a single damaged base by cutting it out.	Lect 4
Base pairing	The rule that DNA bases pair up, A with T and G with C. It is why one strand tells you what the other says.	Ch 4, Lect 1
Bcl-2 family	The group of proteins that decides whether the mitochondria release the signal that starts apoptosis.	Lect 2
Benzo(a)pyrene	A tobacco carcinogen that forms a DNA adduct once the cell has activated it.	Lect 3
Beta sheet (beta-sheet)	A folded shape in which several stretched protein strands lie side by side, held together by hydrogen bonds.	Ch 3
Binucleated cell	One cell containing two nuclei and 4N DNA. It is what you get when cytokinesis is blocked but mitosis proceeds.	Ch 5, Lect 2

Term (technical, English)	Plain-English meaning	Where it appears
Biomarker	A molecule measured in a sample that tells you something about disease or about how a patient will respond.	Ch 2, Lect 1
Biomolecule	A molecule made by a living system: a carbohydrate, a lipid, a protein or a nucleic acid.	Ch 2, Lect 1
Bispecific antibody	An antibody engineered with two different binding heads, so that it can hit two targets at once.	Lect 2
Bisulfite sequencing	The gold-standard methylation test. Unmethylated cytosine is converted to uracil; methylated cytosine is left alone.	Lect 3
BRCA1 and BRCA2	Two genes whose loss leaves a cell unable to repair broken chromosomes by homologous recombination, which is exactly what PARP inhibitors exploit.	Lect 4
BRD4	A protein that reads acetylation marks and recruits the transcriptional machinery. It is a current drug target.	Lect 3
Breathomics	The study of the chemicals in exhaled breath, as a way of looking inside the body.	Lect 1
cAMP (cyclic adenosine monophosphate)	A small second messenger whose level rises when glucose is low, letting CAP help RNA polymerase onto the promoter.	Lect 5
Cancer-associated fibroblast (CAF)	A fibroblast that a tumour has recruited and reprogrammed to build up the surrounding stroma.	Lect 2
CAP	A protein that binds cAMP and helps RNA polymerase onto the lac promoter. Together they make transcription strong.	Lect 5
Cap (5 prime cap)	The chemical cap on the front of an mRNA. It protects the message and tells the ribosome to pick it up.	Lect 5
CAP-cAMP (cAMP-CAP)	The pair of CAP protein and cAMP that sits on the lac promoter and makes transcription strong when glucose is low.	Lect 5
CAR (Chimeric Antigen Receptor)	A synthetic receptor put onto a T-cell so that it recognises a tumour protein and then kills the cell.	Lect 2
Carbohydrate	One of the four biomolecule classes. The main fuel, and also structural material such as cellulose.	Ch 2, Lect 1
CAR-NK	A CAR built on a natural killer cell instead of a T-cell.	Lect 3
Carrier	A person who carries one copy of a faulty gene without showing the disease.	Lect 3
Caspase	A protease that cuts other proteins. The caspases are the executioners of apoptosis.	Ch 6, Lect 2
Caspase 8 and caspase 9	The two upstream executioners that start the extrinsic and the intrinsic apoptosis routes respectively.	Lect 2
Catabolism	The breaking-down half of metabolism. It releases energy from big molecules.	Ch 6
CD28	The co-stimulatory signal in a CAR that tells the T-cell it is properly activated.	Lect 2
CD3 zeta (CD3z)	The signalling tail of the T-cell receptor, used inside a CAR to switch the T-cell on.	Lect 2
CDH17 (Cadherin-17)	The cell-surface protein Dr. Wong's own laboratory studied for twenty years, used as the tumour-targeting half of his bispecific antibody.	Lect 2
CDK4/6 inhibitor	A drug that blocks the kinase that lets a cell leave G1, so the cell cycle stops. Palbociclib is one.	Ch 5, Ch 6, Lect 2
CDKN2A	A gene that encodes two proteins on two reading frames, p16INK4a and p14ARF. When its promoter is methylated the cell loses two brakes at once.	Lect 3

Term (technical, English)	Plain-English meaning	Where it appears
CDR1, CDR2 and CDR3	The three hypervariable loops inside the variable region. Together they form the antigen-binding site.	Ch 3, Lect 1
Cell adhesion molecule (CAM)	A protein on the cell surface that sticks to another cell and controls who may move through tissue.	Lect 2
Cell cycle	The schedule G1, S, G2, M and cytokinesis that lets a cell copy itself.	Ch 5, Lect 2
Cell cycle arrest	The cell stopping at a checkpoint instead of dividing. It is what a CDK4/6 inhibitor causes.	Ch 6, Lect 2
Cell-free DNA (cfDNA)	DNA fragments floating in the blood. A tumour sheds some of its own into the blood.	Lect 3
Cell-type specificity	The fact that one gene is switched on in some cell types and not in others.	Lect 5
Cellular memory	The way a cell keeps a switch turned on long after the original signal has gone.	Lect 3
Cellulose	The structural carbohydrate that makes plant cell walls.	Ch 2
Central dogma	The rule that genetic information runs one way: DNA, then RNA, then protein.	Ch 6, Lect 1
Centromere	The narrow waist of a chromosome where the spindle fibres attach.	Ch 5, Lect 3
Centrosome	The body that organises the microtubules which form the mitotic spindle.	Ch 5, Lect 2
Cetuximab	The anti-EGFR antibody that works in wild-type K-Ras colorectal cancer and fails in K-Ras-mutant disease.	Lect 1
Chaperone	A protein that helps another protein fold correctly and stops it clumping.	Ch 3, Ch 6
Checkpoint	A control point in the cell cycle where the cell checks its work before going on.	Ch 5, Lect 2
Chemokine	A signal protein that makes white blood cells move towards a particular place.	Lect 2
CHO cell	A hamster cell line used to make human proteins for medicine, because it adds the right sugar decorations.	Ch 6, Lect 1, Lect 5
Cholesterol	A lipid built into the membrane that keeps it fluid at low temperature.	Ch 2
Chromatin	DNA plus the proteins that package it. The manual and its packaging.	Ch 4, Lect 3
Chromatin fibre	The 30 nanometre fibre formed when nucleosomes pack together.	Ch 4, Lect 3
Chromatin remodelling complex	An ATP-powered machine that slides nucleosomes along the DNA to expose or cover a gene.	Ch 4, Lect 3
Chromosome	One packaged copy of the genome. The printed manual.	Ch 4, Lect 3
CIMP	CpG island methylator phenotype. A pattern of widespread methylation that marks one group of colorectal cancers.	Lect 3
Cip/Kip	One of the two families of CDK inhibitors, the other being INK4. The Cip/Kip members - p21, p27, p57 - are broad-spectrum and bind the whole Cyclin-CDK complex; the INK4 members - p16/INK4A, p14ARF, p15, p18, p19 - target CDK4/6 only.	Lect 2
Circular DNA	DNA shaped as a ring rather than a line. Mitochondrial DNA is a circle of about 17 kilobases.	Ch 5, Lect 2
Circulating tumour cell (CTC)	A tumour cell that has broken away and is travelling in the blood.	Lect 2
Cis face	The receiving side of the Golgi apparatus, where arriving vesicles dock.	Ch 5
Cis-regulatory element	A DNA sequence such as a promoter, enhancer or silencer that regulates the gene on the same DNA molecule it sits on.	Lect 5
Co-activator (nuclear receptor)	A protein recruited to a switched-on nuclear receptor to help it drive transcription.	Lect 5

Term (technical, English)	Plain-English meaning	Where it appears
Codon	Three letters on the mRNA that name one amino acid.	Ch 6, Lect 1
Codon optimisation	Rewriting a gene so that the host cell prefers its own preferred codons, which makes the protein far easier to make.	Ch 6, Lect 5
Codon-usage bias	Different species prefer different codon spellings for the same amino acid. When a gene is moved into a new host cell it has to be rewritten in the host's preferred spelling - codon optimisation - for the protein to be made easily.	Ch 3, Ch 6, Lect 5
Coenzyme	A small helper molecule that an enzyme needs in order to work.	Ch 1
Cofactor	A non-protein partner an enzyme needs, often a metal ion.	Ch 1
Cold tumour	A tumour carrying few neoantigens. It responds poorly to checkpoint inhibitors.	Ch 6, Lect 5
Co-localisation	Two proteins found in the same place in the same cell, shown by overlapping fluorescent stains.	Lect 2
Companion diagnostic	A test that must be run before a drug is given, to decide whether that patient will benefit from it.	Lect 2
Complement	A group of blood proteins that can kill cells on their own, with no antibody involved.	Lect 2
Compound	A chemical made of two or more elements joined together.	Ch 1
Confidence interval	The range around a measured value that the true value is likely to fall in.	Lect 3
Confocal microscopy	The microscope that focuses one thin optical slice at a time, so a labelled protein can be located precisely.	Lect 2
Consensus sequence	The version of a DNA word that appears most often in real genomes.	Background
Contact inhibition	The behaviour in which healthy cells stop dividing once they touch their neighbours.	Lect 1
Contractile ring	The band of actin and myosin that pinches a dividing cell in two. Blocking it leaves one cell with two nuclei.	Ch 5, Lect 2
Copy number variation (CNV)	A region of DNA present in more, or in fewer, copies than normal.	Lect 3
Core promoter	The stretch of DNA right at the start of a gene where the transcription machinery assembles.	Lect 5
Corepressor	A molecule that helps a repressor do its job. Tryptophan is the corepressor of the trp operon.	Lect 5
Co-repressor (nuclear receptor)	A protein recruited to a nuclear receptor to hold transcription down. It is the opposite of a co-activator.	Lect 5
Co-stimulatory domain	The part of a CAR inside the T-cell that delivers the second signal needed to switch the T-cell fully on.	Lect 2
Covalent bond	Two atoms sharing electrons. The usual way molecules are held together.	Ch 1
CpG island	A short stretch of DNA rich in C followed by G. Methylating one usually silences the gene next to it.	Ch 4, Lect 3, Lect 5
CpG site	A cytosine next to a guanine. Methylating one is the switch that silences a gene.	Lect 3, Lect 5
CRISPR/Cas9	The gene-editing system that cuts DNA at a chosen place, so that the cell's own repair machinery makes the change.	Lect 2, Lect 4
Cristae	The folds of the inner membrane of a mitochondrion. The ATP-making machinery sits on them.	Ch 5
Crossing over	The exchange of DNA between a matched pair of chromosomes during meiosis, which shuffles inherited variants.	Ch 5, Lect 3

Term (technical, English)	Plain-English meaning	Where it appears
Cryptic exon	An exon that is normally skipped but is switched on by a mutation.	Lect 5
C-terminus	The last end of a protein chain, where the amino acid keeps a free carboxyl group.	Ch 3
CTGF	Connective tissue growth factor. One of the target genes YAP switches on through TEAD when the Hippo pathway is off.	Lect 1, Lect 5
CTLA-4	The checkpoint receptor that competes with CD28 for the T-cell's co-stimulatory signal.	Lect 2, Lect 3
Cyclin	A protein that rises and falls through the cell cycle and switches on a matching CDK. The key, not the machine.	Ch 5, Lect 2
Cyclin-dependent kinase (CDK)	The enzyme that drives a cell-cycle step. It only works when a matching cyclin switches it on.	Ch 5, Lect 2
Cytochrome c	A protein stored in mitochondria that is released into the cytoplasm to start the intrinsic apoptosis pathway.	Lect 2
Cytokine	A small protein released by a cell to send a message to nearby cells.	Ch 6, Lect 5
Cytokinesis	The division of the cytoplasm that follows mitosis and splits one cell into two.	Ch 5, Lect 2
Cytoplasm	The watery jelly inside the cell, where most of the chemistry happens.	Ch 5
Cytosine	One of the four DNA bases, and the one that gets methylated.	Ch 4, Lect 3
Cytoskeleton	The network of fibres that gives the cell shape, moves cargo, forms the spindle and pinches the cell in two.	Ch 5, Lect 2
Daraxonrasib (Rasonque)	A RAS inhibitor. It gave a median overall survival of 13.2 months against 6.7 for chemotherapy.	Lect 1
Deacetylation	Removing that acetyl group, which tightens the chromatin and silences the gene.	Ch 3, Lect 3, Lect 5
Deamidation	A slow chemical change in a protein that turns asparagine into aspartate and can change how well the protein works.	Ch 3, Lect 1
Death receptor	A receptor on the cell surface whose activation triggers apoptosis from outside the cell.	Ch 6, Lect 2
Deletion	The loss of a stretch of DNA.	Ch 4, Lect 3
Delta ligand	The molecule on a neighbouring cell that Notch listens to.	Lect 5
Dendritic cell	An immune cell that picks up antigens and presents them to T-cells.	Lect 2, Lect 4
Deoxyribose	The sugar in DNA. It has one oxygen atom fewer than the sugar in RNA.	Ch 4
Developmental specificity	The fact that the same gene is switched on at different times as the body develops.	Lect 5
Diagnostic biomarker	Tells you whether the patient has the disease right now.	Lect 1
Dicer	The enzyme that cuts a long RNA into short pieces, one of which becomes a microRNA.	Lect 1, Lect 5
Differentiation	The commitment of a cell to become one particular specialised type of cell.	Lect 2
Diffusion	The passive spread of molecules from where they are many to where they are few, with no energy spent.	Ch 1
Dimerisation	Two identical or different proteins joining into a pair, which often makes DNA binding far stronger.	Lect 5
Diploid	Having two sets of chromosomes, one from each parent. Human body cells are diploid, with 23 pairs.	Ch 5, Lect 3
Disaccharide	Two sugar units joined together, such as sucrose.	Ch 2, Lect 1

Term (technical, English)	Plain-English meaning	Where it appears
Disulfide bond	A covalent link between two cysteines that helps hold a protein's shape together.	Ch 3, Lect 1
DNA adduct	A chemical molecule stuck onto a DNA base, which can then mispair during copying.	Lect 3
DNA Damage Response (DDR)	The cell's alarm system for DNA damage. It triggers repair, or cell death if the damage is too bad.	Lect 4
DNA (deoxyribonucleic acid)	The master instruction manuals, written in four letters and stored in the nucleus.	Ch 4
DNA ligase	The enzyme that seals the last gap in a newly built DNA strand.	Lect 4
DNA polymerase	The enzyme that builds a new DNA strand, adding nucleotides only in the 5 prime to 3 prime direction.	Lect 4
DNA repair	The set of pathways that fix damage before it becomes a permanent mutation.	Lect 4
DNA-bending protein	A protein that loops DNA so that a distant enhancer can touch its promoter.	Lect 5
DNA-binding domain (DBD)	The highly conserved part of a nuclear receptor that recognises its own DNA word.	Lect 5
DNase	The enzyme that cuts DNA up. It destroyed the transforming activity, which is how DNA was identified.	Ch 4, Lect 1
DNMT1	The maintenance methyltransferase. It copies the methylation pattern onto the new strand after DNA is copied.	Lect 3
DNMT3A and DNMT3B	The de novo methyltransferases. They put the first methylation mark on a piece of DNA.	Lect 3
Dose escalation	The first stage of a trial, in which the dose is raised slowly to find the safe limit.	Lect 2, Lect 4
Double helix	Two DNA strands wound around each other, with the bases pairing across the middle like ladder rungs.	Ch 4
Double-strand break (DSB)	The most dangerous kind of DNA damage: both strands cut at the same place.	Lect 4
Down syndrome	Trisomy 21. Three copies of chromosome 21 instead of two.	Lect 3
Driver	A mutation that gives the cell a growth advantage. A typo in an important word.	Ch 4, Lect 3
Drosha	The enzyme that cuts a primary microRNA transcript in the nucleus into a short hairpin.	Lect 1, Lect 5
Drug resistance	The state in which a tumour no longer responds to a drug it was previously sensitive to. It can be caused by a new mutation, or by an isoform that has lost the drug's binding site.	Lect 4, Lect 5
Drug target	A molecule a drug is designed to act on. Membrane proteins and kinases are prime examples.	Lect 1, Lect 2
Duplication	A copy of a stretch of DNA that is made in addition to the original.	Ch 4, Lect 3
DXd	The drug payload carried by trastuzumab deruxtecan.	Lect 1
E2F	The transcription factor that pRb normally keeps switched off, and releases when the cell is ready to divide.	Lect 2
ECM (extracellular matrix)	The scaffold of protein and sugar that cells sit in, and that they have to dig through to move.	Lect 2
Ectoderm	The outer germ layer. It becomes the skin and the nervous system.	Lect 2
Edman degradation	The standard way of reading a protein's sequence one amino acid at a time, starting from the N-terminus.	Ch 3

Term (technical, English)	Plain-English meaning	Where it appears
EGFR	The growth-factor receptor at the top of the RAS chain. Blocking it is a standard cancer drug strategy.	Ch 6, Lect 1
Electron	A particle with a negative charge that moves around the nucleus and does all the bonding.	Ch 1
Electron shell	A fixed energy layer around the nucleus. The outermost shell decides what an atom can bond to.	Ch 1
ELISA	The standard test that measures how much of one protein is present in a sample of fluid.	Lect 1, Lect 2
Endoderm	The inner germ layer. It becomes the gut lining and the lungs.	Lect 2
Endometrium	The lining of the uterus. Tamoxifen behaves differently there than in breast tissue.	Lect 5
Endoplasmic reticulum (ER)	The membrane network beside the nucleus where new proteins are made, folded, checked and glycosylated.	Ch 5, Lect 2
Endosymbiotic origin	The idea that mitochondria began as bacteria that were engulfed and kept by an ancestral cell.	Ch 5, Lect 2
Endotoxin	The poisonous part of the outer membrane of Gram-negative bacteria. It is the same molecule as LPS.	Ch 6, Lect 1
End-replication problem	The fact that DNA polymerase cannot copy the very end of a linear chromosome, so telomeres shorten at every division.	Ch 4, Lect 3
Energy	The ability to do work. In a cell it comes from the net release of a metabolic reaction - breaking old bonds costs energy, forming new ones returns more, and the difference is what the cell spends (usually banked as ATP).	Ch 1
Enhanceosome	A cluster of proteins sitting together on an enhancer and on its promoter, which is how a far-away enhancer reaches its gene.	Lect 5
Enhancer	A DNA sequence that increases transcription, wherever it happens to sit relative to the gene.	Lect 5
Enzyme	A protein that makes a chemical reaction go faster without being used up itself.	Ch 3, Ch 6
EpCAM	The surface protein used to pick out tumour cells from a blood sample.	Lect 2
Epithelial-to-mesenchymal transition (EMT)	The change in which a cell loses its grip on its neighbours and becomes free to move and invade.	Lect 2
Epitope	The exact patch of an antigen that an antibody actually touches.	Ch 3
EPO	Erythropoietin. The red-blood-cell growth factor that HIF-1alpha switches on.	Lect 5
ERBB2	The gene name for HER2.	Lect 1, Lect 5
E-selectin	One of the selectins that makes white blood cells roll along the wall of a blood vessel.	Lect 2
Estrogen receptor (ER)	The nuclear receptor for estrogen. Tamoxifen acts on it.	Lect 5
Estrogen response element (ERE)	The DNA word the estrogen receptor pair binds to.	Lect 5
Euchromatin	Open, loose chromatin that is being transcribed.	Lect 3
Eukaryote	A cell with a nucleus. Animals, plants, fungi and yeasts are all eukaryotes.	Ch 5, Lect 5
Ex vivo	Outside the body. Cells are taken out, treated, and put back.	Lect 2
Exon	A stretch of a gene that survives splicing and ends up in the mature mRNA.	Lect 5

Term (technical, English)	Plain-English meaning	Where it appears
Exon inclusion and exon skipping	The two opposite choices the splicing machinery makes when it builds a mature message.	Lect 5
Exonuclease	The proofreading activity of DNA polymerase that cuts a wrongly inserted base back out.	Ch 4
Exportin-5	The protein that carries a short hairpin RNA out of the nucleus into the cytoplasm.	Lect 1, Lect 5
Extrinsic apoptosis pathway	The route into apoptosis that starts outside the cell, when a death receptor is triggered by its ligand.	Ch 6, Lect 2
Fab fragment	The arm of an antibody. It is the part that grabs the antigen.	Ch 3, Lect 1
FACS	The machine that sorts cells by their fluorescence into separate tubes.	Lect 2
Fatty acid	A chain of carbons with a small acid end. Saturated or unsaturated depending on whether there are double bonds in the chain.	Ch 2, Lect 1
Fatty-acid chain	The long carbon tail of a fatty acid, which is either saturated or unsaturated.	Ch 2
Fc fragment	The stem of an antibody. It is the part that recruits the rest of the immune system.	Ch 3, Lect 1
FISH	A test that uses DNA probes, not antibodies, to count gene copies inside a cell, such as HER2.	Lect 2
Flow cytometry	The laser machine that counts cells and measures their size, internal complexity and fluorescence.	Lect 2
Fold (protein folding)	The process by which a chain of amino acids twists into the specific three-dimensional shape that gives it its job.	Ch 3
Forward scatter (FSC)	The flow-cytometry signal that reports cell size.	Lect 2
Frameshift	A mutation that adds or removes letters not in multiples of three, so the reading frame shifts and the protein is usually truncated.	Ch 6, Lect 1, Lect 5
Functional group	A small cluster of atoms on a molecule that gives that molecule its chemical personality and its reactivity.	Ch 1
Fv fragment	The smallest piece of an antibody, the variable region alone. About 25 kDa.	Ch 3, Lect 1
G0	The resting side branch of the cell cycle, taken by cells that are alive but not dividing.	Ch 5, Lect 2
G1	The growth phase of the cell cycle, before the DNA is copied.	Ch 5, Lect 2
G2	The checkpoint phase after the DNA has been copied and before the cell divides.	Ch 5, Lect 2
Gating	Drawing a boundary on a flow-cytometry plot to select one population of cells out of many.	Lect 2
G-banding	The pattern of light and dark bands made by staining chromosomes. It reveals large structural changes.	Ch 4, Lect 3
GC-MS	Gas chromatography with mass spectrometry. It separates the chemicals in a sample and then identifies them.	Lect 1
Gene	One recipe. A stretch of DNA that specifies one product.	Ch 4
Gene expression	The whole process of turning a gene on, copying the message and building the protein.	Ch 6, Lect 5
Gene expression programme	The set of genes a cell type switches on and off. It is what makes a T-cell different from a dendritic cell.	Lect 5
Gene knock-in	Putting a working copy of a gene into the genome. It needs the precise repair pathway.	Lect 2, Lect 4

Term (technical, English)	Plain-English meaning	Where it appears
Gene knock-out	Making a gene stop working, usually by letting CRISPR cut it.	Lect 2, Lect 4
General transcription factor	A protein needed to start transcription at any gene, rather than at one particular gene.	Lect 5
Genome	The complete set of DNA belonging to an organism or a cell.	Ch 4
Genotype	The genetic makeup of a cell or person: the versions of each gene it carries.	Ch 4, Lect 3
Germ cell	A cell that will make sperm or eggs, and so passes mutations on.	Lect 3
Glucose	The simple sugar that cells burn for fuel.	Ch 2, Lect 1
GLUT1	The glucose transporter that HIF-1alpha switches on, so that a cell can still work without oxygen.	Lect 5
Glycogen	The storage form of glucose in animals and fungi, a branched chain of sugar units.	Ch 2, Lect 1
Glycolysis	The first step of energy production. Glucose is split into pyruvate without using oxygen.	Ch 6, Lect 1
Glycosidic bond	The chemical link between one sugar unit and the next in a chain of sugars.	Ch 2
Glycosylation	The attaching of sugar groups to a protein, usually in the endoplasmic reticulum.	Ch 5, Ch 6
Golgi apparatus	The packing room. It sorts finished proteins and posts them to their destination.	Ch 5, Lect 2
GPCR (G-protein-coupled receptor)	A large family of membrane receptors that are prime drug targets.	Lect 2, Lect 5
Granzyme	The enzyme that kills a target cell once it is inside.	Lect 2
Griffith	The scientist whose 1928 bacterial work showed that material from dead pneumococcus could turn live harmless bacteria into virulent ones.	Ch 4, Lect 1
gRNA	The guide RNA that tells the CRISPR enzyme where to cut.	Lect 2
Guide (lncRNA)	The way a long non-coding RNA carries a chromatin-remodelling protein to the DNA word it should act on.	Lect 5
H3K4me3	The mark that sits on active promoters.	Ch 3, Lect 3
H3K9me3	The mark that tells the cell a region of DNA is meant to stay shut.	Ch 3, Lect 3
Haploid	Having one set of chromosomes, as gametes do. A human gamete has 23.	Ch 5, Lect 3
Hayflick limit	The number of times a human cell can divide before it has to stop.	Ch 4, Lect 3
HBV	Hepatitis B virus. Its integration into liver DNA is a prognostic marker.	Lect 2
HDAC inhibitor	A drug that blocks histone deacetylase, so acetylation builds up and the chromatin opens.	Ch 3, Lect 3, Lect 5
Heavy chain	The two larger chains of an antibody, which form the stem and part of the arms.	Ch 3, Lect 1
Helicase	The enzyme that unwinds the two DNA strands so that they can be copied.	Lect 4
Helper T cell	The T-cell that releases cytokines and tells the rest of the immune system what to do.	Lect 2
Hepatocellular carcinoma (HCC)	Primary cancer of the liver.	Lect 1, Lect 2
Hepatocyte	A liver cell.	Ch 5, Lect 1
HER1, HER3 and HER4	The other members of the HER receptor family, which HER2 pairs with.	Lect 5

Term (technical, English)	Plain-English meaning	Where it appears
HER2	A receptor whose gene is amplified in some breast and gastric cancers. Trastuzumab blocks it.	Lect 2, Lect 5
Heterochromatin	Tightly packed, silent chromatin. It physically keeps transcription factors away from the promoter.	Lect 3
Heterodimer	Two different proteins joined up, such as HER2 with HER3.	Lect 5
Heteroplasmy	A mixture of normal and mutant mitochondrial DNA inside the same cell.	Lect 2
Heterozygous	Having two different versions of a gene, one from each parent.	Ch 4, Lect 3
HIF-1alpha	The oxygen sensor protein. It is made all the time and destroyed in normal oxygen.	Lect 5
HIF-1beta	The partner HIF-1alpha joins with in the nucleus to make the working hypoxia factor.	Lect 5
HIF-CAR	A CAR whose expression is put under HRE control, so the T-cell only arms itself in a hypoxic tumour.	Lect 5
High specific heat	The amount of energy a substance needs to warm up by one degree. Water's is high, so body temperature stays steady.	Ch 1
Hinge region	The flexible linker between a nuclear receptor's DNA-binding and ligand-binding domains.	Lect 5
Hippo pathway	The pathway that uses cell contact to switch growth off. MST1 activates LATS1, which phosphorylates YAP.	Ch 6, Lect 1, Lect 5
Histone	The protein around which DNA wraps to make a nucleosome.	Ch 4, Lect 3, Ch 3
Histone code	The system of histone marks that tells the cell whether a region of DNA is open or closed.	Lect 3
Histone modification	A chemical tag on a histone that opens or closes the chromatin around it.	Lect 3, Lect 5
Histone variants (H2A, H2B, H3 and H4)	The four protein types that make up the nucleosome core.	Ch 4, Lect 3
Homodimer	Two identical proteins joined up. Dimers bind DNA far more strongly than single proteins do.	Lect 5
Homoeologous recombination	Recombination between chromosomes that are similar but not identical. It is a crop-breeding mechanism in polyploids. The tumour mechanism that produces copy-number variation is NAHR - non-allelic homologous recombination - and the two must not be confused.	Background
Homologous chromosome	One of the matched pair of chromosomes, one from each parent, which separate in the first meiotic division.	Ch 5, Lect 3
Homologous recombination (HR)	The high-fidelity repair of a broken chromosome using the matching sister chromatid as a template. It only works in S and G2.	Lect 4
Homology-directed repair	The precise repair pathway that uses a matching template. It is the pathway a CRISPR knock-in needs.	Lect 2, Lect 4
Homoplasmy	A cell in which every copy of the mitochondrial DNA is the same, normal or mutant.	Lect 2
Homozygous	Having two identical versions of a gene.	Ch 4, Lect 3
Hormone	A chemical messenger released by a gland into the blood, acting on cells far away.	Lect 5
Hot tumour	A tumour carrying many neoantigens. It responds well to immunotherapy.	Ch 6, Lect 4, Lect 5
HSP90	The chaperone that holds the estrogen receptor folded in the cytoplasm until a hormone arrives.	Lect 5
Human Protein Atlas	The public database he used to nominate drug targets and to check where a protein is made.	Lect 1

Term (technical, English)	Plain-English meaning	Where it appears
Hydrogen bond	A weak attraction between a hydrogen attached to an electronegative atom and another electronegative atom nearby.	Ch 1
Hypoxia	Too little oxygen. It is the signal that switches on HIF-1alpha.	Lect 5
Hypoxia-responsive element (HRE)	The DNA word that switches on a gene when HIF-1 is active.	Lect 5
ICAM-1	An adhesion molecule that lets a white blood cell grip the wall of a blood vessel firmly.	Lect 2
IDO	An enzyme that starves T-cells of the amino acid they need, so it switches them off.	Lect 2
IFN-gamma	The third cytokine in the NF-kappaB response, after TNF-alpha and IL-1beta.	Lect 5
IgG4	An antibody class that does not kill by ADCC and stays in the blood longer than IgG1.	Lect 2, Lect 1
IkappaB (I-kappa-B)	The protein that holds NF-kappaB in the cytoplasm until a signal destroys it.	Lect 1, Lect 5
IkappaB kinase (IKK)	The enzyme that adds a phosphate to IkappaB, which marks it for destruction.	Lect 5
IL-15	The cytokine that comes after IFN-gamma in the NF-kappaB response and keeps NK cells alive.	Lect 5
IL-1beta	The second cytokine in the NF-kappaB response, after TNF-alpha.	Lect 5
IL-2	The cytokine that comes after IFN-gamma in the NF-kappaB response and drives T-cell proliferation.	Lect 5
Immune checkpoint	A molecular brake that stops an immune cell from attacking. PD-1 on the T-cell is the important one.	Lect 2
Immune evasion	The state in which a tumour avoids being attacked because the T-cells have been switched off.	Lect 2
Immunofluorescence (IF)	Staining tissue with a fluorescently tagged antibody, so you can see exactly where a protein sits.	Lect 2
Immunohistochemistry (IHC)	Staining tissue with an antibody and an enzyme that leaves a coloured deposit, so you can see where a protein sits.	Lect 2
Inclusion body	A hard, insoluble clump of misfolded protein. It forms when a recombinant protein is made too fast or too early.	Lect 5
IND (Investigational New Drug)	The application a company files before it may test a drug in people.	Lect 2
Indel	An insertion or a deletion of DNA letters. If it is not a multiple of three, it causes a frameshift.	Lect 1, Lect 3
Inducible system	A system that stays switched off until a signal arrives.	Lect 5
Induction	Switching a gene on by adding an inducer, such as lactose in the lac system.	Lect 5
Inhibitor of apoptosis (IAP)	A protein that holds the caspase executioners back.	Lect 2
INK4	A family of inhibitors that block the CDK4/6 kinases and so stop the cell cycle.	Lect 2
Insulator	A DNA sequence that stops an enhancer from acting on the wrong gene.	Lect 5
Integrin	A cell-surface protein that grips other cells and the matrix, and is used by white blood cells to hold on while they move.	Lect 2
Intermembrane space	The gap between the two membranes of a mitochondrion, where protons build up to drive ATP synthase.	Ch 5

Term (technical, English)	Plain-English meaning	Where it appears
Intrinsic apoptosis pathway	The route into apoptosis that starts inside the cell, usually at the mitochondria.	Ch 6, Lect 2
Intron	A stretch of a gene that is cut out during splicing and does not appear in the mature mRNA.	Lect 5
Inversion	A stretch of chromosome cut out and put back in the opposite direction. No DNA is gained or lost, but expression can change.	Ch 4, Lect 3
Ionic bond	The attraction between oppositely charged ions, as in table salt.	Ch 1
Ipilimumab	An anti-CTLA-4 checkpoint inhibitor, which works by releasing the T-cell brake.	Lect 3
iPSC (induced pluripotent stem cell)	A reprogrammed adult cell that has been pushed back to an embryonic-like state, so it can become almost any cell type.	Lect 2
IPTG	The artificial, non-metabolised inducer used to switch on recombinant genes in E. coli.	Lect 5
Isoelectric focusing	A separation method that sorts proteins by the pH at which they carry no charge.	Ch 3, Lect 1
Isoform	A slightly different version of a protein made from the same gene by choosing a different set of exons.	Lect 5
Isotype	A version of an antibody class. IgG1 kills by ADCC; IgG4 does not.	Lect 2
Kaplan-Meier curve	A plot of the fraction of patients still surviving over time. A curve that drops early and steeply means a worse prognosis.	Lect 1, Lect 3
Karyotype	The picture of all the chromosomes of a cell, used to spot large structural abnormalities.	Lect 3
Karyotyping	Staining chromosomes with Giemsa dye to make G-banding, so that large structural changes can be read off.	Ch 4, Lect 3
KIBRA, MER and WILLIN	The proteins at the cell junction that recruit and switch on the Hippo kinases.	Lect 5
Kinase	An enzyme that adds a phosphate group to another protein, usually switching it on.	Ch 6, Lect 1
K-Ras	The gene whose stuck-on mutant drives many cancers, including colorectal and pancreatic cancer.	Ch 4, Ch 6, Lect 1
Lac operon	The classic bacterial switch, in which several genes for lactose use are turned on together by one signal.	Lect 5
Lac repressor	The protein that sits on the operator and blocks the lac genes until allolactose removes it.	Lect 5
Lactate	The molecule a cancer cell makes when it ferments sugar instead of burning it.	Ch 6, Lect 1
Lactose	The milk sugar that the bacterium E. coli can use for energy, and which the lac operon exists to exploit.	Ch 2, Lect 5
Lagging strand	The strand that cannot be copied continuously, so it is built in short Okazaki fragments.	Lect 4
Laser capture microdissection (LCM)	Cutting a pure region out of a tissue slide under the microscope, to avoid contamination from other cells.	Lect 2
LATS1	The kinase in the Hippo pathway that phosphorylates YAP and holds it in the cytoplasm.	Ch 6, Lect 1, Lect 5
LC-MS/MS	The method that cuts a protein into pieces, weighs them and reads their sequences. It tells you which proteins are present and how much of each.	Ch 3, Lect 1
Leading strand	The strand that can be copied continuously, in one smooth run.	Lect 4
Ligand	The molecule that binds a receptor. The message, not the receiver.	Ch 6, Lect 5

Term (technical, English)	Plain-English meaning	Where it appears
Ligand-binding domain (LBD)	The part of a nuclear receptor that holds the hormone or drug and changes shape when it does.	Lect 5
Light chain	The two smaller chains of an antibody, which sit at the tips of the arms.	Ch 3, Lect 1
Lipid	A fatty molecule, used for energy storage, for membranes and for signalling.	Ch 2, Lect 1
Lipid bilayer	The two-molecule-thick sheet that forms cell membranes.	Ch 5
Lipid nanoparticle (LNP)	A fatty bubble that carries mRNA into a cell. Used for in-vivo CAR-T and for mRNA vaccines.	Lect 2, Lect 5
Lipoprotein	A particle that carries fats around the blood, wrapped in protein. It is how cholesterol travels.	Ch 2
Liquid biopsy	Looking for tumour material in a blood sample instead of cutting out a piece of the tumour.	Lect 2
LncRNA (long non-coding RNA)	An RNA of more than 200 nucleotides that makes no protein but guides the chromatin machinery.	Lect 5
Log-rank test	The statistical test that compares two survival curves and produces a p-value.	Lect 1, Lect 3
LPS (lipopolysaccharide)	The bacterial molecule that is detected by TLR4 and sets off the NF-kappaB inflammation cascade.	Ch 6, Lect 1, Lect 5
Luciferase	The reporter gene most often used for promoter experiments, because it makes light.	Lect 5
Lysine	The amino acid that carries most of the histone marks. Its side chain is the handle the mark is hung on.	Ch 3, Lect 3
Lysosome	The acidic recycling and disposal plant of the cell.	Ch 5
M phase	The stage of the cell cycle in which the nucleus divides. Mitosis is the first part of it.	Ch 5, Lect 2
Macromolecule	A very large molecule, such as a protein or a nucleic acid, built from many small units.	Ch 1, Ch 2
Mass spectrometry	The technique that weighs molecules and identifies them from the pattern of fragments they produce.	Ch 3, Lect 1
Maternal inheritance	The rule that a mitochondrial condition passes from an affected mother to all her children and from an affected father to none.	Ch 5, Lect 2
Matrix (mitochondrial matrix)	The fluid interior of a mitochondrion, where the TCA cycle runs and the mitochondrial DNA sits.	Ch 5
MCP-1	A chemokine that draws white blood cells towards a place.	Lect 2
MDSCs	Myeloid-derived suppressor cells. They are tumour-recruited cells that switch the immune response off.	Lect 2
Meiosis	The special division in germ cells that makes four genetically different gametes.	Ch 5, Lect 3
Meiosis I and meiosis II	The first division separates matched chromosomes; the second separates sister chromatids.	Lect 3
Melting temperature (Tm)	The temperature at which half of a DNA double helix comes apart.	Ch 4, Lect 3
Membrane protein	A protein sitting in the plasma membrane, often a receptor. Prime drug targets because drugs can reach them from outside the cell.	Ch 5, Lect 2
Mesoderm	The middle germ layer. It becomes muscle, bone and blood.	Lect 2
Metabolism	All the chemical reactions of the cell: breaking things down for energy and building things up.	Ch 6, Lect 1
Metabolite	A small molecule the cell makes or consumes while it works.	Ch 6, Lect 1

Term (technical, English)	Plain-English meaning	Where it appears
Metabolite fingerprint	The pattern of small molecules that makes one sample different from another.	Lect 1
Metabolomics	The study of all the small molecules in a sample. It is how many biomarkers are found.	Ch 6, Lect 1
Methyl group	The tiny -CH3 tag that DNA methylation adds to a cytosine.	Lect 3
Methylation	Adding a methyl group to DNA, usually in order to silence a gene.	Lect 3, Lect 5
MGMT	A gene whose promoter methylation predicts that a brain tumour will respond well to alkylating chemotherapy.	Lect 3
MHC (major histocompatibility complex)	A surface protein that displays a short protein fragment so that T-cells can inspect the cell.	Ch 6, Lect 5
Microarray	A chip carrying thousands of gene probes, used to measure which genes are switched on.	Lect 1
MicroRNA (miRNA)	A short RNA, about 22 nucleotides long, that blocks the translation of its target mRNA.	Lect 1, Lect 5
Microsatellite	A short repeated stretch of DNA. Its length changes when mismatch repair fails, which is how MSI is measured.	Lect 4
Microsatellite instability (MSI)	The change in short repeated DNA that follows from failed mismatch repair. MSI-high tumours are hot tumours.	Lect 4
Microtubule	A hollow tube of protein that forms the mitotic spindle and acts as a cargo track.	Ch 5
Misfolding	A protein that does not take up its correct three-dimensional shape. It is one of the liabilities of a therapeutic protein.	Ch 3, Lect 1, Lect 5
Mismatch repair (MMR)	The repair pathway that corrects copying errors left behind after replication.	Lect 4
Missense mutation	A single-letter change that swaps one amino acid for another.	Ch 4, Lect 3
Mitochondrial DNA (mtDNA)	The small circular DNA inside mitochondria, about 17 kilobases carrying 37 genes. Inherited from the mother.	Ch 5, Lect 2
Mitochondrial Eve	The single woman from whom all living mitochondrial DNA is thought to descend.	Lect 2
Mitochondrion	The power station. It makes most of the ATP, controls apoptosis, and keeps its own DNA.	Ch 5, Lect 2
Mitosis	The division of the nucleus, in which the duplicated chromosomes separate.	Ch 5, Lect 2
Mitotic clock	The counting mechanism that turns cell division into a limit on how many times a cell may divide.	Lect 3, Lect 4
Mitotic spindle	The ropes of microtubule that pull the duplicated chromosomes apart.	Ch 5, Lect 2
Molecular biology	The study of life at the level of molecules, especially DNA, RNA and protein.	Ch 1
Molecule	Two or more atoms joined together in a fixed arrangement.	Ch 1
Monosaccharide	A single sugar unit, such as glucose or fructose.	Ch 2, Lect 1
Monosomy	One copy of a chromosome instead of two, written as 2n minus 1.	Lect 3
mRNA half-life	How long an mRNA lasts before it is broken down. A short half-life means less protein.	Lect 5
mRNA (messenger RNA)	A photocopy of one recipe, sent from the nucleus to the factory floor.	Ch 4, Ch 6
MST1	The kinase at the start of the Hippo pathway. It activates LATS1.	Ch 6, Lect 1, Lect 5

Term (technical, English)	Plain-English meaning	Where it appears
Multinucleated cell	A cell that has more than one nucleus, because the cytoplasm did not divide.	Lect 2
Multiplex immunofluorescence	Staining several proteins at once, each with a different fluorescent colour.	Lect 2
Multipotent	A cell that can become several cell types, but only within one family.	Lect 2
Multivariate logistic regression	The test used to work out whether a marker adds independent information once several other factors are taken into account.	Lect 3
Mutagen	Anything that causes a mutation, such as a chemical or a form of radiation.	Lect 3
Mutant splice site	A new splice site created by a single-letter change, which makes the RNA join in the wrong place and produce a shorter isoform. In his liver-cancer worked example the point is prognosis - the shorter isoform predicts significantly worse overall survival. Making a neoantigen the immune system has not seen is a different consequence of aberrant splicing, and the two should not be run together.	Lect 5
Mutation	A typo in the master manuals.	Ch 4, Lect 3
Myosin	The motor protein that works with actin to build the contractile ring.	Ch 5, Lect 2
NAHR (non-allelic homologous recombination)	Recombination between two similar but non-matching stretches of DNA, which causes rearrangements.	Lect 3
Natural killer cell (NK cell)	An immune cell that kills tumour cells without needing to be told which target to hit.	Lect 2
Necrosis	Uncontrolled cell death after an injury, which spills cell contents and inflames the surrounding tissue.	Ch 6
Neoantigen	A protein fragment the immune system has never seen, usually made by a frameshift or a new splice site. Ideal immunotherapy fuel.	Lect 1, Lect 4, Lect 5
Neutron	A particle with no charge, sitting in the nucleus. It adds mass without changing the chemistry.	Ch 1
NF-kappaB (nuclear factor kappa B)	The transcription factor that is held in the cytoplasm by I-kappa-B until a signal such as LPS makes it move into the nucleus.	Ch 6, Lect 1, Lect 5
NICD	The intracellular part of Notch, which is cut loose and moves into the nucleus to act as a transcription factor.	Lect 5
Nivolumab	Another anti-PD-1 checkpoint inhibitor.	Lect 3
NMR	Nuclear magnetic resonance. It works out the shape of a molecule in solution.	Lect 1
Non-disjunction	The failure of chromosomes to separate properly during meiosis, which gives the wrong number of chromosomes.	Ch 4, Lect 3
Non-homologous end joining (NHEJ)	The repair pathway for a broken chromosome that needs no template and is therefore error-prone. It builds the antibody repertoire.	Lect 4
Nonsense mutation	A single-letter change that creates an early stop codon, so the protein is truncated.	Ch 4, Lect 3
Notch	The receptor that lets one cell tell its neighbour to change its gene expression.	Lect 5
NOTCH2	One of the target genes YAP switches on through TEAD when the Hippo pathway is off.	Lect 1, Lect 5
NPM1	The gene in his worked example for telling a driver from a passenger: you have to integrate the CNV copy-number profile with the mRNA expression data before you can call a gene such as NPM1 a driver. He does not teach NPM1's own biology.	Lect 3

Term (technical, English)	Plain-English meaning	Where it appears
N-terminal domain (NTD)	The regulatory region at the start of a nuclear receptor. It is not the same thing as a signal peptide.	Lect 5
N-terminal signal peptide	The short postal address at the very start of a protein that routes it into the endoplasmic reticulum and so into the secretory pathway.	Ch 5, Lect 2, Lect 5
N-terminus	The first end of a protein chain, where the amino acid keeps a free amino group.	Ch 3
Nuclear envelope	The double membrane that seals the nucleus.	Ch 5, Lect 3
Nuclear pore	A doorway in the nuclear envelope that lets messages out and proteins in.	Ch 5
Nuclear receptor	A protein that binds a hormone or drug inside the cell and then switches genes on in the nucleus.	Lect 5
Nuclear transcription factor	A protein that enters the nucleus and switches a gene on or off there.	Lect 5
Nucleic acid	The biomolecule class that includes DNA and RNA. It carries the information.	Ch 2, Ch 4
Nucleoid	The region of a bacterial cell where the DNA sits, with no membrane around it.	Ch 5
Nucleolus	The dense region inside the nucleus where the parts of ribosomes are made.	Ch 5
Nucleoside	A base plus a sugar, with no phosphate. The building block before the phosphate is added.	Ch 2, Lect 1
Nucleosome	DNA wrapped around a ring of histone proteins. The basic unit of chromatin packing.	Ch 4, Lect 3
Nucleotide	A base plus a sugar plus a phosphate. The building block of DNA and RNA.	Ch 2, Ch 4
Nucleotide excision repair (NER)	The repair pathway for bulky damage such as UV dimers, which cuts out a whole stretch of the strand.	Lect 4
Nucleus	The control room. It holds the DNA and decides which genes are read.	Ch 5
Odds ratio	A way of expressing how much more likely something is in one group than in another.	Lect 3
Okazaki fragment	The short piece of DNA made on the lagging strand before the pieces are joined together.	Lect 4
Oncogene	A gene that, when switched on too hard, pushes a cell to divide. A mutated driver.	Ch 4, Lect 1
Operator	The DNA word the repressor sits on. While the repressor is on it, the genes behind it cannot be copied.	Lect 5
Operon	A group of neighbouring genes that are switched on and off together by one shared control region.	Lect 5
Organelle	A small compartment inside a cell with one job, like a room in a factory.	Ch 5
Organoid	A miniature, grown-from-stem-cell version of a piece of tissue, used to test drugs on.	Lect 2
Oxidation	Losing electrons, or gaining oxygen. It releases energy.	Ch 1
Oxidative phosphorylation	The mitochondrial process that uses oxygen to make most of the cell's ATP.	Ch 6, Lect 2
p16	The tumour-suppressor protein encoded by CDKN2A.	Lect 3
p300/CBP	The proteins that HIF-1 recruits to help it drive transcription of its target genes.	Lect 5
p50 and p65	The two halves of the NF-kappaB pair that enters the nucleus and reads inflammatory gene promoters.	Lect 5

Term (technical, English)	Plain-English meaning	Where it appears
p53	The tumour-suppressor protein that stops a cell with damaged DNA from dividing. Mutated in about 40 per cent of cancers.	Ch 5, Ch 6, Lect 4
Palbociclib	A CDK4/6 inhibitor that arrests the cell cycle in G1.	Ch 5, Lect 2
PAM	The short DNA sequence next to the target that the CRISPR enzyme requires before it will cut.	Lect 2
p-arm	The short arm of a chromosome, above the centromere.	Ch 4, Lect 3
PARP inhibitor	A drug that blocks a DNA repair enzyme and so kills BRCA-deficient tumour cells while normal cells survive.	Lect 4
Passenger	A mutation that rides along with a driver and does no harm itself. A typo on page 900.	Ch 4, Lect 3
P-body	The part of the cytoplasm where silenced messenger RNAs are gathered and destroyed.	Lect 5
PCA (principal component analysis)	A way of compressing thousands of measurements down to a few axes so cells can be plotted.	Lect 1
PCR	The polymerase chain reaction, which makes millions of copies of a chosen DNA stretch so it can be measured.	Lect 4
PD-1	The checkpoint receptor on T-cells. When PD-L1 binds it, the T-cell is switched off.	Ch 6, Lect 2
PD-L1	The protein on tumour cells that binds PD-1 and switches the T-cell off.	Ch 6, Lect 2
PECAM-1	The adhesion molecule a white blood cell grips to squeeze between two cells and leave a blood vessel.	Lect 2
Pembrolizumab	An anti-PD-1 checkpoint inhibitor. It is the drug for MSI-high and other hot tumours.	Lect 3
Penetrance	How likely a person who carries a mutation is to show the disease. It is never 100 per cent.	Lect 3
Peptide	A short chain of amino acids joined by peptide bonds.	Ch 3
Peptide bond	The chemical link between one amino acid and the next in a protein chain.	Ch 3
Perforin	The protein that punches holes in a target cell so a granzyme can get in.	Lect 2
pH	A scale of acidity. Below 7 is acidic, above 7 is alkaline, and 7 is neutral.	Ch 1
PHD and FIH	The enzymes that add an oxygen group to HIF-1alpha, which is the step that marks it for destruction.	Lect 5
Phenotype	What a cell or person actually looks and does. It is the genotype plus the environment plus the gene regulation.	Ch 4, Lect 5
Phosphatase	An enzyme that removes a phosphate group, switching a protein back off.	Ch 6, Lect 1
Phosphodiester bond	The chemical link that joins neighbouring nucleotides together in a DNA strand.	Ch 4
Phospholipid	A molecule with a water-loving head and a water-hating tail. That is what lets membranes form themselves.	Ch 2, Ch 5
Phosphorylation	The attaching of a phosphate group, usually switching a protein on.	Ch 6, Lect 1
PKC (protein kinase C)	A signalling enzyme that can start inflammatory and growth pathways.	Lect 1, Lect 5
Plasma membrane	The cell's outer skin. A lipid bilayer that controls entry and exit and carries receptors.	Ch 5
Pluripotent	A cell that can become almost any cell type in the body.	Lect 2
PNCTR	A long non-coding RNA whose YYUCUY motifs bind PTBP1 and so control splicing.	Lect 5

Term (technical, English)	Plain-English meaning	Where it appears
Point mutation	A change of a single letter in the DNA. It may be harmless, or it may be a driver.	Ch 4, Lect 3
Polarity	The way a molecule has one slightly positive end and one slightly negative end, like a small magnet.	Ch 1
Poly-A tail	The string of adenosines at the end of an mRNA. A longer tail means more translation; a shorter one means the message decays faster.	Lect 5
Polyadenylation	Adding the poly-A tail to the end of an mRNA. Choosing where to do it is itself a regulatory step.	Lect 5
Polyadenylation site (PAS)	The place in the RNA where the poly-A tail is added. Choosing an early one or a late one changes the message.	Lect 5
Polycistronic mRNA	One RNA message carrying several genes at once, as in a bacterial operon.	Lect 5
Polypeptide	A long chain of amino acids. It becomes a protein when it folds.	Ch 3
Polysaccharide	A long chain of sugar units, such as glycogen, starch or cellulose.	Ch 2, Lect 1
Polyubiquitination	Tagging a protein with a chain of ubiquitin molecules. In this course it is the tag that sends HIF-1alpha to the proteasome once VHL has recognised it. Not every polyubiquitination means degradation, so do not treat it as a general rule.	Lect 5
Positive-feedback loop	When the product of a reaction switches its own reaction on, so the response amplifies quickly.	Lect 5
Post-translational modification	A chemical change made to a protein after the chain has been built.	Ch 6, Lect 5
pRb	The protein that holds the cell cycle at G1 until the cell is ready to copy its DNA.	Lect 2
Predictive biomarker	A marker that predicts how a patient will respond to a therapy.	Lect 1
Prognostic biomarker	Tells you how the disease will go, whatever treatment you give.	Lect 1
Pre-initiation complex	The assembled set of general transcription factors and RNA polymerase that sits on a promoter before transcription starts.	Lect 5
Pre-mRNA	The first RNA copy of a gene, before splicing has removed the introns.	Lect 5
Primary structure	The plain sequence of amino acids in a protein, read from one end to the other.	Ch 3
Primase	The enzyme that lays down the short RNA primer that DNA polymerase must start from.	Lect 4
Prokaryote	A cell with no nucleus. Bacteria and archaea are prokaryotes.	Ch 5, Lect 5
Proliferative signal	A message that tells a cell to start copying its DNA.	Lect 2
Promoter	The DNA word that tells RNA polymerase where a gene starts.	Lect 5
Proofreading	The back-checking activity of DNA polymerase, which removes wrongly inserted bases as it copies.	Lect 4
Propidium iodide (PI)	A dye that binds DNA. Staining with it and running a flow cytometer tells you how much DNA each cell has, and so which phase it is in.	Ch 5, Lect 2
Protease	An enzyme that cuts proteins.	Ch 3
Proteasome	The protein-digesting machine that breaks down proteins the cell has tagged for destruction.	Ch 3
Protein	A folded machine made from a bead necklace of amino acids.	Ch 3, Ch 6
Protein domain	A stretch of a protein that folds on its own and has its own job.	Ch 3, Lect 1
Protein kinase	A kinase. An enzyme that adds phosphate groups to other proteins.	Lect 1

Term (technical, English)	Plain-English meaning	Where it appears
Proteolytic cleavage	Cutting a protein in two with an enzyme. It is how Notch and many receptors are switched on.	Lect 5
Proteome	The complete set of proteins a cell or a tissue makes.	Ch 3, Lect 1
Proteomics	The study of the whole set of proteins in a cell or a tissue.	Ch 3, Lect 1
Proton	A particle with a positive charge, sitting in the nucleus. The number of protons is what makes an element what it is.	Ch 1
P-selectin	One of the selectins that makes white blood cells roll along the wall of a blood vessel.	Lect 2
P-value	How surprising the data would be if the null hypothesis were true. It is not the chance that the result is a fluke. Below 0.05 is the usual line.	Lect 1, Lect 3
Pyrogen	A substance that raises body temperature and causes fever.	Lect 1
Pyruvate	The molecule that glucose is broken down into. It goes to lactate in a cancer cell, or into the mitochondrion in a normal one.	Ch 6, Lect 1
q-arm	The long arm of a chromosome, below the centromere.	Ch 4, Lect 3
qPCR	Quantitative PCR. It measures how much of a DNA sequence is present in a sample.	Lect 4
Quaternary structure	The assembly of several protein chains into one working complex.	Ch 3
Quiescent	Resting and not dividing, but still alive and ready to re-enter the cycle. This is G0.	Ch 5, Lect 2
Reactive oxygen species (ROS)	Chemically reactive molecules made by the cell's own metabolism that can damage DNA.	Ch 1, Lect 3
Receptor	A protein that catches one specific signal molecule and changes shape. Usually sits in the membrane.	Ch 5, Ch 6
Receptor occupancy	The fraction of receptors that must be bound by drug before there is a clinical effect.	Lect 2, Lect 5
Receptor tyrosine kinase (RTK)	A membrane receptor with its own kinase activity. Trastuzumab works by blocking HER2 dimerisation.	Lect 2, Lect 5
Recombinant protein	A protein made by a host cell that has been given a human gene.	Lect 5
Reduction	Gaining electrons, or losing oxygen. It stores energy.	Ch 1
Regulatory gene	A gene that makes the switch protein that controls an operon, such as lacI.	Lect 5
Replication bubble	The opened bubble of DNA where copying starts, with a fork running in each direction.	Ch 4
Reporter assay	The experiment that measures promoter strength by reading out a reporter gene.	Lect 5
Reporter gene	A gene put in place of the real one to measure how active a promoter is.	Lect 5
Repressible system	A system that stays switched on until its own end product switches it off.	Lect 5
Repressor	A protein that sits on DNA and blocks transcription.	Lect 5
Reverse transcriptase	An enzyme that copies RNA into DNA. Telomerase is one, and so are the retroviruses.	Lect 3, Lect 4
R-group (side chain)	The variable side piece of an amino acid. It is what decides how the chain folds.	Ch 3
Ribose	The sugar in RNA.	Ch 4, Ch 2
Ribosome	The machine that reads mRNA and joins amino acids to build a protein.	Ch 5, Ch 6
RISC	The RNA-induced silencing complex. It is the machine that uses a microRNA to switch a target gene off.	Lect 1, Lect 5

Term (technical, English)	Plain-English meaning	Where it appears
RNA interference (RNAi)	Using small RNA molecules to switch a gene off. The older siRNA and shRNA methods.	Lect 2
RNA polymerase	The enzyme that copies a gene into RNA.	Lect 5
RNA polymerase II	The main enzyme that copies protein-coding genes into RNA.	Lect 5
RNA (ribonucleic acid)	A family of molecules, including mRNA, tRNA and rRNA, that read and carry out the instructions in DNA.	Ch 4
RNA vaccine	A vaccine that delivers mRNA, which the patient's own ribosomes translate into antigen.	Lect 5
RNA-binding protein (RBP)	A protein that grabs a particular RNA and changes how stable it is or how well it is read.	Lect 5
Rough ER	The part of the endoplasmic reticulum that has ribosomes stuck to it, where new proteins are made and folded.	Ch 5, Lect 2
rRNA	Ribosomal RNA. It makes up most of the ribosome and lines up the mRNA for reading.	Ch 4
RXR	The partner receptor that forms a heterodimer with Type II nuclear receptors.	Lect 5
S phase	The stage of the cell cycle in which the DNA is copied.	Ch 5, Lect 2
S-adenosyl methionine (SAMe)	The molecule that carries the methyl group when a gene is methylated.	Lect 3
Sanger sequencing	The older, gold-standard method of reading DNA one base at a time from a chain of coloured peaks.	Lect 3
SAV	The protein that carries the signal from MST to LATS in the Hippo pathway.	Lect 5
Scaffold	A molecule that holds two or more proteins together at the right spot, so they can work as a team.	Ch 3, Lect 5
scFv	The smallest useful piece of an antibody, one variable region from each chain. It is the standard warhead in a CAR.	Lect 2
SCID (severe combined immunodeficiency)	The clinical result of being unable to build a working antibody and T-cell repertoire.	Lect 2, Lect 4
SDS-PAGE	The standard gel test that sorts proteins by their size. Four bands means four sizes, not four identities.	Ch 3, Lect 1
Second messenger	A small molecule inside the cell that carries a signal onward, such as cAMP.	Lect 5
Secondary structure	The local folding of a chain into an alpha helix or a beta sheet, held by hydrogen bonds.	Ch 3
Secretory pathway	The route rough ER, Golgi, vesicle, plasma membrane, and out of the cell.	Ch 5, Lect 5
Seed sequence	The short stretch of a microRNA, positions 2 to 8, that does the target matching.	Lect 5
Selectin	A cell-surface protein that makes white blood cells roll along the wall of a blood vessel.	Lect 2
Selective Estrogen Receptor Modulator (SERM)	Tamoxifen. It blocks estrogen in breast tissue but weakly activates it in the endometrium.	Lect 5
Semi-conservative replication	The rule that every new DNA double helix ends up with one old strand and one new one.	Ch 4
Senescence	The state a cell enters when it stops dividing for good but does not die.	Lect 3
Sensitivity	Of all the people who truly have the disease, the percentage the test correctly picks out.	Lect 1

Term (technical, English)	Plain-English meaning	Where it appears
Septic shock	The collapse that happens when sepsis overwhelms the body's own inflammation response.	Ch 6, Lect 1
Serial deletion	Deleting a promoter piece by piece and measuring what is left, to find which piece did the work.	Lect 5
Serum starvation	Dropping the growth factor in the medium from 10 per cent to 0.5 to 1 per cent to park cells in G1, so they can be studied in step.	Ch 5, Lect 2
Signal peptide	A short address at the start of a protein that routes it into the ER and so into the secretory pathway.	Ch 5, Ch 6
Signal transduction	The relay of a message from the cell surface to somewhere inside the cell.	Ch 6, Lect 5
Silencer	A DNA sequence that reduces transcription, wherever it sits relative to the gene.	Lect 5
Sister chromatid	One of the two identical copies of a chromosome after the DNA has been copied.	Ch 5, Lect 3
Small molecule	A drug-sized chemical, usually made by chemistry rather than by biology.	Ch 2, Lect 4
Smooth ER	The part of the endoplasmic reticulum that makes lipids rather than proteins.	Ch 5
SN-38	A topoisomerase-I inhibitor used as the cytotoxic payload in an antibody-drug conjugate. It is also the active metabolite of the small-molecule drug irinotecan.	Lect 1
snRNP	A small nuclear RNA plus its proteins. Several of these make up the spliceosome.	Lect 5
Solvent	The liquid that something else dissolves in. Water is the cell's solvent.	Ch 1
Specificity	Of all the people who truly do not have the disease, the percentage the test correctly clears.	Lect 1
Spliceosome	The machine that cuts out introns and joins exons together.	Lect 5
Splicing	The cutting out of introns and joining together of exons to make a mature mRNA.	Lect 5
Splicing factor	A protein that helps choose which exons are joined together.	Lect 5
Starch	The plant storage form of glucose.	Ch 2
Start codon	AUG, which starts translation and codes for methionine.	Ch 6, Lect 1
STAT3	A transcription factor. On his Lecture 3 slide, HDAC6 activates STAT3, STAT3 binds the PD-L1 promoter, and PD-L1 expression goes up - which is what makes the anti-PD-L1 antibody work.	Lect 3
Stem cell	A cell that can copy itself and also turn into more specialised cells.	Lect 2
Stop codon	One of UAA, UAG or UGA. It ends translation.	Ch 6, Lect 1
Strand	One of the two long chains that make up a DNA double helix.	Ch 4
Structural gene	A gene that makes a product the cell needs, such as an enzyme for using lactose.	Lect 5
Structural variant	A large rearrangement of the DNA, such as a deletion, duplication, inversion or translocation.	Lect 3
Substitution	A structural variant in which a segment of one chromosome is removed and inserted into another chromosome (his Lecture 3 slide 40 classification). Keep it distinct from the mutation sense of "substitution", which is a single base swapped for another - the sense he uses when talking about a single-nucleotide substitution creating a novel splice site.	Lect 3
Substrate	The molecule an enzyme acts on.	Ch 1, Ch 3
Subunit	One chain of a protein that is built from several chains working together.	Ch 3

Term (technical, English)	Plain-English meaning	Where it appears
Sugar (carbohydrate)	One of the four biomolecule classes. The main fuel, and also structural material.	Ch 2, Lect 1
Sugar-phosphate backbone	The structural framework of a DNA strand, made of alternating sugar and phosphate groups.	Ch 4
synNotch	An engineered version of Notch that lets one cell's presence switch a gene on in its neighbour.	Lect 5
Synthetic biology	The engineering of new biological parts and switches. synNotch and HIF-controlled CARs come from it.	Lect 5
Tamoxifen	The drug that blocks estrogen in breast tissue but weakly activates it in the endometrium.	Lect 5
TATA box	The DNA word about 30 bases upstream of a gene where the transcription machinery assembles.	Lect 5
TBP (TATA-box-binding protein)	The part of TFIID that recognises and sits on the TATA box.	Lect 5
TCA cycle	The cyclic set of reactions in the mitochondrion that extracts energy from fuel and loads it into NADH and FADH2.	Ch 5, Ch 6, Lect 2
T-cell	A lymphocyte of the immune system. There are three kinds: CD4+ helper cells, CD8+ cytotoxic cells (the ones that actually kill), and Tregs, which act as the brakes. A CAR-T cell is a T cell that has been engineered.	Lect 2, Lect 4
TCR (T-cell receptor)	The protein on a T-cell that recognises a foreign fragment displayed by another cell.	Lect 4
TEAD	The DNA-binding protein that YAP works with in the nucleus to switch growth genes on.	Ch 6, Lect 5
Telomerase	The unusual enzyme that can rebuild chromosome ends. Most cancer cells switch it back on.	Ch 4, Lect 3
Telomere	The repetitive DNA cap at the end of a chromosome. It protects the end and shortens with each division.	Lect 3, Lect 4
Template strand	The DNA strand that is read when a fresh copy is being made.	Ch 4, Lect 1
Teratoma assay	The gold-standard test that a reprogrammed cell is truly pluripotent: inject it into a mouse and look for tissue from all three germ layers.	Lect 2
Tertiary structure	The overall three-dimensional fold of one protein chain.	Ch 3
TET	The enzyme family that removes methylation marks by oxidising methylated cytosine.	Lect 3
TFIID	The first general transcription factor to arrive at the promoter. It contains the TATA-box-binding protein.	Lect 5
TFIIH	The general transcription factor that opens the DNA and then starts RNA polymerase II moving.	Lect 5
TGFbeta	A cytokine that the tumour uses to suppress the immune attack around it.	Lect 2
Therapeutic window	The gap between the smallest dose that works and the largest dose that is still tolerable. A good drug has a wide one.	Lect 4
Threshold effect	Disease appears only when the proportion of mutant mitochondrial DNA rises above what that tissue can tolerate.	Ch 5, Lect 2
TINCR	A long non-coding RNA that base-pairs with an mRNA and binds Staufen 1 to control how stable that message is.	Lect 5
TLR4	The cell-surface receptor that detects LPS. It is the first step of the sepsis cascade.	Ch 6, Lect 1, Lect 5
TNF-alpha (TNF-alpha)	The cytokine that comes first in the NF-kappaB inflammatory response, peaking at 4 to 6 hours.	Lect 1, Lect 5

Term (technical, English)	Plain-English meaning	Where it appears
Topoisomerase	An enzyme that relieves the twisting strain ahead of a replication fork by cutting and rejoining DNA.	Lect 4
Toxic pre-expression	Making the recombinant protein before the host cell is ready, which can damage or kill the culture.	Lect 5
Trans face	The shipping side of the Golgi apparatus, where finished proteins are sorted into vesicles.	Ch 5
Trans-acting factor	A diffusible protein, such as a transcription factor, that can act on a gene sitting on a different chromosome.	Lect 5
Transcription	Copying a recipe out of the DNA into a working message.	Ch 6, Lect 5
Transcription factor	A protein that binds a specific DNA word and helps or blocks transcription there.	Lect 5
Transcription start site	The exact base at which a new RNA copy begins.	Lect 5
Transducer	A protein that converts one kind of signal into another. RAS is one.	Ch 4, Lect 1
Transforming principle	The name Griffith gave to whatever in dead bacteria could change live bacteria. It turned out to be DNA.	Ch 4, Lect 1
Translation	Reading the mRNA and building the protein that it describes.	Ch 6, Lect 1
Translocation	The swapping of a piece of one chromosome with a piece of another.	Ch 4, Lect 3
Transmembrane domain	The stretch of a protein that sits in the membrane. Remove it and the protein becomes soluble and can be measured in blood.	Lect 2, Lect 5
Transport vesicle	A small membrane bubble that carries cargo from one compartment to the next.	Ch 5
Trastuzumab (Herceptin)	The anti-HER2 antibody used in HER2-positive breast and gastric cancer.	Lect 2, Lect 5
Tregs	Regulatory T-cells. They are the brakes on the immune system.	Lect 2
Triglyceride	Three fatty acids joined to a glycerol. The body's way of storing fuel.	Ch 2, Lect 1
Trisomy	Three copies of a chromosome instead of two, written as 2n plus 1. Trisomy 21 is Down syndrome.	Lect 3
Trisomy 21	Down syndrome. Three copies of chromosome 21 instead of two.	Lect 3
tRNA (transfer RNA)	The adaptor molecule that carries one amino acid to the ribosome and reads the codon for it.	Ch 4
Trop2	A cell-surface protein used as an antibody target in lung and breast cancer.	Lect 1
Trp repressor	The protein that blocks the trp operon when tryptophan binds it.	Lect 5
Trp (tryptophan) operon	The mirror image of the lac operon: when tryptophan is plentiful it switches its own synthesis off.	Lect 5
Tumor microenvironment	The whole neighbourhood of a tumour: cancer cells, immune cells, fibroblasts and matrix together.	Lect 2
Tumor suppressor	A gene whose normal job is to restrain division. Lose it and the cell runs away.	Ch 4, Lect 3
Turner syndrome (45,X)	One X chromosome in a female instead of two. One X was lost during meiosis.	Lect 3
Type I nuclear receptor	A receptor held in the cytoplasm by a chaperone until its ligand arrives, such as the estrogen receptor.	Lect 5
Type II nuclear receptor	A receptor already bound to DNA with co-repressors until its ligand arrives, such as the thyroid hormone receptor.	Lect 5
Ubiquitin	A small protein that is chained onto another protein to mark it for destruction.	Lect 5

Term (technical, English)	Plain-English meaning	Where it appears
Undraggable target	A target that no drug can currently hit. K-Ras used to be the famous example.	Lect 1
Unsaturated fatty acid	A fatty acid with one or more double bonds in its chain, so it is kinked and does not pack tight.	Ch 2
Untranslated region (UTR)	The part of an mRNA before the start codon or after the stop codon. It carries regulatory information.	Lect 5
Variable region	The tip of an antibody arm, which differs from antibody to antibody and does the grabbing.	Ch 3
VCAM-1	An adhesion molecule that lets a white blood cell grip the wall of a blood vessel firmly.	Lect 2
V(D)J recombination	The rearrangement of the V, D and J gene segments that builds antibody and T-cell receptor diversity.	Lect 2, Lect 4
VEGF	The blood-vessel growth factor that HIF-1alpha switches on. It is how a tumour makes its own blood supply.	Lect 5
Vesicle	A small membrane-bound bubble that carries cargo between compartments.	Ch 5
VHL	The protein that recognises the marked HIF-1alpha and hands it to the proteasome.	Lect 5
Vimentin	A structural protein inside cells that leaks into blood in liver cancer. Dr. Wong's worked biomarker example.	Ch 3, Lect 1
VIP (vasoactive intestinal peptide)	A neuropeptide that can switch T cells off and so help a tumour hide. It is background knowledge. Note that VIP > 1.0 on his metabolomics slide is something different - variable importance in projection, a feature-scoring threshold. A student looking at his slides will hit the metabolomics one first.	Background
Warburg effect	Cancer cells ferment glucose to lactate even when oxygen is available, because they want the raw material.	Ch 6, Lect 1, Lect 2
Water	The solvent of the cell. Its unusual shape gives it a high melting point and lets it dissolve many molecules.	Ch 1
Watson, Crick and Franklin	The three who built the double-helix model, using Rosalind Franklin's X-ray images.	Ch 4, Lect 1
Western blot	The standard test that shows whether a particular protein is present in a sample.	Ch 3, Lect 1
Wild type	The normal, unmutated version of a gene or of its product.	Ch 4, Lect 1
XIST	The long non-coding RNA that silences one whole X chromosome. It is the classic lncRNA.	Lect 5
X-linked dominant	An inheritance pattern carried on the X chromosome that appears in every generation.	Lect 2
X-linked recessive	An inheritance pattern carried on the X chromosome that mainly affects males.	Lect 2
X-ray crystallography	The method that works out the exact three-dimensional atomic shape of a protein.	Ch 3
Yamanaka factors	Oct3/4, Sox2, Klf4 and c-Myc. The four factors that reprogram a fibroblast into an iPSC.	Lect 2
YAP	The protein at the end of the Hippo pathway. Unphosphorylated, it enters the nucleus and drives growth.	Ch 6, Lect 1, Lect 5
ZFN	Zinc finger nuclease. An older, protein-based way of cutting DNA at a chosen place.	Lect 2

Term (technical, English)	Plain-English meaning	Where it appears
Zinc finger	The small folded shape, held together by a zinc ion, that lets a DNA-binding domain grip DNA.	Lect 5

13.2 The five questions he named as exam-type

The five he named outright are 13.2.1 to 13.2.5. **13.2.6 is a Tier 2 concept** - he says you must understand it, and it is case 16 in the scenario bank, but it is not one of the five he named as exam-type.

These are the five highest-confidence items in the whole course. He said, in his own words, that these are the question types he puts in exams. Learn the chain in each one, not just the answer.

13.2.1 Reading a Kaplan-Meier curve - Lecture 1

> **Lecture 1**
> *"And I ask this single question every, every time when I talk about the KM plot, yeah. And only a single student give me a good answer last year, yeah."*

> **Lecture 1**
> *"So again, this is the kind of the question that I would include in the exam, yeah. Just give you a taste about the midterm exam, yeah."*

> **Lecture 1**
> *"But the key is to let you understand that from a single plot, we can really, really find out many, many useful informations for the future study."*

The core skill is reading one curve and deducing early versus late involvement of a gene. The x-axis is months; the y-axis is the probability of survival or of progression-free survival. The curve starts at 1.0 and falls in steps. Flat segments mean no events in that interval. Downward steps mean events. Small tick marks on the curve are censored patients. A curve that drops steeply and early means the gene is associated with early involvement and a poor prognosis. A curve that drops late means late involvement. Two curves that separate mean the gene separates the two groups. Curves that cross or overlap mean the gene does not separate prognosis. The lecture's worked example is that the same target can be pro-tumour in one cancer and anti-tumour in another: a receptor tyrosine kinase predicts poor survival in pancreatic cancer but better survival in renal cell carcinoma. ERBB2 in the Human Protein Atlas runs in parallel, being prognostic in renal cancer, where it is favourable, and in endometrial and pancreatic cancer, where it is unfavourable, all at p less than 0.001.

13.2.2 Inhibiting the contractile ring gives you a binucleated cell - Lecture 2

> **Lecture 2**
> *"Again, like I - I remember that I asked this question during the midterm in the first year, yeah. I think only half student can answer it correctly, yeah... Actually quite a surprise to me, yeah. So again this is the type of the question that I would try to include in the midterm, yeah. So when you do the revision, try to think in this way, yeah."*

The question is what you see if you treat dividing cells with an inhibitor of the contractile ring, or knock out its expression. The contractile ring, made of actin and myosin, drives

cytokinesis, the division of the cytoplasm. **Mitosis**, the separation of the chromosomes, is independent of cytokinesis and proceeds normally. So the genetic material replicates and separates, but the cell cannot cleave into two daughter cells. The result is one cell with two nuclei, a binucleated cell, with 4N DNA content, which on a flow-cytometry DNA-content plot gives a 4N peak instead of separate 2N and 4N peaks. The whole point of the question is the distinction between mitosis and cytokinesis.

13.2.3 A defect in NHEJ gives you SCID - Lecture 4

> **Lecture 4**
> *"Therefore, if I ask you, if the people showing a defective in this non-homologous end joining, what's the outcome? Definitely the outcome is the severe immune deficiency, right? Because he cannot generate a sufficient amount of the antibody pool to recognize the thousand copies of the foreign antigen. Yeah, that's the clinical outcome of this genetic defect."*

The chain you must be able to draw: NHEJ is co-opted by V(D)J recombination to assemble the V, D and J gene segments, which generates antibody diversity in B-cells and T-cell receptor diversity in T-cells. A defect in NHEJ therefore means the repertoire cannot be built, and the clinical outcome is SCID, severe combined immunodeficiency. The segment copy numbers he gave are V of about 39 copies, D equal to 27 copies and J equal to 6 copies.

13.2.4 Reading a mitochondrial pedigree - Lecture 2

> **Lecture 2**
> *"So, this is you'll have to a 100% understand, right? Not only the mutation within the mitochondrial DNA can cause the mitochondrial disease. Some mitochondrial disease can actually result from the mutation within the nuclear DNA."*

> **Lecture 2**
> *"Just make sure you understand that different mutation can cause the mitochondrial diseases, and all this mutation can cause the different inheritance patterns when you analyze this pedigree tree, yeah."*

This is a pedigree-reading question, so it is a scenario question. The three hallmarks of maternal, mitochondrial, inheritance are that an affected mother passes the condition to all of her children, that an affected father passes it to none of his children because sperm mitochondria are destroyed after fertilisation, and that the condition appears in every generation. The twist that catches people is that mitochondrial disease is not always mtDNA. Mutations in nuclear genes encoding mitochondrial proteins cause the same phenotype with a completely different inheritance pattern, namely Mendelian rather than maternal. A pedigree of mitochondrial-looking disease can therefore be nuclear. The answer must mention both. The threshold effect, or heteroplasmy, is the other half: the presence of mutant mtDNA does not guarantee disease, because disease appears only when the proportion of mutant mtDNA exceeds a tissue-specific threshold. High-energy tissues such as contracting skeletal muscle have a low threshold and are affected first; low-demand tissues such as hearing have a high threshold.

13.2.5 Telling meiosis I from meiosis II - Lecture 3

> **Lecture 3**
> *"And you might need to distinguish the M1 or the M2 failure, because if there's a M1 failure, there's a formation of the germ cell without a losing copy of the chromosome after the M1..."*

> **Lecture 3**
> *"And I think you need to distinguish these two situations here, yeah."*

	Meiosis I (M1)	Meiosis II (M2)
What fails to separate	Homologous chromosomes	Sister chromatids
Gametes from one meiosis	2 normal and 2 abnormal, all abnormal of the same type	1 normal and 3 abnormal, two of one type and one of another
Recombination	Occurs between homologues	Does not occur

The related point about inversions is that an inversion changes no genetic material and no gene dosage, and yet it can change expression drastically by moving a gene away from its enhancer, or by putting an enhancer next to a different promoter. As he says, "you need to understand that some of these abnormalities can cause the loss of the genetic materials, but some does not."

13.2.6 Lecture 5's own item - the lac operon (Tier 2, not one of the five)

The lac operon is one of the four Lecture 5 topics in the exam-focus document's Tier 2 list, and its entry heads "The lac operon - two conditions, not one". It is also scenario case 16 in the case-study bank, "Reading the lac operon logic correctly". The four things the marker looks for are the two conditions, allolactose, cAMP-CAP and IPTG.

The two conditions, again. Strong transcription needs lactose present, so that allolactose binds the Lac repressor and it can no longer block the operator, and it needs glucose low, so that cAMP rises and the cAMP-CAP complex binds the promoter and recruits the copying machinery. Both must be true at once. The trp operon is the mirror image: tryptophan binds the Trp repressor as a corepressor and shuts down tryptophan biosynthesis. IPTG is the artificial, non-metabolised inducer used in E. coli recombinant systems. And CHO cells, the mammalian cells used when glycosylation is required, need codon optimisation, in his words: "So remember that when you use the CHO cell, you need to do the codon optimization."

13.3 The key numbers

Every number in this course is exam content. Each row below gives the number exactly as Dr. Wong gave it, says in plain English what it means, and names the lecture he said it in.

The exam itself

Number	What it refers to	What it actually means	Where he said it
30 %	The midterm's share of the final grade	The midterm is not a formality. It is a third of the course mark.	Lect 1
40 %	The final exam's share	The final paper is worth slightly more than the midterm.	Lect 1
30 %	The share of two "super-easy" assignments	He called them 'super-easy'. Together they carry more of the course grade than the midterm, so do not skip them.	Lect 1
20 MCQs (20 marks)	The multiple-choice part	Pure recall. This is the passing floor, not the place to win marks.	Lect 5
2 of 3 short questions (10 marks)	The scenario part	The place the marks are. He said the short questions "emphasize on data analysis and also the scenarios questions".	Lect 5
1.5 hours (90 minutes)	The length of the midterm	Budget roughly 55 minutes for the MCQs, 30 for the short question, 5 to review.	Lect 5
25 questions, 25 %	The earlier statement, from Lecture 4	Conflicting with the line above. Lecture 5 is later and is the version written on his slide, so plan for the Lecture 5 version. The ratio is the same either way.	Lect 4

Gene expression: the lac operon and recombinant production

Number	What it refers to	What it actually means	Where he said it
2 conditions	Lactose present and glucose low	Strong transcription of the lac genes needs both. Naming only lactose is the classic wrong answer.	Lect 5
3 genes	lacZ, lacY, lacA	The three lactose-use genes, copied together as one polycistronic message.	Lect 5
5 genes	trpE, trpD, trpC, trpB, trpA	The five tryptophan-biosynthesis genes, which are switched off when tryptophan is plentiful.	Lect 5
100 micromolar	The optimal IPTG concentration on his slide	The tested range was 0 to 1000 micromolar, and the integrated peak area peaked at 100.	Lect 5
200 nucleotides	The length above which an RNA is a long non-coding RNA	Below that it is a small RNA. Above it, it is an lncRNA, which makes no protein.	Lect 5
22 nucleotides	The length of a microRNA	Short enough to bind a seed sequence by partial base-pairing.	Lect 5
425 and 250 base pairs	The two band sizes in the liver-cancer splicing gel	425 is the full-length exon 5-6-7-8 form, 250 is the exon 7-skipped form.	Lect 5
23 and 19	The two patient groups in the liver-cancer survival plot	23 without alternative splicing, 19 with it, P less than 0.001 on the log-rank test.	Lect 5
220 and 97	The microRNA screen in liver cancer	220 microRNAs measured in 97 paired tumour and adjacent non-tumour tissues.	Lect 5
14q32.2	The over-expressed microRNA cluster	Correlated with log10 of AFP at P equal to 4.23 times 10 to the minus 10, and with survival at P equal to 0.009 and Chi squared equal to 6.81.	Lect 5

Nuclear receptors, tamoxifen and receptor occupancy

Number	What it refers to	What it actually means	Where he said it
1 %	Receptor occupancy for a typical antibody	The drug only has to fill one receptor in every hundred to work.	Lect 5
2 %	Receptor occupancy for Herceptin in breast cancer	Two in every hundred. This is why a low dose still works.	Lect 5
10 %	Receptor occupancy for Herceptin in gastric cancer	Ten in every hundred, so gastric disease needs a different dosing strategy from the same drug.	Lect 5
3 domains	N-terminal domain, DNA-binding domain, ligand-binding domain	The variable regulatory end, the conserved DNA-recognising middle, and the ligand-catching end.	Lect 5
4 months	The work a postdoc lost by leaving off a signal peptide	The reason Dr. Wong says signal peptides are "really, really important for your study".	Lect 2

The NF-kappaB cytokine clock

Number	What it refers to	What it actually means	Where he said it
4 to 6 hours	When TNF-alpha peaks	It is the first cytokine of the response, and it peaks six hours after the signal.	Lect 5
IL-1beta	The second cytokine	It comes after TNF-alpha and before IFN-gamma.	Lect 5
IFN-gamma	The third cytokine	It comes after IL-1beta.	Lect 5
IL-2 and IL-15	The fourth cytokines	They come last in the sequence and drive T-cell and natural killer cell proliferation.	Lect 5

He also put this exact sequence on his not-examined list: "don't worry about all these cytokines. I'm just want to show you that different responses can be observed in different times." So know the shape, the time course and the positive-feedback loop, not the list.

Cis versus trans regulation

Number	What it refers to	What it actually means	Where he said it
Chromosome 1 and chromosome 13	A repressor encoded on chromosome 1 acting on a target gene on chromosome 13	The clearest possible example of trans-regulation: the regulator and the regulated gene are on different chromosomes.	Lect 5

Cancer genetics, biomarkers and statistics

Number	What it refers to	What it actually means	Where he said it
15 %	The share of colorectal cancers that are MSI-high	Mismatch-repair failure makes a tumour hot and immunotherapy-responsive, and it happens in about one colorectal cancer in seven.	Lect 4
39.6	Odds ratio for CIMP-high versus CIMP-0	With a 95 % confidence interval of 20.6 to 76.1 and p less than 0.0001. A very strong association with methylation-defined colorectal cancer.	Lect 3
12.7	Odds ratio for detected methylation	With a confidence interval of 7.84 to 20.7 and p less than 0.0001.	Lect 3
V of about 39, D equal to 27, J equal to 6	V(D)J gene segment copies	How much raw material the immune system has to build antibody and T-cell receptor diversity from.	Lect 4
3 or more	HBV integration sites	Three or more integration sites is associated with significantly poorer survival.	Lect 2
Top 10 %	The longest telomeres in hepatocellular carcinoma	The longest 10 % of telomeres had worse recurrence-free survival, which is counter-intuitive and worth remembering.	Lect 4
ERBB2 prognostic in renal (favourable), endometrial (unfavourable), pancreatic (unfavourable), all p < 0.001	Human Protein Atlas	The same target can point in opposite directions in different tumour types, so never carry the reading across tumour types.	Lect 1
13.2 versus 6.7 months	Median overall survival with daraxonrasib versus chemotherapy	The K-Ras G12C inhibitor, which showed that K-Ras is no longer undruggable.	Lect 1
176.0 plus or minus 61.3 ng/mL	Serum vimentin in controls	The baseline against which the tumour groups are read.	Lect 1
271.5 plus or minus 165.9 ng/mL	Serum vimentin in hepatocellular carcinoma at or below 2 cm	The smallest tumours gave the biggest rise.	Lect 1
231.3 plus or minus 132.3 ng/mL	Serum vimentin in hepatocellular carcinoma above 2 cm	Slightly lower than the small-tumour group.	Lect 1
2 of 12, p equal to 0.01	Anti-TERT antisense oligonucleotide, 3 weeks pretreatment, in the Lecture 4 mouse experiment	Tumour-take rate 2 of 12 mice, against 9 of 12 untreated (Fisher's p = 0.01). It is a result about telomerase, not about vimentin.	Lect 4
0.981	The AUC of the miR-15b plus miR-130b classifier	With sensitivity 98.25 %, specificity 91.53 % and accuracy 94.83 %. Two weak signals make one strong test.	Lect 1

Chemistry, molecules and methods

Number	What it refers to	What it actually means	Where he said it
3 and 2	Hydrogen bonds in G-C and A-T base pairs	G-C-rich DNA needs a higher melting temperature, which is why GC content drives primer design.	Lect 1, Lect 3
About 40 degrees	The melting-temperature example in his notes - **the source is inconsistent here**	His slide gives only the direction: more GC content means a higher Tm. The notes also carry a '~40 deg C drop' and a '60 deg C vs 55 deg C' board example, and those two do not reconcile. Learn the direction, not a number.	Lect 3
2.8 angstrom	The nucleosome crystal structure	Below 2.0 angstrom he calls "very excellent".	Lect 3
About 10 nm	The diameter of a nucleosome	Which is why DNA wrapped around histones looks like beads on a string.	Lect 3
3 to 20 kb	The TTAGGG repeat block at a telomere	The repetitive cap at the end of a chromosome.	Lect 3
100 to 300 kb	The subtelomeric repetitive region	Just inside the telomere.	Lect 3
23 pairs	Human chromosome pairs	Plus trisomy 21 and Turner syndrome, 45,X.	Lect 3
0.5 to 1 %	The serum level used to starve cells	Serum starvation arrests cells in G1 so that they can be studied in step.	Lect 2, Lect 4

Replication, repair and the Hippo pathway

Number	What it refers to	What it actually means	Where he said it
5 prime to 3 prime	The direction nucleotides are added	Elongation always goes this way.	Lect 4
3 prime to 5 prime	The direction proofreading and excision go	It is the polymerase's exonuclease activity that checks its own work.	Lect 4
Thousands	The number of replication origins	Each opens a bubble with a fork running in each direction.	Lect 4
50 to 200 base pairs	Telomere loss per somatic cell division	This is the end-replication problem, and it is what the Hayflick limit counts.	Lect 3
About 90 %	The share of cancers that up-regulate telomerase	Telomerase is how a cancer cell divides forever.	Lect 3
2	The kinases MST1 and LATS1 in the Hippo cascade	MST1 activates LATS1, which phosphorylates YAP and keeps it out of the nucleus.	Lect 1, Lect 5

Clinical trials

Number	What it refers to	What it actually means	Where he said it
About US$200,000 per patient	The cost of running a clinical trial, per patient enrolled	Clinical trials are expensive, which is why biomarker-driven enrolment and surrogate endpoints such as CEA matter.	Lect 2, Lect 4
Phase 1A, 1B, 2 and 3	The trial phases	1A is dose escalation to find the safe dose, 1B is expansion, and 2 and 3 test efficacy.	Lect 2, Lect 4

Clinical and manufacturing

Number	What it refers to	What it actually means	Where he said it
USD 370,000 to 475,000	The quoted cost range for one course of therapy	The number that makes platform choice a real clinical issue.	Lect 2
5 or more	Circulating tumour cells, the CTC threshold	More than this is a meaningful signal.	Lect 2
11.3 % and 41.6 %	The CD3+CD4+ quadrant percentages in the Lecture 2 flow-cytometry gating example	They are the fraction of cells in one quadrant of one FACS plot - not population statistics, and not part of the circulating-tumour-cell study.	Lect 2
About 50 %	The share of studied solid tumours with hypoxic tissue	It is the reason an HRE-controlled CAR is a workable idea.	Lect 5
Under 0.2 within 2 to 3 hours	How fast the HIF-controlled CAR falls when oxygen returns	The constitutive CAR stays at 0.8 to 1.0. The switch is genuinely reversible.	Lect 5

13.4 Will NOT be examined

Each of these he explicitly put out of scope. Read this section carefully, because knowing what to skip is part of knowing the course. But read it honestly: in every one of these he still expects the concept. Not memorising the glycosylase chemistry does not mean not knowing what base excision repair does.

Excluded	His words	Lecture
Chemical structures and metabolic pathway details	"Please don't memorize all these chemical structure... They would not be included in my exam."	1
The amino-acid R-group table	"don't memorize this... definitely not include in the exam"	1
FISH protocol parameters, denaturation and temperature	"Definitely not include in the exam. I will not ask you about what's the denaturations, what's the temperature"	2
Specific cyclin and cyclin-dependent kinase pairings	"you don't need to memorize all this cyclin or cyclin-dependent kinase, CDK"	2
The detailed signalling cascades on the slide	"I don't memorize all this thing, I just highlight here in case some student are really interested in molecular biology"	2
ADC structures and dual-payload ADCs	"all this would not be included in the midterm. I'm just want you to know more"	4
The pharma target nomination and druggability workflow	"it's not included in exam, but I think it's really more interesting"	4
DNA glycosylase and abasic site chemistry	"I can ensure that they will not be included in the coming midterm"	4
Copy number variation mechanisms in depth	"I'm just give you a taste... don't be scared with all these mechanism"	3
Electron-microscopy chromatin images	"don't memorize all these figures, yeah. It's not necessary"	3
The exact cytokine sequence	"don't worry about all these cytokines. I'm just want to show you that different responses can be observed in different times."	5

Two of these are worth reading twice. The exact cytokine sequence is a Lecture 5 exclusion, so do not memorise the list; know the time course and the positive-feedback loop. The other is the chemical-structure exclusion, which covers the kind of detail that a beginner might assume is central to a molecular biology course.

13.5 The 19 distractor pairs

These are the classic "which one is different?" structures. He returns to each pair, and each pair has exactly one discriminating fact. Learn the fact in the right-hand column, not just the two names.

Pair	The one fact that tells them apart
BER versus NER	BER fixes a single damaged base; NER cuts out a bulky multi-nucleotide lesion such as a UV dimer.
MMR versus HR	MMR corrects replication mismatches and small indels; HR is high-fidelity repair of a broken chromosome.
HR versus NHEJ	HR uses a template and works only in S and G2; NHEJ needs no template, works in all phases, and is error-prone.
Cis versus trans	Cis acts on the same DNA molecule it sits on; trans is a diffusible factor encoded elsewhere, and can reach another chromosome.
IgG1 versus IgG4	IgG1 kills by ADCC; IgG4 does not, and it lasts longer in the blood.
Autologous versus allogeneic CAR-T	Autologous is taken from the patient; allogeneic comes from a donor or from iPSCs.
In vivo versus ex vivo CAR-T	In vivo means the CAR is delivered into the body as mRNA in lipid nanoparticles; ex vivo means cells are taken out, engineered, and put back.
Driver versus passenger	A driver gives the tumour a growth advantage; a passenger is merely carried along.
Euchromatin versus heterochromatin	Euchromatin is open and being transcribed; heterochromatin is condensed and silent.
DBD versus LBD	The DBD binds DNA; the LBD binds the ligand and changes shape.
De novo versus maintenance methylation	De novo establishes a new mark; maintenance copies the mark onto the daughter strand after replication.
Predictive versus prognostic biomarker	Predictive predicts the response to therapy; prognostic predicts the outcome of the disease.
Dimerisation versus monomer	Monomers bind DNA poorly; dimers bind strongly.
5 prime to 3 prime versus 3 prime to 5 prime	Elongation goes 5 prime to 3 prime; proofreading and excision go 3 prime to 5 prime.
M1 versus M2 non-disjunction	M1 fails to separate homologues and gives 2 normal and 2 abnormal gametes; M2 fails to separate sister chromatids and gives 1 normal and 3 abnormal.
mtDNA versus nuclear DNA	mtDNA gives maternal inheritance; nuclear DNA gives Mendelian inheritance and a different threshold logic.
ADCC versus CDC	ADCC is antibody-dependent cellular cytotoxicity; CDC is complement-dependent cytotoxicity.
Signal peptide versus NTD	The signal peptide routes the protein into the ER and the secretory pathway; the N-terminal domain is the regulatory end of a nuclear receptor.
Exon 1 versus Exon 2 methylation	Exon 2 is the diagnostically informative region.

13.6 A self-test

Twenty questions, four from each of the five lectures. Answer in writing, in English, in full sentences, before you look at the key. The key below gives the reasoning, not just the fact, because the marks are for reasoning.

Q1. (Lecture 1) A Kaplan-Meier curve for a new biomarker starts at 1.0, falls to about 0.5 within the first six months, and then stays flat. The p-value against the comparison group is 0.01. What do you tell the clinician?

Q2. (Lecture 1) Serum vimentin in a patient with a 1.5 cm liver lesion is 271.5 plus or minus 165.9 ng/mL, and the control group is 176.0 plus or minus 61.3 ng/mL. Is this a positive test? What can you not conclude?

Q3. (Lecture 1) A colorectal cancer is found to carry a K-Ras mutation. The oncologist proposes cetuximab. What happens, and what would you do instead?

Q4. (Lecture 1) What does the Warburg effect tell you about a cancer cell's priorities, and why is it not a mistake by the tumour?

Q5. (Lecture 2) You treat dividing cells with an inhibitor of the contractile ring. Under the microscope you see one cell with two nuclei, and the flow-cytometry DNA-content plot shows a single 4N peak. Explain the result.

Q6. (Lecture 2) A pedigree shows a mother with a muscle disorder whose son and daughter are both affected, while her affected husband's children from a previous marriage are unaffected. What is the pattern, and what must you still check before you commit to it?

Q7. (Lecture 2) You want to know whether a protein sits on the plasma membrane or inside the cytoplasm. Which assay do you use, and what does it actually tell you?

Q8. (Lecture 2) A patient with HER2-amplified breast cancer is being considered for trastuzumab. Which test decides, and what does that test measure?

Q9. (Lecture 3) A patient's tumour shows promoter hypermethylation of a tumour-suppressor gene. What is the consequence, and what kind of drug might you expect to help?

Q10. (Lecture 3) A karyotype is normal but the tumour has a known activating point mutation. Why can the karyotype miss it, and what resolution is needed?

Q11. (Lecture 3) A chromosome has an inversion that changes no DNA and no gene dosage. How can it still change gene expression?

Q12. (Lecture 3) A tumour has a high number of copy number variations. How do you decide which of them are drivers and which are passengers?

Q13. (Lecture 4) A patient has a BRCA1 mutation and is being considered for a PARP inhibitor. Explain the logic, and explain what makes the therapeutic window wide.

Q14. (Lecture 4) A tumour is mismatch-repair deficient. What does that produce, and which therapy does it predict?

Q15. (Lecture 4) A gene-editing experiment needs a precise change at one exact base. Which repair pathway will do it, and in which cell-cycle phases can it work?

Q16. (Lecture 4) A child has severe combined immunodeficiency. The clinical scientist suspects a defect in non-homologous end joining. Explain how that defect produces the disease.

Q17. (Lecture 5) A culture of E. coli is on lactose and glucose. The lac genes are being transcribed slowly. A second culture has glucose and no lactose and transcribes nothing at all. Explain both, and say what happens to the second culture if you add lactose.

Q18. (Lecture 5) The same antibody is used to treat HER2-positive breast cancer and HER2-positive gastric cancer, but at different doses. Give the receptor occupancy numbers and explain the difference.

Q19. (Lecture 5) Why is tamoxifen an anti-tumour drug in breast tissue and a risk factor in the endometrium?

Q20. (Lecture 5) A single-nucleotide substitution in a liver-cancer gene creates a new splice site. Give the protein-level, immune-level and drug-level consequences.

Answer key

A1. The observation first. The curve falls from 1.0 to about 0.5 in the first six months and then stays flat, and the two groups are separated at p equal to 0.01. By his rule, a steep and early drop means early involvement and a poor prognosis for that group, so the marker is associated with events that happen early. The mechanism would be that the marker sits in, or upstream of, something that occurs before the event. The clinical meaning is that measuring this marker separates patients who will need different treatment intensity, and the flat segment after the drop means the early events are concentrated in the first six months. The action is to use the marker for early risk assessment and to name the limitation. As he says, the same target can behave in opposite directions in different tumour types, with ERBB2 favourable in renal cancer and unfavourable in endometrial and pancreatic cancer, all at p less than 0.001, so this reading must not be generalised past the tumour type it came from.

A2. The observation is that the value is above the control group: 271.5 plus or minus 165.9 in the hepatocellular carcinoma group at or below 2 cm, against 176.0 plus or minus 61.3 in controls. This patient's value is that <= 2 cm group's own value, so it cannot sit above that group - it is the group with the biggest rise of all. The mechanism is that the tumour releases vimentin, and the smallest tumours give the largest signal, which is what makes this promising for early detection. The clinical meaning is that a blood test could support surveillance, which matters because hepatocellular carcinoma is caught late - but it does not replace imaging and biopsy. It is the marker CEA, not serum vimentin, that the source gives the avoid-repeated-scans role. The action and the limit: you cannot call this a positive test from one number. The spread is wide at 165.9, the groups overlap, and the figure came from a cohort rather than from a diagnostic rule with a pre-set cut-off. A single elevated value needs a diagnostic threshold, a second test and imaging. The related result also stops you over-claiming: in Lecture 4 he showed an anti-TERT antisense oligonucleotide given for 3 weeks before tumour implantation cut the tumour-take rate to 2 of 12 mice, against 9 of 12 untreated, at p equal to 0.01. That is a result about telomerase and TERT, not about vimentin.

A3. The observation is that the tumour is K-Ras-mutant. The mechanism is that cetuximab is an anti-EGFR antibody, so it works by blocking the receptor at the top of the chain, EGFR, RAS, RAF, MEK, ERK. Wild-type K-Ras is only active when EGFR fires, so blocking EGFR switches the whole chain off. A mutant K-Ras is constitutively active, stuck on, sending growth signals whatever EGFR does, so blocking upstream cannot switch off a signal that is already on downstream and the drug fails. The clinical meaning is that K-Ras status is a predictive biomarker, so you must test before you treat, and a K-Ras-mutant patient should not be given cetuximab on the expectation of benefit. The action is to choose a different strategy, and K-Ras is no longer undruggable: daraxonrasib, a RAS inhibitor, gave a median overall survival of 13.2 months against 6.7 months for chemotherapy in previously treated metastatic pancreatic cancer. The limit is that this is a statement about cetuximab in colorectal cancer, not about whether K-Ras-mutant cancer is treatable.

A4. This is the Warburg effect, and the naive reading is that the tumour has made a mistake. The mechanism says otherwise. Oxidative phosphorylation makes energy but consumes the sugar intermediate, whereas fermentation leaves the half-broken sugar molecule in the cell as pyruvate. That leftover is raw material for nucleotides, lipids and amino acids, which is exactly what a cell about to divide a billion times needs. The clinical meaning is that the apparently wasteful metabolism is a growth strategy, and it leaves a fingerprint: lactate is raised, which is why lactate and other metabolites became candidate biomarkers, and metabolomics became a way to find them. The limit on the conclusion is that this is not a claim that the tumour can do without mitochondria, and his own slide still shows about 5 % of pyruvate going into them.

A5. The observation is one cell with two nuclei and a single 4N peak on the DNA-content plot. The mechanism is that the contractile ring, made of actin and myosin, drives cytokinesis, the cytoplasmic division, and the drug blocks it. Mitosis, the separation of the chromosomes, is independent of cytokinesis and proceeds normally, so the DNA replicates and divides while the cytoplasm does not cleave. The clinical meaning is that you have separated two processes that are often confused, and that is the entire content of the question. The action is to interpret the 4N peak as a binucleated cell rather than as a tetraploid population. The limit is that this is not the same as a cell that never replicated its DNA, because a 4N peak on its own cannot distinguish a blocked cytokinesis from a genuine tetraploid cell without a nuclear count.

A6. The observation is that an affected mother transmits to all of her children while an affected father transmits to none, and the condition appears in every generation. That is maternal, mitochondrial, inheritance, and it is

supported by all three hallmarks: an affected mother passes it to all her children, an affected father passes it to none, because sperm mitochondria are destroyed after fertilisation, and it appears in every generation. But you must still check the nuclear DNA, and he says so explicitly: "Not only the mutation within the mitochondrial DNA can cause the mitochondrial disease. Some mitochondrial disease can actually result from the mutation within the nuclear DNA." Mutations in nuclear genes encoding mitochondrial proteins give the same phenotype with a completely different, Mendelian, inheritance pattern, so a pedigree of mitochondrial-looking disease can be nuclear. The confirmation step is to sequence both the mitochondrial genome and the relevant nuclear genes. The recurrence risk is a probability rather than a certainty, because of the threshold effect: disease appears only when the proportion of mutant mtDNA exceeds what that tissue can tolerate, so contracting skeletal muscle has a low threshold and is affected first, while hearing, with a low energy demand, has a high threshold.

A7. You use immunofluorescence. You stain with an antibody tagged with a fluorescent dye, or label the antibody and the protein of interest with two different fluorophores and look for overlap. The assay tells you where the protein sits inside the cell, which is the co-localisation question, and with two colours you can show that two proteins are in the same place at the same time. The clinical meaning is that location is evidence. A receptor on the membrane is reachable by a drug from outside the cell, which is why membrane proteins are prime drug targets, and the same protein inside the nucleus is doing something completely different. The limit is that co-localisation is not the same as direct interaction. Two proteins can sit in the same compartment without binding each other, so co-localisation is evidence for association, not proof of binding.

A8. The test is FISH, fluorescent in-situ hybridisation, which uses DNA probes, not antibodies, to count the number of copies of the HER2 gene inside the tumour cells. It measures gene amplification, that is, how many copies of the gene the cell carries, and the standard result that matters is a ratio of HER2 copies to chromosome 17 copies. The clinical meaning is that HER2 amplification predicts response to trastuzumab, which is why the test is a companion diagnostic that must be done before the drug is offered, not after. The limit is that FISH counts copies and needs enough cells to interpret, and that the assay parameters themselves, the denaturation and the temperature, are explicitly on his not-examined list.

A9. The consequence is silencing. Methylation of the CpG island in the promoter stops the transcription machinery from getting in, so the tumour-suppressor gene is present and at normal copy number but is not making RNA. That is the "genotype is not phenotype" idea, and it is why a mutation is not the only way to lose a tumour suppressor. The clinical meaning is that this is targetable, because the gene is still there and could be switched back on. The drug you would expect to help is a histone deacetylase inhibitor, which blocks the removal of acetyl groups and lets the chromatin loosen again. Two numbers are attached to the methylation story in general: an odds ratio of 39.6 for CIMP-high against CIMP-0, and 12.7 for detected methylation. The limit is that these drugs are not tumour-specific, so the therapeutic window is narrower than for a targeted agent, and a methylation result is a probability, not a certainty about individual outcome.

A10. The karyotype misses it because of resolution. A karyotype, which is G-banding, detects gross structural change at roughly 0.5 to 10 megabases, and he quotes both 5 to 10 megabases and 0.5 to 1 megabase in different chunks of the lecture, so quote the range rather than one number. A point mutation, a single-nucleotide variant, changes no chromosome structure and produces no band, so it is invisible to the stain. The resolution you need is base-level, which means sequencing, either Sanger for a single site or next-generation sequencing for a panel or a genome. The clinical meaning is that a normal karyotype does not exclude a cancer-driving mutation, so the test you choose depends on the clinical question. The limit is the cost and time, which is why a karyotype is still the right first test for large structural rearrangements and a sequencing panel is the right test for small variants.

A11. The mechanism is that expression depends on where a gene sits relative to its regulatory DNA. An inversion cuts out a stretch and puts it back in the opposite direction, which does not change the amount of DNA and does not change the gene dosage, but it does change the physical relationship between the gene and its enhancer. If the inversion moves a gene away from its enhancer, the gene is under-expressed. If it puts an enhancer next to a different promoter, that gene is over-expressed. The clinical meaning is that gene expression can change with no change in DNA content, so expression data and copy-number data are not the same measurement and one does not substitute for the other. The limit is that the rearrangement can still be constitutional, inherited, and may have nothing to do with the tumour at all.

A12. You integrate the copy number profile with the mRNA expression data, and you look for concordance: a region that is both gained and over-expressed is a candidate driver, whereas a region that is gained but silent, or amplified because it sits next to an amplified gene, is probably a passenger. He makes this point repeatedly, most sharply in Lecture 3, where a driver is a change that gives the tumour a growth advantage and a passenger is simply carried along by chromosomal proximity or instability. His worked example is NPM1 in leukaemia. The clinical meaning is that target nomination depends on this integration, because a region can look impressive on a copy-number plot and mean nothing. The limit is that concordance raises the probability but does not prove the gene is a driver, and that the CNV mechanisms themselves are explicitly not to be memorised.

A13. The logic is synthetic lethality. A BRCA1 or BRCA2 loss leaves the tumour unable to repair broken chromosomes by homologous recombination, so it depends on the backup pathway, and a PARP inhibitor blocks exactly that backup. The normal cells around it still have intact homologous recombination and survive, which is what makes the therapeutic window wide, meaning the gap between the smallest effective dose and the largest tolerable dose. The clinical meaning is that the same drug that kills the tumour spares the patient. The action is to test for the BRCA lesion before offering the drug, because a tumour with intact HR will not respond. The limit is that resistance develops, and that PARP inhibitors are not for every patient and carry their own toxicity.

A14. Mismatch-repair deficiency means copying errors and small indels are not corrected. The consequence is that short repeated sequences, the microsatellites, change length, which is measured as **microsatellite instability**, and the tumour is called MSI-high when the changes are frequent. The chain from there is the important part: the indels shift the reading frame, frameshifts produce protein fragments the immune system has never seen, those fragments are neoantigens, and a high neoantigen burden makes a tumour "hot". So MSI-high predicts response to checkpoint blockade, specifically an anti-PD-1 antibody, and within the anti-PD-1 cohorts he quotes, the MMR-deficient group did clearly better on progression-free survival. The clinical meaning is that a mismatch-repair or MSI result is both a prognostic and a predictive biomarker, so you should test for it. The limit is that about 15 % of colorectal cancers are MSI-high, so a non-MSI result says nothing bad about the other 85 %, and CEA remains the routine marker for the rest.

A15. Homologous recombination does it, because it is the high-fidelity pathway that uses a template. It only works when a sister chromatid is available, which means S phase and G2 phase. The action follows from that constraint: you knock the edit into a dividing cell, you deliver the donor template as well, and you time the experiment so that the cells are in S or G2. The alternative, non-homologous end joining, needs no template, works in all phases, and is error-prone, so it is used for knock-outs, where an imperfect edit is acceptable, and not for knock-ins. The clinical and research meaning is that CRISPR knock-in and knock-out are not interchangeable, and that his own point in Lecture 2 is that CRISPR has largely replaced the older siRNA and shRNA knockdown methods.

A16. The chain is the one you must be able to draw. NHEJ is co-opted by the immune system for V(D)J recombination, which assembles the variable, diversity and joining gene segments to build the enormous repertoire of antibodies and T-cell receptors. The segment copy numbers he gives are V of about 39 copies, D equal to 27 copies and J equal to 6 copies, and that is the raw material. If NHEJ is defective, the repertoire cannot be assembled, so neither B-cells nor T-cells can mature properly, and the clinical outcome is severe combined immunodeficiency. The clinical meaning is that the immune deficiency is not a failure of antibody action but a failure of antibody production, which is why a defect in one repair pathway produces a whole disease. The limit is that the diagnosis is a genetic one and needs confirmation, because SCID has other causes.

A17. The observation is that the first culture transcribes slowly and the second does not transcribe at all. The mechanism is the two-condition rule. Lactose present means allolactose is made by the lacZ enzyme and binds the Lac repressor, knocking it off the operator. Glucose low means cAMP rises, binds CAP, and the cAMP-CAP complex recruits the copying machinery onto the promoter and turns slow basal transcription into strong transcription. The first culture has lactose but not low glucose, so it is at basal level only. The second has neither condition, so the repressor is on the operator and nothing is transcribed, and CAP cannot help because the master switch is closed. If you add lactose to the second culture, allolactose removes the repressor and transcription starts, but only slowly, because glucose is still high and there is still no cAMP-CAP. The limit on the answer is that CAP can only add to an open system, so even with cAMP-CAP present, a repressor on the operator means blocked.

A18. The numbers are 1 % for a typical antibody, 2 % for Herceptin in breast cancer, and 10 % for Herceptin in gastric cancer. In plain English: in breast cancer the drug only has to occupy two receptors in every hundred, while in gastric cancer it has to occupy ten in every hundred. The mechanism is twofold. Occupancy is the fraction of receptors that must be bound before there is a clinical effect, and it is a tissue-specific number, because the receptor density and the strength of the downstream signal differ between tissues. And HER2 does not signal on its own, because it cannot homodimerise efficiently; it pairs with HER1, HER3 or HER4, and that pairing drives the growth signal, which is exactly what trastuzumab blocks. The clinical meaning is that the same antibody is dosed differently in the two diseases, so the dose is a property of the disease and the tissue, not of the drug. The limit is that occupancy says nothing about toxicity, so the therapeutic window is a separate calculation.

A19. The mechanism is tissue context, not the drug. The estrogen receptor is the same protein in both tissues, and tamoxifen is the same molecule in both. What differs is what the tissue supplies. In breast tissue, tamoxifen occupies the receptor and blocks estrogen, so no co-activator is recruited and the cell does not proliferate, so tamoxifen is an antagonist and an anti-tumour drug. In the endometrium, the tamoxifen-receptor complex does recruit a co-activator, so the cell proliferates, so tamoxifen is a partial agonist there and long-term use carries a small increased risk of endometrial cancer. The clinical meaning is that the outcome of a drug is decided by the cell it lands in, which is why tissue-selective drugs exist at all, and why the SERM class was designed. As he puts it, "This is the concept that you have to bear in mind." The limit is that this is a partial agonism, not a full one, so the risk is small but real and is not zero.

A20. The protein-level consequence is that the new splice site makes the splicing machinery build a shorter isoform, one that excludes an exon. His slide shows the two band sizes, 425 base pairs for the full-length exon 5-6-7-8 form and 250 base pairs for the exon 7-skipped form, and the survival comparison of 23 patients without alternative splicing against 19 with it, at P less than 0.001 on the log-rank test. The immune-level consequence is that a protein fragment the body has never made becomes a neoantigen, which is what makes the tumour hot in his terms: "for the tumor that present many neoantigen and can be treated with the immunotherapy, we call this is a hot tumor, yeah. And not all the solid tumor is a hot tumor." The drug-level consequence has two sides. On the favourable side, a neoantigen is something the immune system can be pointed at, which is the logic of checkpoint blockade in neoantigen-rich disease. On the unfavourable side, splicing can delete the very part of a protein that a drug binds to, so an aberrant isoform makes the tumour resistant to that drug, and the resistance is then inherited by every daughter cell. The limit is that this is mechanism and association, not a treatment rule. You cannot treat a patient on the strength of a splice variant alone; you need the drug, the tumour type and the evidence that this particular isoform in this particular tumour carries the variant.

13.7 Where to go next

Four documents in this project sit alongside this guide, and each one does a different job. A fifth, the Terminology Deep Dive, is listed at the end.

- **The exam-focus document.** The file GNBF5020_Exam_Focus_Prediction.md in the _audio folder is the single most useful document for the midterm, because every Tier 1 question, every key number, every exclusion and every distractor pair in it is derived from Dr. Wong's own words rather than inferred. Section 3 of that document holds the five questions he named as exam-type, section 4 the concepts he says you must understand, section 5 the numbers, section 6 the exclusions, section 7 the nineteen pairs, and section 10 the English answer templates you should practise with. A Traditional Chinese mirror of the same document sits beside it, so that the explanation can be read in Chinese while every technical term stays in English.

- **The Today plan.** GNBF5020_Today_Study_Plan.md in the same folder turns the exam-focus document into a timed study schedule, with six blocks covering the last 72 hours, from the Tier 1 questions through to a one-page "will NOT be asked" sheet. Use it when the content is understood and you need to convert it into marks.

- **The case-study bank.** GNBF5020_Case_Studies.md holds sixteen worked clinical scenarios with model answers, and they are where the short questions come from. The highest-yield cases for you are 1 to 6 and 10 to 13. Three of them belong to this lecture: case 14, the compound that switches itself off in the right cells, which is the HIF-controlled CAR; case 15, the isoform that predicts drug resistance, which is alternative splicing; and case 16, reading the lac operon logic correctly.

- **The one-page cheat sheet and the flashcards.** GNBF5020_CheatSheet.md is the one-page summary of all five lectures, including a "key terms to never confuse" list that is the same list as the distractor pairs in section 13.5 above. GNBF5020_Flashcards.md is the drill set for the numbers.

- **The Terminology Deep Dive.** GNBF5020_Terminology_Deep_Dive.pdf - 《GNBF5020 術語詳解》 - is the full reference for this course: 905 entries, every medical and biology term you meet gets a one-line summary, the mechanism, why the course mentions it and where it appears, and the twelve domain chapters each end with a table of the terms they cover, marked 考 / 不考 / 排除. It ships in English, with the technical terms kept in English, and the Traditional Chinese edition of the same book is GNBF5020_術語詳解_繁體中文.pdf. Use it when a word stops you in the middle of a chapter; use the exam-focus document to decide what to memorise.

One last instruction, and it is the one he said three times. Every answer, in every part of the paper, must be in English, with the technical terms in English inside full English sentences. "I have no choice that I give the zero mark, because everything should be in English."